

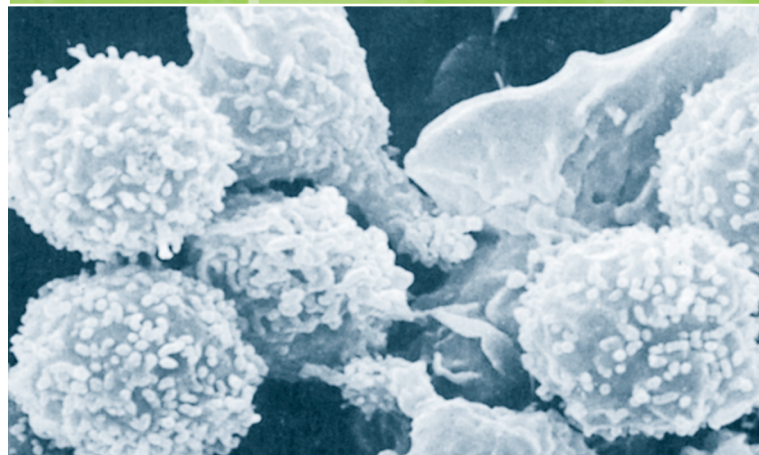
Overview of the Immune System

THE IMMUNE SYSTEM IS A REMARKABLY VERSATILE defense system that has evolved to protect animals from invading pathogenic microorganisms and cancer. It is able to generate an enormous variety of cells and molecules capable of specifically recognizing and eliminating an apparently limitless variety of foreign invaders. These cells and molecules act together in a dynamic network whose complexity rivals that of the nervous system.

Functionally, an immune response can be divided into two related activities—recognition and response. Immune recognition is remarkable for its specificity. The immune system is able to recognize subtle chemical differences that distinguish one foreign pathogen from another. Furthermore, the system is able to discriminate between foreign molecules and the body's own cells and proteins. Once a foreign organism has been recognized, the immune system recruits a variety of cells and molecules to mount an appropriate response, called an **effector response**, to eliminate or neutralize the organism. In this way the system is able to convert the initial recognition event into a variety of effector responses, each uniquely suited for eliminating a particular type of pathogen. Later exposure to the same foreign organism induces a **memory response**, characterized by a more rapid and heightened immune reaction that serves to eliminate the pathogen and prevent disease.

This chapter introduces the study of immunology from an historical perspective and presents a broad overview of the cells and molecules that compose the immune system, along with the mechanisms they use to protect the body against foreign invaders. Evidence for the presence of very simple immune systems in certain invertebrate organisms then gives an evolutionary perspective on the mammalian immune system, which is the major subject of this book. Elements of the primitive immune system persist in vertebrates as *innate immunity* along with a more highly evolved system of specific responses termed *adaptive immunity*. These two systems work in concert to provide a high degree of protection for vertebrate species. Finally, in some circumstances, the immune system fails to act as protector because of some deficiency in its components; at other times, it becomes an aggressor and turns its awesome powers against its own host. In this introductory chapter, our description of immunity is simplified to reveal the essential structures and function of the immune system. Substantive discussions, experimental approaches, and in-depth definitions are left to the chapters that follow.

chapter 1



Numerous T Lymphocytes Interacting with a Single Macrophage

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Like the later chapters covering basic topics in immunology, this one includes a section called “Clinical Focus” that describes human disease and its relation to immunity. These sections investigate the causes, consequences, or treatments of diseases rooted in impaired or hyperactive immune function.

Historical Perspective

The discipline of immunology grew out of the observation that individuals who had recovered from certain infectious diseases were thereafter protected from the disease. The Latin term *immunis*, meaning “exempt,” is the source of the English word immunity, meaning the state of protection from infectious disease.

Perhaps the earliest written reference to the phenomenon of immunity can be traced back to Thucydides, the great historian of the Peloponnesian War. In describing a plague in Athens, he wrote in 430 BC that only those who had recovered from the plague could nurse the sick because they would not contract the disease a second time. Although early societies recognized the phenomenon of immunity, almost

two thousand years passed before the concept was successfully converted into medically effective practice.

The first recorded attempts to induce immunity deliberately were performed by the Chinese and Turks in the fifteenth century. Various reports suggest that the dried crusts derived from smallpox pustules were either inhaled into the nostrils or inserted into small cuts in the skin (a technique called *variolation*). In 1718, Lady Mary Wortley Montagu, the wife of the British ambassador to Constantinople, observed the positive effects of variolation on the native population and had the technique performed on her own children. The method was significantly improved by the English physician Edward Jenner, in 1798. Intrigued by the fact that milkmaids who had contracted the mild disease cowpox were subsequently immune to smallpox, which is a disfiguring and often fatal disease, Jenner reasoned that introducing fluid from a cowpox pustule into people (i.e., inoculating them) might protect them from smallpox. To test this idea, he inoculated an eight-year-old boy with fluid from a cowpox pustule and later intentionally infected the child with smallpox. As predicted, the child did not develop smallpox.

Jenner's technique of inoculating with cowpox to protect against smallpox spread quickly throughout Europe. However, for many reasons, including a lack of obvious disease targets and knowledge of their causes, it was nearly a hundred years before this technique was applied to other diseases. As so often happens in science, serendipity in combination with astute observation led to the next major advance in immunology, the induction of immunity to cholera. Louis Pasteur had succeeded in growing the bacterium thought to cause fowl cholera in culture and then had shown that chickens injected with the cultured bacterium developed cholera. After returning from a summer vacation, he injected some chickens with an old culture. The chickens became ill, but, to Pasteur's surprise, they recovered. Pasteur then grew a fresh culture of the bacterium with the intention of injecting it into some fresh chickens. But, as the story goes, his supply of chickens was limited, and therefore he used the previously injected chickens. Again to his surprise, the chickens were completely protected from the disease. Pasteur hypothesized and proved that aging had weakened the virulence of the pathogen and that such an attenuated strain might be administered to protect against the disease. He called this attenuated strain a **vaccine** (from the Latin *vacca*, meaning "cow"), in honor of Jenner's work with cowpox inoculation.

Pasteur extended these findings to other diseases, demonstrating that it was possible to **attenuate**, or weaken, a pathogen and administer the attenuated strain as a vaccine. In a now classic experiment at Pouilly-le-Fort in 1881, Pasteur first vaccinated one group of sheep with heat-attenuated anthrax bacillus (*Bacillus anthracis*); he then challenged the vaccinated sheep and some unvaccinated sheep with a virulent culture of the bacillus. All the vaccinated sheep lived, and all the unvaccinated animals died. These experiments marked the beginnings of the discipline of immunology. In



FIGURE 1-1 Wood engraving of Louis Pasteur watching Joseph Meister receive the rabies vaccine. [From Harper's Weekly 29:836; courtesy of the National Library of Medicine.]

1885, Pasteur administered his first vaccine to a human, a young boy who had been bitten repeatedly by a rabid dog (Figure 1-1). The boy, Joseph Meister, was inoculated with a series of attenuated rabies virus preparations. He lived and later became a custodian at the Pasteur Institute.

Early Studies Revealed Humoral and Cellular Components of the Immune System

Although Pasteur proved that vaccination worked, he did not understand how. The experimental work of Emil von Behring and Shibasaburo Kitasato in 1890 gave the first insights into the mechanism of immunity, earning von Behring the Nobel prize in medicine in 1901 (Table 1-1). Von Behring and Kitasato demonstrated that **serum** (the liquid, noncellular component of coagulated blood) from animals previously immunized to diphtheria could transfer the immune state to unimmunized animals. In search of the protective agent, various researchers during the next decade demonstrated that an active component from immune serum could neutralize toxins, precipitate toxins, and agglutinate (clump) bacteria. In each case, the active agent was named for the activity it exhibited: antitoxin, precipitin, and agglutinin, respectively.

TABLE 1-1 Nobel Prizes for immunologic research

Year	Recipient	Country	Research
1901	Emil von Behring	Germany	Serum antitoxins
1905	Robert Koch	Germany	Cellular immunity to tuberculosis
1908	Elie Metchnikoff Paul Ehrlich	Russia Germany	Role of phagocytosis (Metchnikoff) and antitoxins (Ehrlich) in immunity
1913	Charles Richet	France	Anaphylaxis
1919	Jules Border	Belgium	Complement-mediated bacteriolysis
1930	Karl Landsteiner	United States	Discovery of human blood groups
1951	Max Theiler	South Africa	Development of yellow fever vaccine
1957	Daniel Bovet	Switzerland	Antihistamines
1960	F. Macfarlane Burnet Peter Medawar	Australia Great Britain	Discovery of acquired immunological tolerance
1972	Rodney R. Porter Gerald M. Edelman	Great Britain United States	Chemical structure of antibodies
1977	Rosalyn R. Yalow	United States	Development of radioimmunoassay
1980	George Snell Jean Dausset Baruj Benacerraf	United States France United States	Major histocompatibility complex
1984	Cesar Milstein Georges E. Köhler Niels K. Jerne	Great Britain Germany Denmark	Monoclonal antibody Immune regulatory theories
1987	Susumu Tonegawa	Japan	Gene rearrangement in antibody production
1991	E. Donnall Thomas Joseph Murray	United States United States	Transplantation immunology
1996	Peter C. Doherty Rolf M. Zinkernagel	Australia Switzerland	Role of major histocompatibility complex in antigen recognition by T cells

Initially, a different serum component was thought to be responsible for each activity, but during the 1930s, mainly through the efforts of Elvin Kabat, a fraction of serum first called gamma-globulin (now **immunoglobulin**) was shown to be responsible for all these activities. The active molecules in the immunoglobulin fraction are called **antibodies**. Because immunity was mediated by antibodies contained in body fluids (known at the time as humors), it was called humoral immunity.

In 1883, even before the discovery that a serum component could transfer immunity, Elie Metchnikoff demonstrated that cells also contribute to the immune state of an animal. He observed that certain white blood cells, which he termed **phagocytes**, were able to ingest (phagocytose) microorganisms and other foreign material. Noting that these phagocytic cells were more active in animals that had been immunized, Metchnikoff hypothesized that cells, rather than serum components, were the major effector of immunity. The active phagocytic cells identified by Metchnikoff were likely blood monocytes and neutrophils (see Chapter 2).

In due course, a controversy developed between those who held to the concept of humoral immunity and those who agreed with Metchnikoff's concept of **cell-mediated immunity**. It was later shown that both are correct—immunity requires both cellular and humoral responses. It was difficult to study the activities of immune cells before the development of modern tissue culture techniques, whereas studies with serum took advantage of the ready availability of blood and established biochemical techniques. Because of these technical problems, information about cellular immunity lagged behind findings that concerned humoral immunity.

In a key experiment in the 1940s, Merrill Chase succeeded in transferring immunity against the tuberculosis organism by transferring white blood cells between guinea pigs. This demonstration helped to rekindle interest in cellular immunity. With the emergence of improved cell culture techniques in the 1950s, the **lymphocyte** was identified as the cell responsible for both cellular and humoral immunity. Soon thereafter, experiments with chickens pioneered by Bruce Glick at Mississippi State University indicated that there were

two types of lymphocytes: T lymphocytes derived from the thymus mediated cellular immunity, and B lymphocytes from the bursa of Fabricius (an outgrowth of the cloaca in birds) were involved in humoral immunity. The controversy about the roles of humoral and cellular immunity was resolved when the two systems were shown to be intertwined, and that both systems were necessary for the immune response.

Early Theories Attempted to Explain the Specificity of the Antibody–Antigen Interaction

One of the greatest enigmas facing early immunologists was the specificity of the antibody molecule for foreign material, or **antigen** (the general term for a substance that binds with a specific antibody). Around 1900, Jules Bordet at the Pasteur Institute expanded the concept of immunity by demonstrating specific immune reactivity to nonpathogenic substances, such as red blood cells from other species. Serum from an animal inoculated previously with material that did not cause infection would react with this material in a specific manner, and this reactivity could be passed to other animals by transferring serum from the first. The work of Karl Landsteiner and those who followed him showed that injecting an animal with almost any organic chemical could induce production of antibodies that would bind specifically to the chemical. These studies demonstrated that antibodies have a capacity for an almost unlimited range of reactivity, including responses to compounds that had only recently been synthesized in the laboratory and had not previously existed in nature. In addition, it was shown that molecules differing in the smallest detail could be distinguished by their reactivity with different antibodies. Two major theories were proposed to account for this specificity: the selective theory and the instructional theory.

The earliest conception of the *selective theory* dates to Paul Ehrlich in 1900. In an attempt to explain the origin of serum antibody, Ehrlich proposed that cells in the blood expressed a variety of receptors, which he called “side-chain receptors,” that could react with infectious agents and inactivate them. Borrowing a concept used by Emil Fischer in 1894 to explain the interaction between an enzyme and its substrate, Ehrlich proposed that binding of the receptor to an infectious agent was like the fit between a lock and key. Ehrlich suggested that interaction between an infectious agent and a cell-bound receptor would induce the cell to produce and release more receptors with the same specificity. According to Ehrlich’s theory, the specificity of the receptor was determined before its exposure to antigen, and the antigen selected the appropriate receptor. Ultimately all aspects of Ehrlich’s theory would be proven correct with the minor exception that the “receptor” exists as both a soluble antibody molecule and as a cell-bound receptor; it is the soluble form that is secreted rather than the bound form released.

In the 1930s and 1940s, the selective theory was challenged by various *instructional theories*, in which antigen played a central role in determining the specificity of the antibody molecule. According to the instructional theories, a particular antigen would serve as a template around which antibody would fold. The antibody molecule would thereby assume a configuration complementary to that of the antigen template. This concept was first postulated by Friedrich Breinl and Felix Haurowitz about 1930 and redefined in the 1940s in terms of protein folding by Linus Pauling. The instructional theories were formally disproved in the 1960s, by which time information was emerging about the structure of DNA, RNA, and protein that would offer new insights into the vexing problem of how an individual could make antibodies against almost anything.

In the 1950s, selective theories resurfaced as a result of new experimental data and, through the insights of Niels Jerne, David Talmadge, and F. Macfarlane Burnet, were refined into a theory that came to be known as the **clonal-selection theory**. According to this theory, an individual lymphocyte expresses membrane receptors that are specific for a distinct antigen. This unique receptor specificity is determined before the lymphocyte is exposed to the antigen. Binding of antigen to its specific receptor activates the cell, causing it to proliferate into a clone of cells that have the same immunologic specificity as the parent cell. The clonal-selection theory has been further refined and is now accepted as the underlying paradigm of modern immunology.

The Immune System Includes Innate and Adaptive Components

Immunity—the state of protection from infectious disease—has both a less specific and more specific component. The less specific component, **innate immunity**, provides the first line of defense against infection. Most components of innate immunity are present before the onset of infection and constitute a set of disease-resistance mechanisms that are not specific to a particular pathogen but that include cellular and molecular components that recognize classes of molecules peculiar to frequently encountered pathogens. Phagocytic cells, such as macrophages and neutrophils, barriers such as skin, and a variety of antimicrobial compounds synthesized by the host all play important roles in innate immunity. In contrast to the broad reactivity of the innate immune system, which is uniform in all members of a species, the specific component, **adaptive immunity**, does not come into play until there is an antigenic challenge to the organism. Adaptive immunity responds to the challenge with a high degree of specificity as well as the remarkable property of “memory.” Typically, there is an adaptive immune response against an antigen within five or six days after the initial exposure to that antigen. Exposure to the same antigen some time in the future results in a memory response: the immune response to the second challenge occurs more quickly than

the first, is stronger, and is often more effective in neutralizing and clearing the pathogen. The major agents of adaptive immunity are lymphocytes and the antibodies and other molecules they produce.

Because adaptive immune responses require some time to marshal, innate immunity provides the first line of defense during the critical period just after the host's exposure to a pathogen. In general, most of the microorganisms encountered by a healthy individual are readily cleared within a few days by defense mechanisms of the innate immune system before they activate the adaptive immune system.

Innate Immunity

Innate immunity can be seen to comprise four types of defensive barriers: anatomic, physiologic, phagocytic, and inflammatory (Table 1-2).

The Skin and the Mucosal Surfaces Provide Protective Barriers Against Infection

Physical and anatomic barriers that tend to prevent the entry of pathogens are an organism's first line of defense against infection. The skin and the surface of mucous membranes are included in this category because they are effective barriers to the entry of most microorganisms. The skin consists of two

distinct layers: a thinner outer layer—the **epidermis**—and a thicker layer—the **dermis**. The epidermis contains several layers of tightly packed epithelial cells. The outer epidermal layer consists of dead cells and is filled with a waterproofing protein called keratin. The dermis, which is composed of connective tissue, contains blood vessels, hair follicles, sebaceous glands, and sweat glands. The sebaceous glands are associated with the hair follicles and produce an oily secretion called **sebum**. Sebum consists of lactic acid and fatty acids, which maintain the pH of the skin between 3 and 5; this pH inhibits the growth of most microorganisms. A few bacteria that metabolize sebum live as commensals on the skin and sometimes cause a severe form of acne. One acne drug, isotretinoin (Accutane), is a vitamin A derivative that prevents the formation of sebum.

Breaks in the skin resulting from scratches, wounds, or abrasion are obvious routes of infection. The skin may also be penetrated by biting insects (e.g., mosquitoes, mites, ticks, fleas, and sandflies); if these harbor pathogenic organisms, they can introduce the pathogen into the body as they feed. The protozoan that causes malaria, for example, is deposited in humans by mosquitoes when they take a blood meal. Similarly, bubonic plague is spread by the bite of fleas, and Lyme disease is spread by the bite of ticks.

The conjunctivae and the alimentary, respiratory, and urogenital tracts are lined by mucous membranes, not by the dry, protective skin that covers the exterior of the body. These

TABLE 1-2 Summary of nonspecific host defenses

Type	Mechanism
<i>Anatomic barriers</i>	
Skin	Mechanical barrier retards entry of microbes. Acidic environment (pH 3–5) retards growth of microbes.
Mucous membranes	Normal flora compete with microbes for attachment sites and nutrients. Mucus entraps foreign microorganisms. Cilia propel microorganisms out of body.
<i>Physiologic barriers</i>	
Temperature	Normal body temperature inhibits growth of some pathogens. Fever response inhibits growth of some pathogens.
Low pH	Acidity of stomach contents kills most ingested microorganisms.
Chemical mediators	Lysozyme cleaves bacterial cell wall. Interferon induces antiviral state in uninfected cells. Complement lyses microorganisms or facilitates phagocytosis. Toll-like receptors recognize microbial molecules, signal cell to secrete immunostimulatory cytokines. Collectins disrupt cell wall of pathogen.
<i>Phagocytic/endocytic barriers</i>	
	Various cells internalize (endocytose) and break down foreign macromolecules. Specialized cells (blood monocytes, neutrophils, tissue macrophages) internalize (phagocytose), kill, and digest whole microorganisms.
<i>Inflammatory barriers</i>	
	Tissue damage and infection induce leakage of vascular fluid, containing serum proteins with antibacterial activity, and influx of phagocytic cells into the affected area.

membranes consist of an outer epithelial layer and an underlying layer of connective tissue. Although many pathogens enter the body by binding to and penetrating mucous membranes, a number of nonspecific defense mechanisms tend to prevent this entry. For example, saliva, tears, and mucous secretions act to wash away potential invaders and also contain antibacterial or antiviral substances. The viscous fluid called mucus, which is secreted by epithelial cells of mucous membranes, entraps foreign microorganisms. In the lower respiratory tract, the mucous membrane is covered by **cilia**, hairlike protrusions of the epithelial-cell membranes. The synchronous movement of cilia propels mucus-entrapped microorganisms from these tracts. In addition, nonpathogenic organisms tend to colonize the epithelial cells of mucosal surfaces. These *normal flora* generally outcompete pathogens for attachment sites on the epithelial cell surface and for necessary nutrients.

Some organisms have evolved ways of escaping these defense mechanisms and thus are able to invade the body through mucous membranes. For example, influenza virus (the agent that causes flu) has a surface molecule that enables it to attach firmly to cells in mucous membranes of the respiratory tract, preventing the virus from being swept out by the ciliated epithelial cells. Similarly, the organism that causes gonorrhea has surface projections that allow it to bind to epithelial cells in the mucous membrane of the urogenital tract. Adherence of bacteria to mucous membranes is due to interactions between hairlike protrusions on a bacterium, called **fimbriae** or **pili**, and certain glycoproteins or glycolipids that are expressed only by epithelial cells of the mucous membrane of particular tissues (Figure 1-2). For this reason, some

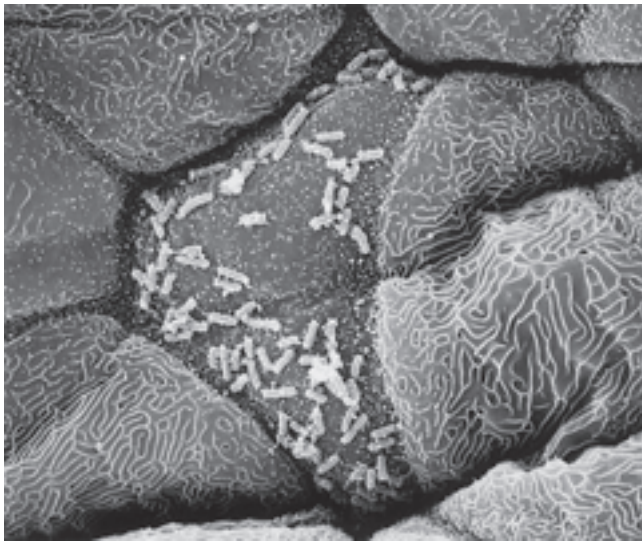


FIGURE 1-2 Electron micrograph of rod-shaped *Escherichia coli* bacteria adhering to surface of epithelial cells of the urinary tract. [From N. Sharon and H. Lis, 1993, *Sci. Am.* **268**(1):85; photograph courtesy of K. Fujita.]

tissues are susceptible to bacterial invasion, whereas others are not.

Physiologic Barriers to Infection Include General Conditions and Specific Molecules

The physiologic barriers that contribute to innate immunity include temperature, pH, and various soluble and cell-associated molecules. Many species are not susceptible to certain diseases simply because their normal body temperature inhibits growth of the pathogens. Chickens, for example, have innate immunity to anthrax because their high body temperature inhibits the growth of the bacteria. Gastric acidity is an innate physiologic barrier to infection because very few ingested microorganisms can survive the low pH of the stomach contents. One reason newborns are susceptible to some diseases that do not afflict adults is that their stomach contents are less acid than those of adults.

A variety of soluble factors contribute to innate immunity, among them the soluble proteins lysozyme, interferon, and complement. **Lysozyme**, a hydrolytic enzyme found in mucous secretions and in tears, is able to cleave the peptidoglycan layer of the bacterial cell wall. **Interferon** comprises a group of proteins produced by virus-infected cells. Among the many functions of the interferons is the ability to bind to nearby cells and induce a generalized antiviral state. **Complement**, examined in detail in Chapter 13, is a group of serum proteins that circulate in an inactive state. A variety of specific and nonspecific immunologic mechanisms can convert the inactive forms of complement proteins into an active state with the ability to damage the membranes of pathogenic organisms, either destroying the pathogens or facilitating their clearance. Complement may function as an effector system that is triggered by binding of antibodies to certain cell surfaces, or it may be activated by reactions between complement molecules and certain components of microbial cell walls. Reactions between complement molecules or fragments of complement molecules and cellular receptors trigger activation of cells of the innate or adaptive immune systems. Recent studies on **collectins** indicate that these surfactant proteins may kill certain bacteria directly by disrupting their lipid membranes or, alternatively, by aggregating the bacteria to enhance their susceptibility to phagocytosis.

Many of the molecules involved in innate immunity have the property of **pattern recognition**, the ability to recognize a given class of molecules. Because there are certain types of molecules that are unique to microbes and never found in multicellular organisms, the ability to immediately recognize and combat invaders displaying such molecules is a strong feature of innate immunity. Molecules with pattern recognition ability may be soluble, like lysozyme and the complement components described above, or they may be cell-associated receptors. Among the class of receptors designated the **toll-like receptors** (TLRs), TLR2 recognizes the lipopolysaccharide (LPS) found on Gram-negative bacteria. It has long been recognized that

FIGURE 1-3 (a) Electronmicrograph of macrophage (pink) attacking *Escherichia coli* (green). The bacteria are phagocytized as described in part b and breakdown products secreted. The monocyte (purple) has been recruited to the vicinity of the encounter by soluble factors secreted by the macrophage. The red sphere is an erythrocyte. (b) Schematic diagram of the steps in phagocytosis of a bacterium. [Part a, Dennis Kunkel Microscopy, Inc./Dennis Kunkel.]

systemic exposure of mammals to relatively small quantities of purified LPS leads to an acute inflammatory response (see below). The mechanism for this response is via a TLR on macrophages that recognizes LPS and elicits a variety of molecules in the inflammatory response upon exposure. When the TLR is exposed to the LPS upon local invasion by a Gram-negative bacterium, the contained response results in elimination of the bacterial challenge.

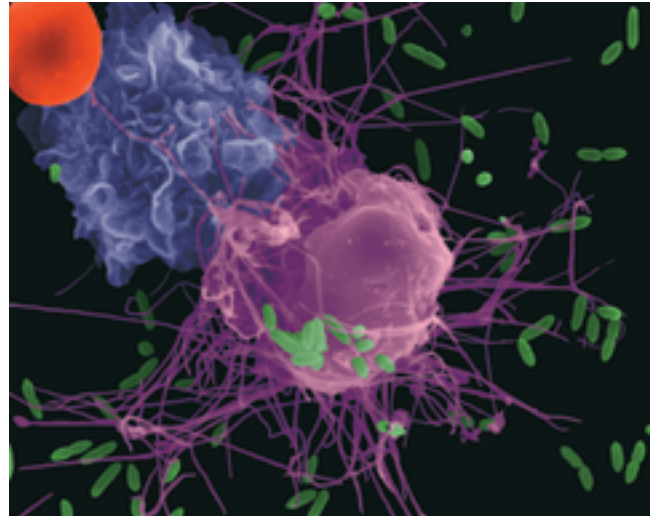
Cells That Ingest and Destroy Pathogens Make Up a Phagocytic Barrier to Infection

Another important innate defense mechanism is the ingestion of extracellular particulate material by **phagocytosis**. Phagocytosis is one type of **endocytosis**, the general term for the uptake by a cell of material from its environment. In phagocytosis, a cell's plasma membrane expands around the particulate material, which may include whole pathogenic microorganisms, to form large vesicles called **phagosomes** (Figure 1-3). Most phagocytosis is conducted by specialized cells, such as blood monocytes, neutrophils, and tissue macrophages (see Chapter 2). Most cell types are capable of other forms of endocytosis, such as *receptor-mediated endocytosis*, in which extracellular molecules are internalized after binding by specific cellular receptors, and *pinocytosis*, the process by which cells take up fluid from the surrounding medium along with any molecules contained in it.

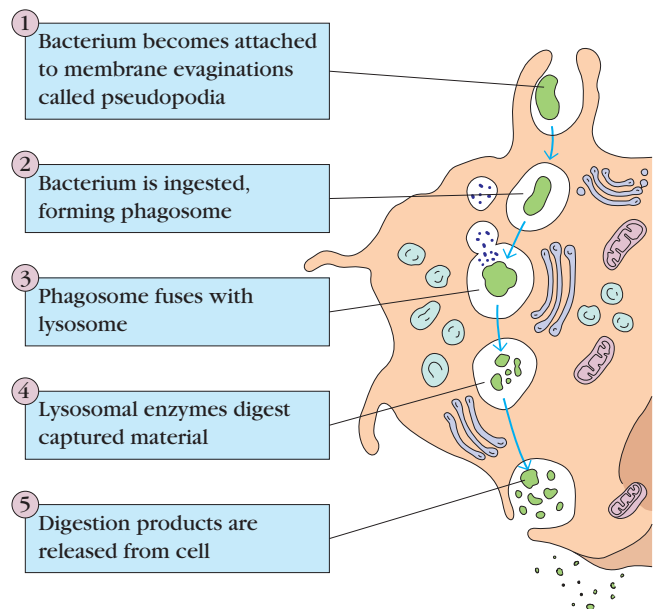
Inflammation Represents a Complex Sequence of Events That Stimulates Immune Responses

Tissue damage caused by a wound or by an invading pathogenic microorganism induces a complex sequence of events collectively known as the **inflammatory response**. As described above, a molecular component of a microbe, such as LPS, may trigger an inflammatory response via interaction with cell surface receptors. The end result of inflammation may be the marshalling of a specific immune response to the invasion or clearance of the invader by components of the innate immune system. Many of the classic features of the inflammatory response were described as early as 1600 BC, in Egyptian papyrus writings. In the first century AD, the Roman physician Celsus described the “four cardinal signs

(a)



(b)



of inflammation” as *rubor* (redness), *tumor* (swelling), *calor* (heat), and *dolor* (pain). In the second century AD, another physician, Galen, added a fifth sign: *functio laesa* (loss of function). The cardinal signs of inflammation reflect the three major events of an inflammatory response (Figure 1-4):

1. **Vasodilation**—an increase in the diameter of blood vessels—of nearby capillaries occurs as the vessels that carry blood away from the affected area constrict, resulting in engorgement of the capillary network. The engorged capillaries are responsible for tissue redness (*erythema*) and an increase in tissue temperature.

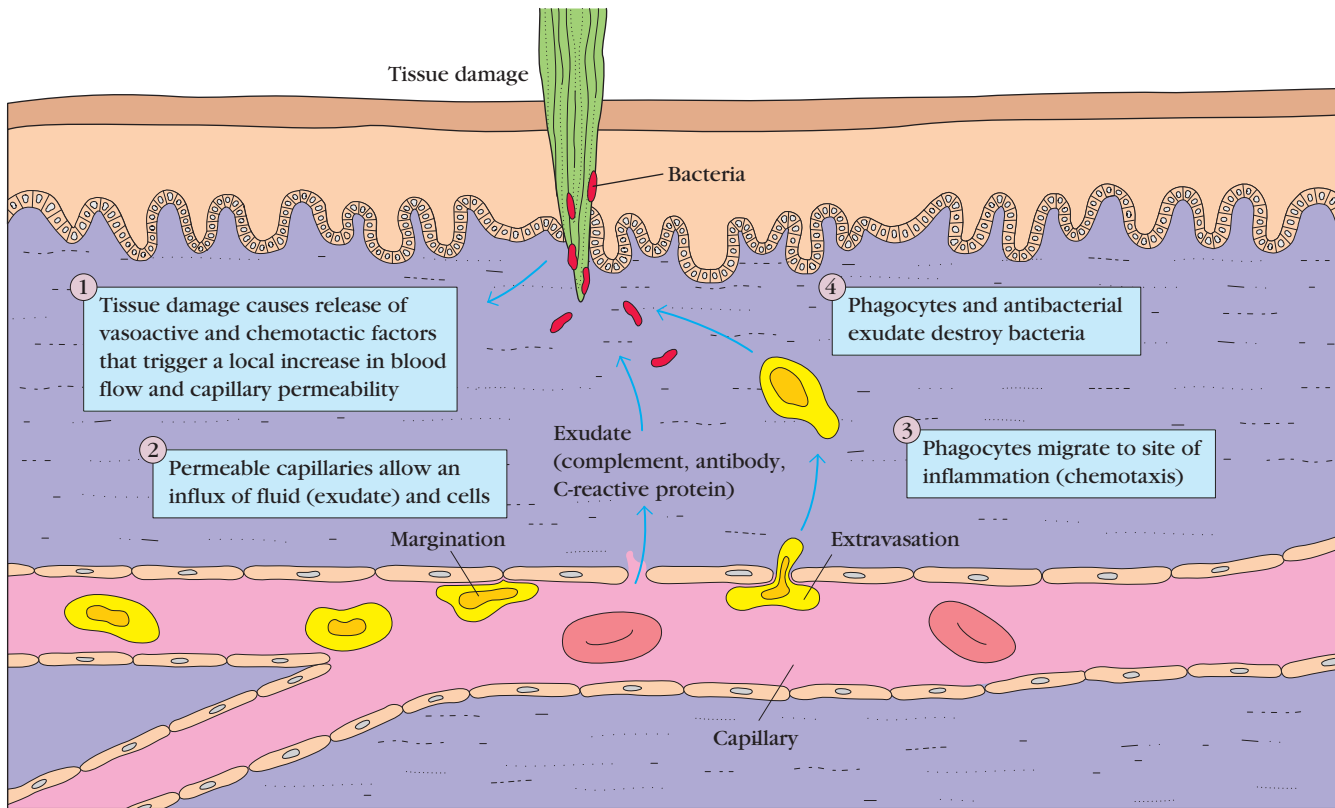


FIGURE 1-4 Major events in the inflammatory response. A bacterial infection causes tissue damage with release of various vasoactive and chemotactic factors. These factors induce increased blood flow to the area, increased capillary permeability, and an influx of white

blood cells, including phagocytes and lymphocytes, from the blood into the tissues. The serum proteins contained in the exudate have antibacterial properties, and the phagocytes begin to engulf the bacteria, as illustrated in Figure 1-3.

2. An *increase in capillary permeability* facilitates an influx of fluid and cells from the engorged capillaries into the tissue. The fluid that accumulates (**exudate**) has a much higher protein content than fluid normally released from the vasculature. Accumulation of exudate contributes to tissue swelling (**edema**).
3. *Influx of phagocytes* from the capillaries into the tissues is facilitated by the increased permeability of the capillaries. The emigration of phagocytes is a multistep process that includes adherence of the cells to the endothelial wall of the blood vessels (**margination**), followed by their emigration between the capillary-endothelial cells into the tissue (**diapedesis** or **extravasation**), and, finally, their migration through the tissue to the site of the invasion (**chemotaxis**). As phagocytic cells accumulate at the site and begin to phagocytose bacteria, they release lytic enzymes, which can damage nearby healthy cells. The accumulation of dead cells, digested material, and fluid forms a substance called pus.

The events in the inflammatory response are initiated by a complex series of events involving a variety of chemical mediators whose interactions are only partly understood. Some of these mediators are derived from invading microorgan-

isms, some are released from damaged cells in response to tissue injury, some are generated by several plasma enzyme systems, and some are products of various white blood cells participating in the inflammatory response.

Among the chemical mediators released in response to tissue damage are various serum proteins called **acute-phase proteins**. The concentrations of these proteins increase dramatically in tissue-damaging infections. C-reactive protein is a major acute-phase protein produced by the liver in response to tissue damage. Its name derives from its pattern-recognition activity: C-reactive protein binds to the C-polysaccharide cell-wall component found on a variety of bacteria and fungi. This binding activates the complement system, resulting in increased clearance of the pathogen either by complement-mediated lysis or by a complement-mediated increase in phagocytosis.

One of the principal mediators of the inflammatory response is **histamine**, a chemical released by a variety of cells in response to tissue injury. Histamine binds to receptors on nearby capillaries and venules, causing vasodilation and increased permeability. Another important group of inflammatory mediators, small peptides called **kinins**, are normally present in blood plasma in an inactive form. Tissue injury activates these peptides, which then cause vasodilation and in-

creased permeability of capillaries. A particular kinin, called bradykinin, also stimulates pain receptors in the skin. This effect probably serves a protective role, because pain normally causes an individual to protect the injured area.

Vasodilation and the increase in capillary permeability in an injured tissue also enable enzymes of the blood-clotting system to enter the tissue. These enzymes activate an enzyme cascade that results in the deposition of insoluble strands of **fibrin**, which is the main component of a blood clot. The fibrin strands wall off the injured area from the rest of the body and serve to prevent the spread of infection.

Once the inflammatory response has subsided and most of the debris has been cleared away by phagocytic cells, tissue repair and regeneration of new tissue begins. Capillaries grow into the fibrin of a blood clot. New connective tissue cells, called fibroblasts, replace the fibrin as the clot dissolves. As fibroblasts and capillaries accumulate, scar tissue forms. The inflammatory response is described in more detail in Chapter 15.

Adaptive Immunity

Adaptive immunity is capable of recognizing and selectively eliminating specific foreign microorganisms and molecules (i.e., foreign antigens). Unlike innate immune responses, adaptive immune responses are not the same in all members of a species but are reactions to specific antigenic challenges. Adaptive immunity displays four characteristic attributes:

- Antigenic specificity
- Diversity
- Immunologic memory
- Self/nonself recognition

The **antigenic specificity** of the immune system permits it to distinguish subtle differences among antigens. Antibodies can distinguish between two protein molecules that differ in only a single amino acid. The immune system is capable of generating tremendous *diversity* in its recognition molecules, allowing it to recognize billions of unique structures on foreign antigens. Once the immune system has recognized and responded to an antigen, it exhibits *immunologic memory*; that is, a second encounter with the same antigen induces a heightened state of immune reactivity. Because of this attribute, the immune system can confer life-long immunity to many infectious agents after an initial encounter. Finally, the immune system normally responds only to foreign antigens, indicating that it is capable of *self/nonself recognition*. The ability of the immune system to distinguish self from nonself and respond only to nonself molecules is essential, for, as described below, the outcome of an inappropriate response to self molecules can be fatal.

Adaptive immunity is not independent of innate immunity. The phagocytic cells crucial to nonspecific immune re-

sponses are intimately involved in activating the specific immune response. Conversely, various soluble factors produced by a specific immune response have been shown to augment the activity of these phagocytic cells. As an inflammatory response develops, for example, soluble mediators are produced that attract cells of the immune system. The immune response will, in turn, serve to regulate the intensity of the inflammatory response. Through the carefully regulated interplay of adaptive and innate immunity, the two systems work together to eliminate a foreign invader.

The Adaptive Immune System Requires Cooperation Between Lymphocytes and Antigen-Presenting Cells

An effective immune response involves two major groups of cells: *T lymphocytes* and *antigen-presenting cells*. Lymphocytes are one of many types of white blood cells produced in the bone marrow by the process of hematopoiesis (see Chapter 2). Lymphocytes leave the bone marrow, circulate in the blood and lymphatic systems, and reside in various lymphoid organs. Because they produce and display antigen-binding cell-surface receptors, lymphocytes mediate the defining immunologic attributes of specificity, diversity, memory, and self/nonself recognition. The two major populations of lymphocytes—**B lymphocytes (B cells)** and **T lymphocytes (T cells)**—are described briefly here and in greater detail in later chapters.

B LYMPHOCYTES

B lymphocytes mature within the bone marrow; when they leave it, each expresses a unique antigen-binding receptor on its membrane (Figure 1-5a). This antigen-binding or B-cell receptor is a membrane-bound **antibody molecule**. Antibodies are glycoproteins that consist of two identical heavy polypeptide chains and two identical light polypeptide chains. Each heavy chain is joined with a light chain by disulfide bonds, and additional disulfide bonds hold the two pairs together. The amino-terminal ends of the pairs of heavy and light chains form a cleft within which antigen binds. When a naive B cell (one that has not previously encountered antigen) first encounters the antigen that matches its membrane-bound antibody, the binding of the antigen to the antibody causes the cell to divide rapidly; its progeny differentiate into **memory B cells** and **effector B cells** called **plasma cells**. Memory B cells have a longer life span than naive cells, and they express the same membrane-bound antibody as their parent B cell. Plasma cells produce the antibody in a form that can be secreted and have little or no membrane-bound antibody. Although plasma cells live for only a few days, they secrete enormous amounts of antibody during this time. It has been estimated that a single plasma cell can secrete more than 2000 molecules of antibody per second. Secreted antibodies are the major effector molecules of humoral immunity.

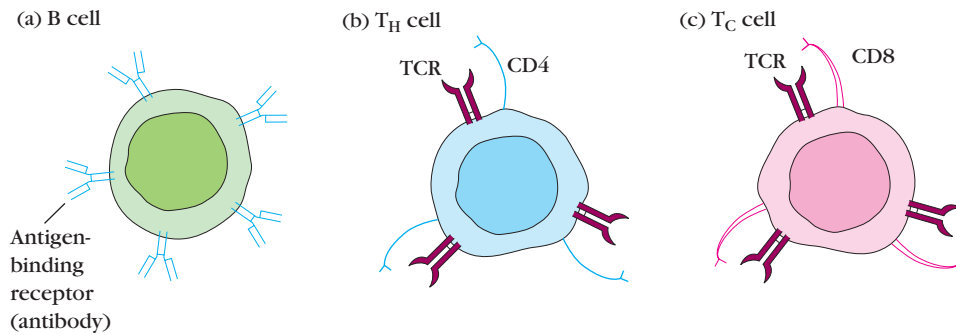


FIGURE 1-5 Distinctive membrane molecules on lymphocytes. (a) B cells have about 10^5 molecules of membrane-bound antibody per cell. All the antibody molecules on a given B cell have the same antigenic specificity and can interact directly with antigen. (b) T cells bearing CD4 ($CD4^+$ cells) recognize only antigen bound to class II MHC molecules. (c) T cells bearing CD8 ($CD8^+$ cells) recognize only

antigen associated with class I MHC molecules. In general, $CD4^+$ cells act as helper cells and $CD8^+$ cells act as cytotoxic cells. Both types of T cells express about 10^5 identical molecules of the antigen-binding T-cell receptor (TCR) per cell, all with the same antigenic specificity.

T LYMPHOCYTES

T lymphocytes also arise in the bone marrow. Unlike B cells, which mature within the bone marrow, T cells migrate to the thymus gland to mature. During its maturation within the thymus, the T cell comes to express a unique antigen-binding molecule, called the **T-cell receptor**, on its membrane. Unlike membrane-bound antibodies on B cells, which can recognize antigen alone, T-cell receptors can recognize only antigen that is bound to cell-membrane proteins called **major histocompatibility complex (MHC) molecules**. MHC molecules that function in this recognition event, which is termed “antigen presentation,” are polymorphic (genetically diverse) glycoproteins found on cell membranes (see Chapter 7). There are two major types of MHC molecules: Class I MHC molecules, which are expressed by nearly all nucleated cells of vertebrate species, consist of a heavy chain linked to a small invariant protein called β_2 -microglobulin. Class II MHC molecules, which consist of an alpha and a beta glycoprotein chain, are expressed only by antigen-presenting cells. When a naive T cell encounters antigen combined with a MHC molecule on a cell, the T cell proliferates and differentiates into memory T cells and various effector T cells.

There are two well-defined subpopulations of T cells: **T helper (T_H)** and **T cytotoxic (T_C) cells**. Although a third type of T cell, called a T suppressor (T_S) cell, has been postulated, recent evidence suggests that it may not be distinct from T_H and T_C subpopulations. T helper and T cytotoxic cells can be distinguished from one another by the presence of either **CD4** or **CD8** membrane glycoproteins on their surfaces (Figure 1-5b,c). T cells displaying CD4 generally function as T_H cells, whereas those displaying CD8 generally function as T_C cells (see Chapter 2).

After a T_H cell recognizes and interacts with an antigen–MHC class II molecule complex, the cell is activated—it becomes an effector cell that secretes various growth factors known collectively as **cytokines**. The secreted cytokines play

an important role in activating B cells, T_C cells, macrophages, and various other cells that participate in the immune response. Differences in the pattern of cytokines produced by activated T_H cells result in different types of immune response.

Under the influence of T_H -derived cytokines, a T_C cell that recognizes an antigen–MHC class I molecule complex proliferates and differentiates into an effector cell called a **cytotoxic T lymphocyte (CTL)**. In contrast to the T_C cell, the CTL generally does not secrete many cytokines and instead exhibits cell-killing or cytotoxic activity. The CTL has a vital function in monitoring the cells of the body and eliminating any that display antigen, such as virus-infected cells, tumor cells, and cells of a foreign tissue graft. Cells that display foreign antigen complexed with a class I MHC molecule are called *altered self-cells*; these are targets of CTLs.

ANTIGEN-PRESENTING CELLS

Activation of both the humoral and cell-mediated branches of the immune system requires cytokines produced by T_H cells. It is essential that activation of T_H cells themselves be carefully regulated, because an inappropriate T-cell response to self-components can have fatal autoimmune consequences. To ensure carefully regulated activation of T_H cells, they can recognize only antigen that is displayed together with class MHC II molecules on the surface of antigen-presenting cells (APCs). These specialized cells, which include macrophages, B lymphocytes, and dendritic cells, are distinguished by two properties: (1) they express class II MHC molecules on their membranes, and (2) they are able to deliver a co-stimulatory signal that is necessary for T_H -cell activation.

Antigen-presenting cells first internalize antigen, either by phagocytosis or by endocytosis, and then display a part of that antigen on their membrane bound to a class II MHC molecule. The T_H cell recognizes and interacts with the

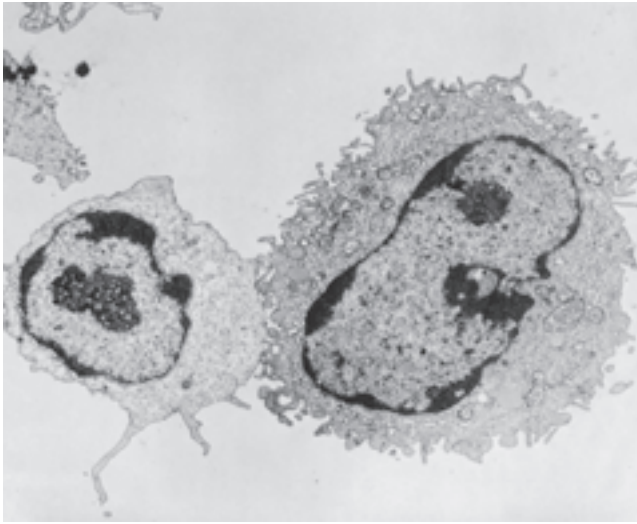


FIGURE 1-6 Electron micrograph of an antigen-presenting macrophage (right) associating with a T lymphocyte. [From A. S. Rosenthal *et al.*, 1982, in *Phagocytosis—Past and Future*, Academic Press, p. 239.]

antigen–class II MHC molecule complex on the membrane of the antigen-presenting cell (Figure 1-6). An additional costimulatory signal is then produced by the antigen-presenting cell, leading to activation of the T_H cell.

Humoral Immunity But Not Cellular Immunity Is Transferred with Antibody

As mentioned earlier, immune responses can be divided into humoral and cell-mediated responses. Humoral immunity refers to immunity that can be conferred upon a nonimmune individual by administration of serum antibodies from an immune individual. In contrast, cell-mediated immunity can be transferred only by administration of T cells from an immune individual.

The humoral branch of the immune system is at work in the interaction of B cells with antigen and their subsequent proliferation and differentiation into antibody-secreting plasma cells (Figure 1-7). Antibody functions as the effector of the humoral response by binding to antigen and neutralizing it or facilitating its elimination. When an antigen is coated with antibody, it can be eliminated in several ways. For example, antibody can cross-link several antigens, forming clusters that are more readily ingested by phagocytic cells. Binding of antibody to antigen on a microorganism can also activate the complement system, resulting in lysis of the foreign organism. Antibody can also neutralize toxins or viral particles by coating them, which prevents them from binding to host cells.

Effector T cells generated in response to antigen are responsible for cell-mediated immunity (see Figure 1-7). Both

activated T_H cells and cytotoxic T lymphocytes (CTLs) serve as effector cells in cell-mediated immune reactions. Cytokines secreted by T_H cells can activate various phagocytic cells, enabling them to phagocytose and kill microorganisms more effectively. This type of cell-mediated immune response is especially important in ridding the host of bacteria and protozoa contained by infected host cells. CTLs participate in cell-mediated immune reactions by killing altered self-cells; they play an important role in the killing of virus-infected cells and tumor cells.

Antigen Is Recognized Differently by B and T Lymphocytes

Antigens, which are generally very large and complex, are not recognized in their entirety by lymphocytes. Instead, both B and T lymphocytes recognize discrete sites on the antigen called **antigenic determinants**, or **epitopes**. Epitopes are the immunologically active regions on a complex antigen, the regions that actually bind to B-cell or T-cell receptors.

Although B cells can recognize an epitope alone, T cells can recognize an epitope only when it is associated with an MHC molecule on the surface of a self-cell (either an antigen-presenting cell or an altered self-cell). Each branch of the immune system is therefore uniquely suited to recognize antigen in a different milieu. The humoral branch (B cells) recognizes an enormous variety of epitopes: those displayed on the surfaces of bacteria or viral particles, as well as those displayed on soluble proteins, glycoproteins, polysaccharides, or lipopolysaccharides that have been released from invading pathogens. The cell-mediated branch (T cells) recognizes protein epitopes displayed together with MHC molecules on self-cells, including altered self-cells such as virus-infected self-cells and cancerous cells.

Thus, four related but distinct cell-membrane molecules are responsible for antigen recognition by the immune system:

- Membrane-bound antibodies on B cells
- T-cell receptors
- Class I MHC molecules
- Class II MHC molecules

Each of these molecules plays a unique role in antigen recognition, ensuring that the immune system can recognize and respond to the different types of antigen that it encounters.

B and T Lymphocytes Utilize Similar Mechanisms To Generate Diversity in Antigen Receptors

The antigenic specificity of each B cell is determined by the membrane-bound antigen-binding receptor (i.e., antibody) expressed by the cell. As a B cell matures in the bone marrow, its specificity is created by random rearrangements of a series



VISUALIZING CONCEPTS

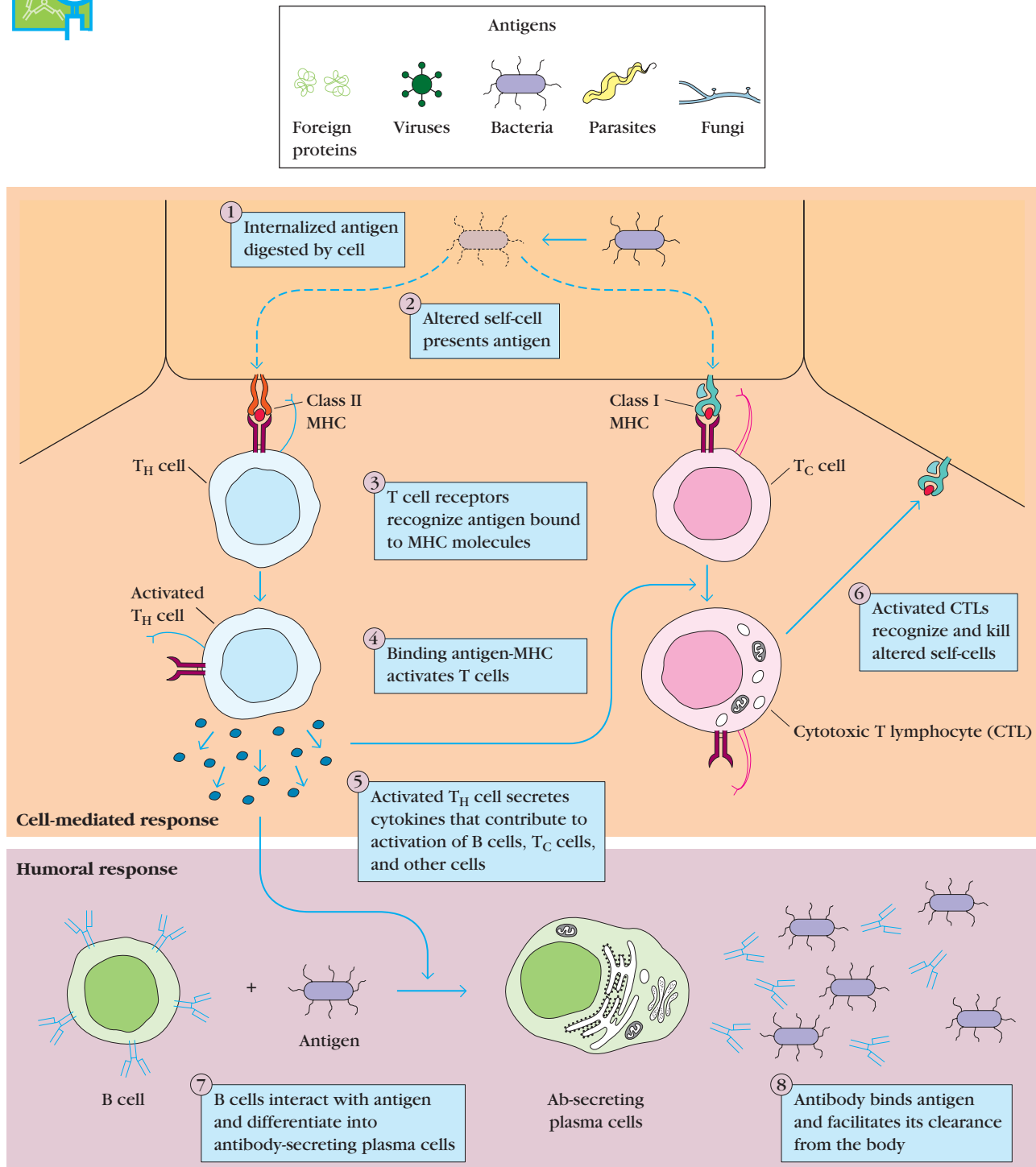


FIGURE 1-7 Overview of the humoral and cell-mediated branches of the immune system. In the humoral response, B cells interact with antigen and then differentiate into antibody-secreting plasma cells. The secreted antibody binds to the antigen and facilitates its clearance from the body. In the cell-mediated re-

sponse, various subpopulations of T cells recognize antigen presented on self-cells. T_H cells respond to antigen by producing cytokines. T_C cells respond to antigen by developing into cytotoxic T lymphocytes (CTLs), which mediate killing of altered self-cells (e.g., virus-infected cells).

of gene segments that encode the antibody molecule (see Chapter 5). As a result of this process, each mature B cell possesses a single functional gene encoding the antibody heavy chain and a single functional gene encoding the antibody light chain; the cell therefore synthesizes and displays antibody with one specificity on its membrane. All antibody molecules on a given B lymphocyte have identical specificity, giving each B lymphocyte, and the clone of daughter cells to which it gives rise, a distinct specificity for a single epitope on an antigen. The mature B lymphocyte is therefore said to be **antigenically committed**.

The random gene rearrangements during B-cell maturation in the bone marrow generate an enormous number of different antigenic specificities. The resulting B-cell population, which consists of individual B cells each expressing a unique antibody, is estimated to exhibit collectively more than 10^{10} different antigenic specificities. The enormous diversity in the mature B-cell population is later reduced by a selection process in the bone marrow that eliminates any B cells with membrane-bound antibody that recognizes self-components. The selection process helps to ensure that self-reactive antibodies (auto-antibodies) are not produced.

The attributes of specificity and diversity also characterize the antigen-binding T-cell receptor (TCR) on T cells. As in B-cell maturation, the process of T-cell maturation includes random rearrangements of a series of gene segments that encode the cell's antigen-binding receptor (see Chapter 9). Each T lymphocyte cell expresses about 10^5 receptors, and all of the receptors on the cell and its clonal progeny have identical specificity for antigen. The random rearrangement of the

TCR genes is capable of generating on the order of 10^9 unique antigenic specificities. This enormous potential diversity is later diminished through a selection process in the thymus that eliminates any T cell with self-reactive receptors and ensures that only T cells with receptors capable of recognizing antigen associated with MHC molecules will be able to mature (see Chapter 10).

The Major Histocompatibility Molecules Bind Antigenic Peptides

The major histocompatibility complex (MHC) is a large genetic complex with multiple loci. The MHC loci encode two major classes of membrane-bound glycoproteins: **class I** and **class II MHC molecules**. As noted above, T_H cells generally recognize antigen combined with class II molecules, whereas T_C cells generally recognize antigen combined with class I molecules (Figure 1-8).

MHC molecules function as antigen-recognition molecules, but they do not possess the fine specificity for antigen characteristic of antibodies and T-cell receptors. Rather, each MHC molecule can bind to a spectrum of **antigenic peptides** derived from the intracellular degradation of antigen molecules. In both class I and class II MHC molecules the distal regions (farthest from the membrane) of different alleles display wide variation in their amino acid sequences. These variable regions form a cleft within which the antigenic peptide sits and is presented to T lymphocytes (see Figure 1-8). Different allelic forms of the genes encoding class I and class

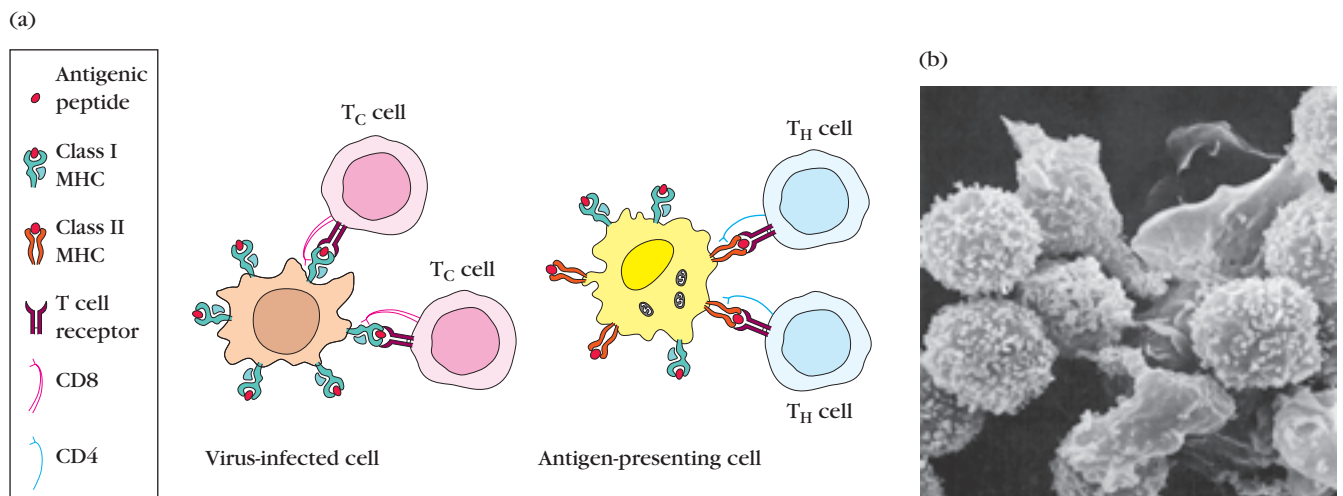


FIGURE 1-8 The role of MHC molecules in antigen recognition by T cells. (a) Class I MHC molecules are expressed on nearly all nucleated cells. Class II MHC molecules are expressed only on antigen-presenting cells. T cells that recognize only antigenic peptides displayed with a class II MHC molecule generally function as T helper (T_H) cells. T cells that recognize only antigenic peptides displayed with a class I MHC molecule generally function as T cytotoxic (T_C)

cells. (b) This scanning electron micrograph reveals numerous T lymphocytes interacting with a single macrophage. The macrophage presents processed antigen combined with class II MHC molecules to the T cells. [Photograph from W. E. Paul (ed.), 1991, *Immunology: Recognition and Response*, W. H. Freeman and Company, New York; micrograph courtesy of M. H. Nielsen and O. Werdelin.]

II molecules confer different structures on the antigen-binding cleft with different specificity. Thus the ability to present an antigen to T lymphocytes is influenced by the particular set of alleles that an individual inherits.

Complex Antigens Are Degraded (Processed) and Displayed (Presented) with MHC Molecules on the Cell Surface

In order for a foreign protein antigen to be recognized by a T cell, it must be degraded into small antigenic peptides that form complexes with class I or class II MHC molecules. This conversion of proteins into MHC-associated peptide fragments is called *antigen processing and presentation*. Whether a particular antigen will be processed and presented together with class I MHC or class II MHC molecules appears to be determined by the route that the antigen takes to enter a cell (Figure 1-9).

Exogenous antigen is produced outside of the host cell and enters the cell by endocytosis or phagocytosis. Antigen-presenting cells (macrophages, dendritic cells, and B cells) degrade ingested exogenous antigen into peptide fragments within the endocytic processing pathway. Experiments suggest that class II MHC molecules are expressed within the endocytic processing pathway and that peptides produced by degradation of antigen in this pathway bind to the cleft within the class II MHC molecules. The MHC molecules bearing the peptide are then exported to the cell surface.

Since expression of class II MHC molecules is limited to antigen-presenting cells, presentation of exogenous peptide–class II MHC complexes is limited to these cells. T cells displaying CD4 recognize antigen combined with class II MHC molecules and thus are said to be *class II MHC restricted*. These cells generally function as T helper cells.

Endogenous antigen is produced within the host cell itself. Two common examples are viral proteins synthesized within virus-infected host cells and unique proteins synthesized by cancerous cells. Endogenous antigens are degraded into peptide fragments that bind to class I MHC molecules within the endoplasmic reticulum. The peptide–class I MHC complex is then transported to the cell membrane. Since all nucleated cells express class I MHC molecules, all cells producing endogenous antigen use this route to process the antigen. T cells displaying CD8 recognize antigen associated with class I MHC molecules and thus are said to be *class I MHC restricted*. These cytotoxic T cells attack and kill cells displaying the antigen–MHC class I complexes for which their receptors are specific.

Antigen Selection of Lymphocytes Causes Clonal Expansion

A mature immunocompetent animal contains a large number of antigen-reactive clones of T and B lymphocytes; the antigenic specificity of each of these clones is determined by the specificity of the antigen-binding receptor on the mem-

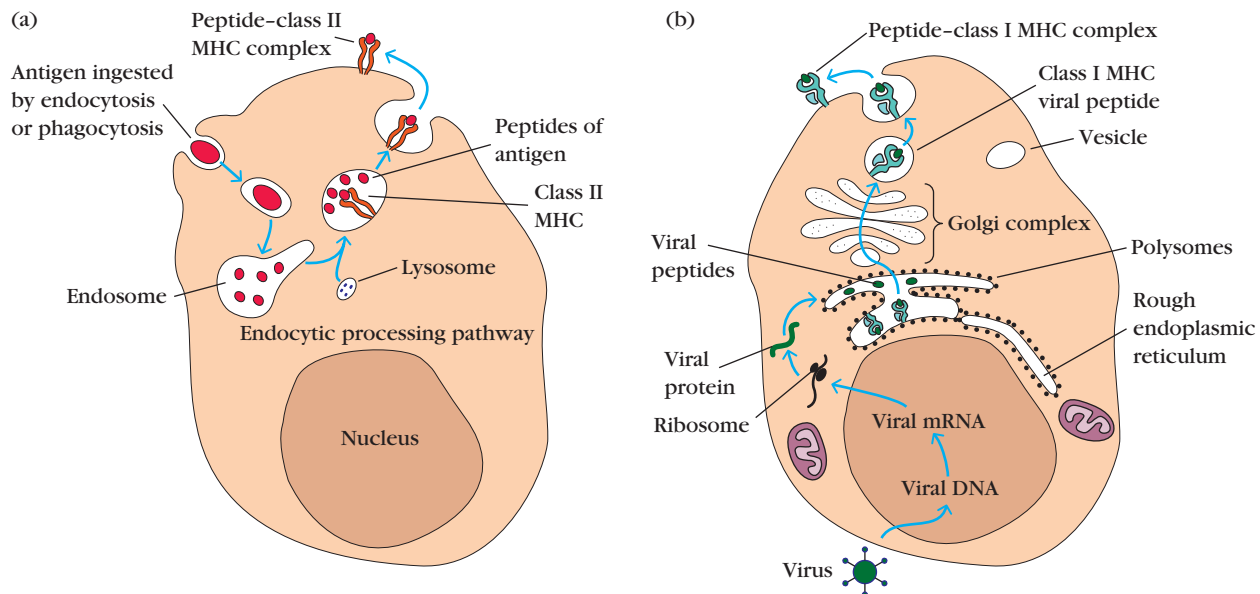


FIGURE 1-9 Processing and presentation of exogenous and endogenous antigens. (a) Exogenous antigen is ingested by endocytosis or phagocytosis and then enters the endocytic processing pathway. Here, within an acidic environment, the antigen is degraded into small peptides, which then are presented with class II MHC molecules on the membrane of the antigen-presenting cell. (b) Endoge-

nous antigen, which is produced within the cell itself (e.g., in a virus-infected cell), is degraded within the cytoplasm into peptides, which move into the endoplasmic reticulum, where they bind to class I MHC molecules. The peptide–class I MHC complexes then move through the Golgi complex to the cell surface.

brane of the clone's lymphocytes. As noted above, the specificity of each T and B lymphocyte is determined before its contact with antigen by random gene rearrangements during maturation in the thymus or bone marrow.

The role of antigen becomes critical when it interacts with and activates mature, antigenically committed T and B lymphocytes, bringing about expansion of the population of cells with a given antigenic specificity. In this process of **clonal selection**, an antigen binds to a particular T or B cell and stimulates it to divide repeatedly into a clone of cells with the same antigenic specificity as the original parent cell (Figure 1-10).

Clonal selection provides a framework for understanding the specificity and self/nonself recognition that is character-

istic of adaptive immunity. Specificity is shown because only lymphocytes whose receptors are specific for a given epitope on an antigen will be clonally expanded and thus mobilized for an immune response. Self/nonself discrimination is accomplished by the elimination, during development, of lymphocytes bearing self-reactive receptors or by the functional suppression of these cells in adults.

Immunologic memory also is a consequence of clonal selection. During clonal selection, the number of lymphocytes specific for a given antigen is greatly amplified. Moreover, many of these lymphocytes, referred to as memory cells, appear to have a longer life span than the naive lymphocytes from which they arise. The initial encounter of a naive immunocompetent lymphocyte with an antigen induces a

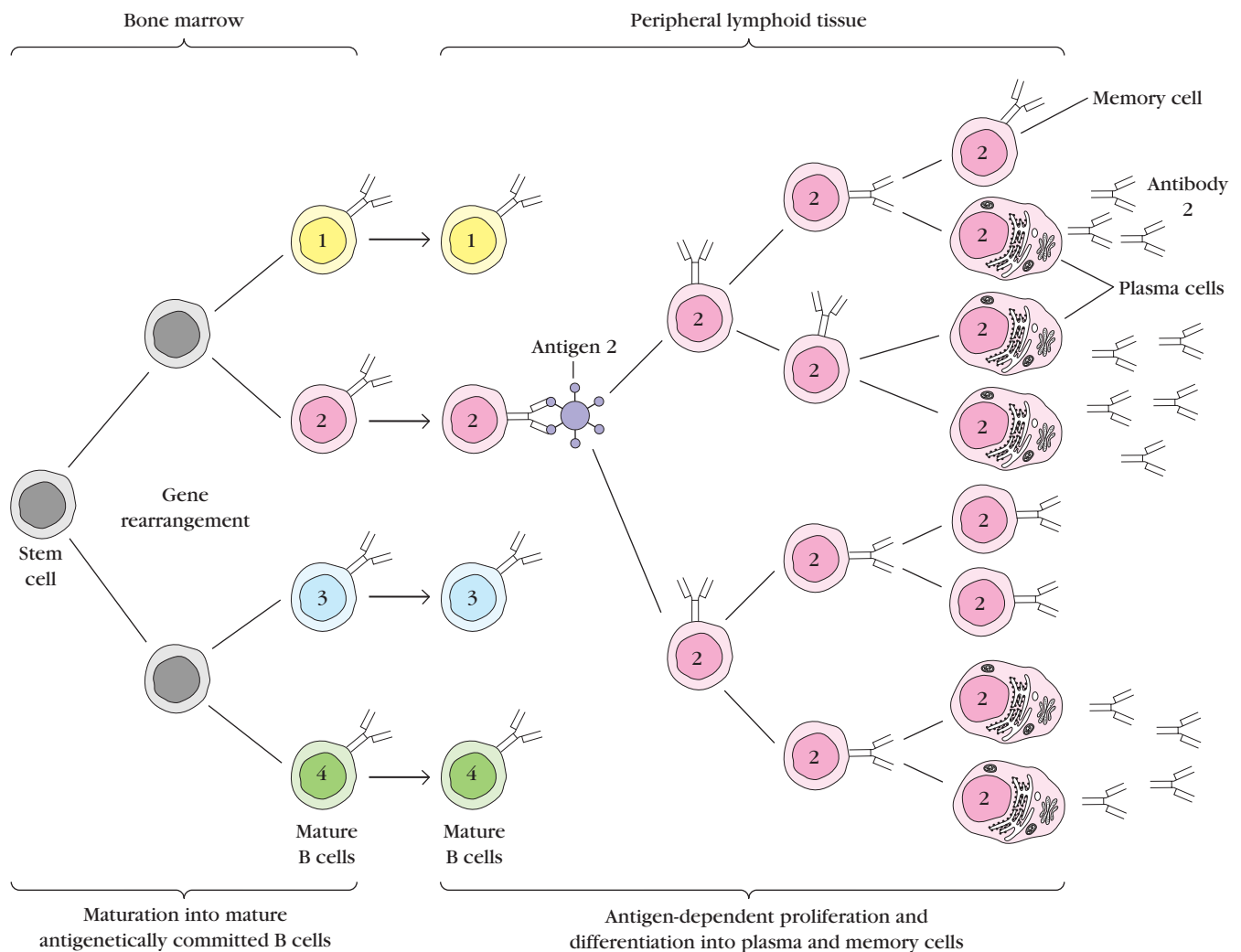


FIGURE 1-10 Maturation and clonal selection of B lymphocytes. Maturation, which occurs in the absence of antigen, produces antigenically committed B cells, each of which expresses antibody with a single antigenic specificity (indicated by 1, 2, 3, and 4). Clonal selection occurs when an antigen binds to a B cell whose membrane-bound antibody molecules are specific for epitopes on that antigen. Clonal expansion of an antigen-activated B cell (number 2 in this ex-

ample) leads to a clone of memory B cells and effector B cells, called plasma cells; all cells in the expanded clone are specific for the original antigen. The plasma cells secrete antibody reactive with the activating antigen. Similar processes take place in the T-lymphocyte population, resulting in clones of memory T cells and effector T cells; the latter include activated T_H cells, which secrete cytokines, and cytotoxic T lymphocytes (CTLs).

primary response; a later contact of the host with antigen will induce a more rapid and heightened **secondary response.** The amplified population of memory cells accounts for the rapidity and intensity that distinguishes a secondary response from the primary response.

In the humoral branch of the immune system, antigen induces the clonal proliferation of B lymphocytes into anti-

body-secreting plasma cells and memory B cells. As seen in Figure 1-11a, the primary response has a lag of approximately 5–7 days before antibody levels start to rise. This lag is the time required for activation of naive B cells by antigen and T_H cells and for the subsequent proliferation and differentiation of the activated B cells into plasma cells. Antibody levels peak in the primary response at about day 14 and then begin to drop off as the plasma cells begin to die. In the secondary response, the lag is much shorter (only 1–2 days), antibody levels are much higher, and they are sustained for much longer. The secondary response reflects the activity of the clonally expanded population of memory B cells. These memory cells respond to the antigen more rapidly than naive B cells; in addition, because there are many more memory cells than there were naive B cells for the primary response, more plasma cells are generated in the secondary response, and antibody levels are consequently 100- to 1000-fold higher.

In the cell-mediated branch of the immune system, the recognition of an antigen-MHC complex by a specific mature T lymphocyte induces clonal proliferation into various T cells with effector functions (T_H cells and CTLs) and into memory T cells. The cell-mediated response to a skin graft is illustrated in Figure 1-11b by a hypothetical transplantation experiment. When skin from strain C mice is grafted onto strain A mice, a primary response develops and all the grafts are rejected in about 10–14 days. If these same mice are again grafted with strain C skin, it is rejected much more vigorously and rapidly than the first grafts. However, if animals previously engrafted with strain C skin are next given skin from an unrelated strain, strain B, the response to strain B is typical of the primary response and is rejected in 10–14 days. That is, graft rejection is a specific immune response. The same mice that showed a secondary response to graft C will show a primary response to graft B. The increased speed of rejection of graft C reflects the presence of a clonally expanded population of memory T_H and T_C cells specific for the antigens of the foreign graft. This expanded memory population generates more effector cells, resulting in faster graft rejection.

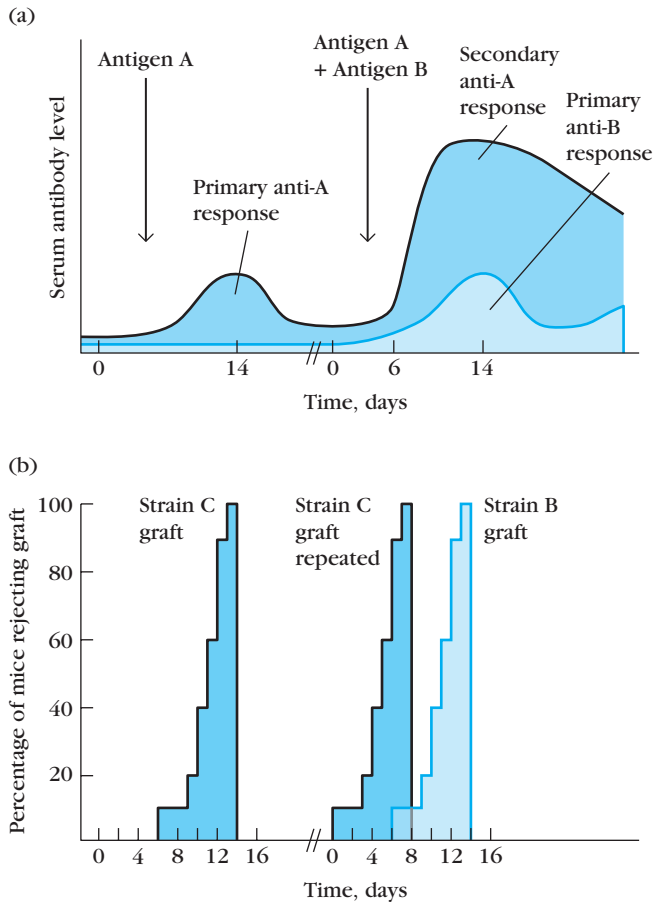


FIGURE 1-11 Differences in the primary and secondary response to injected antigen (humoral response) and to a skin graft (cell-mediated response) reflect the phenomenon of immunologic memory. (a) When an animal is injected with an antigen, it produces a primary serum antibody response of low magnitude and short duration, peaking at about 10–17 days. A second immunization with the same antigen results in a secondary response that is greater in magnitude, peaks in less time (2–7 days), and lasts longer (months to years) than the primary response. Compare the secondary response to antigen A with the primary response to antigen B administered to the same mice. (b) Results from a hypothetical experiment in which skin grafts from strain C mice are transplanted to 20 mice of strain A; the grafts are rejected in about 10–14 days. The 20 mice are rested for 2 months and then 10 are given strain C grafts and the other 10 are given skin from strain B. Mice previously exposed to strain C skin reject C grafts much more vigorously and rapidly than the grafts from strain B. Note that the rejection of the B graft follows a time course similar to that of the first strain C graft.

The Innate and Adaptive Immune Systems Collaborate, Increasing the Efficiency of Immune Responsiveness

It is important to appreciate that adaptive and innate immunity do not operate independently—they function as a highly interactive and cooperative system, producing a combined response more effective than either branch could produce by itself. Certain immune components play important roles in both types of immunity.

An example of cooperation is seen in the encounter between macrophages and microbes. Interactions between receptors on macrophages and microbial components generate soluble proteins that stimulate and direct adaptive immune responses, facilitating the participation of the adap-

TABLE 1-3

Comparison of adaptive and innate immunity

	Innate	Adaptive
Response time	Hours	Days
Specificity	Limited and fixed	Highly diverse, improves during the course of immune response
Response to repeat infection	Identical to primary response	Much more rapid than primary response

tive immune system in the elimination of the pathogen. Stimulated macrophages also secrete cytokines that can direct adaptive immune responses against particular intracellular pathogens.

Just as important, the adaptive immune system produces signals and components that stimulate and increase the effectiveness of innate responses. Some T cells, when they encounter appropriately presented antigen, synthesize and secrete cytokines that increase the ability of macrophages to kill the microbes they have ingested. Also, antibodies produced against an invader bind to the pathogen, marking it as a target for attack by complement and serving as a potent activator of the attack.

A major difference between adaptive and innate immunity is the rapidity of the innate immune response, which utilizes a pre-existing but limited repertoire of responding components. Adaptive immunity compensates for its slower onset by its ability to recognize a much wider repertoire of foreign substances, and also by its ability to improve during a response, whereas innate immunity remains constant. It may also be noted that secondary adaptive responses are considerably faster than primary responses. Principle characteristics of the innate and adaptive immune systems are compared in Table 1-3. With overlapping roles, the two systems together form a highly effective barrier to infection.

Comparative Immunity

The field of immunology is concerned mostly with how innate and adaptive mechanisms collaborate to protect vertebrates from infection. Although many cellular and molecular actors have important roles, antibodies and lymphocytes are considered to be the principal players. Yet despite their prominence in vertebrate immune systems, it would be a mistake to conclude that these extraordinary molecules and versatile cells are essential for immunity. In fact, a determined search for antibodies, T cells, and B cells in organisms of the nonvertebrate phyla has failed to find them. The interior spaces of organisms as diverse as fruit flies, cockroaches, and plants do not contain unchecked microbial populations,

however, which implies that some sort of immunity exists in most, possibly all, multicellular organisms, including those with no components of adaptive immunity.

Insects and plants provide particularly clear and dramatic examples of innate immunity that is not based on lymphocytes. The invasion of the interior body cavity of the fruit fly, *Drosophila melanogaster*, by bacteria or molds triggers the synthesis of small peptides that have strong antibacterial or antifungal activity. The effectiveness of these antimicrobial peptides is demonstrated by the fate of mutants that are unable to produce them. For example, a fungal infection overwhelms a mutant fruit fly that is unable to trigger the synthesis of drosomycin, an antifungal peptide (Figure 1-12). Further evidence for immunity in the fruit fly is given by the recent findings that cell receptors recognizing various classes of microbial molecules (the toll-like receptors) were first found in *Drosophila*.

Plants respond to infection by producing a wide variety of antimicrobial proteins and peptides, as well as small



FIGURE 1-12 Severe fungal infection in a fruit fly (*Drosophila melanogaster*) with a disabling mutation in a signal-transduction pathway required for the synthesis of the antifungal peptide drosomycin. [From B. Lemaitre et al., 1996, *Cell* **86**:973; courtesy of J. A. Hoffman, University of Strasbourg.]

nonpeptide organic molecules that have antibiotic activity. Among these agents are enzymes that digest microbial cell walls, peptides and a protein that damages microbial membranes, and the small organic molecules phytoalexins. The importance of the phytoalexins is shown by the fact that mutations that alter their biosynthetic pathways result in loss of resistance to many plant pathogens. In some cases, the response of plants to pathogens goes beyond this chemical assault to include an architectural response, in which the plant isolates cells in the infected area by strengthening the walls of surrounding cells. Table 1-4 compares the capabilities of immune systems in a wide range of multicellular organisms, both animals and plants.

Immune Dysfunction and Its Consequences

The above overview of innate and adaptive immunity depicts a multicomponent interactive system that protects the host from infectious diseases and from cancer. This overview would not be complete without mentioning that the immune system can function improperly. Sometimes the immune system fails to protect the host adequately or misdirects its activities to cause discomfort, debilitating disease, or even death. There are several common manifestations of immune dysfunction:

- Allergy and asthma
- Graft rejection and graft-versus-host disease
- Autoimmune disease
- Immunodeficiency

Allergy and asthma are results of inappropriate immune responses, often to common antigens such as plant pollen, food, or animal dander. The possibility that certain substances increased sensitivity rather than protection was recognized in about 1902 by Charles Richet, who attempted to immunize dogs against the toxins of a type of jellyfish, *Physalia*. He and his colleague Paul Portier observed that dogs exposed to sublethal doses of the toxin reacted almost instantly, and fatally, to subsequent challenge with minute amounts of the toxin. Richet concluded that a successful immunization or vaccination results in *phylaxis*, or protection, and that an opposite result may occur—**anaphylaxis**—in which exposure to antigen can result in a potentially lethal sensitivity to the antigen if the exposure is repeated. Richet received the Nobel Prize in 1913 for his discovery of the anaphylactic response.

Fortunately, most allergic reactions in humans are not rapidly fatal. A specific allergic or anaphylactic response usually involves one antibody type, called IgE. Binding of IgE to its specific antigen (allergen) releases substances that cause irritation and inflammation. When an allergic individual is exposed to an allergen, symptoms may include sneezing,

wheezing, and difficulty in breathing (asthma); dermatitis or skin eruptions (hives); and, in more extreme cases, strangulation due to blockage of airways by inflammation. A significant fraction of our health resources is expended to care for those suffering from allergy and asthma. The frequency of allergy and asthma in the United States place these complaints among the most common reasons for a visit to the doctor's office or to the hospital emergency room (see Clinical Focus).

When the immune system encounters foreign cells or tissue, it responds strongly to rid the host of the invaders. However, in some cases, the transplantation of cells or an organ from another individual, although viewed by the immune system as a foreign invasion, may be the only possible treatment for disease. For example, it is estimated that more than 60,000 persons in the United States alone could benefit from a kidney transplant. Because the immune system will attack and reject any transplanted organ that it does not recognize as self, it is a serious barrier to this potentially life-saving treatment. An additional danger in transplantation is that any transplanted cells with immune function may view the new host as nonself and react against it. This reaction, which is termed graft-versus-host disease, can be fatal. The rejection reaction and graft-versus-host disease can be suppressed by drugs, but this type of treatment suppresses all immune function, so that the host is no longer protected by its immune system and becomes susceptible to infectious diseases. Transplantation studies have played a major role in the development of immunology. A Nobel prize was awarded to Karl Landsteiner, in 1930, for the discovery of human blood groups, a finding that allowed blood transfusions to be carried out safely. In 1980, G. Snell, J. Dausset, and B. Benacerraf were recognized for discovery of the major histocompatibility complex, and, in 1991, E. D. Thomas and J. Murray were awarded Nobel Prizes for advances in transplantation immunity. To enable a foreign organ to be accepted without suppressing immunity to all antigens remains a challenge for immunologists today.

In certain individuals, the immune system malfunctions by losing its sense of self and nonself, which permits an immune attack upon the host. This condition, **autoimmunity**, can cause a number of chronic debilitating diseases. The symptoms of autoimmunity differ depending on which tissues and organs are under attack. For example, multiple sclerosis is due to an autoimmune attack on the brain and central nervous system, Crohn's disease is an attack on the tissues in the gut, and rheumatoid arthritis is an attack on joints of the arms and legs. The genetic and environmental factors that trigger and sustain autoimmune disease are very active areas of immunologic research, as is the search for improved treatments.

If any of the many components of innate or specific immunity is defective because of genetic abnormality, or if any immune function is lost because of damage by chemical, physical, or biological agents, the host suffers from **immunodeficiency**. The severity of the immunodeficiency disease

TABLE 1-4 Immunity in multicellular organisms

Taxonomic group	Innate immunity (nonspecific)	Adaptive immunity (specific)	Invasion-induced protective enzymes and enzyme cascades	Phagocytosis	Antimicrobial peptides	Pattern-recognition receptors	Graft rejection	T and B cells	Antibodies
<i>Higher plants</i>	+	—	+	—	+	+	—	—	—
<i>Invertebrate animals</i>									
Porifera (sponges)	+	—	?	+	?	?	+	—	—
Annelids (earthworms)	+	—	?	+	?	?	+	—	—
Arthropods (insects, crustaceans)	+	—	+	+	+	+	?	—	—
<i>Vertebrate animals</i>									
Elasmobranchs (cartilaginous fish; e.g., sharks, rays)	+	+	+	+	equivalent agents	+	+	+	+
Teleost fish and bony fish (e.g., salmon, tuna)	+	+	+	+	probable	+	+	+	+
Amphibians	+	+	+	+	+	+	+	+	+
Reptiles	+	+	+	+	?	+	+	+	+
Birds	+	+	+	+	?	+	+	+	+
Mammals	+	+	+	+	+	+	+	+	+

KEY: + = definitive demonstration; — = failure to demonstrate thus far; ? = presence or absence remains to be established.

SOURCES: L. Du Pasquier and M. Flajnik, 1999, "Origin and Evolution of the Vertebrate Immune System," in *Fundamental Immunology*, 4th ed. W. E. Paul (ed.), Lippincott, Philadelphia; B. Fritig, T. Heitz, and M. Legrand, 1998, *Curr. Opin. Immunol.* 10:16; K. Soderhall and L. Cerenius, 1998, *Curr. Opin. Immunol.* 10:23.



CLINICAL FOCUS

Allergy and Asthma as Serious Public Health Problems

Although the immune system serves to protect the host from infection and cancer, inappropriate responses of this system can lead to disease. Common among the results of immune dysfunction are allergies and asthma, both serious public health prob-

lems. Details of the mechanisms that underlie allergic and asthmatic responses to environmental antigens (or allergens) will be considered in Chapter 16. Simply stated, allergic reactions are responses to antigenic stimuli that result in immunity based mainly on the IgE class of immunoglobulin. Exposure to the antigen

(or allergen) triggers an IgE-mediated release of molecules that cause symptoms ranging from sneezing and dermatitis to inflammation of the lungs in an asthmatic attack. The sequence of events in an allergic response is depicted in the accompanying figure.

The discomfort from common allergies such as plant pollen allergy (often called ragweed allergy) consists of a week or two of sneezing and runny nose, which may seem trivial compared with health problems such as cancer, cardiac arrest, or life-threatening infections. A more serious allergic reaction is asthma,

(continued)



CLINICAL FOCUS (continued)

Allergy and Asthma as Serious Public Health Problems

a chronic disease of the lungs in which inflammation, mediated by environmental antigens or infections, causes severe difficulty in breathing. Approximately 15 million persons in the United States suffer from asthma, and it causes about 5000 deaths per year. In the past twenty years, the prevalence of asthma in the Western World has doubled.*

Data on the frequency of care sought for the most common medical complaints in the United States show that asthma and allergy together resulted in more than 28 million visits to the doctor in 1995. The importance of allergy as a public health problem is underscored by the fact that the annual number of doctor visits for hypertension, routine medical examinations, or normal pregnancy, are each fewer than the number of visits for allergic conditions. In fact, the most common reason for a visit to a hospital emergency room is an asthma attack, accounting for one third of all visits. In addition to those treated in the ER, there were about 160,000 hospitalizations for asthma in the past year, with an average stay of 3 to 4 days.

Although all ages and races are affected, deaths from asthma are 3.5 times more common among African-American children. The reasons for the increases in number of asthma cases and for the higher death rate in African-American children remain unknown, although some clues may have been uncovered by recent

studies of genetic factors in allergic disease (see Clinical Focus in Chapter 16).

An increasingly serious health problem is food allergy, especially to peanuts and tree nuts (almonds, cashews, and walnuts).† Approximately 3 million Americans are allergic to these foods and they are the leading causes of fatal and near-fatal food allergic (anaphylactic) reactions. While avoidance of these foods can prevent harmful consequences, the ubiquitous use of peanut protein and other nut products in a variety of foods makes this very difficult for the allergic individual. At least 50% of serious reactions are caused by accidental exposures to peanuts, tree nuts, or their products. This has led to controversial movements to ban peanuts from schools and airplanes.

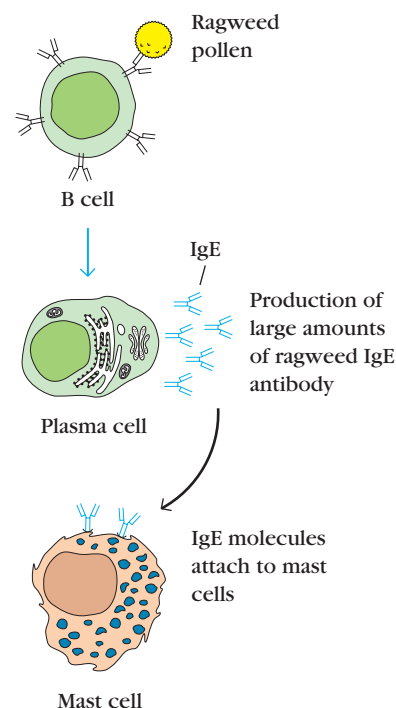
Anaphylaxis generally occurs within an hour of ingesting the food allergen and the most effective treatment is injection of the drug epinephrine. Those prone to anaphylactic attacks often carry injectable epinephrine to be used in case of exposure.

In addition to the suffering and anxiety caused by inappropriate immune responses or allergies to environmental antigens, there is a staggering cost in terms of lost work time for those affected and for caregivers. These costs well justify the extensive efforts by basic and clinical immunologists and allergists to relieve the suffering caused by these disorders.

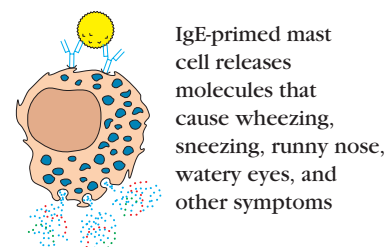
*Holgate, S. T. 1999. The epidemic of allergy and asthma, *Nature Supp.* to vol. 402, B2.

†Hughes, D. A., and C. Mills. 2001. Food allergy: A problem on the rise. *Biologist* (London) 48:201.

First contact with an allergen (ragweed)



Subsequent contact with allergen



Sequence of events leading to an allergic response. When the antibody produced upon contact with an allergen is IgE, this class of antibody reacts via its constant region with a mast cell. Subsequent reaction of the antibody binding site with the allergen triggers the mast cell to which the IgE is bound to secrete molecules that cause the allergic symptoms.

depends on the number of affected components. A common type of immunodeficiency in North America is a selective immunodeficiency in which only one type of immunoglobulin, IgA, is lacking; the symptoms may be minor or even go unnoticed. In contrast, a rarer immunodeficiency called

severe combined immunodeficiency (SCID), which affects both B and T cells, if untreated, results in death from infection at an early age. Since the 1980s, the most common form of immunodeficiency has been acquired immune deficiency syndrome, or AIDS, which results from infection with the

retrovirus human immunodeficiency virus, or HIV. In AIDS, T helper cells are infected and destroyed by HIV, causing a collapse of the immune system. It is estimated that 35 million persons worldwide suffer from this disease, which is usually fatal within 8 to 10 years after infection. Although certain treatments can prolong the life of AIDS patients, there is no known cure for this disease.

This chapter has been a brief introduction to the immune system, and it has given a thumbnail sketch of how this complex system functions to protect the host from disease. The following chapters will concern the structure and function of the individual cells, organs, and molecules that make up this system. They will describe our current understanding of how the components of immunity interact and the experiments that allowed discovery of these mechanisms. Specific areas of applied immunology, such as immunity to infectious diseases, cancer, and current vaccination practices are the subject matter of later chapters. Finally, to complete the description of the immune system in all of its activities, a chapter addresses each of the major types of immune dysfunction.

SUMMARY

- Immunity is the state of protection against foreign organisms or substances (antigens). Vertebrates have two types of immunity, innate and adaptive.
- Innate immunity is not specific to any one pathogen but rather constitutes a first line of defense, which includes anatomic, physiologic, endocytic and phagocytic, and inflammatory barriers.
- Innate and adaptive immunity operate in cooperative and interdependent ways. The activation of innate immune responses produces signals that stimulate and direct subsequent adaptive immune responses.
- Adaptive immune responses exhibit four immunologic attributes: specificity, diversity, memory, and self/nonself recognition.
- The high degree of specificity in adaptive immunity arises from the activities of molecules (antibodies and T-cell receptors) that recognize and bind specific antigens.
- Antibodies recognize and interact directly with antigen. T-cell receptors recognize only antigen that is combined with either class I or class II major histocompatibility complex (MHC) molecules.
- The two major subpopulations of T lymphocytes are the CD4⁺ T helper (T_H) cells and CD8⁺ T cytotoxic (T_C) cells. T_H cells secrete cytokines that regulate immune response upon recognizing antigen combined with class II MHC. T_C cells recognize antigen combined with class I MHC and give rise to cytotoxic T cells (CTLs), which display cytotoxic ability.
- Exogenous (extracellular) antigens are internalized and degraded by antigen-presenting cells (macrophages, B cells, and dendritic cells); the resulting antigenic peptides complexed with class II MHC molecules are then displayed on the cell surface.
- Endogenous (intracellular) antigens (e.g., viral and tumor proteins produced in altered self-cells) are degraded in the cytoplasm and then displayed with class I MHC molecules on the cell surface.
- The immune system produces both humoral and cell-mediated responses. The humoral response is best suited for elimination of exogenous antigens; the cell-mediated response, for elimination of endogenous antigens.
- While an adaptive immune system is found only in vertebrates, innate immunity has been demonstrated in organisms as different as insects, earthworms, and higher plants.
- Dysfunctions of the immune system include common maladies such as allergy or asthma. Loss of immune function leaves the host susceptible to infection; in autoimmunity, the immune system attacks host cells or tissues.

References

- Akira, S., K. Takeda, and T. Kaisho. 2001. Toll-like receptors: Critical proteins linking innate and acquired immunity. *Nature Immunol.* **2**:675.
- Burnet, F. M. 1959. *The Clonal Selection Theory of Acquired Immunity*. Cambridge University Press, Cambridge.
- Cohen, S. G., and M. Samter. 1992. *Excerpts from Classics in Allergy*. Symposia Foundation, Carlsbad, California.
- Desour, L. 1922. *Pasteur and His Work* (translated by A. F. and B. H. Wedd). T. Fisher Unwin Ltd., London.
- Fritig, B., T. Heitz, and M. Legrand. 1998. Antimicrobial proteins in induced plant defense. *Curr. Opin. Immunol.* **10**:12.
- Kimbrell, D. A., and B. Beutler. 2001. The evolution and genetics of innate immunity. *Nature Rev. Genet.* **2**:256.
- Kindt, T. J., and J. D. Capra. 1984. *The Antibody Enigma*. Plenum Press, New York.
- Landsteiner, K. 1947. *The Specificity of Serologic Reactions*. Harvard University Press, Cambridge, Massachusetts.
- Lawson, P. R., and K. B. Reid. 2000. The roles of surfactant proteins A and D in innate immunity. *Immunologic Reviews* **173**:66.
- Medawar, P. B. 1958. *The Immunology of Transplantation. The Harvey Lectures 1956–1957*. Academic Press, New York.
- Medzhitov, R., and C. A. Janeway. 2000. Innate immunity. *N. Eng. J. Med.* **343**:338.
- Metchnikoff, E. 1905. *Immunity in the Infectious Diseases*. MacMillan, New York.
- Otvos, L. 2000. Antibacterial peptides isolated from insects. *J. Peptide Sci.* **6**:497.
- Paul, W., ed. 1999. *Fundamental Immunology*, 4th ed. Lippincott-Raven, Philadelphia.

Roitt, I. M., and P. J. Delves, eds. 1998. *An Encyclopedia of Immunology*, 2nd ed., vols. 1–4. Academic Press, London.



USEFUL WEB SITES

<http://www.aaaai.org/>

The American Academy of Allergy Asthma and Immunology site includes an extensive library of information about allergic diseases.

<http://12.17.12.70/aa/default.asp>

The Web site of the American Association of Immunologists contains a good deal of information of interest to immunologists.

<http://www.ncbi.nlm.nih.gov/PubMed/>

PubMed, the National Library of Medicine database of more than 9 million publications, is the world's most comprehensive bibliographic database for biological and biomedical literature. It is also a highly user-friendly site.

Study Questions

CLINICAL FOCUS QUESTION You have a young nephew who has developed a severe allergy to tree nuts. What precautions would you advise for him and for his parents? Should school officials be aware of this condition?

- Indicate to which branch(es) of the immune system the following statements apply, using **H** for the humoral branch and **CM** for the cell-mediated branch. Some statements may apply to both branches.
 - _____ Involves class I MHC molecules
 - _____ Responds to viral infection
 - _____ Involves T helper cells
 - _____ Involves processed antigen
 - _____ Most likely responds following an organ transplant
 - _____ Involves T cytotoxic cells
 - _____ Involves B cells
 - _____ Involves T cells
 - _____ Responds to extracellular bacterial infection
 - _____ Involves secreted antibody
 - _____ Kills virus-infected self-cells
- Specific immunity exhibits four characteristic attributes, which are mediated by lymphocytes. List these four attributes and briefly explain how they arise.
- Name three features of a secondary immune response that distinguish it from a primary immune response.
- Compare and contrast the four types of antigen-binding molecules used by the immune system—antibodies, T-cell receptors, class I MHC molecules, and class II MHC molecules—in terms of the following characteristics:
 - Specificity for antigen
 - Cellular expression
 - Types of antigen recognized

- Fill in the blanks in the following statements with the most appropriate terms:
 - _____, _____, and _____ all function as antigen-presenting cells.
 - Antigen-presenting cells deliver a _____ signal to _____ cells.
 - Only antigen-presenting cells express class _____ MHC molecules, whereas nearly all cells express class _____ MHC molecules.
 - _____ antigens are internalized by antigen-presenting cells, degraded in the _____, and displayed with class _____ MHC molecules on the cell surface.
 - _____ antigens are produced in altered self-cells, degraded in the _____, and displayed with class _____ MHC molecules on the cell surface.
- Briefly describe the three major events in the inflammatory response.
- The T cell is said to be class I restricted. What does this mean?
- Match each term related to innate immunity (a–p) with the most appropriate description listed below (1–19). Each description may be used once, more than once, or not at all.

Terms

- _____ Fimbriae or pili
- _____ Exudate
- _____ Sebum
- _____ Margination
- _____ Dermis
- _____ Lysosome
- _____ Histamine
- _____ Macrophage
- _____ Lysozyme
- _____ Bradykinin
- _____ Interferon
- _____ Edema
- _____ Complement
- _____ Extravasation
- _____ C-reactive protein
- _____ Phagosome

Descriptions

- Thin outer layer of skin
- Layer of skin containing blood vessels and sebaceous glands
- One of several acute-phase proteins
- Hydrolytic enzyme found in mucous secretions
- Migration of a phagocyte through the endothelial wall into the tissues
- Acidic antibacterial secretion found on the skin
- Has antiviral activity
- Induces vasodilation
- Accumulation of fluid in intercellular space, resulting in swelling
- Large vesicle containing ingested particulate material
- Accumulation of dead cells, digested material, and fluid
- Adherence of phagocytic cells to the endothelial wall



- (13) Structures involved in microbial adherence to mucous membranes
 - (14) Stimulates pain receptors in the skin
 - (15) Phagocytic cell found in the tissues
 - (16) Phagocytic cell found in the blood
 - (17) Group of serum proteins involved in cell lysis and clearance of antigen
 - (18) Cytoplasmic vesicle containing degradative enzymes
 - (19) Protein-rich fluid that leaks from the capillaries into the tissues
9. Innate and adaptive immunity act in cooperative and interdependent ways to protect the host. Discuss the collaboration of these two forms of immunity.
10. How might an arthropod, such as a cockroach or beetle, protect itself from infection? In what ways might the innate immune responses of an arthropod be similar to those of a plant and how might they differ?
11. Give examples of mild and severe consequences of immune dysfunction. What is the most common cause of immunodeficiency throughout the world today?
12. Adaptive immunity has evolved in vertebrates but they have also retained innate immunity. What would be the disadvantages of having only an adaptive immune system? Comment on how possession of both types of immunity enhances protection against infection.

Cells and Organs of the Immune System

THE IMMUNE SYSTEM CONSISTS OF MANY DIFFERENT organs and tissues that are found throughout the body. These organs can be classified functionally into two main groups. The *primary lymphoid organs* provide appropriate microenvironments for the development and maturation of lymphocytes. The *secondary lymphoid organs* trap antigen from defined tissues or vascular spaces and are sites where mature lymphocytes can interact effectively with that antigen. Blood vessels and lymphatic systems connect these organs, uniting them into a functional whole.

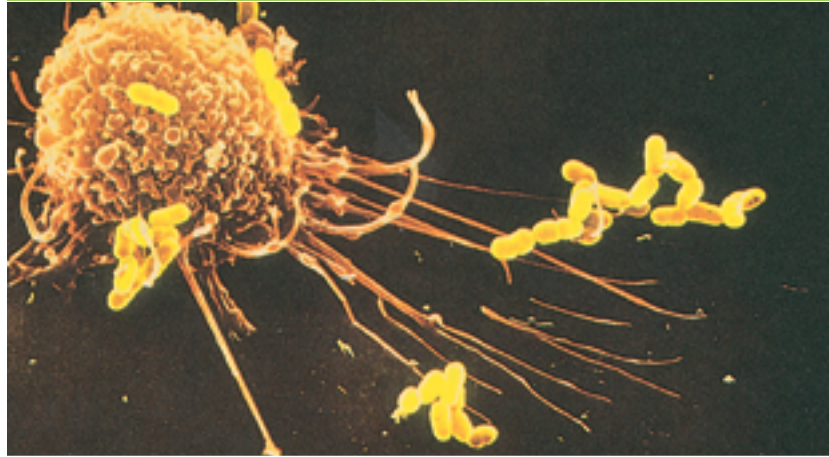
Carried within the blood and lymph and populating the lymphoid organs are various white blood cells, or **leukocytes**, that participate in the immune response. Of these cells, only the lymphocytes possess the attributes of diversity, specificity, memory, and self/nonself recognition, the hallmarks of an adaptive immune response. All the other cells play accessory roles in adaptive immunity, serving to activate lymphocytes, to increase the effectiveness of antigen clearance by phagocytosis, or to secrete various immune-effector molecules. Some leukocytes, especially T lymphocytes, secrete various protein molecules called cytokines. These molecules act as immunoregulatory hormones and play important roles in the regulation of immune responses. This chapter describes the formation of blood cells, the properties of the various immune-system cells, and the functions of the lymphoid organs.

Hematopoiesis

All blood cells arise from a type of cell called the **hematopoietic stem cell (HSC)**. **Stem cells** are cells that can differentiate into other cell types; they are self-renewing—they maintain their population level by cell division. In humans, **hematopoiesis**, the formation and development of red and white blood cells, begins in the embryonic yolk sac during the first weeks of development. Here, yolk-sac stem cells differentiate into primitive erythroid cells that contain embryonic hemoglobin. In the third month of gestation, hematopoietic stem cells migrate from the yolk sac to the fetal liver and then to the spleen; these two organs have major roles in hematopoiesis from the third to the seventh months of gestation. After that, the differentiation of HSCs in the bone marrow becomes the major factor in hematopoiesis, and by birth there is little or no hematopoiesis in the liver and spleen.

It is remarkable that every functionally specialized, mature blood cell is derived from the same type of stem cell. In

chapter 2



Macrophage Interacting with Bacteria

- Hematopoiesis
- Cells of the Immune System
- Organs of the Immune System
- Systemic Function of the Immune System
- Lymphoid Cells and Organs—Evolutionary Comparisons

contrast to a *unipotent* cell, which differentiates into a single cell type, a hematopoietic stem cell is *multipotent*, or *pluripotent*, able to differentiate in various ways and thereby generate erythrocytes, granulocytes, monocytes, mast cells, lymphocytes, and megakaryocytes. These stem cells are few, normally fewer than one HSC per 5×10^4 cells in the bone marrow.

The study of hematopoietic stem cells is difficult both because of their scarcity and because they are hard to grow in vitro. As a result, little is known about how their proliferation and differentiation are regulated. By virtue of their capacity for self-renewal, hematopoietic stem cells are maintained at stable levels throughout adult life; however, when there is an increased demand for hematopoiesis, HSCs display an enormous proliferative capacity. This can be demonstrated in mice whose hematopoietic systems have been completely destroyed by a lethal dose of x-rays (950 rads; one rad represents the absorption by an irradiated target of an amount of radiation corresponding to 100 ergs/gram of target). Such irradiated mice will die within 10 days unless they are infused with normal bone-marrow cells from a syngeneic (genetically identical) mouse. Although a normal mouse has 3×10^8 bone-marrow cells, infusion of only 10^4 – 10^5 bone-marrow cells (i.e., 0.01%–0.1% of the normal amount) from a donor is sufficient to completely restore the hematopoietic system,

which demonstrates the enormous proliferative and differentiative capacity of the stem cells.

Early in hematopoiesis, a multipotent stem cell differentiates along one of two pathways, giving rise to either a common **lymphoid progenitor cell** or a common **myeloid progenitor cell**

progenitor cell (Figure 2-1). The types and amounts of growth factors in the microenvironment of a particular stem cell or progenitor cell control its differentiation. During the development of the lymphoid and myeloid lineages, stem cells differentiate into **progenitor cells**, which have lost the

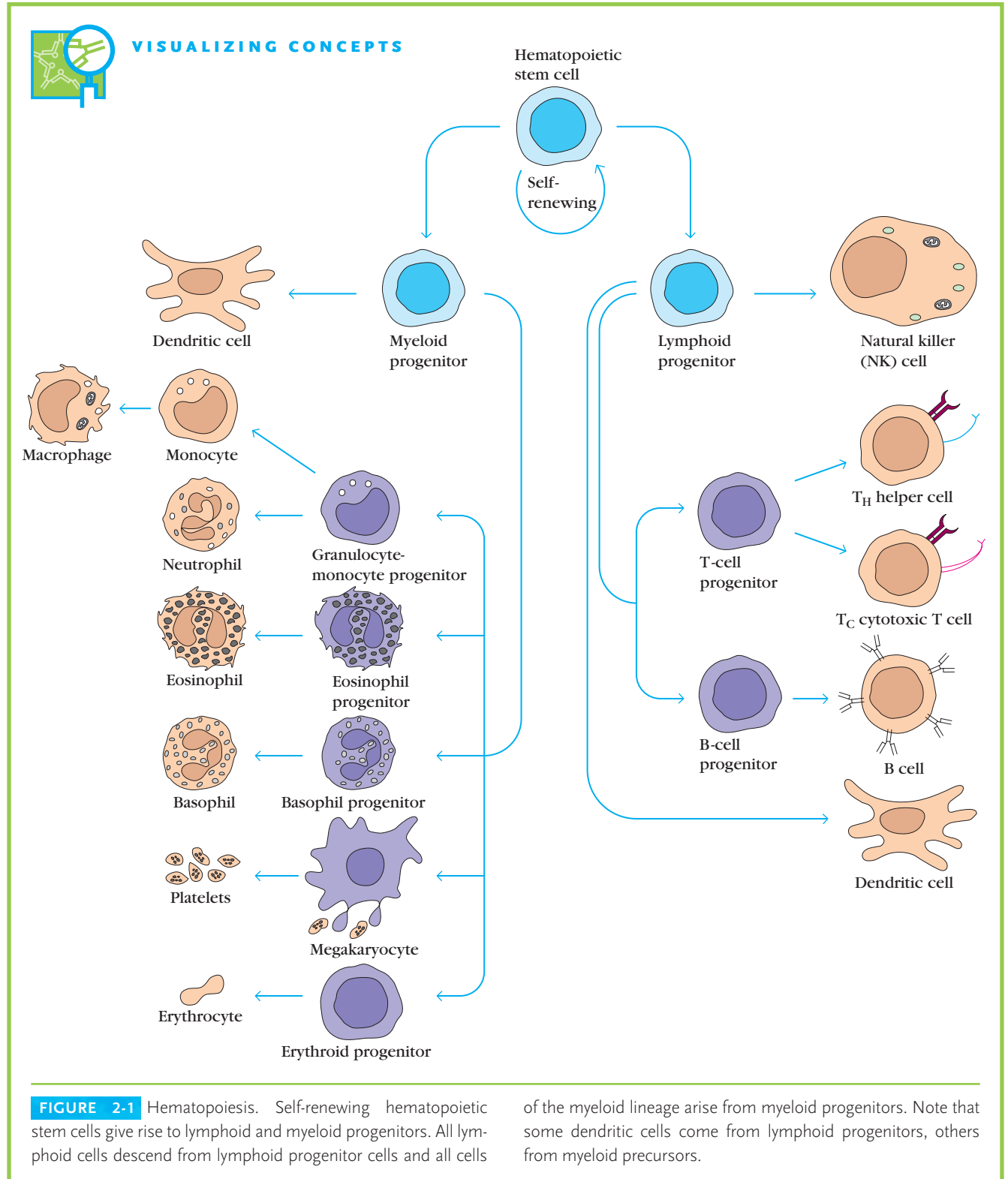


FIGURE 2-1 Hematopoiesis. Self-renewing hematopoietic stem cells give rise to lymphoid and myeloid progenitors. All lymphoid cells descend from lymphoid progenitor cells and all cells

of the myeloid lineage arise from myeloid progenitors. Note that some dendritic cells come from lymphoid progenitors, others from myeloid precursors.

capacity for self-renewal and are committed to a particular cell lineage. Common lymphoid progenitor cells give rise to B, T, and NK (natural killer) cells and some dendritic cells. Myeloid stem cells generate progenitors of red blood cells (erythrocytes), many of the various white blood cells (neutrophils, eosinophils, basophils, monocytes, mast cells, dendritic cells), and platelets. Progenitor commitment depends on the acquisition of responsiveness to particular growth factors and **cytokines**. When the appropriate factors and cytokines are present, progenitor cells proliferate and differentiate into the corresponding cell type, either a mature erythrocyte, a particular type of leukocyte, or a platelet-generating cell (the megakaryocyte). Red and white blood cells pass into bone-marrow channels, from which they enter the circulation.

In bone marrow, hematopoietic cells grow and mature on a meshwork of **stromal cells**, which are nonhematopoietic cells that support the growth and differentiation of hematopoietic cells. Stromal cells include fat cells, endothelial cells, fibroblasts, and macrophages. Stromal cells influence the differentiation of hematopoietic stem cells by providing a **hematopoietic-inducing microenvironment (HIM)** consisting of a cellular matrix and factors that promote growth and differentiation. Many of these hematopoietic growth factors are soluble agents that arrive at their target cells by diffusion, others are membrane-bound molecules on the surface of stromal cells that require cell-to-cell contact between the responding cells and the stromal cells. During infection, hematopoiesis is stimulated by the production of hematopoietic growth factors by activated macrophages and T cells.

Hematopoiesis Can Be Studied In Vitro

Cell-culture systems that can support the growth and differentiation of lymphoid and myeloid stem cells have made it

possible to identify many hematopoietic growth factors. In these in vitro systems, bone-marrow stromal cells are cultured to form a layer of cells that adhere to a petri dish; freshly isolated bone-marrow hematopoietic cells placed on this layer will grow, divide, and produce large visible colonies (Figure 2-2). If the cells have been cultured in semisolid agar, their progeny will be immobilized and can be analyzed for cell types. Colonies that contain stem cells can be replated to produce mixed colonies that contain different cell types, including progenitor cells of different cell lineages. In contrast, progenitor cells, while capable of division, cannot be replated and produce lineage-restricted colonies.

Various growth factors are required for the survival, proliferation, differentiation, and maturation of hematopoietic cells in culture. These growth factors, the hematopoietic cytokines, are identified by their ability to stimulate the formation of hematopoietic cell colonies in bone-marrow cultures. Among the cytokines detected in this way was a family of acidic glycoproteins, the **colony-stimulating factors (CSFs)**, named for their ability to induce the formation of distinct hematopoietic cell lines. Another important hematopoietic cytokine detected by this method was the glycoprotein **erythropoietin (EPO)**. Produced by the kidney, this cytokine induces the terminal development of erythrocytes and regulates the production of red blood cells. Further studies showed that the ability of a given cytokine to signal growth and differentiation is dependent upon the presence of a receptor for that cytokine on the surface of the target cell—commitment of a progenitor cell to a particular differentiation pathway is associated with the expression of membrane receptors that are specific for particular cytokines. Many cytokines and their receptors have since been shown to play essential roles in hematopoiesis. This topic is explored much more fully in the chapter on cytokines (Chapter 11).

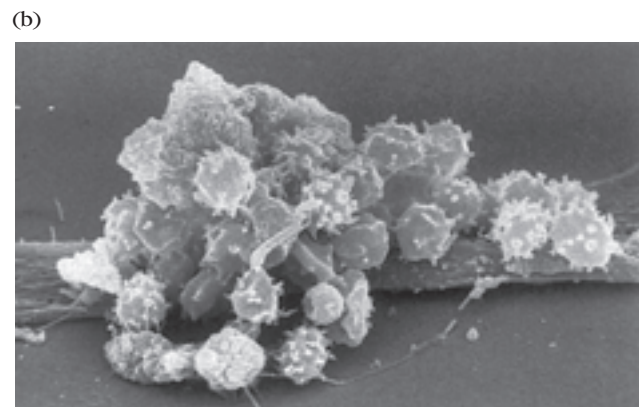
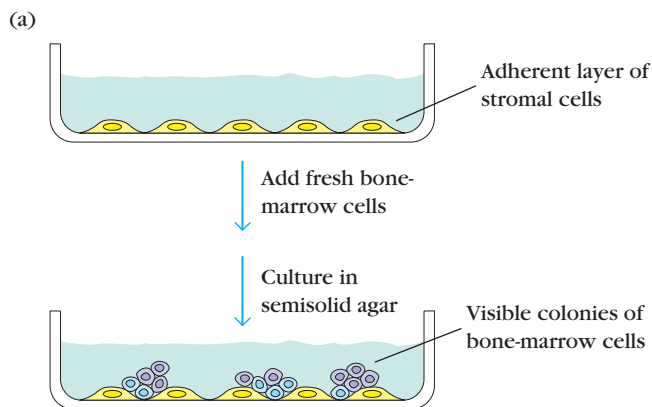


FIGURE 2-2 (a) Experimental scheme for culturing hematopoietic cells. Adherent bone-marrow stromal cells form a matrix on which the hematopoietic cells proliferate. Single cells can be transferred to semisolid agar for colony growth and the colonies analyzed for differentiated cell types. (b) Scanning electron micrograph of cells

in long-term culture of human bone marrow. [Photograph from M. J. Cline and D. W. Golde, 1979, *Nature* **277**:180; reprinted by permission; © 1979 Macmillan Magazines Ltd., micrograph courtesy of S. Quan.]

Hematopoiesis Is Regulated at the Genetic Level

The development of pluripotent hematopoietic stem cells into different cell types requires the expression of different sets of lineage-determining and lineage-specific genes at appropriate times and in the correct order. The proteins specified by these genes are critical components of regulatory networks that direct the differentiation of the stem cell and its descendants. Much of what we know about the dependence of hematopoiesis on a particular gene comes from studies of mice in which a gene has been inactivated or “knocked out” by targeted disruption, which blocks the production of the protein that it encodes (see Targeted Disruption of Genes, in Chapter 23). If mice fail to produce red cells or particular white blood cells when a gene is knocked out, we conclude that the protein specified by the gene is necessary for development of those cells. Knockout technology is one of the most powerful tools available for determining the roles of particular genes in a broad range of processes and it has made important contributions to the identification of many genes that regulate hematopoiesis.

Although much remains to be done, targeted disruption and other approaches have identified a number of transcription factors (Table 2-1) that play important roles in hematopoiesis. Some of these transcription factors affect many different hematopoietic lineages, and others affect only a single lineage, such as the developmental pathway that leads to lymphocytes. One transcription factor that affects multiple lineages is GATA-2, a member of a family of transcription factors that recognize the tetranucleotide sequence GATA, a nucleotide motif in target genes. A functional **GATA-2 gene**, which specifies this transcription factor, is essential for the development of the lymphoid, erythroid, and myeloid lineages. As might be expected, animals in which this gene is disrupted die during embryonic development. In contrast to GATA-2, another transcription factor, **Ikaros**, is required only for the development of cells of the lymphoid lineage. Although Ikaros knockout mice do not produce significant

numbers of B, T, and NK cells, their production of erythrocytes, granulocytes, and other cells of the myeloid lineage is unimpaired. Ikaros knockout mice survive embryonic development, but they are severely compromised immunologically and die of infections at an early age.

Hematopoietic Homeostasis Involves Many Factors

Hematopoiesis is a continuous process that generally maintains a steady state in which the production of mature blood cells equals their loss (principally from aging). The average erythrocyte has a life span of 120 days before it is phagocytosed and digested by macrophages in the spleen. The various white blood cells have life spans ranging from a few days, for neutrophils, to as long as 20–30 years for some T lymphocytes. To maintain steady-state levels, the average human being must produce an estimated 3.7×10^{11} white blood cells per day.

Hematopoiesis is regulated by complex mechanisms that affect all of the individual cell types. These regulatory mechanisms ensure steady-state levels of the various blood cells, yet they have enough built-in flexibility so that production of blood cells can rapidly increase tenfold to twentyfold in response to hemorrhage or infection. Steady-state regulation of hematopoiesis is accomplished in various ways, which include:

- Control of the levels and types of cytokines produced by bone-marrow stromal cells
- The production of cytokines with hematopoietic activity by other cell types, such as activated T cells and macrophages
- The regulation of the expression of receptors for hematopoietically active cytokines in stem cells and progenitor cells
- The removal of some cells by the controlled induction of cell death

A failure in one or a combination of these regulatory mechanisms can have serious consequences. For example, abnormalities in the expression of hematopoietic cytokines or their receptors could lead to unregulated cellular proliferation and may contribute to the development of some leukemias. Ultimately, the number of cells in any hematopoietic lineage is set by a balance between the number of cells removed by cell death and the number that arise from division and differentiation. Any one or a combination of regulatory factors can affect rates of cell reproduction and differentiation. These factors can also determine whether a hematopoietic cell is induced to die.

Programmed Cell Death Is an Essential Homeostatic Mechanism

Programmed cell death, an induced and ordered process in which the cell actively participates in bringing about its own demise, is a critical factor in the homeostatic regulation of

TABLE 2-1 Some transcription factors essential for hematopoietic lineages	
Factor	Dependent lineage
GATA-1	Erythroid
GATA-2	Erythroid, myeloid, lymphoid
PU.1	Erythroid (maturational stages), myeloid (later stages), lymphoid
BM11	Myeloid, lymphoid
Ikaros	Lymphoid
Oct-2	B lymphoid (differentiation of B cells into plasma cells)

many types of cell populations, including those of the hematopoietic system.

Cells undergoing programmed cell death often exhibit distinctive morphologic changes, collectively referred to as **apoptosis** (Figures 2-3, 2-4). These changes include a pronounced decrease in cell volume, modification of the cytoskeleton that results in membrane blebbing, a condensation of the chromatin, and degradation of the DNA into smaller fragments. Following these morphologic changes, an apoptotic cell sheds tiny membrane-bounded apoptotic bodies containing intact organelles. Macrophages quickly phagocytose apoptotic bodies and cells in the advanced stages of apoptosis. This ensures that their intracellular contents, including proteolytic and other lytic enzymes, cationic proteins, and oxidizing molecules are not released into the surrounding tissue. In this way, apoptosis does not induce a local inflammatory response. Apoptosis differs markedly from **necrosis**, the changes associated with cell death arising from injury. In necrosis the injured cell swells and bursts, re-

leasing its contents and possibly triggering a damaging inflammatory response.

Each of the leukocytes produced by hematopoiesis has a characteristic life span and then dies by programmed cell death. In the adult human, for example, there are about 5×10^{10} neutrophils in the circulation. These cells have a life span of only a few days before programmed cell death is initiated. This death, along with constant neutrophil production, maintains a stable number of these cells. If programmed cell death fails to occur, a leukemic state may develop. Programmed cell death also plays a role in maintaining proper numbers of hematopoietic progenitor cells. For example, when colony-stimulating factors are removed, progenitor cells undergo apoptosis. Beyond hematopoiesis, apoptosis is important in such immunological processes as tolerance and the killing of target cells by cytotoxic T cells or natural killer cells. Details of the mechanisms underlying apoptosis are emerging; Chapter 13 describes them in detail.

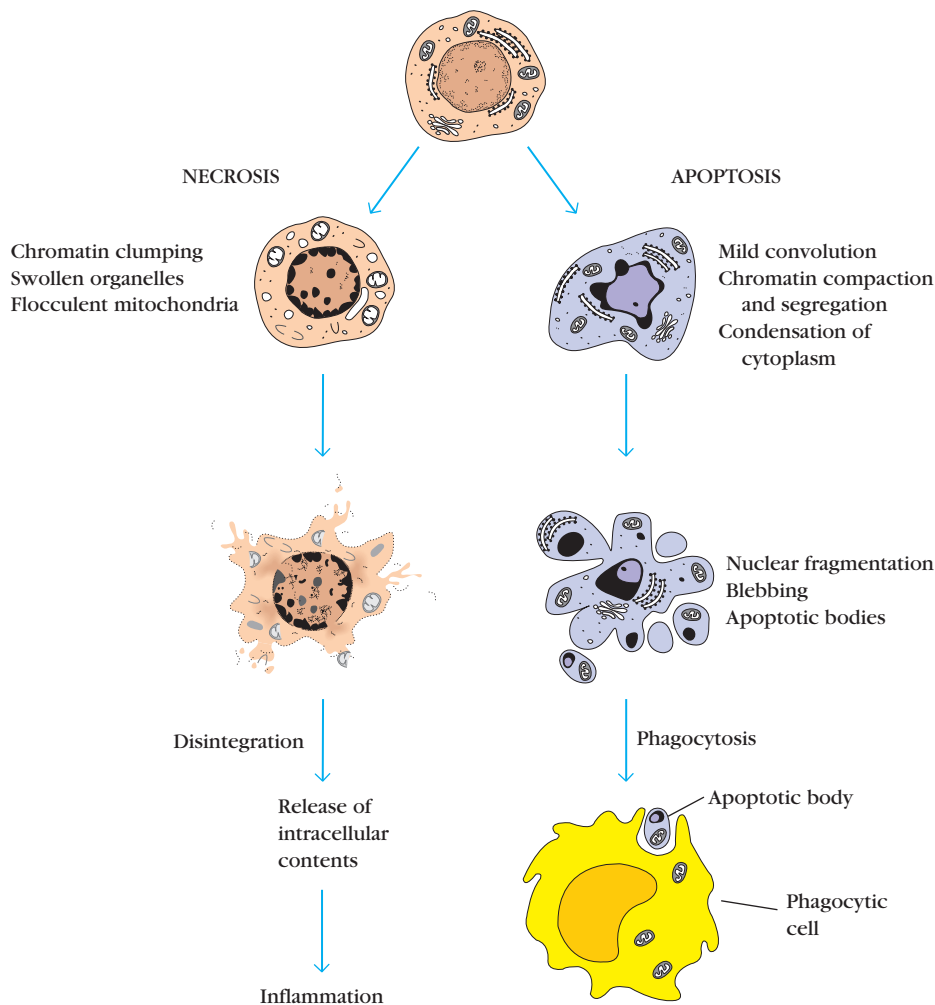


FIGURE 2-3 Comparison of morphologic changes that occur in apoptosis and necrosis. Apoptosis, which results in the programmed cell death of hematopoietic cells, does not induce a local inflamma-

tory response. In contrast, necrosis, the process that leads to death of injured cells, results in release of the cells' contents, which may induce a local inflammatory response.



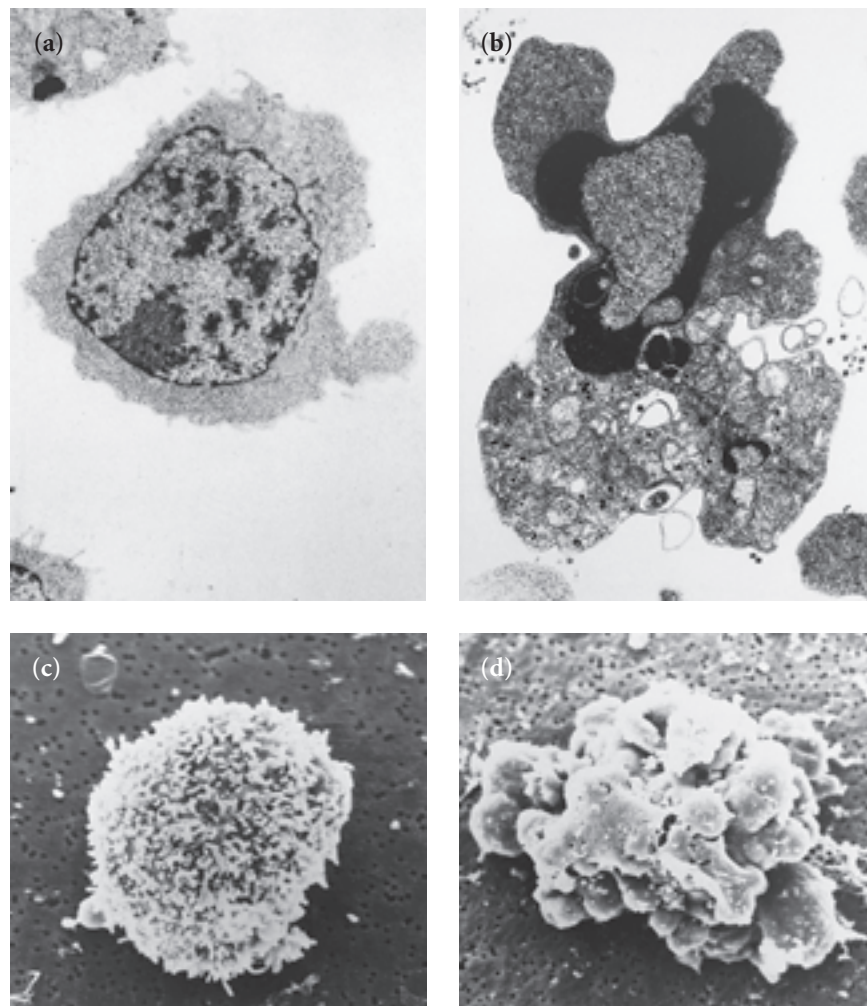


FIGURE 2-4 Apoptosis. Light micrographs of (a) normal thymocytes (developing T cells in the thymus) and (b) apoptotic thymocytes. Scanning electron micrographs of (c) normal and (d)

apoptotic thymocytes. [From B. A. Osborne and S. Smith, 1997, *Journal of NIH Research* **9**:35; courtesy B. A. Osborne, University of Massachusetts at Amherst.]

The expression of several genes accompanies apoptosis in leukocytes and other cell types (Table 2-2). Some of the proteins specified by these genes induce apoptosis, others are critical during apoptosis, and still others inhibit apoptosis. For example, apoptosis can be induced in thymocytes by radiation, but only if the protein p53 is present; many cell deaths are induced by signals from Fas, a molecule present on the surface of many cells; and proteases known as caspases take part in a cascade of reactions that lead to apoptosis. On the other hand, members of the *bcl-2* (B-cell lymphoma 2) family of genes, *bcl-2* and *bcl-X_L*, encode protein products that inhibit apoptosis. Interestingly, the first member of this gene family, *bcl-2*, was found in studies that were concerned not with cell death but with the uncontrolled proliferation of B cells in a type of cancer known as B-lymphoma. In this case, the *bcl-2* gene was at the breakpoint of a chromosomal translocation in a human B-cell lymphoma. The translocation moved the *bcl-2* gene into the immunoglobulin heavy-chain locus, resulting in tran-

scriptional activation of the *bcl-2* gene and overproduction of the encoded Bcl-2 protein by the lymphoma cells. The resulting high levels of Bcl-2 are thought to help transform lymphoid cells into cancerous lymphoma cells by inhibiting the signals that would normally induce apoptotic cell death.

Bcl-2 levels have been found to play an important role in regulating the normal life span of various hematopoietic cell lineages, including lymphocytes. A normal adult has about 5 L of blood with about 2000 lymphocytes/mm³ for a total of about 10¹⁰ lymphocytes. During acute infection, the lymphocyte count increases 4- to 15-fold, giving a total lymphocyte count of 40–50 × 10⁹. Because the immune system cannot sustain such a massive increase in cell numbers for an extended period, the system needs a means to eliminate unneeded activated lymphocytes once the antigenic threat has passed. Activated lymphocytes have been found to express lower levels of Bcl-2 and therefore are more susceptible to the induction of apoptotic death than are naive lymphocytes or

TABLE 2-2 Genes that regulate apoptosis

Gene	Function	Role in apoptosis
<i>bcl-2</i>	Prevents apoptosis	Inhibits
<i>bax</i>	Opposes <i>bcl-2</i>	Promotes
<i>bcl-X_L</i> (<i>bcl-Long</i>)	Prevents apoptosis	Inhibits
<i>bcl-X_S</i> (<i>bcl-Short</i>)	Opposes <i>bcl-X_L</i>	Promotes
caspase (several different ones)	Protease	Promotes
<i>fas</i>	Induces apoptosis	Initiates

memory cells. However, if the lymphocytes continue to be activated by antigen, then the signals received during activation block the apoptotic signal. As antigen levels subside, so does activation of the block and the lymphocytes begin to die by apoptosis (Figure 2-5).

Hematopoietic Stem Cells Can Be Enriched

I. L. Weissman and colleagues developed a novel way of enriching the concentration of mouse hematopoietic stem cells, which normally constitute less than 0.05% of all bone-marrow cells in mice. Their approach relied on the use of antibodies specific for molecules known as **differentiation antigens**, which are expressed only by particular cell types. They exposed bone-marrow samples to antibodies that had been labeled with a fluorescent compound and were specific for the differentiation antigens expressed on the surface of mature red and white blood cells (Figure 2-6). The labeled cells were then removed by flow cytometry with a fluorescence-activated cell sorter (see Chapter 6). After each sorting, the remaining cells were assayed to determine the number needed for restoration of hematopoiesis in a lethally x-irradiated mouse. As the pluripotent stem cells were becoming relatively more numerous in the remaining population, fewer and fewer cells were needed to restore hematopoiesis in this system. Because stem cells do not express differentiation antigens

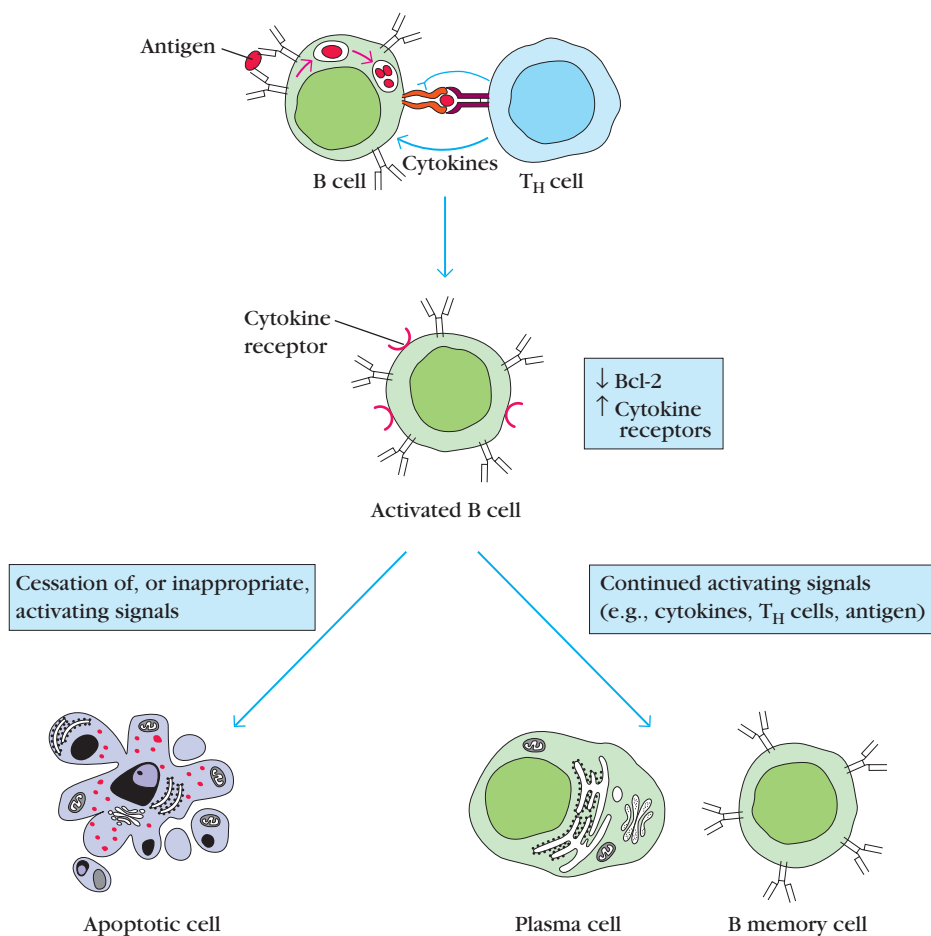


FIGURE 2-5 Regulation of activated B-cell numbers by apoptosis. Activation of B cells induces increased expression of cytokine receptors and decreased expression of Bcl-2. Because Bcl-2 prevents apoptosis, its reduced level in activated B cells is an important factor in

making activated B cells more susceptible to programmed cell death than either naive or memory B cells. A reduction in activating signals quickly leads to destruction of excess activated B cells by apoptosis. Similar processes occur in T cells.

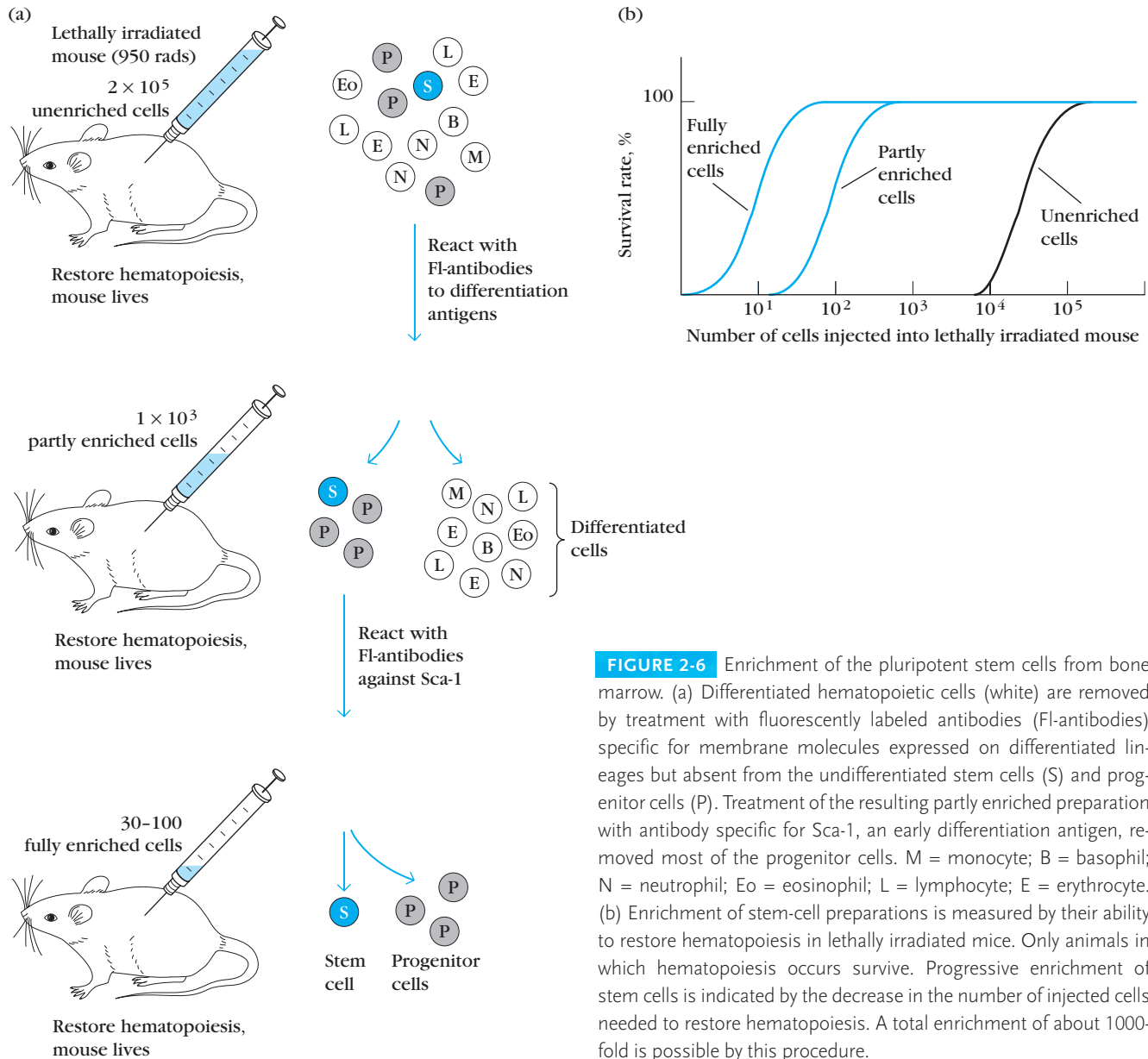


FIGURE 2-6 Enrichment of the pluripotent stem cells from bone marrow. (a) Differentiated hematopoietic cells (white) are removed by treatment with fluorescently labeled antibodies (FI-antibodies) specific for membrane molecules expressed on differentiated lineages but absent from the undifferentiated stem cells (S) and progenitor cells (P). Treatment of the resulting partly enriched preparation with antibody specific for Sca-1, an early differentiation antigen, removed most of the progenitor cells. M = monocyte; B = basophil; N = neutrophil; Eo = eosinophil; L = lymphocyte; E = erythrocyte. (b) Enrichment of stem-cell preparations is measured by their ability to restore hematopoiesis in lethally irradiated mice. Only animals in which hematopoiesis occurs survive. Progressive enrichment of stem cells is indicated by the decrease in the number of injected cells needed to restore hematopoiesis. A total enrichment of about 1000-fold is possible by this procedure.

known to be on developing and mature hematopoietic cells, by removing those hematopoietic cells that express known differentiation antigens, these investigators were able to obtain a 50- to 200-fold enrichment of pluripotent stem cells. To further enrich the pluripotent stem cells, the remaining cells were incubated with various antibodies raised against cells likely to be in the early stages of hematopoiesis. One of these antibodies recognized a differentiation antigen called stem-cell antigen 1 (Sca-1). Treatment with this antibody aided capture of undifferentiated stem cells and yielded a preparation so enriched in pluripotent stem cells that an aliquot containing only 30–100 cells routinely restored hematopoiesis in a lethally x-irradiated mouse, whereas

more than 10^4 nonenriched bone-marrow cells were needed for restoration. Using a variation of this approach, H. Nakauchi and his colleagues have devised procedures that allow them to show that, in 1 out of 5 lethally irradiated mice, a single hematopoietic cell can give rise to both myeloid and lymphoid lineages (Table 2-3).

It has been found that CD34, a marker found on about 1% of hematopoietic cells, while not actually unique to stem cells, is found on a small population of cells that contains stem cells. By exploiting the association of this marker with stem cell populations, it has become possible to routinely enrich preparations of human stem cells. The administration of human-cell populations suitably enriched for CD34⁺ cells

TABLE 2-3 Reconstitution of hematopoiesis by HSCs

Number of enriched HSCs	Number of mice reconstituted (%)
1	9 of 41 (21.9%)
2	5 of 21 (23.8%)
5	9 of 17 (52.9%)
10	10 of 11 (90.9%)
20	4 of 4 (100%)

SOURCE: Adapted from M. Osawa, et al. 1996. *Science* 273:242.

(the “+” indicates that the factor is present on the cell membrane) can reconstitute a patient’s entire hematopoietic system (see Clinical Focus).

A major tool in studies to identify and characterize the human hematopoietic stem cell is the use of **SCID (severe combined immunodeficiency) mice** as in vivo assay systems for the presence and function of HSCs. SCID mice do not have B and T lymphocytes and are unable to mount adaptive immune responses such as those that act in the normal rejection of foreign cells, tissues, and organs. Consequently, these animals do not reject transplanted human cell populations containing HSCs or tissues such as thymus and bone marrow. It is necessary to use immunodeficient mice as surrogate or alternative hosts in human stem-cell research because there is no human equivalent of the irradiated mouse. SCID mice implanted with fragments of human thymus and bone marrow support the differentiation of human hematopoietic stem cells into mature hematopoietic cells. Different subpopulations of CD34⁺ human bone-marrow cells are injected into these SCID-human mice, and the development of various lineages of human cells in the bone-marrow fragment is subsequently assessed. In the absence of human growth factors, only low numbers of granulocyte-macrophage progenitors develop. However, when appropriate cytokines such as erythropoietin and others are administered along with CD34⁺ cells, progenitor and mature cells of the myeloid, lymphoid, and erythroid lineages develop. This system has enabled the study of subpopulations of CD34⁺ cells and the effect of human growth factors on the differentiation of various hematopoietic lineages.

Cells of the Immune System

Lymphocytes are the central cells of the immune system, responsible for adaptive immunity and the immunologic attributes of diversity, specificity, memory, and self/nonself recognition. The other types of white blood cells play impor-

tant roles, engulfing and destroying microorganisms, presenting antigens, and secreting cytokines.

Lymphoid Cells

Lymphocytes constitute 20%–40% of the body’s white blood cells and 99% of the cells in the lymph (Table 2-4). There are approximately 10^{11} (range depending on body size and age: $\sim 10^{10}$ – 10^{12}) lymphocytes in the human body. These lymphocytes continually circulate in the blood and lymph and are capable of migrating into the tissue spaces and lymphoid organs, thereby integrating the immune system to a high degree.

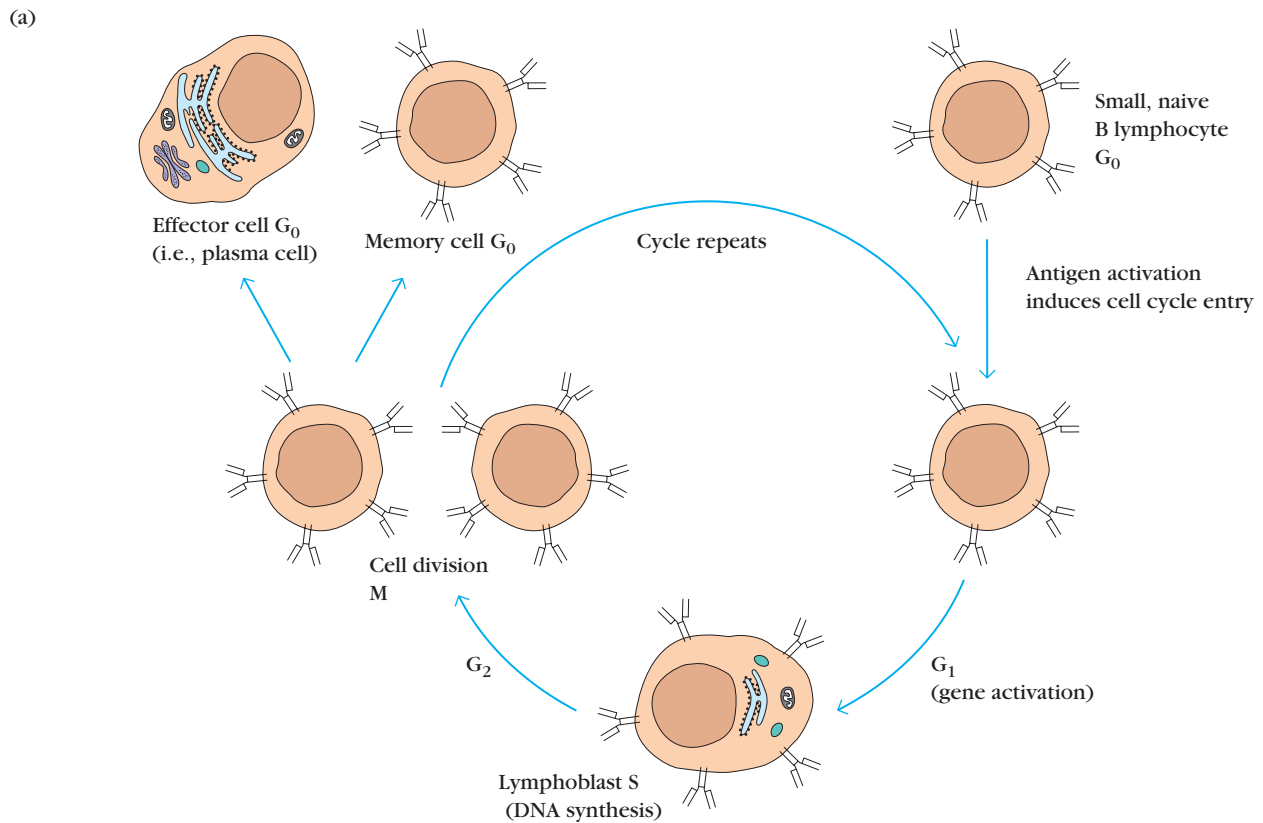
The lymphocytes can be broadly subdivided into three populations—B cells, T cells, and natural killer cells—on the basis of function and cell-membrane components. **Natural killer cells (NK cells)** are large, granular lymphocytes that do not express the set of surface markers typical of B or T cells. Resting B and T lymphocytes are small, motile, nonphagocytic cells, which cannot be distinguished morphologically. B and T lymphocytes that have not interacted with antigen—referred to as **naïve**, or unprimed—are resting cells in the G₀ phase of the cell cycle. Known as small lymphocytes, these cells are only about 6 μm in diameter; their cytoplasm forms a barely discernible rim around the nucleus. Small lymphocytes have densely packed chromatin, few mitochondria, and a poorly developed endoplasmic reticulum and Golgi apparatus. The naïve lymphocyte is generally thought to have a short life span. Interaction of small lymphocytes with antigen, in the presence of certain cytokines discussed later, induces these cells to enter the cell cycle by progressing from G₀ into G₁ and subsequently into S, G₂, and M (Figure 2-7a). As they progress through the cell cycle, lymphocytes enlarge into 15 μm -diameter blast cells, called **lymphoblasts**; these cells have a higher cytoplasm:nucleus ratio and more organelle complexity than small lymphocytes (Figure 2-7b).

Lymphoblasts proliferate and eventually differentiate into **effector cells** or into **memory cells**. Effector cells function in various ways to eliminate antigen. These cells have short life

TABLE 2-4 Normal adult blood-cell counts

Cell type	Cells/mm ³	%
Red blood cells	5.0×10^6	
Platelets	2.5×10^5	
Leukocytes	7.3×10^3	
Neutrophil		50–70
Lymphocyte		20–40
Monocyte		1–6
Eosinophil		1–3
Basophil		<1





(b)

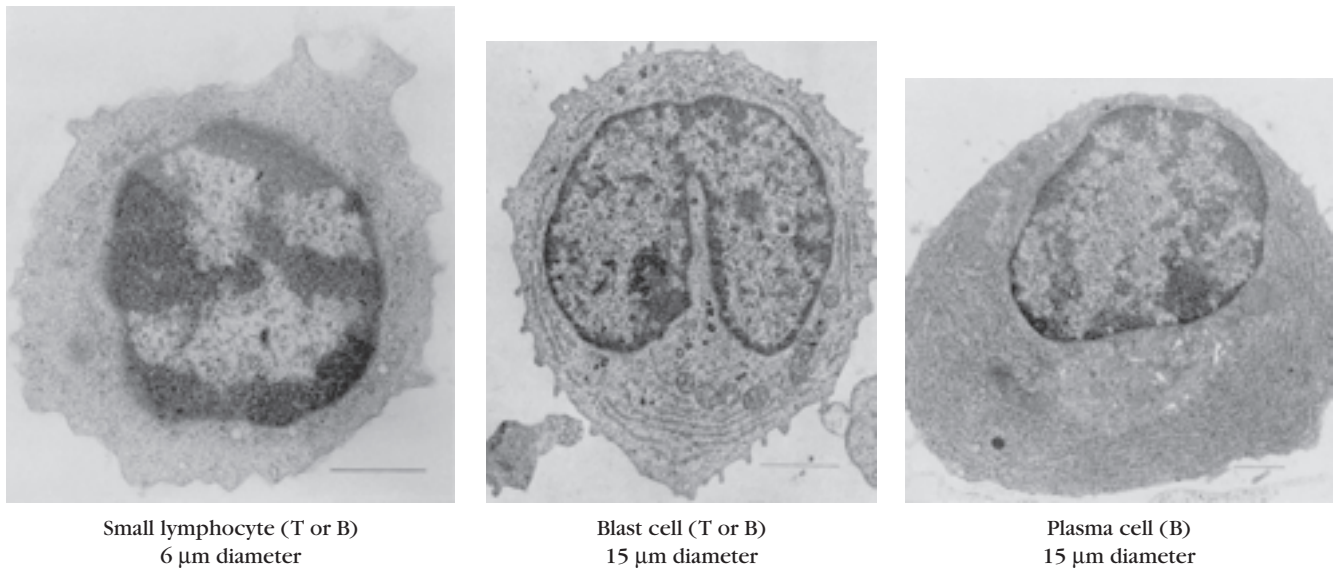


FIGURE 2-7 Fate of antigen-activated small lymphocytes. (a) A small resting (naive or unprimed) lymphocyte resides in the G_0 phase of the cell cycle. At this stage, B and T lymphocytes cannot be distinguished morphologically. After antigen activation, a B or T cell enters the cell cycle and enlarges into a lymphoblast, which undergoes several rounds of cell division and, eventually, generates effector cells and memory cells. Shown here are cells of the B-cell lineage. (b) Electron micrographs of a small lymphocyte (left) showing con-

densed chromatin indicative of a resting cell, an enlarged lymphoblast (center) showing decondensed chromatin, and a plasma cell (right) showing abundant endoplasmic reticulum arranged in concentric circles and a prominent nucleus that has been pushed to a characteristically eccentric position. The three cells are shown at different magnifications. [Micrographs courtesy of Dr. J. R. Goodman, Dept. of Pediatrics, University of California at San Francisco.]



CLINICAL FOCUS

Stem Cells—Clinical Uses and Potential

Stem-cell transplantation holds great promise for the regeneration of diseased, damaged, or defective tissue. Hematopoietic stem cells are already used to restore hematopoietic cells, and their use is described in the clinic below. However, rapid advances in stem-cell research have raised the possibility that other stem-cell types, too, may soon be routinely employed for replacement of other cells and tissues. Two properties of stem cells underlie their utility and promise. They have the capacity to give rise to more differentiated cells, and they are self-renewing, because each division of a stem cell creates at least one stem cell. If stem cells are classified according to their descent and developmental potential, four levels of stem cells can be recognized: totipotent, pluripotent, multipotent, and unipotent.

Totipotent cells can give rise to an entire organism. A fertilized egg, the zygote, is a totipotent cell. In humans the initial divisions of the zygote and its descendants produce cells that are also totipotent. In fact, identical twins, each with its own placenta, develop when totipotent cells separate and develop into genetically identical fetuses. Pluripotent stem cells arise from totipotent cells and can give rise to most but not all of the cell types necessary for fetal development. For example, human pluripotent stem cells can give rise to all of the cells of the body but cannot generate a placenta. Further differentiation of pluripotent stem cells leads to the formation of multipotent and unipotent stem cells. Multipotent stem cells can give rise to only a limited number of cell types, and unipotent cells to a single cell type. Pluripotent cells, called embryonic stem cells, or simply ES cells, can be isolated from early embryos, and for many years it has been possible to grow mouse ES cells as cell

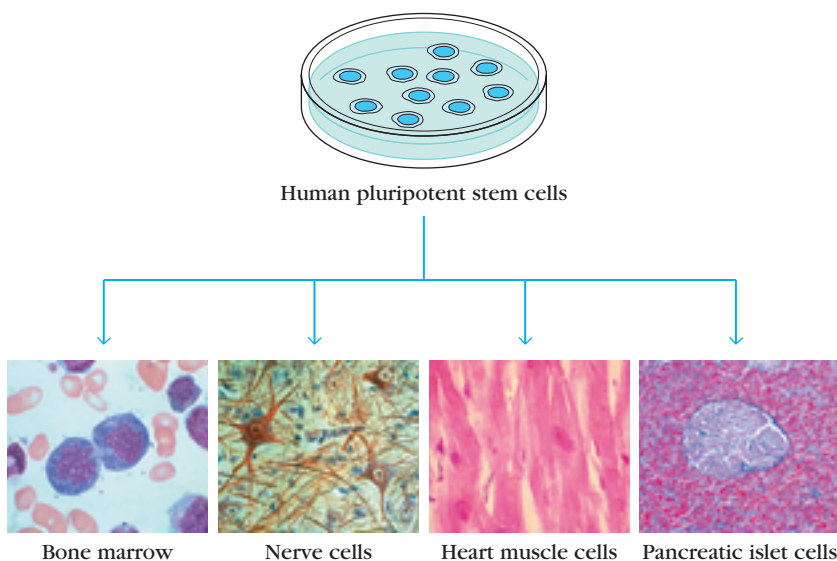
lines in the laboratory. Strikingly, these ES cells can be induced to generate many different types of cells. Mouse ES cells have been shown to give rise to muscle cells, nerve cells, liver cells, pancreatic cells, and, of course, hematopoietic cells.

Recent advances have made it possible to grow lines of human pluripotent cells. This is a development of considerable importance to the understanding of human development, and it has great therapeutic potential. In vitro studies of the factors that determine or influence the development of human pluripotent stem cells along one developmental path as opposed to another will provide considerable insight into the factors that affect the differentiation of cells into specialized types. There is also great interest in exploring the use of pluripotent

stem cells to generate cells and tissues that could be used to replace diseased or damaged ones. Success in this endeavor would be a major advance because transplantation medicine now depends totally upon donated organs and tissues. Unfortunately, the need far exceeds the number of donations and is increasing. Success in deriving practical quantities of cells, tissues, and organs from pluripotent stem cells would provide skin replacement for burn patients, heart muscle cells for those with chronic heart disease, pancreatic islet cells for patients with diabetes, and neurons for use in Parkinson's disease or Alzheimer's disease.

The transplantation of hematopoietic stem cells (HSCs) is an important therapy for patients whose hematopoietic systems must be replaced. It has three major applications:

1. Providing a functional immune system to individuals with a genetically determined immunodeficiency, such as severe



Human pluripotent stem cells can differentiate into a variety of different cell types, some of which are shown here. [Adapted from Stem Cells: A Primer, NIH web site <http://www.nih.gov/news/stemcell/primer.htm>. Micrographs (left to right): Biophoto Associates/Science Source/Photo Researchers; Biophoto Associates/Photo Researchers; AFIP/Science Source/Photo Researchers; Astrid & Hanns-Frieder Michler/Science Photo Library/Photo Researchers.]

combined immunodeficiency (SCID).

2. Replacing a defective hematopoietic system with a functional one to cure some patients who have a life-threatening nonmalignant genetic disorder in hematopoiesis, such as sickle-cell anemia or thalassemia.
3. Restoring the hematopoietic system of cancer patients after treatment with doses of chemotherapeutic agents and radiation so high that they destroy the system. These high-dose regimens can be much more effective at killing tumor cells than are therapies that use more conventional doses of cytotoxic agents. Stem-cell transplantation makes it possible to recover from such drastic treatment. Also, certain cancers, such as some cases of acute myeloid leukemia, can be cured only by destroying the source of the leukemia cells, the patient's own hematopoietic system.

Restoration of the hematopoietic system by transplanting stem cells is facilitated by several important technical considerations. First, HSCs have extraordinary powers of regeneration. Experiments in mice indicate that only a few—perhaps, on occasion, a single HSC—can completely restore the erythroid population and the immune system. In humans it is necessary to administer as little as 10% of a donor's total volume of bone marrow to provide enough HSCs to completely restore the hematopoietic system. Once injected into a vein, HSCs enter the circulation and find their own way to the bone marrow, where they begin the process of engraftment. There is no need for a surgeon to directly inject the cells into bones. In addition, HSCs can be preserved by freezing. This means that hematopoietic cells can be “banked.” After collection, the cells are treated with a cryopreservative, frozen, and then stored for later use. When needed, the frozen preparation is thawed and infused into the patient, where it reconstitutes the hematopoietic system. This cell-freezing technology even makes it pos-

sible for individuals to store their own hematopoietic cells for transplantation to themselves at a later time. Currently, this procedure is used to allow cancer patients to donate cells before undergoing chemotherapy and radiation treatments and then to reconstitute their hematopoietic system from their own stem cells. Hematopoietic stem cells are found in cell populations that display distinctive surface antigens. One of these antigens is CD34, which is present on only a small percentage (~1%) of the cells in adult bone marrow. An antibody specific for CD34 is used to select cells displaying this antigen, producing a population enriched in CD34⁺ stem cells. Various versions of this selection procedure have been used to enrich populations of stem cells from a variety of sources.

Transplantation of stem cell populations may be **autologous** (the recipient is also the donor), **syngeneic** (the donor is genetically identical, i.e., an identical twin of the recipient), or **allogeneic** (the donor and recipient are not genetically identical). In any transplantation procedure, genetic differences between donor and recipient can lead to immune-based rejection reactions. Aside from host rejection of transplanted tissue (host versus graft), lymphocytes in the graft can attack the recipient's tissues, thereby causing **graft-versus-host disease (GVHD)**, a life-threatening affliction. In order to suppress rejection reactions, powerful immunosuppressive drugs must be used. Unfortunately, these drugs have serious side effects, and immunosuppression increases the patient's risk of infection and further growth of tumors. Consequently, HSC transplantation has fewest complications when there is genetic identity between donor and recipient.

At one time, bone-marrow transplantation was the only way to restore the hematopoietic system. However, the essential element of bone-marrow transplantation is really stem-cell transplantation. Fortunately, significant numbers of stem cells can be obtained from other tissues, such as peripheral blood and umbilical-cord blood (“cord blood”). These alternative sources of HSCs are attractive because the

donor does not have to undergo anesthesia and the subsequent highly invasive procedure that extracts bone marrow. Many in the transplantation community believe that peripheral blood will replace marrow as the major source of hematopoietic stem cells for many applications. To obtain HSC-enriched preparations from peripheral blood, agents are used to induce increased numbers of circulating HSCs, and then the HSC-containing fraction is separated from the plasma and red blood cells in a process called leukopheresis. If necessary, further purification can be done to remove T cells and to enrich the CD34⁺ population.

Umbilical cord blood already contains a significant number of hematopoietic stem cells. Furthermore, it is obtained from placental tissue (the “afterbirth”) which is normally discarded. Consequently, umbilical cord blood has become an attractive source of cells for HSC transplantation. Although HSCs from cord blood fail to engraft somewhat more often than do cells from peripheral blood, grafts of cord blood cells produce GVHD less frequently than do marrow grafts, probably because cord blood has fewer mature T cells.

Beyond its current applications in cancer treatment, many researchers feel that autologous stem-cell transplantation will be useful for gene therapy, the introduction of a normal gene to correct a disorder caused by a defective gene. Rapid advances in genetic engineering may soon make gene therapy a realistic treatment for genetic disorders of blood cells, and hematopoietic stem cells are attractive vehicles for such an approach. The therapy would entail removing a sample of hematopoietic stem cells from a patient, inserting a functional gene to compensate for the defective one, and then reinjecting the engineered stem cells into the donor. The advantage of using stem cells in gene therapy is that they are self renewing. Consequently, at least in theory, patients would have to receive only a single injection of engineered stem cells. In contrast, gene therapy with engineered mature lymphocytes or other blood cells would require periodic injections because these cells are not capable of self renewal.

spans, generally ranging from a few days to a few weeks. **Plasma cells**—the antibody-secreting effector cells of the B-cell lineage—have a characteristic cytoplasm that contains abundant endoplasmic reticulum (to support their high rate of protein synthesis) arranged in concentric layers and also many Golgi vesicles (see Figure 2-7). The effector cells of the T-cell lineage include the cytokine-secreting T helper cell (T_H cell) and the T cytotoxic lymphocyte (T_C cell). Some of the progeny of B and T lymphoblasts differentiate into memory cells. The persistence of this population of cells is responsible for life-long immunity to many pathogens. Memory cells look like small lymphocytes but can be distinguished from naive cells by the presence or absence of certain cell-membrane molecules.

Different lineages or maturational stages of lymphocytes can be distinguished by their expression of membrane molecules recognized by particular monoclonal antibodies (antibodies that are specific for a single epitope of an antigen; see Chapter 4 for a description of monoclonal antibodies). All of the monoclonal antibodies that react with a particular membrane molecule are grouped together as a **cluster of differentiation (CD)**. Each new monoclonal antibody that recognizes a leukocyte membrane molecule is analyzed for whether it falls within a recognized CD designation; if it does

not, it is given a new CD designation reflecting a new membrane molecule. Although the CD nomenclature was originally developed for the membrane molecules of human leukocytes, the homologous membrane molecules of other species, such as mice, are commonly referred to by the same CD designations. Table 2-5 lists some common CD molecules (often referred to as CD markers) found on human lymphocytes. However, this is only a partial listing of the more than 200 CD markers that have been described. A complete list and description of known CD markers is in the appendix at the end of this book.

The general characteristics and functions of B and T lymphocytes were described in Chapter 1 and are reviewed briefly in the next sections. These central cells of the immune system will be examined in more detail in later chapters.

B LYMPHOCYTES

The B lymphocyte derived its letter designation from its site of maturation, in the *bursa* of Fabricius in birds; the name turned out to be apt, for *bone marrow* is its major site of maturation in a number of mammalian species, including humans and mice. Mature B cells are definitively distinguished from other lymphocytes by their synthesis and display of membrane-bound immunoglobulin (antibody) molecules,

TABLE 2-5 Common CD markers used to distinguish functional lymphocyte subpopulations

CD designation*	Function	T CELL			
		B cell	T_H	T_C	NK cell
CD2	Adhesion molecule; signal transduction	—	+	+	+
CD3	Signal-transduction element of T-cell receptor	—	+	+	—
CD4	Adhesion molecule that binds to class II MHC molecules; signal transduction	—	+	—	—
			(usually)	(usually)	
CD5	Unknown	+	+	+	—
			(subset)		
CD8	Adhesion molecule that binds to class I MHC molecules; signal transduction	—	—	+	+
			(usually)	(usually)	(variable)
CD16 (Fc γ RIII)	Low-affinity receptor for Fc region of IgG	—	—	—	+
CD21 (CR2)	Receptor for complement (C3d) and Epstein-Barr virus	+	—	—	—
CD28	Receptor for co-stimulatory B7 molecule on antigen-presenting cells	—	+	+	—
CD32 (Fc γ RII)	Receptor for Fc region of IgG	+	—	—	—
CD35 (CR1)	Receptor for complement (C3b)	+	—	—	—
CD40	Signal transduction	+	—	—	—
CD45	Signal transduction	+	+	+	+
CD56	Adhesion molecule	—	—	—	+

*Synonyms are shown in parentheses.

which serve as receptors for antigen. Each of the approximately 1.5×10^5 molecules of antibody on the membrane of a single B cell has an identical binding site for antigen. Among the other molecules expressed on the membrane of mature B cells are the following:

- **B220** (a form of CD45) is frequently used as a marker for B cells and their precursors. However, unlike antibody, it is not expressed uniquely by B-lineage cells.
- **Class II MHC molecules** permit the B cell to function as an antigen-presenting cell (APC).
- **CR1** (CD35) and **CR2** (CD21) are receptors for certain complement products.
- **FcγRII** (CD32) is a receptor for IgG, a type of antibody.
- **B7-1** (CD80) and **B7-2** (CD86) are molecules that interact with CD28 and CTLA-4, important regulatory molecules on the surface of different types of T cells, including T_H cells.
- **CD40** is a molecule that interacts with CD40 ligand on the surface of helper T cells. In most cases this interaction is critical for the survival of antigen-stimulated B cells and for their development into antibody-secreting plasma cells or memory B cells.

Interaction between antigen and the membrane-bound antibody on a mature naive B cell, as well as interactions with T cells and macrophages, selectively induces the activation and differentiation of B-cell clones of corresponding specificity. In this process, the B cell divides repeatedly and differentiates over a 4- to 5-day period, generating a population of plasma cells and memory cells. Plasma cells, which have lower levels of membrane-bound antibody than B cells, synthesize and secrete antibody. All clonal progeny from a given B cell secrete antibody molecules with the same antigen-binding specificity. Plasma cells are terminally differentiated cells, and many die in 1 or 2 weeks.

T LYMPHOCYTES

T lymphocytes derive their name from their site of maturation in the thymus. Like B lymphocytes, these cells have membrane receptors for antigen. Although the antigen-binding T-cell receptor is structurally distinct from immunoglobulin, it does share some common structural features with the immunoglobulin molecule, most notably in the structure of its antigen-binding site. Unlike the membrane-bound antibody on B cells, though, the T-cell receptor (TCR) does not recognize free antigen. Instead the TCR recognizes only antigen that is bound to particular classes of self-molecules. Most T cells recognize antigen only when it is bound to a self-molecule encoded by genes within the major histocompatibility complex (MHC). Thus, as explained in Chapter 1, a fundamental difference between the humoral and cell-mediated branches of the immune system is that the B cell is capable of binding soluble antigen, whereas the T cell

is restricted to binding antigen displayed on self-cells. To be recognized by most T cells, this antigen must be displayed together with MHC molecules on the surface of antigen-presenting cells or on virus-infected cells, cancer cells, and grafts. The T-cell system has developed to eliminate these altered self-cells, which pose a threat to the normal functioning of the body.

Like B cells, T cells express distinctive membrane molecules. All T-cell subpopulations express the T-cell receptor, a complex of polypeptides that includes CD3; and most can be distinguished by the presence of one or the other of two membrane molecules, CD4 and CD8. In addition, most mature T cells express the following membrane molecules:

- **CD28**, a receptor for the co-stimulatory B7 family of molecules present on B cells and other antigen-presenting cells
- **CD45**, a signal-transduction molecule

T cells that express the membrane glycoprotein molecule CD4 are restricted to recognizing antigen bound to class II MHC molecules, whereas T cells expressing CD8, a dimeric membrane glycoprotein, are restricted to recognition of antigen bound to class I MHC molecules. Thus the expression of CD4 versus CD8 corresponds to the MHC restriction of the T cell. In general, expression of CD4 and of CD8 also defines two major functional subpopulations of T lymphocytes. $CD4^+$ T cells generally function as T helper (T_H) cells and are class-II restricted; $CD8^+$ T cells generally function as T cytotoxic (T_C) cells and are class-I restricted. Thus the ratio of T_H to T_C cells in a sample can be approximated by assaying the number of $CD4^+$ and $CD8^+$ T cells. This ratio is approximately 2:1 in normal human peripheral blood, but it may be significantly altered by immunodeficiency diseases, autoimmune diseases, and other disorders.

The classification of $CD4^+$ class II-restricted cells as T_H cells and $CD8^+$ class I-restricted cells as T_C cells is not absolute. Some $CD4^+$ cells can act as killer cells. Also, some T_C cells have been shown to secrete a variety of cytokines and exert an effect on other cells comparable to that exerted by T_H cells. The distinction between T_H and T_C cells, then, is not always clear; there can be ambiguous functional activities. However, because these ambiguities are the exception and not the rule, the generalization of T helper (T_H) cells as being $CD4^+$ and class-II restricted and of T cytotoxic cells (T_C) as being $CD8^+$ and class-I restricted is assumed throughout this text, unless otherwise specified.

T_H cells are activated by recognition of an antigen–class II MHC complex on an antigen-presenting cell. After activation, the T_H cell begins to divide and gives rise to a clone of effector cells, each specific for the same antigen–class II MHC complex. These T_H cells secrete various cytokines, which play a central role in the activation of B cells, T cells, and other cells that participate in the immune response. Changes in the pattern of cytokines produced by T_H cells can change the type of immune response that develops among

other leukocytes. The **T_H1 response** produces a cytokine profile that supports inflammation and activates mainly certain T cells and macrophages, whereas the **T_H2 response** activates mainly B cells and immune responses that depend upon antibodies. T_C cells are activated when they interact with an antigen–class I MHC complex on the surface of an altered self-cell (e.g., a virus-infected cell or a tumor cell) in the presence of appropriate cytokines. This activation, which results in proliferation, causes the T_C cell to differentiate into an effector cell called a **cytotoxic T lymphocyte (CTL)**. In contrast to T_H cells, most CTLs secrete few cytokines. Instead, CTLs acquire the ability to recognize and eliminate altered self-cells.

Another subpopulation of T lymphocytes—called **T suppressor (T_S) cells**—has been postulated. It is clear that some T cells help to suppress the humoral and the cell-mediated branches of the immune system, but the actual isolation and cloning of normal T_S cells is a matter of controversy and dispute among immunologists. For this reason, it is uncertain whether T_S cells do indeed constitute a separate functional subpopulation of T cells. Some immunologists believe that the suppression mediated by T cells observed in some systems is simply the consequence of activities of T_H or T_C subpopulations whose end results are suppressive.

NATURAL KILLER CELLS

The natural killer cell was first described in 1976, when it was shown that the body contains a small population of large, granular lymphocytes that display cytotoxic activity against a wide range of tumor cells in the absence of any previous immunization with the tumor. NK cells were subsequently shown to play an important role in host defense both against tumor cells and against cells infected with some, though not all, viruses. These cells, which constitute 5%–10% of lymphocytes in human peripheral blood, do not express the membrane molecules and receptors that distinguish T- and B-cell lineages. Although NK cells do not have T-cell receptors or immunoglobulin incorporated in their plasma membranes, they can recognize potential target cells in two different ways. In some cases, an NK cell employs NK cell receptors to distinguish abnormalities, notably a reduction in the display of class I MHC molecules and the unusual profile of surface antigens displayed by some tumor cells and cells infected by some viruses. Another way in which NK cells recognize potential target cells depends upon the fact that some tumor cells and cells infected by certain viruses display antigens against which the immune system has made an antibody response, so that antitumor or antiviral antibodies are bound to their surfaces. Because NK cells express CD16, a membrane receptor for the carboxyl-terminal end of the IgG molecule, called the Fc region, they can attach to these antibodies and subsequently destroy the targeted cells. This is an example of a process known as **antibody-dependent cell-mediated cytotoxicity (ADCC)**. The exact mechanism of NK-cell cytotoxicity, the focus of much current experimental study, is described further in Chapter 14.

Several observations suggest that NK cells play an important role in host defense against tumors. For example, in humans the **Chediak-Higashi syndrome**—an autosomal recessive disorder—is associated with impairment in neutrophils, macrophages, and NK cells and an increased incidence of lymphomas. Likewise, mice with an autosomal mutation called *beige* lack NK cells; these mutants are more susceptible than normal mice to tumor growth following injection with live tumor cells.

There has been growing recognition of a cell type, the **NK1-T cell**, that has some of the characteristics of both T cells and NK cells. Like T cells, NK1-T cells have T cell receptors (TCRs). Unlike most T cells, the TCRs of NK1-T cells interact with MHC-like molecules called CD1 rather than with class I or class II MHC molecules. Like NK cells, they have variable levels of CD16 and other receptors typical of NK cells, and they can kill cells. A population of triggered NK1-T cells can rapidly secrete large amounts of the cytokines needed to support antibody production by B cells as well as inflammation and the development and expansion of cytotoxic T cells. Some immunologists view this cell type as a kind of rapid response system that has evolved to provide early help while conventional T_H responses are still developing.

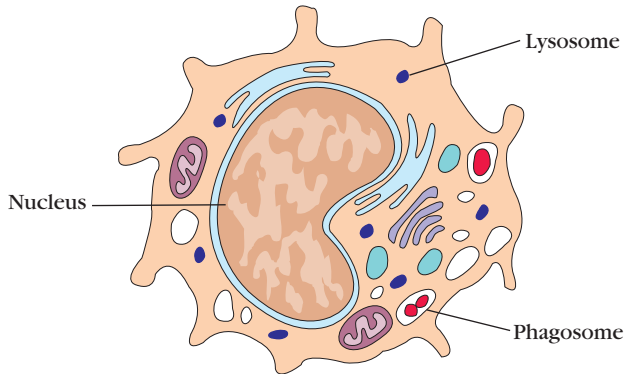
Mononuclear Phagocytes

The mononuclear phagocytic system consists of **monocytes** circulating in the blood and **macrophages** in the tissues (Figure 2-8). During hematopoiesis in the bone marrow, granulocyte-monocyte progenitor cells differentiate into promonocytes, which leave the bone marrow and enter the blood, where they further differentiate into mature monocytes. Monocytes circulate in the bloodstream for about 8 h, during which they enlarge; they then migrate into the tissues and differentiate into specific tissue macrophages or, as discussed later, into dendritic cells.

Differentiation of a monocyte into a tissue macrophage involves a number of changes: The cell enlarges five- to tenfold; its intracellular organelles increase in both number and complexity; and it acquires increased phagocytic ability, produces higher levels of hydrolytic enzymes, and begins to secrete a variety of soluble factors. Macrophages are dispersed throughout the body. Some take up residence in particular tissues, becoming fixed macrophages, whereas others remain motile and are called free, or wandering, macrophages. Free macrophages travel by amoeboid movement throughout the tissues. Macrophage-like cells serve different functions in different tissues and are named according to their tissue location:

- **Alveolar macrophages** in the lung
- **Histiocytes** in connective tissues
- **Kupffer cells** in the liver
- **Mesangial cells** in the kidney

(a) Monocyte



(b) Macrophage

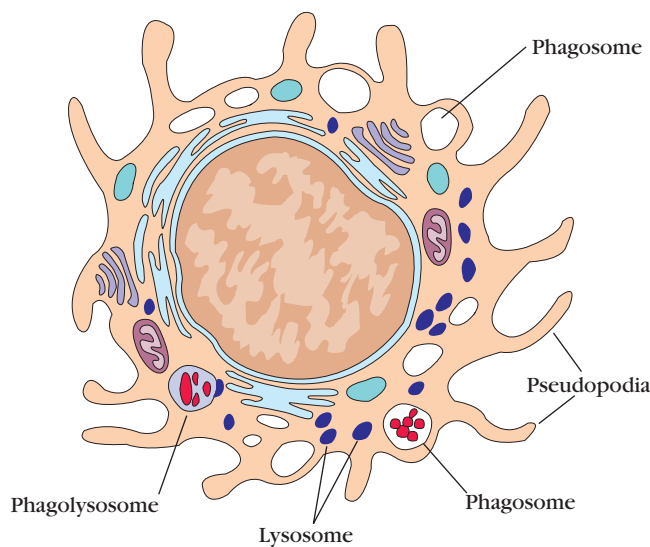


FIGURE 2-8 Typical morphology of a monocyte and a macrophage. Macrophages are five- to tenfold larger than monocytes and contain more organelles, especially lysosomes.

- **Microglial cells** in the brain
- **Osteoclasts** in bone

Although normally in a resting state, macrophages are activated by a variety of stimuli in the course of an immune response. Phagocytosis of particulate antigens serves as an initial activating stimulus. However, macrophage activity can be further enhanced by cytokines secreted by activated T_H cells, by mediators of the inflammatory response, and by components of bacterial cell walls. One of the most potent activators of macrophages is interferon gamma ($IFN-\gamma$) secreted by activated T_H cells.

Activated macrophages are more effective than resting ones in eliminating potential pathogens, because they exhibit greater phagocytic activity, an increased ability to kill ingested microbes, increased secretion of inflammatory mediators, and an increased ability to activate T cells. In addition,

activated macrophages, but not resting ones, secrete various cytotoxic proteins that help them eliminate a broad range of pathogens, including virus-infected cells, tumor cells, and intracellular bacteria. Activated macrophages also express higher levels of class II MHC molecules, allowing them to function more effectively as antigen-presenting cells. Thus, macrophages and T_H cells facilitate each other's activation during the immune response.

PHAGOCYTOSIS

Macrophages are capable of ingesting and digesting exogenous antigens, such as whole microorganisms and insoluble particles, and endogenous matter, such as injured or dead host cells, cellular debris, and activated clotting factors. In the first step in phagocytosis, macrophages are attracted by and move toward a variety of substances generated in an immune response; this process is called **chemotaxis**. The next step in phagocytosis is adherence of the antigen to the macrophage cell membrane. Complex antigens, such as whole bacterial cells or viral particles, tend to adhere well and are readily phagocytosed; isolated proteins and encapsulated bacteria tend to adhere poorly and are less readily phagocytosed. Adherence induces membrane protrusions, called **pseudopodia**, to extend around the attached material (Figure 2-9a). Fusion of the pseudopodia encloses the material within a membrane-bound structure called a **phagosome**, which then enters the endocytic processing pathway (Figure 2-9b). In this pathway, a phagosome moves toward the cell interior, where it fuses with a **lysosome** to form a **phagolysosome**. Lysosomes contain lysozyme and a variety of other hydrolytic enzymes that digest the ingested material. The digested contents of the phagolysosome are then eliminated in a process called **exocytosis** (see Figure 2-9b).

The macrophage membrane has receptors for certain classes of antibody. If an antigen (e.g., a bacterium) is coated with the appropriate antibody, the complex of antigen and antibody binds to antibody receptors on the macrophage membrane more readily than antigen alone and phagocytosis is enhanced. In one study, for example, the rate of phagocytosis of an antigen was 4000-fold higher in the presence of specific antibody to the antigen than in its absence. Thus, antibody functions as an **opsonin**, a molecule that binds to both antigen and macrophage and enhances phagocytosis. The process by which particulate antigens are rendered more susceptible to phagocytosis is called **opsonization**.

ANTIMICROBIAL AND CYTOTOXIC ACTIVITIES

A number of antimicrobial and cytotoxic substances produced by activated macrophages can destroy phagocytosed microorganisms (Table 2-6). Many of the mediators of cytotoxicity listed in Table 2-6 are reactive forms of oxygen.

OXYGEN-DEPENDENT KILLING MECHANISMS Activated phagocytes produce a number of **reactive oxygen intermediates (ROIs)** and **reactive nitrogen intermediates** that have

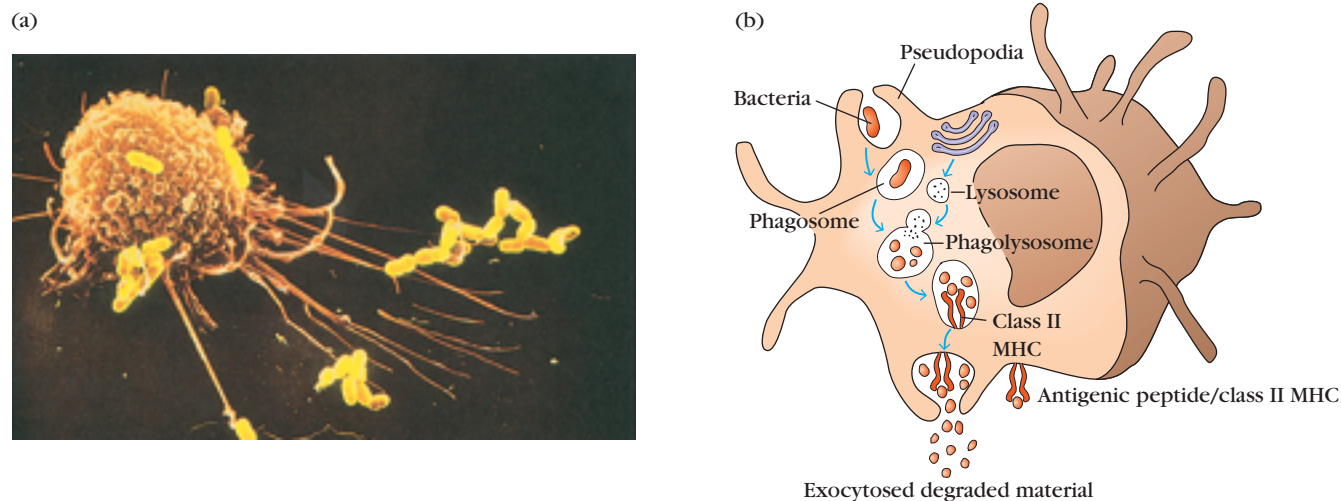
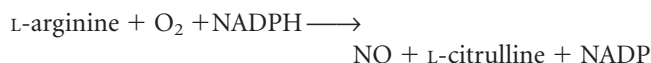


FIGURE 2-9 Macrophages can ingest and degrade particulate antigens, including bacteria. (a) Scanning electron micrograph of a macrophage. Note the long pseudopodia extending toward and making contact with bacterial cells, an early step in phagocytosis. (b) Phagocytosis and processing of exogenous antigen by macrophages.

potent antimicrobial activity. During phagocytosis, a metabolic process known as the **respiratory burst** occurs in activated macrophages. This process results in the activation of a membrane-bound oxidase that catalyzes the reduction of oxygen to superoxide anion, a reactive oxygen intermediate that is extremely toxic to ingested microorganisms. The superoxide anion also generates other powerful oxidizing agents, including hydroxyl radicals and hydrogen peroxide. As the lysosome fuses with the phagosome, the activity of myeloperoxidase produces hypochlorite from hydrogen per-

Most of the products resulting from digestion of ingested material are exocytosed, but some peptide products may interact with class II MHC molecules, forming complexes that move to the cell surface, where they are presented to T_H cells. [Photograph by L. Nilsson, © Boehringer Ingelheim International GmbH.]

oxide and chloride ions. Hypochlorite, the active agent of household bleach, is toxic to ingested microbes. When macrophages are activated with bacterial cell-wall components such as lipopolysaccharide (LPS) or, in the case of mycobacteria, muramyl dipeptide (MDP), together with a T-cell–derived cytokine (IFN- γ), they begin to express high levels of **nitric oxide synthetase (NOS)**, an enzyme that oxidizes L-arginine to yield L-citrulline and **nitric oxide (NO)**, a gas:



Nitric oxide has potent antimicrobial activity; it also can combine with the superoxide anion to yield even more potent antimicrobial substances. Recent evidence suggests that much of the antimicrobial activity of macrophages against bacteria, fungi, parasitic worms, and protozoa is due to nitric oxide and substances derived from it.

TABLE 2-6

Mediators of antimicrobial and cytotoxic activity of macrophages and neutrophils

Oxygen-dependent killing	Oxygen-independent killing
Reactive oxygen intermediates	Defensins
O_2^- (superoxide anion)	Tumor necrosis factor α
$\text{OH}^\cdot$ (hydroxyl radicals)	(macrophage only)
H_2O_2 (hydrogen peroxide)	Lysozyme
ClO^- (hypochlorite anion)	Hydrolytic enzymes
Reactive nitrogen intermediates	
NO (nitric oxide)	
NO_2 (nitrogen dioxide)	
HNO_2 (nitrous acid)	
Others	
NH_2Cl (monochloramine)	

OXYGEN-INDEPENDENT KILLING MECHANISMS Activated macrophages also synthesize **lysozyme** and various hydrolytic enzymes whose degradative activities do not require oxygen. In addition, activated macrophages produce a group of antimicrobial and cytotoxic peptides, commonly known as **defensins**. These molecules are cysteine-rich cationic peptides containing 29–35 amino-acid residues. Each peptide, which contains six invariant cysteines, forms a circular molecule that is stabilized by intramolecular disulfide bonds. These circularized defensin peptides have been shown to form ion-permeable channels in bacterial cell membranes. Defensins can kill a variety of bacteria, including *Staphylococcus aureus*, *Streptococcus pneumoniae*, *Escherichia coli*,

Pseudomonas aeruginosa, and *Haemophilus influenzae*. Activated macrophages also secrete **tumor necrosis factor α (TNF- α)**, a cytokine that has a variety of effects and is cytotoxic for some tumor cells.

ANTIGEN PROCESSING AND PRESENTATION

Although most of the antigen ingested by macrophages is degraded and eliminated, experiments with radiolabeled antigens have demonstrated the presence of antigen peptides on the macrophage membrane. As depicted in Figure 2-9b, phagocytosed antigen is digested within the endocytic processing pathway into peptides that associate with class II MHC molecules; these peptide–class II MHC complexes then move to the macrophage membrane. Activation of macrophages induces increased expression of both class II MHC molecules and the co-stimulatory B7 family of membrane molecules, thereby rendering the macrophages more effective in activating T_H cells. This processing and presentation of antigen, examined in detail in Chapter 7, are critical to T_H-cell activation, a central event in the development of both humoral and cell-mediated immune responses.

SECRETION OF FACTORS

A number of important proteins central to development of immune responses are secreted by activated macrophages (Table 2-7). These include a collection of cytokines, such as **interleukin 1 (IL-1)**, TNF- α and **interleukin 6 (IL-6)**, that promote inflammatory responses. Typically, each of these agents has a variety of effects. For example, IL-1 activates lymphocytes; and IL-1, IL-6, and TNF- α promote fever by affecting the thermoregulatory center in the hypothalamus.

TABLE 2-7

Some factors secreted by activated macrophages

Factor	Function
Interleukin 1 (IL-1)	Promotes inflammatory responses and fever
Interleukin 6 (IL-6) } TNF- α	Promote innate immunity and elimination of pathogens
Complement proteins	Promote inflammatory response and elimination of pathogens
Hydrolytic enzymes	Promote inflammatory response
Interferon alpha (IFN- α)	Activates cellular genes, resulting in the production of proteins that confer an antiviral state on the cell
Tumor necrosis factor (TNF- α)	Kills tumor cells
GM-CSF } G-CSF } M-CSF }	Promote inducible hematopoiesis

Activated macrophages secrete a variety of factors involved in the development of an inflammatory response. The **complement proteins** are a group of proteins that assist in eliminating foreign pathogens and in promoting the ensuing inflammatory reaction. The major site of synthesis of complement proteins is the liver, although these proteins are also produced in macrophages. The hydrolytic enzymes contained within the lysosomes of macrophages also can be secreted when the cells are activated. The buildup of these enzymes within the tissues contributes to the inflammatory response and can, in some cases, contribute to extensive tissue damage. Activated macrophages also secrete soluble factors, such as TNF- α , that can kill a variety of cells. The secretion of these cytotoxic factors has been shown to contribute to tumor destruction by macrophages. Finally, as mentioned earlier, activated macrophages secrete a number of cytokines that stimulate inducible hematopoiesis.

Granulocytic Cells

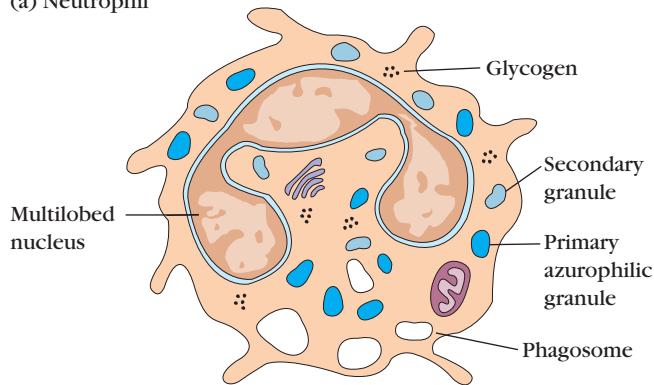
The **granulocytes** are classified as neutrophils, eosinophils, or basophils on the basis of cellular morphology and cytoplasmic staining characteristics (Figure 2-10). The **neutrophil** has a multilobed nucleus and a granulated cytoplasm that stains with both acid and basic dyes; it is often called a polymorphonuclear leukocyte (PMN) for its multilobed nucleus. The **eosinophil** has a bilobed nucleus and a granulated cytoplasm that stains with the acid dye eosin red (hence its name). The **basophil** has a lobed nucleus and heavily granulated cytoplasm that stains with the basic dye methylene blue. Both neutrophils and eosinophils are phagocytic, whereas basophils are not. Neutrophils, which constitute 50%–70% of the circulating white blood cells, are much more numerous than eosinophils (1%–3%) or basophils (<1%).

NEUTROPHILS

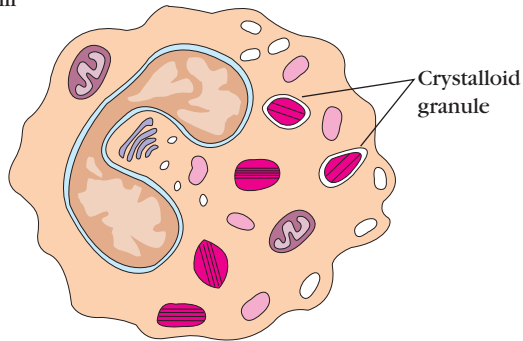
Neutrophils are produced by hematopoiesis in the bone marrow. They are released into the peripheral blood and circulate for 7–10 h before migrating into the tissues, where they have a life span of only a few days. In response to many types of infections, the bone marrow releases more than the usual number of neutrophils and these cells generally are the first to arrive at a site of inflammation. The resulting transient increase in the number of circulating neutrophils, called **leukocytosis**, is used medically as an indication of infection.

Movement of circulating neutrophils into tissues, called **extravasation**, takes several steps: the cell first adheres to the vascular endothelium, then penetrates the gap between adjacent endothelial cells lining the vessel wall, and finally penetrates the vascular basement membrane, moving out into the tissue spaces. (This process is described in detail in Chapter 15.) A number of substances generated in an inflammatory reaction serve as **chemotactic factors** that promote accumulation of neutrophils at an inflammatory site. Among these chemotactic factors are some of the complement

(a) Neutrophil



(b) Eosinophil



(c) Basophil

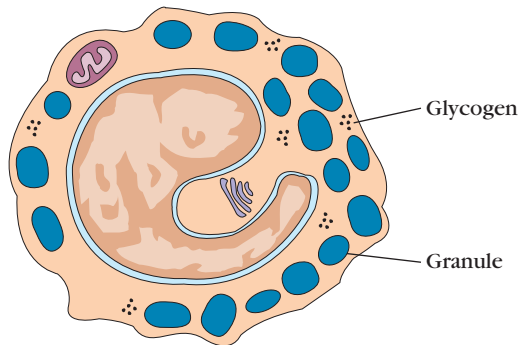


FIGURE 2-10 Drawings showing typical morphology of granulocytes. Note differences in the shape of the nucleus and in the number and shape of cytoplasmic granules.

components, components of the blood-clotting system, and several cytokines secreted by activated T_H cells and macrophages.

Like macrophages, neutrophils are active phagocytic cells. Phagocytosis by neutrophils is similar to that described for macrophages, except that the lytic enzymes and bactericidal substances in neutrophils are contained within primary and secondary granules (see Figure 2-10a). The larger, denser primary granules are a type of lysosome containing peroxidase, lysozyme, and various hydrolytic enzymes. The smaller secondary granules contain collagenase, lactoferrin, and lysozyme. Both primary and secondary granules fuse with phagosomes, whose contents are then digested and eliminated much as they are in macrophages.

Neutrophils also employ both oxygen-dependent and oxygen-independent pathways to generate antimicrobial substances. Neutrophils are in fact much more likely than macrophages to kill ingested microorganisms. Neutrophils exhibit a larger respiratory burst than macrophages and consequently are able to generate more reactive oxygen intermediates and reactive nitrogen intermediates (see Table 2-6). In addition, neutrophils express higher levels of defensins than macrophages do.

EOSINOPHILS

Eosinophils, like neutrophils, are motile phagocytic cells that can migrate from the blood into the tissue spaces. Their phagocytic role is significantly less important than that of neutrophils, and it is thought that they play a role in the defense against parasitic organisms (see Chapter 17). The secreted contents of eosinophilic granules may damage the parasite membrane.

BASOPHILS

Basophils are nonphagocytic granulocytes that function by releasing pharmacologically active substances from their cytoplasmic granules. These substances play a major role in certain allergic responses.

MAST CELLS

Mast-cell precursors, which are formed in the bone marrow by hematopoiesis, are released into the blood as undifferentiated cells; they do not differentiate until they leave the blood and enter the tissues. Mast cells can be found in a wide variety of tissues, including the skin, connective tissues of various organs, and mucosal epithelial tissue of the respiratory, genitourinary, and digestive tracts. Like circulating basophils, these cells have large numbers of cytoplasmic granules that contain histamine and other pharmacologically active substances. Mast cells, together with blood basophils, play an important role in the development of allergies.

DENDRITIC CELLS

The **dendritic cell (DC)** acquired its name because it is covered with long membrane extensions that resemble the dendrites of nerve cells. Dendritic cells can be difficult to isolate because the conventional procedures for cell isolation tend to damage their long extensions. The development of isolation techniques that employ enzymes and gentler dispersion has facilitated isolation of these cells for study *in vitro*. There are many types of dendritic cells, although most mature dendritic cells have the same major function, the presentation of antigen to T_H cells. Four types of dendritic cells are known: Langerhans cells, interstitial dendritic cells, myeloid cells, and lymphoid dendritic cells. Each arises from hematopoietic stem cells via different pathways and in different locations. Figure 2-11 shows that they descend through both the myeloid and lymphoid lineages. Despite their differences,

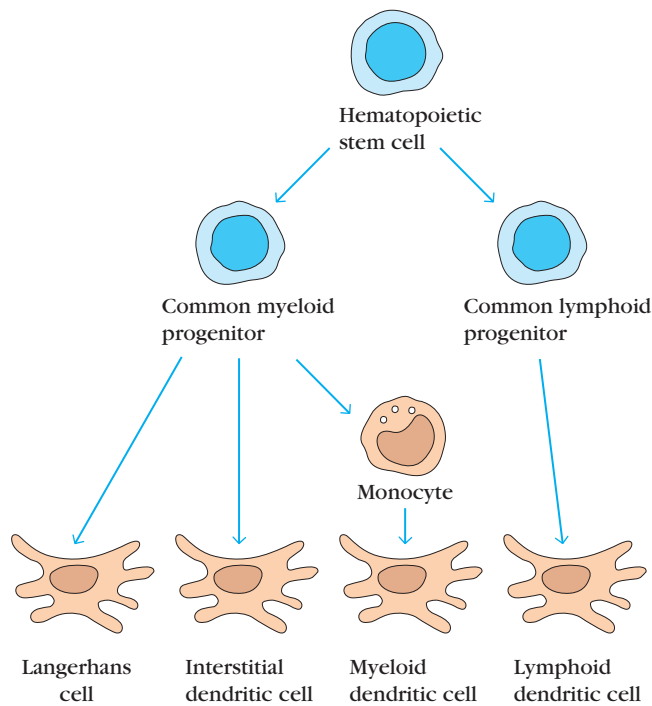


FIGURE 2-11 Dendritic cells arise from both the myeloid and lymphoid lineages. The myeloid pathway that gives rise to the monocyte/macrophage cell type also gives rise to dendritic cells. Some dendritic cells also arise from the lymphoid lineage. These considerations do not apply to follicular dendritic cells, which are not derived from bone marrow.

they all constitutively express high levels of both class II MHC molecules and members of the co-stimulatory B7 family. For this reason, they are more potent antigen-presenting cells than macrophages and B cells, both of which need to be activated before they can function as antigen-presenting cells (APCs). Immature or precursor forms of each of these types of dendritic cells acquire antigen by phagocytosis or endocytosis; the antigen is processed, and mature dendritic cells present it to T_H cells. Following microbial invasion or during inflammation, mature and immature forms of Langerhans cells and interstitial dendritic cells migrate into draining lymph nodes, where they make the critical presentation of antigen to T_H cells that is required for the initiation of responses by those key cells.

Another type of dendritic cell, the **follicular dendritic cell** (Figure 2-12), does not arise in bone marrow and has a different function from the antigen-presenting dendritic cells described above. Follicular dendritic cells do not express class II MHC molecules and therefore do not function as antigen-presenting cells for T_H -cell activation. These dendritic cells were named for their exclusive location in organized structures of the lymph node called lymph follicles, which are rich in B cells. Although they do not express class II molecules, follicular dendritic cells express high levels of membrane receptors for antibody, which allows the binding of antigen-an-

tibody complexes. The interaction of B cells with this bound antigen can have important effects on B cell responses.

Organs of the Immune System

A number of morphologically and functionally diverse organs and tissues have various functions in the development of immune responses. These can be distinguished by function as the **primary** and **secondary lymphoid organs** (Figure 2-13). The thymus and bone marrow are the primary (or central) lymphoid organs, where maturation of lymphocytes takes place. The lymph nodes, spleen, and various mucosal-associated lymphoid tissues (MALT) such as gut-associated lymphoid tissue (GALT) are the secondary (or peripheral) lymphoid organs, which trap antigen and provide sites for mature lymphocytes to interact with that antigen. In addition, **tertiary lymphoid tissues**, which normally contain fewer lymphoid cells than secondary lymphoid organs, can import lymphoid cells during an inflammatory response. Most prominent of these are cutaneous-associated lymphoid tissues. Once mature lymphocytes have been generated in the primary lymphoid organs, they circulate in the blood and **lymphatic system**, a network of vessels that collect fluid that has escaped into the tissues from capillaries of the circulatory system and ultimately return it to the blood.

Primary Lymphoid Organs

Immature lymphocytes generated in hematopoiesis mature and become committed to a particular antigenic specificity within the primary lymphoid organs. Only after a lympho-

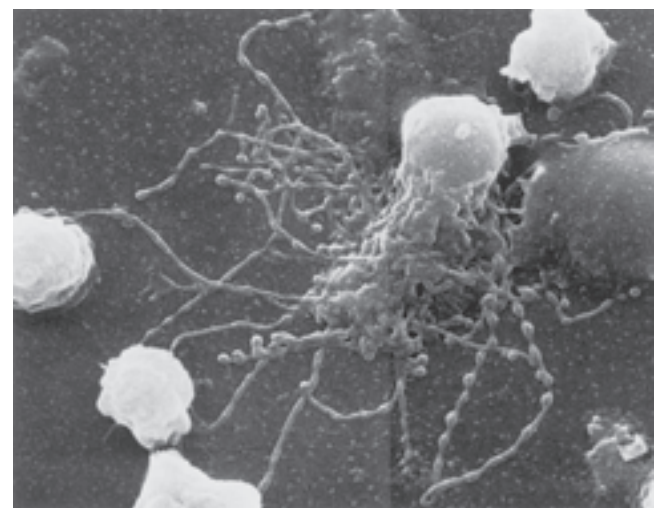


FIGURE 2-12 Scanning electron micrograph of follicular dendritic cells showing long, beaded dendrites. The beads are coated with antigen-antibody complexes. The dendrites emanate from the cell body. [From A. K. Szakal et al., 1985, *J. Immunol.* **134**:1353; © 1996 by American Association of Immunologists, reprinted with permission.]

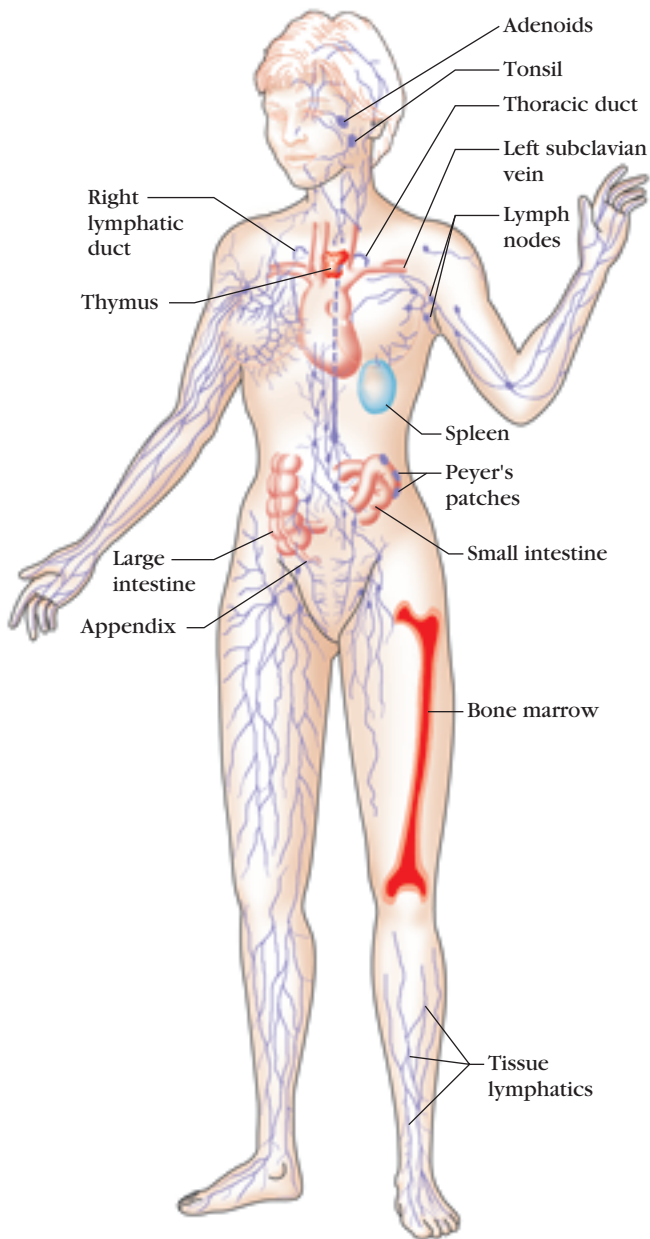


FIGURE 2-13 The human lymphoid system. The primary organs (bone marrow and thymus) are shown in red; secondary organs and tissues, in blue. These structurally and functionally diverse lymphoid organs and tissues are interconnected by the blood vessels (not shown) and lymphatic vessels (purple) through which lymphocytes circulate. Only one bone is shown, but all major bones contain marrow and thus are part of the lymphoid system. [Adapted from H. Lodish et al., 1995, *Molecular Cell Biology*, 3rd ed., Scientific American Books.]

cyte has matured within a primary lymphoid organ is the cell **immunocompetent** (capable of mounting an immune response). T cells arise in the **thymus**, and in many mammals—humans and mice for example—B cells originate in **bone marrow**.

THYMUS

The thymus is the site of T-cell development and maturation. It is a flat, bilobed organ situated above the heart. Each lobe is surrounded by a capsule and is divided into lobules, which are separated from each other by strands of connective tissue called trabeculae. Each lobule is organized into two compartments: the outer compartment, or *cortex*, is densely packed with immature T cells, called thymocytes, whereas the inner compartment, or *medulla*, is sparsely populated with thymocytes.

Both the cortex and medulla of the thymus are crisscrossed by a three-dimensional stromal-cell network composed of epithelial cells, dendritic cells, and macrophages, which make up the framework of the organ and contribute to the growth and maturation of thymocytes. Many of these stromal cells interact physically with the developing thymocytes (Figure 2-14). Some thymic epithelial cells in the outer cortex, called **nurse cells**, have long membrane extensions that surround as many as 50 thymocytes, forming large multicellular complexes. Other cortical epithelial cells have long interconnecting cytoplasmic extensions that form a network and have been shown to interact with numerous thymocytes as they traverse the cortex.

The function of the thymus is to generate and select a repertoire of T cells that will protect the body from infection. As thymocytes develop, an enormous diversity of T-cell receptors is generated by a random process (see Chapter 9) that produces some T cells with receptors capable of recognizing antigen-MHC complexes. However, most of the T-cell receptors produced by this random process are incapable of recognizing antigen-MHC complexes and a small portion react with combinations of self antigen-MHC complexes. Using mechanisms that are discussed in Chapter 10, the thymus induces the death of those T cells that cannot recognize antigen-MHC complexes and those that react with self-antigen-MHC and pose a danger of causing autoimmune disease. More than 95% of all thymocytes die by apoptosis in the thymus without ever reaching maturity.

THE THYMUS AND IMMUNE FUNCTION The role of the thymus in immune function can be studied in mice by examining the effects of neonatal thymectomy, a procedure in which the thymus is surgically removed from newborn mice. These thymectomized mice show a dramatic decrease in circulating lymphocytes of the T-cell lineage and an absence of cell-mediated immunity. Other evidence of the importance of the thymus comes from studies of a congenital birth defect in humans (**DiGeorge's syndrome**) and in certain mice (**nude mice**) in which the thymus fails to develop. In both cases, there is an absence of circulating T cells and of cell-mediated immunity and an increase in infectious disease.

Aging is accompanied by a decline in thymic function. This decline may play some role in the decline in immune function during aging in humans and mice. The thymus reaches its maximal size at puberty and then atrophies, with a significant decrease in both cortical and medullary cells and

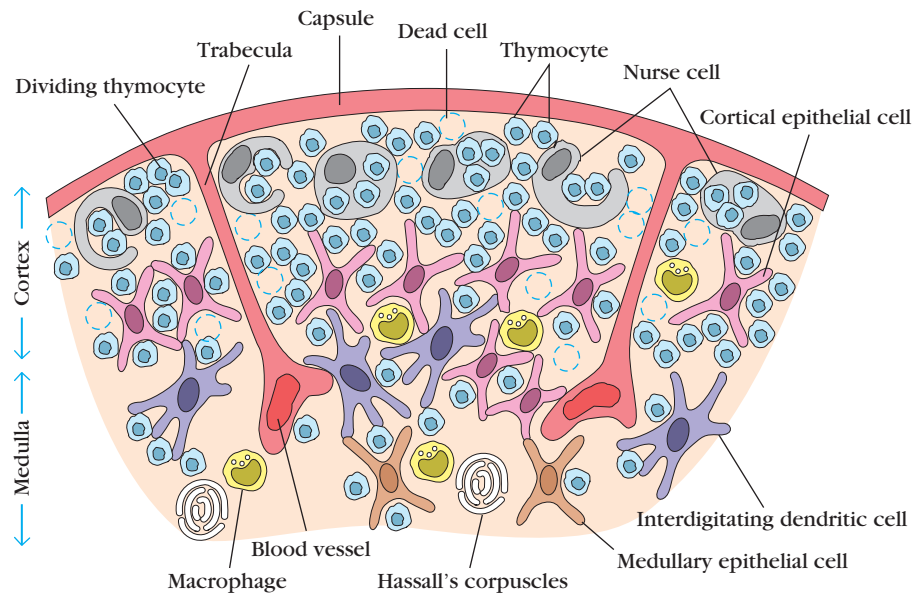


FIGURE 2-14 Diagrammatic cross section of a portion of the thymus, showing several lobules separated by connective tissue strands (trabeculae). The densely populated outer cortex is thought to contain many immature thymocytes (blue), which undergo rapid proliferation coupled with an enormous rate of cell death. Also present in the outer cortex are thymic nurse cells (gray), which are specialized epithelial cells with long membrane extensions that surround as many as 50 thymocytes. The medulla is sparsely populated and is thought to contain thymocytes that are more mature. During their

stay within the thymus, thymocytes interact with various stromal cells, including cortical epithelial cells (light red), medullary epithelial cells (tan), interdigitating dendritic cells (purple), and macrophages (yellow). These cells produce thymic hormones and express high levels of class I and class II MHC molecules. Hassall's corpuscles, found in the medulla, contain concentric layers of degenerating epithelial cells. [Adapted, with permission, from W. van Ewijk, 1991, *Annu. Rev. Immunol.* **9**:591, © 1991 by Annual Reviews.]

an increase in the total fat content of the organ. Whereas the average weight of the thymus is 70 g in infants, its age-dependent involution leaves an organ with an average weight of only 3 g in the elderly (Figure 2-15).

A number of experiments have been designed to look at the effect of age on the immune function of the thymus. In one experiment, the thymus from a 1-day-old or 33-month-old mouse was grafted into thymectomized adults. (For most laboratory mice, 33 months is very old.) Mice receiving the newborn thymus graft showed a significantly larger improvement in immune function than mice receiving the 33-month-old thymus.

BONE MARROW

In humans and mice, bone marrow is the site of B-cell origin and development. Arising from lymphoid progenitors, immature B cells proliferate and differentiate within the bone marrow, and stromal cells within the bone marrow interact directly with the B cells and secrete various cytokines that are required for development. Like thymic selection during T-cell maturation, a selection process within the bone marrow eliminates B cells with self-reactive antibody receptors. This process is explained in detail in Chapter 11. Bone marrow is not the site of B-cell development in all species. In birds, a lymphoid organ called the bursa of Fabricius, a lymphoid

tissue associated with the gut, is the primary site of B-cell maturation. In mammals such as primates and rodents, there is no bursa and no single counterpart to it as a primary lymphoid organ. In cattle and sheep, the primary lymphoid tissue hosting the maturation, proliferation, and diversification of B cells early in gestation is the fetal spleen. Later in gestation, this function is assumed by a patch of tissue embedded

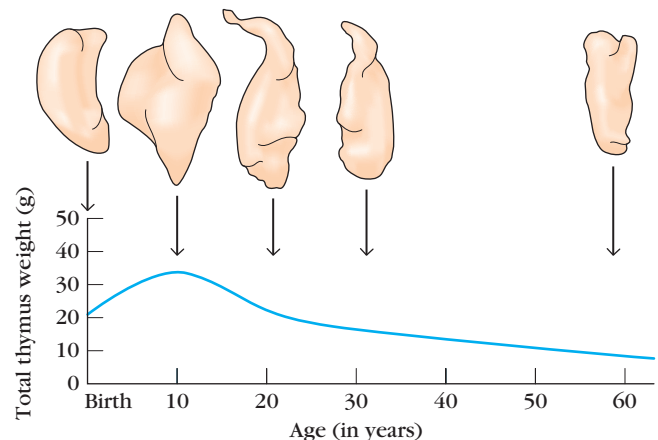


FIGURE 2-15 Changes in the thymus with age. The thymus decreases in size and cellularity after puberty.

in the wall of the intestine called the ileal Peyer's patch, which contains a large number ($>10^{10}$) B cells. The rabbit, too, uses gut-associated tissues such as the appendix as primary lymphoid tissue for important steps in the proliferation and diversification of B cells.

Lymphatic System

As blood circulates under pressure, its fluid component (**plasma**) seeps through the thin wall of the capillaries into the surrounding tissue. Much of this fluid, called **interstitial fluid**, returns to the blood through the capillary membranes. The remainder of the interstitial fluid, now called **lymph**, flows from the spaces in connective tissue into a network of tiny open lymphatic capillaries and then into a series of pro-

gressively larger collecting vessels called **lymphatic vessels** (Figure 2-16).

The largest lymphatic vessel, the **thoracic duct**, empties into the left subclavian vein near the heart (see Figure 2-13). In this way, the lymphatic system captures fluid lost from the blood and returns it to the blood, thus ensuring steady-state levels of fluid within the circulatory system. The heart does not pump the lymph through the lymphatic system; instead the flow of lymph is achieved as the lymph vessels are squeezed by movements of the body's muscles. A series of one-way valves along the lymphatic vessels ensures that lymph flows only in one direction.

When a foreign antigen gains entrance to the tissues, it is picked up by the lymphatic system (which drains all the tissues of the body) and is carried to various organized lymphoid tissues such as lymph nodes, which trap the foreign antigen. As lymph passes from the tissues to lymphatic vessels, it becomes progressively enriched in lymphocytes. Thus, the lymphatic system also serves as a means of transporting lymphocytes and antigen from the connective tissues to organized lymphoid tissues where the lymphocytes may interact with the trapped antigen and undergo activation.

Secondary Lymphoid Organs

Various types of organized lymphoid tissues are located along the vessels of the lymphatic system. Some lymphoid tissue in the lung and lamina propria of the intestinal wall consists of diffuse collections of lymphocytes and macrophages. Other lymphoid tissue is organized into structures called lymphoid follicles, which consist of aggregates of lymphoid and nonlymphoid cells surrounded by a network of draining lymphatic capillaries. Until it is activated by antigen, a lymphoid follicle—called a **primary follicle**—comprises a network of follicular dendritic cells and small resting B cells. After an antigenic challenge, a primary follicle becomes a larger **secondary follicle**—a ring of concentrically packed B lymphocytes surrounding a center (the **germinal center**) in which one finds a focus of proliferating B lymphocytes and an area that contains nondividing B cells, and some helper T cells interspersed with macrophages and follicular dendritic cells (Figure 2-17).

Most antigen-activated B cells divide and differentiate into antibody-producing plasma cells in lymphoid follicles, but only a few B cells in the antigen-activated population find their way into germinal centers. Those that do undergo one or more rounds of cell division, during which the genes that encode their antibodies mutate at an unusually high rate. Following the period of division and mutation, there is a rigorous selection process in which more than 90% of these B cells die by apoptosis. In general, those B cells producing antibodies that bind antigen more strongly have a much better chance of surviving than do their weaker companions. The small number of B cells that survive the germinal center's rigorous selection differentiate into plasma cells or memory

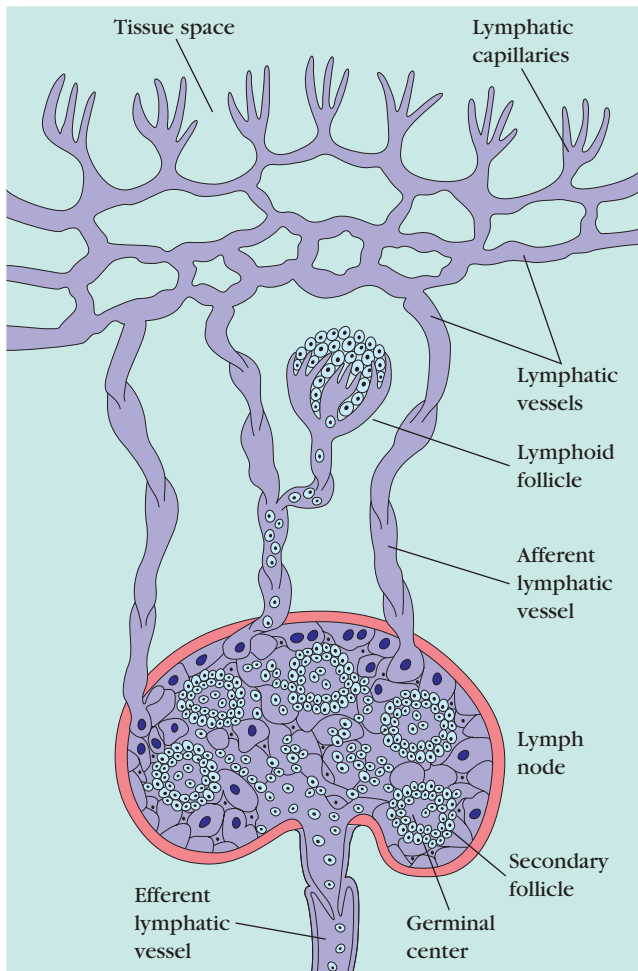


FIGURE 2-16 Lymphatic vessels. Small lymphatic capillaries opening into the tissue spaces pick up interstitial tissue fluid and carry it into progressively larger lymphatic vessels, which carry the fluid, now called lymph, into regional lymph nodes. As lymph leaves the nodes, it is carried through larger efferent lymphatic vessels, which eventually drain into the circulatory system at the thoracic duct or right lymph duct (see Figure 2-13).

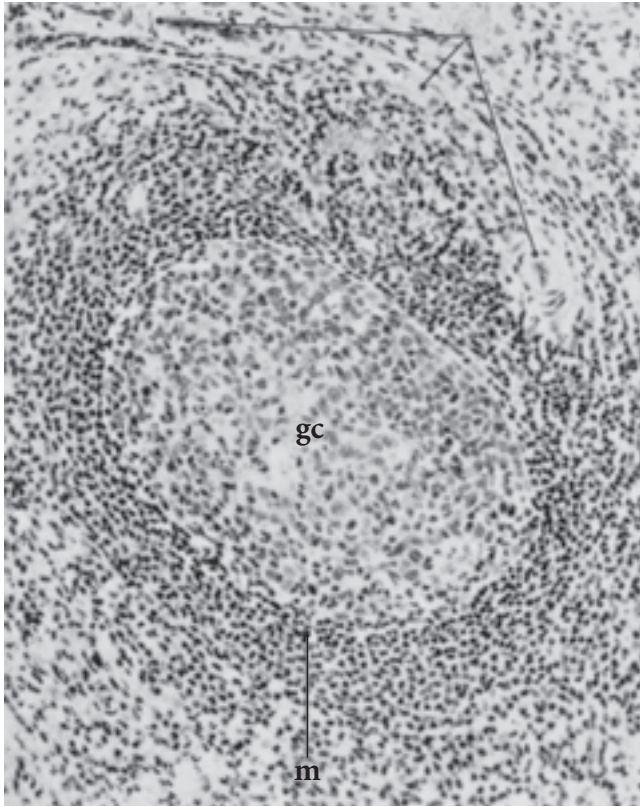


FIGURE 2-17 A secondary lymphoid follicle consisting of a large germinal center (gc) surrounded by a dense mantle (m) of small lymphocytes. [From W. Bloom and D. W. Fawcett, 1975, *Textbook of Histology*, 10th ed., © 1975 by W. B. Saunders Co.]

cells and emerge. The process of B-cell proliferation, mutation, and selection in germinal centers is described more fully in Chapter 11.

Lymph nodes and the **spleen** are the most highly organized of the secondary lymphoid organs; they comprise not only lymphoid follicles, but additional distinct regions of T-cell and B-cell activity, and they are surrounded by a fibrous capsule. Less-organized lymphoid tissue, collectively called mucosal-associated lymphoid tissue (MALT), is found in various body sites. MALT includes Peyer's patches (in the small intestine), the tonsils, and the appendix, as well as numerous lymphoid follicles within the lamina propria of the intestines and in the mucous membranes lining the upper airways, bronchi, and genital tract.

LYMPH NODES

Lymph nodes are the sites where immune responses are mounted to antigens in lymph. They are encapsulated bean-shaped structures containing a reticular network packed with lymphocytes, macrophages, and dendritic cells. Clustered at junctions of the lymphatic vessels, lymph nodes are

the first organized lymphoid structure to encounter antigens that enter the tissue spaces. As lymph percolates through a node, any particulate antigen that is brought in with the lymph will be trapped by the cellular network of phagocytic cells and dendritic cells (follicular and interdigitating). The overall architecture of a lymph node supports an ideal microenvironment for lymphocytes to effectively encounter and respond to trapped antigens.

Morphologically, a lymph node can be divided into three roughly concentric regions: the cortex, the paracortex, and the medulla, each of which supports a distinct microenvironment (Figure 2-18). The outermost layer, the **cortex**, contains lymphocytes (mostly B cells), macrophages, and follicular dendritic cells arranged in primary follicles. After antigenic challenge, the primary follicles enlarge into secondary follicles, each containing a germinal center. In children with B-cell deficiencies, the cortex lacks primary follicles and germinal centers. Beneath the cortex is the **paracortex**, which is populated largely by T lymphocytes and also contains interdigitating dendritic cells thought to have migrated from tissues to the node. These interdigitating dendritic cells express high levels of class II MHC molecules, which are necessary for presenting antigen to T_H cells. Lymph nodes taken from neonatally thymectomized mice have unusually few cells in the paracortical region; the paracortex is therefore sometimes referred to as a **thymus-dependent area** in contrast to the cortex, which is a **thymus-independent area**. The innermost layer of a lymph node, the **medulla**, is more sparsely populated with lymphoid-lineage cells; of those present, many are plasma cells actively secreting antibody molecules.

As antigen is carried into a regional node by the lymph, it is trapped, processed, and presented together with class II MHC molecules by interdigitating dendritic cells in the paracortex, resulting in the activation of T_H cells. The initial activation of B cells is also thought to take place within the T-cell-rich paracortex. Once activated, T_H and B cells form small foci consisting largely of proliferating B cells at the edges of the paracortex. Some B cells within the foci differentiate into plasma cells secreting IgM and IgG. These foci reach maximum size within 4–6 days of antigen challenge. Within 4–7 days of antigen challenge, a few B cells and T_H cells migrate to the primary follicles of the cortex. It is not known what causes this migration. Within a primary follicle, cellular interactions between follicular dendritic cells, B cells, and T_H cells take place, leading to development of a secondary follicle with a central germinal center. Some of the plasma cells generated in the germinal center move to the medullary areas of the lymph node, and many migrate to bone marrow.

Afferent lymphatic vessels pierce the capsule of a lymph node at numerous sites and empty lymph into the subcapsular sinus (see Figure 2-18b). Lymph coming from the tissues percolates slowly inward through the cortex, paracortex, and medulla, allowing phagocytic cells and dendritic cells to trap any bacteria or particulate material (e.g., antigen-antibody complexes) carried by the lymph. After infection or the

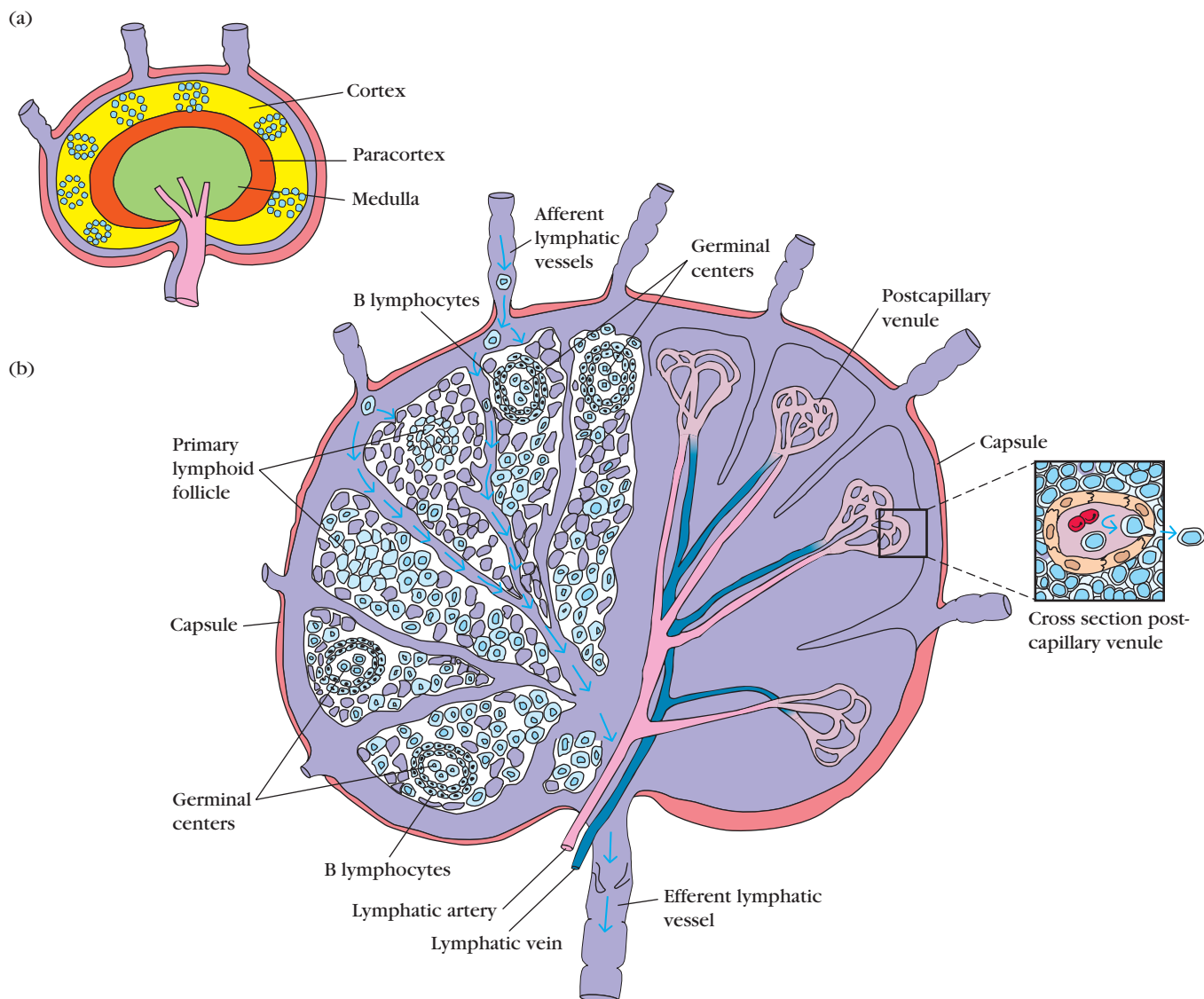


FIGURE 2-18 Structure of a lymph node. (a) The three layers of a lymph node support distinct microenvironments. (b) The left side depicts the arrangement of reticulum and lymphocytes within the various regions of a lymph node. Macrophages and dendritic cells, which trap antigen, are present in the cortex and paracortex. T_H cells are concentrated in the paracortex; B cells are located primarily in the cortex, within follicles and germinal centers. The medulla is popu-

lated largely by antibody-producing plasma cells. Lymphocytes circulating in the lymph are carried into the node by afferent lymphatic vessels; they either enter the reticular matrix of the node or pass through it and leave by the efferent lymphatic vessel. The right side of (b) depicts the lymphatic artery and vein and the postcapillary venules. Lymphocytes in the circulation can pass into the node from the postcapillary venules by a process called extravasation (*inset*).

introduction of other antigens into the body, the lymph leaving a node through its single efferent lymphatic vessel is enriched with antibodies newly secreted by medullary plasma cells and also has a fiftyfold higher concentration of lymphocytes than the afferent lymph.

The increase in lymphocytes in lymph leaving a node is due in part to lymphocyte proliferation within the node in response to antigen. Most of the increase, however, represents blood-borne lymphocytes that migrate into the node by passing between specialized endothelial cells that line the

postcapillary venules of the node. Estimates are that 25% of the lymphocytes leaving a lymph node have migrated across this endothelial layer and entered the node from the blood. Because antigenic stimulation within a node can increase this migration tenfold, the concentration of lymphocytes in a node that is actively responding can increase greatly, and the node swells visibly. Factors released in lymph nodes during antigen stimulation are thought to facilitate this increased migration.

SPLEEN

The spleen plays a major role in mounting immune responses to antigens in the blood stream. It is a large, ovoid secondary lymphoid organ situated high in the left abdominal cavity. While lymph nodes are specialized for trapping antigen from local tissues, the spleen specializes in filtering blood and trapping blood-borne antigens; thus, it can respond to systemic infections. Unlike the lymph nodes, the spleen is not supplied by lymphatic vessels. Instead, blood-borne antigens and lymphocytes are carried into the spleen through the splenic artery. Experiments with radioactively labeled lymphocytes show that more recirculating lymphocytes pass daily through the spleen than through all the lymph nodes combined.

The spleen is surrounded by a capsule that extends a number of projections (trabeculae) into the interior to form a compartmentalized structure. The compartments are of two types, the red pulp and white pulp, which are separated by a diffuse marginal zone (Figure 2-19). The splenic **red pulp** consists of a network of sinusoids populated by macrophages and numerous red blood cells (erythrocytes) and few lymphocytes; it is the site where old and defective red blood cells are destroyed and removed. Many of the macrophages within the red pulp contain engulfed red blood cells or iron pigments from degraded hemoglobin. The splenic **white pulp** surrounds the branches of the splenic artery, forming a **periarteriolar lymphoid sheath (PALS)** populated mainly by T lymphocytes. Primary lymphoid follicles are attached to the

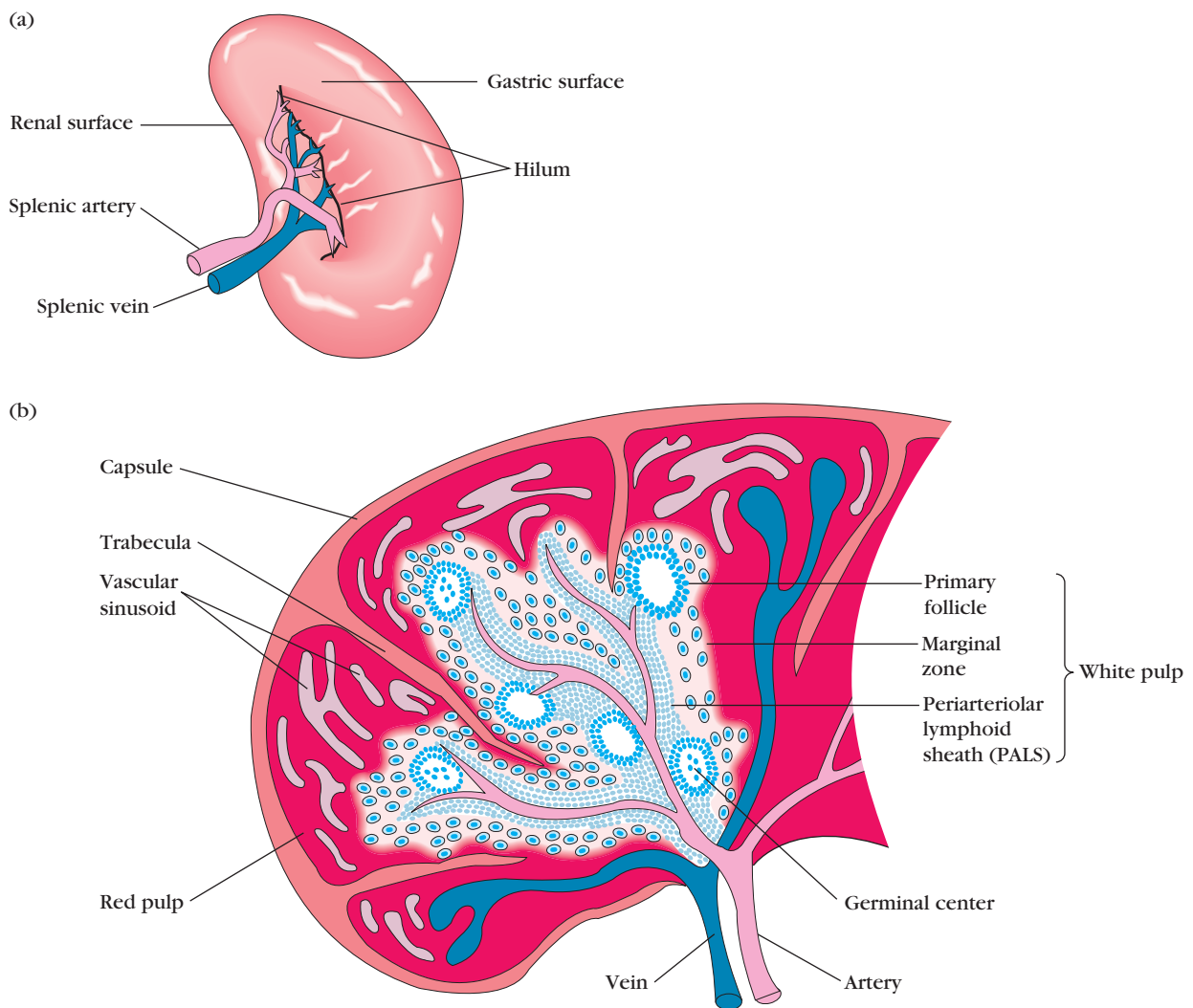


FIGURE 2-19 Structure of the spleen. (a) The spleen, which is about 5 inches long in adults, is the largest secondary lymphoid organ. It is specialized for trapping blood-borne antigens. (b) Diagrammatic cross section of the spleen. The splenic artery pierces the capsule and divides into progressively smaller arterioles, ending in vascular sinusoids that drain back into the splenic vein. The erythro-

cyte-filled red pulp surrounds the sinusoids. The white pulp forms a sleeve, the periarteriolar lymphoid sheath (PALS), around the arterioles; this sheath contains numerous T cells. Closely associated with the PALS is the marginal zone, an area rich in B cells that contains lymphoid follicles that can develop into secondary follicles containing germinal centers.

PALS. These follicles are rich in B cells and some of them contain germinal centers. The **marginal zone**, located peripheral to the PALS, is populated by lymphocytes and macrophages.

Blood-borne antigens and lymphocytes enter the spleen through the splenic artery, which empties into the marginal zone. In the marginal zone, antigen is trapped by interdigitating dendritic cells, which carry it to the PALS. Lymphocytes in the blood also enter sinuses in the marginal zone and migrate to the PALS.

The initial activation of B and T cells takes place in the T-cell-rich PALS. Here interdigitating dendritic cells capture antigen and present it combined with class II MHC molecules to T_H cells. Once activated, these T_H cells can then activate B cells. The activated B cells, together with some T_H cells, then migrate to primary follicles in the marginal zone. Upon antigenic challenge, these primary follicles develop into characteristic secondary follicles containing germinal centers (like those in the lymph nodes), where rapidly dividing B cells (centroblasts) and plasma cells are surrounded by dense clusters of concentrically arranged lymphocytes.

The effects of splenectomy on the immune response depends on the age at which the spleen is removed. In children, splenectomy often leads to an increased incidence of bacterial sepsis caused primarily by *Streptococcus pneumoniae*, *Neisseria meningitidis*, and *Haemophilus influenzae*. Splenectomy in adults has less adverse effects, although it leads to some increase in blood-borne bacterial infections (**bacteremia**).

MUCOSAL-ASSOCIATED LYMPHOID TISSUE

The mucous membranes lining the digestive, respiratory, and urogenital systems have a combined surface area of about

400 m² (nearly the size of a basketball court) and are the major sites of entry for most pathogens. These vulnerable membrane surfaces are defended by a group of organized lymphoid tissues mentioned earlier and known collectively as **mucosal-associated lymphoid tissue (MALT)**. Structurally, these tissues range from loose, barely organized clusters of lymphoid cells in the lamina propria of intestinal villi to well-organized structures such as the familiar tonsils and appendix, as well as Peyer's patches, which are found within the submucosal layer of the intestinal lining. The functional importance of MALT in the body's defense is attested to by its large population of antibody-producing plasma cells, whose number far exceeds that of plasma cells in the spleen, lymph nodes, and bone marrow combined.

The **tonsils** are found in three locations: lingual at the base of the tongue; palatine at the sides of the back of the mouth; and pharyngeal (adenoids) in the roof of the nasopharynx (Figure 2-20). All three tonsil groups are nodular structures consisting of a meshwork of reticular cells and fibers interspersed with lymphocytes, macrophages, granulocytes, and mast cells. The B cells are organized into follicles and germinal centers; the latter are surrounded by regions showing T-cell activity. The tonsils defend against antigens entering through the nasal and oral epithelial routes.

The best studied of the mucous membranes is the one that lines the gastrointestinal tract. This tissue, like that of the respiratory and urogenital tracts, has the capacity to endocytose antigen from the lumen. Immune reactions are initiated against pathogens and antibody can be generated and exported to the lumen to combat the invading organisms. As shown in Figures 2-21 and 2-22, lymphoid cells are found in various regions within this tissue. The outer mucosal epithe-

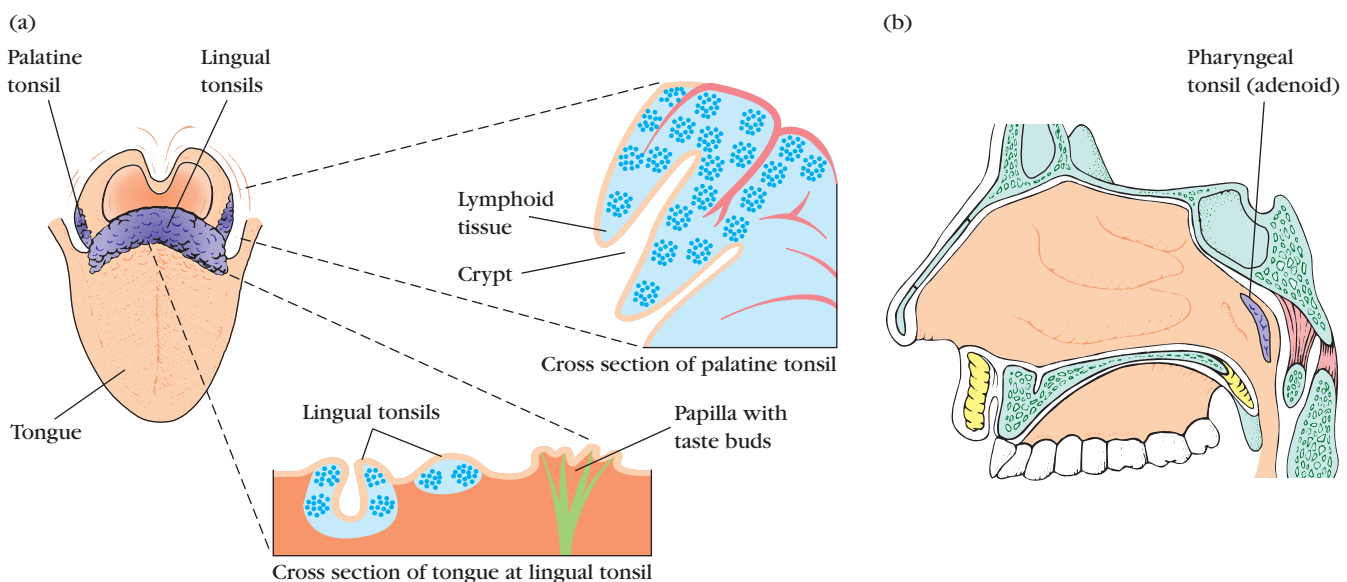


FIGURE 2-20 Three types of tonsils. (a) The position and internal features of the palatine and lingual tonsils; (b) a view of the position of the nasopharyngeal tonsils (adenoids). [Part b adapted from

J. Klein, 1982, Immunology, The Science of Self-Nonself Discrimination, © 1982 by John Wiley and Sons, Inc.]

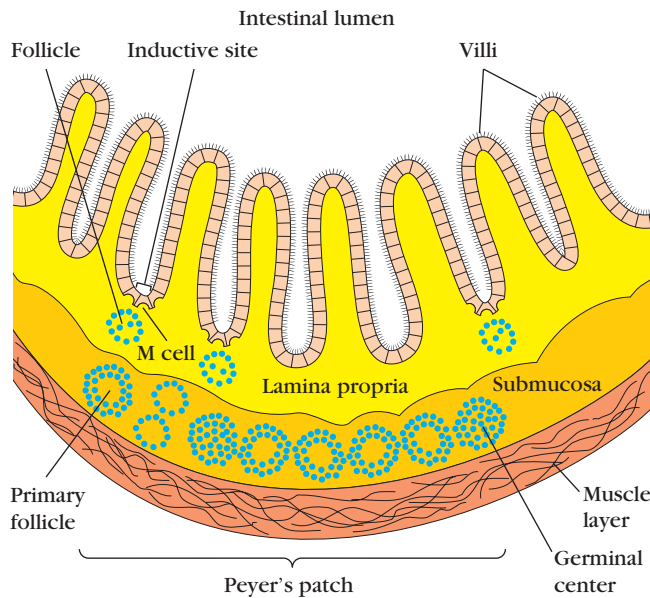


FIGURE 2-21 Cross-sectional diagram of the mucous membrane lining the intestine showing a nodule of lymphoid follicles that constitutes a Peyer's patch in the submucosa. The intestinal lamina propria contains loose clusters of lymphoid cells and diffuse follicles.

lial layer contains so-called **intraepithelial lymphocytes (IELs)**. Many of these lymphocytes are T cells that express unusual receptors ($\gamma\delta$ T-cell receptors, or $\gamma\delta$ TCRs), which exhibit limited diversity for antigen. Although this population of T cells is well situated to encounter antigens that enter through the intestinal mucous epithelium, their actual

function remains largely unknown. The lamina propria, which lies under the epithelial layer, contains large numbers of B cells, plasma cells, activated T_H cells, and macrophages in loose clusters. Histologic sections have revealed more than 15,000 lymphoid follicles within the intestinal lamina propria of a healthy child. The submucosal layer beneath the lamina propria contains Peyer's patches, nodules of 30–40 lymphoid follicles. Like lymphoid follicles in other sites, those that compose Peyer's patches can develop into secondary follicles with germinal centers.

The epithelial cells of mucous membranes play an important role in promoting the immune response by delivering small samples of foreign antigen from the lumina of the respiratory, digestive, and urogenital tracts to the underlying mucosal-associated lymphoid tissue. This antigen transport is carried out by specialized **M cells**. The structure of the M cell is striking: these are flattened epithelial cells lacking the microvilli that characterize the rest of the mucous epithelium. In addition, M cells have a deep invagination, or pocket, in the basolateral plasma membrane; this pocket is filled with a cluster of B cells, T cells, and macrophages (Figure 2-22a). Luminal antigens are endocytosed into vesicles that are transported across the luminal membrane to the underlying pocket membrane. The vesicles then fuse with the pocket membrane, delivering the potentially response-activating antigens to the clusters of lymphocytes contained within the pocket.

M cells are located in so-called **inductive sites**—small regions of a mucous membrane that lie over organized lymphoid follicles (Figure 2-22b). Antigens transported across the mucous membrane by M cells can activate B cells within

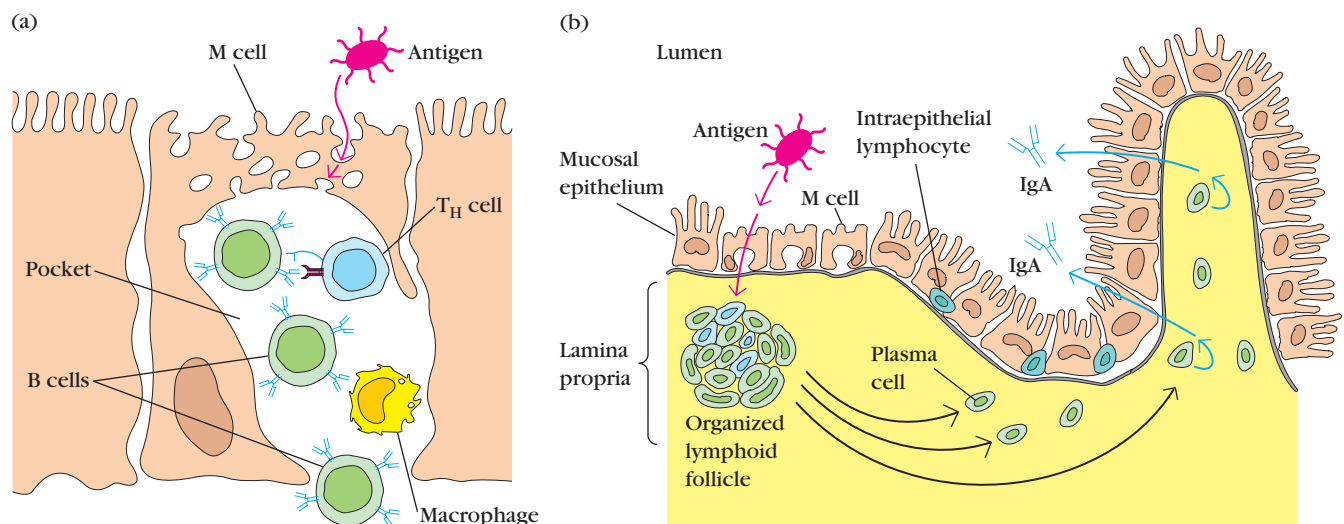


FIGURE 2-22 Structure of M cells and production of IgA at inductive sites. (a) M cells, located in mucous membranes, endocytose antigen from the lumen of the digestive, respiratory, and urogenital tracts. The antigen is transported across the cell and released into the large basolateral pocket. (b) Antigen transported across the epithelial layer by M cells at an inductive site activates B cells in the underlying

lymphoid follicles. The activated B cells differentiate into IgA-producing plasma cells, which migrate along the submucosa. The outer mucosal epithelial layer contains intraepithelial lymphocytes, of which many are $CD8^+$ T cells that express $\gamma\delta$ TCRs with limited receptor diversity for antigen.

these lymphoid follicles. The activated B cells differentiate into plasma cells, which leave the follicles and secrete the IgA class of antibodies. These antibodies then are transported across the epithelial cells and released as **secretory IgA** into the lumen, where they can interact with antigens.

As described in Chapter 1, mucous membranes are an effective barrier to the entrance of most pathogens, which thereby contributes to nonspecific immunity. One reason for this is that the mucosal epithelial cells are cemented to one another by tight junctions that make it difficult for pathogens to penetrate. Interestingly, some enteric pathogens, including both bacteria and viruses, have exploited the M cell as an entry route through the mucous-membrane barrier. In some cases, the pathogen is internalized by the M cell and transported into the pocket. In other cases, the pathogen binds to the M cell and disrupts the cell, thus allowing entry of the pathogen. Among the pathogens that use M cells in these ways are several invasive *Salmonella* species, *Vibrio cholerae*, and the polio virus.

Cutaneous-Associated Lymphoid Tissue

The skin is an important anatomic barrier to the external environment, and its large surface area makes this tissue important in nonspecific (innate) defenses. The epidermal (outer) layer of the skin is composed largely of specialized epithelial cells called keratinocytes. These cells secrete a number of cytokines that may function to induce a local inflammatory reaction. In addition, keratinocytes can be induced to express class II MHC molecules and may function as antigen-presenting cells. Scattered among the epithelial-cell matrix of the epidermis are Langerhans cells, a type of dendritic cell, which internalize antigen by phagocytosis or endocytosis. The Langerhans cells then migrate from the epidermis to regional lymph nodes, where they differentiate into interdigitating dendritic cells. These cells express high levels of class II MHC molecules and function as potent activators of naive T_H cells.

The epidermis also contains so-called *intraepidermal lymphocytes*. These are similar to the intraepithelial lymphocytes of MALT in that most of them are $CD8^+$ T cells, many of which express $\gamma\delta$ T-cell receptors, which have limited diversity for antigen. These intraepidermal T cells are well situated to encounter antigens that enter through the skin and some immunologists believe that they may play a role in combating antigens that enter through the skin. The underlying dermal layer of the skin contains scattered $CD4^+$ and $CD8^+$ T cells and macrophages. Most of these dermal T cells were either previously activated cells or are memory cells.

Systemic Function of the Immune System

The many different cells, organs, and tissues of the immune system are dispersed throughout the body, yet the various components communicate and collaborate to produce an ef-

fective response to an infection. An infection that begins in one area of the body initiates processes that eventually involve cells, organs, and tissues distant from the site of pathogen invasion. Consider what happens when the skin is broken, allowing bacteria to enter the body and initiate infection.

The tissue damage associated with the injury and infection results in an inflammatory response that causes increased blood flow, vasodilation, and an increase in capillary permeability. Chemotactic signals are generated that can cause phagocytes and lymphocytes to leave the blood stream and enter the affected area. Factors generated during these early stages of the infection stimulate the capacity of the adaptive immune system to respond. Langerhans cells (dendritic cells found throughout the epithelial layers of the skin and the respiratory, gastrointestinal, urinary, and genital tracts) can capture antigens from invading pathogens and migrate into a nearby lymphatic vessel, where the flow of lymph carries them to nearby lymph nodes. In the lymph nodes these class II MHC-bearing cells can become members of the interdigitating dendritic-cell population and initiate adaptive immune responses by presenting antigen to T_H cells. The recognition of antigen by T_H cells can have important consequences, including the activation and proliferation of T_H cells within the node as the T_H cells recognize the antigen, and the secretion by the activated T cells of factors that support T-cell-dependent antibody production by B cells that may already have been activated by antigen delivered to the lymph node by lymph. The antigen-stimulated T_H cells also release chemotactic factors that cause lymphocytes to leave the blood circulation and enter the lymph node through the endothelium of the postcapillary venules. Lymphocytes that respond to the antigen are retained in the lymph node for 48 hours or so as they undergo activation and proliferation before their release via the node's efferent lymphatic vessel.

Once in the lymph, the newly released activated lymphocytes can enter the bloodstream via the subclavian vein. Eventually, the circulation carries them to blood vessels near the site of the infection, where the inflammatory process makes the vascular endothelium of the nearby blood vessels more adherent for activated T cells and other leukocytes (see Chapter 15). Chemotactic factors that attract lymphocytes, macrophages, and neutrophils are also generated during the inflammatory process, promoting leukocyte adherence to nearby vascular epithelium and leading leukocytes to the site of the infection. Later in the course of the response, pathogen-specific antibodies produced in the node are also carried to the bloodstream. Inflammation aids the delivery of the anti-pathogen antibody by promoting increased vascular permeability, which increases the flow of antibody-containing plasma from the blood circulation to inflamed tissue. The result of this network of interactions among diffusible molecules, cells, organs, the lymphatic system, and the circulatory system is an effective and focused immune response to an infection.

Lymphoid Cells and Organs— Evolutionary Comparisons

While innate systems of immunity are seen in invertebrates and even in plants, the evolution of lymphoid cells and organs evolved only in the phylum Vertebrata. Consequently,

adaptive immunity, which is mediated by antibodies and T cells, is only seen in this phylum. However, as shown in Figure 2-23, the kinds of lymphoid tissues seen in different orders of vertebrates differ.

As one considers the spectrum from the earliest vertebrates, the jawless fishes (Agnatha), to the birds and mammals, evolution has added organs and tissues with immune

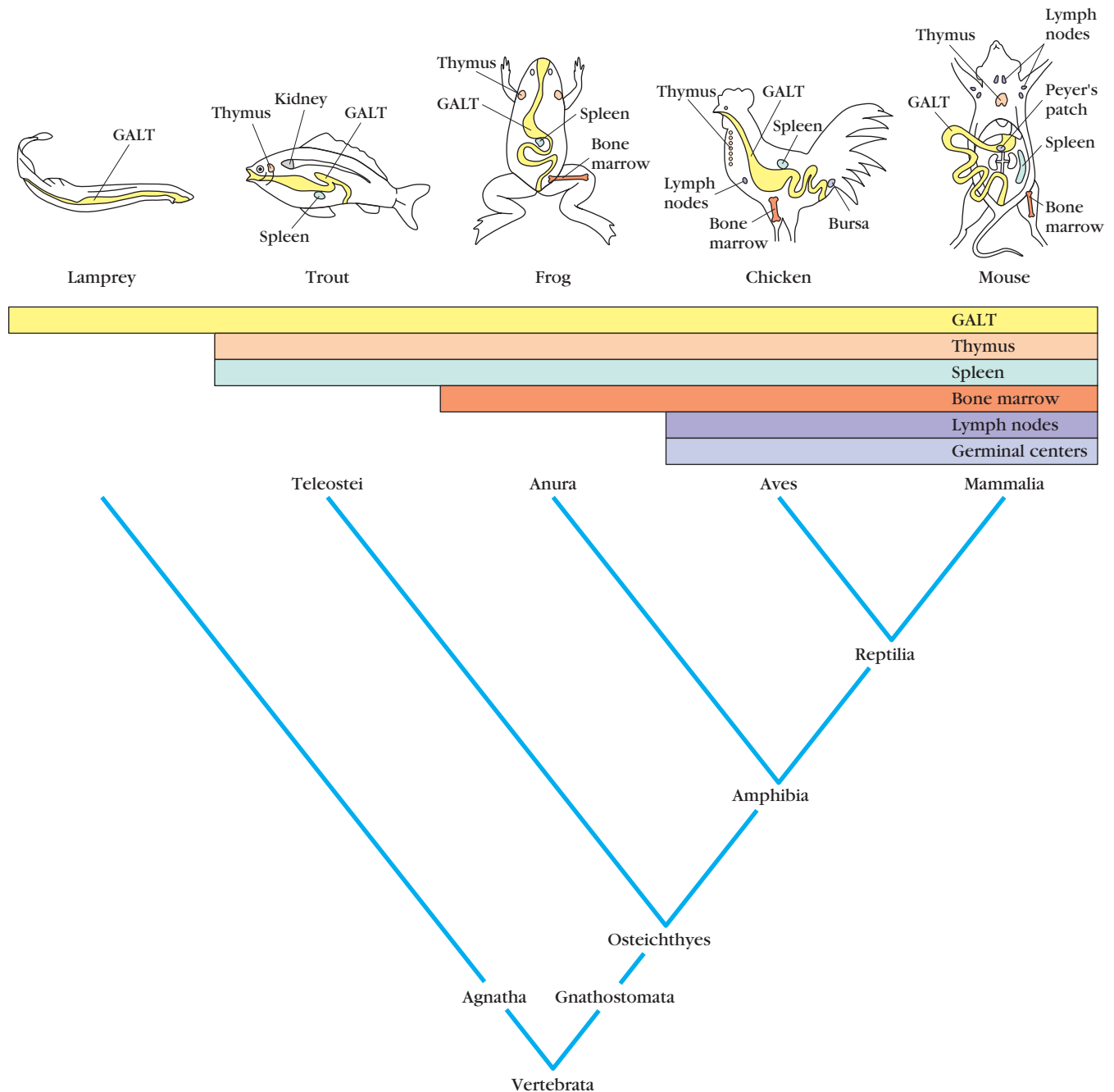


FIGURE 2-23 Evolutionary distribution of lymphoid tissues. The presence and location of lymphoid tissues in several major orders of vertebrates are shown. Although they are not shown in the diagram, cartilaginous fish such as sharks and rays have GALT, thymus, and a spleen. Reptiles also have GALT, thymus, and spleen and they also

may have lymph nodes that participate in immunological reactions. Whether bone marrow is involved in the generation of lymphocytes in reptiles is under investigation. [Adapted from Dupasquier and M. Flajnik, 1999. In *Fundamental Immunology* 4th ed., W. E. Paul, ed., Lippincott-Raven, Philadelphia.]

functions but has tended to retain those evolved by earlier orders. While all have gut-associated lymphoid tissue (GALT) and most have some version of a spleen and thymus, not all have blood-cell-forming bone marrow or lymph nodes, and the ability to form germinal centers is not shared by all. The differences seen at the level of organs and tissues are also reflected at the cellular level. Lymphocytes that express antigen-specific receptors on their surfaces are necessary to mount an adaptive immune response. So far, it has not been possible to demonstrate the presence of T or B lymphocytes in the jawless fishes, and attempts to demonstrate an adaptive immune response in lampreys and hagfish, members of the order Agnatha, have failed. In fact, only jawed vertebrates (Gnathostomata), of which the cartilaginous fish (sharks, rays) are the earliest example, have B and T lymphocytes and support adaptive immune responses.

SUMMARY

- The cells that participate in the immune response are white blood cells, or leukocytes. The lymphocyte is the only cell to possess the immunologic attributes of specificity, diversity, memory, and self/nonself recognition.
 - Many of the body's cells, tissues, and organs arise from the progeny of different stem-cell populations. The division of a stem cell can result in the production of another stem cell and a differentiated cell of a specific type or group.
 - All leukocytes develop from a common multipotent hematopoietic stem cell during hematopoiesis. Various hematopoietic growth factors (cytokines) induce proliferation and differentiation of the different blood cells. The differentiation of stem cells into different cell types requires the expression of different lineage-determining genes. A number of transcription factors play important roles in this regard.
 - Hematopoiesis is closely regulated to assure steady-state levels of each of the different types of blood cell. Cell division and differentiation of each of the lineages is balanced by programmed cell death.
 - There are three types of lymphocytes: B cells, T cells, and natural killer cells (NK cells). NK cells are much less abundant than B and T cells, and most lack a receptor that is specific for a particular antigen. However, a subtype of NK cells, NK1-T cells, have both T-cell receptors and many of the markers characteristic of NK cells. The three types of lymphoid cells are best distinguished on the basis of function and the presence of various membrane molecules.
 - Naive B and T lymphocytes (those that have not encountered antigen) are small resting cells in the G_0 phase of the cell cycle. After interacting with antigen, these cells enlarge into lymphoblasts that proliferate and eventually differentiate into effector cells and memory cells.
 - Macrophages and neutrophils are specialized for the phagocytosis and degradation of antigens (see Figure 2-9).
- Phagocytosis is facilitated by opsonins such as antibody, which increase the attachment of antigen to the membrane of the phagocyte.
- Activated macrophages secrete various factors that regulate the development of the adaptive immune response and mediate inflammation (see Table 2-7). Macrophages also process and present antigen bound to class II MHC molecules, which can then be recognized by T_H cells.
 - Basophils and mast cells are nonphagocytic cells that release a variety of pharmacologically active substances and play important roles in allergic reactions.
 - Dendritic cells capture antigen. With the exception of follicular dendritic cells, these cells express high levels of class II MHC molecules. Along with macrophages and B cells, dendritic cells play an important role in T_H -cell activation by processing and presenting antigen bound to class II MHC molecules and by providing the required co-stimulatory signal. Follicular dendritic cells, unlike the others, facilitate B-cell activation but play no role in T-cell activation.
 - The primary lymphoid organs provide sites where lymphocytes mature and become antigenically committed. T lymphocytes mature within the thymus, and B lymphocytes arise and mature within the bone marrow of humans, mice, and several other animals, but not all vertebrates.
 - Primary lymphoid organs are also places of selection where many lymphocytes that react with self antigens are eliminated. Furthermore, the thymus eliminates thymocytes that would mature into useless T cells because their T-cell receptors are unable to recognize self-MHC.
 - The lymphatic system collects fluid that accumulates in tissue spaces and returns this fluid to the circulation via the left subclavian vein. It also delivers antigens to the lymph nodes, which interrupt the course of lymphatic vessels.
 - Secondary lymphoid organs capture antigens and provide sites where lymphocytes become activated by interaction with antigens. Activated lymphocytes undergo clonal proliferation and differentiation into effector cells.
 - There are several types of secondary lymphoid tissue: lymph nodes, spleen, the loose clusters of follicles, and Peyer's patches of the intestine, and cutaneous-associated lymphoid tissue. Lymph nodes trap antigen from lymph, spleen traps blood-borne antigens, intestinal-associated lymphoid tissues (as well as other secondary lymphoid tissues) interact with antigens that enter the body from the gastrointestinal tract, and cutaneous-associated lymphoid tissue protects epithelial tissues.
 - An infection that begins in one area of the body eventually involves cells, organs, and tissues that may be distant from the site of pathogen invasion. Antigen from distant sites can arrive at lymph nodes via lymph and dendritic cells, thereby assuring activation of T cells and B cells and release of these cells and their products to the circulation. Inflammatory processes bring lymphocytes and other leukocytes to the site of infection. Thus, although dispersed through-

out the body, the components of the immune system communicate and collaborate to produce an effective response to infection.

- Vertebrate orders differ greatly in the kinds of lymphoid organs, tissues, and cells they possess. The most primitive vertebrates, the jawless fishes, have only gut-associated lymphoid tissues, lack B and T cells, and cannot mount adaptive immune responses. Jawed vertebrates possess a greater variety of lymphoid tissues, have B and T cells, and display adaptive immunity.

References

- Appelbaum, F. R. 1996. Hematopoietic stem cell transplantation. In *Scientific American Medicine*. D. Dale and D. Federman, eds. Scientific American Publishers, New York.
- Banchereau J., F. Briere, C. Caux, J. Davoust, S. Lebecque, Y. J. Liu, B. Pulendran, and K. Palucka. 2000. Immunobiology of dendritic cells. *Annu. Rev. Immunol.* **18**:767.
- Bendelac, A., M. N. Rivera, S-H. Park, and J. H. Roark. 1997. Mouse CD1-specific NK1 T cells: Development, specificity and function. *Annu. Rev. Immunol.* **15**:535.
- Clevers, H. C., and R. Grosschedl. 1996. Transcriptional control of lymphoid development: lessons from gene targeting. *Immunol. Today* **17**:336.
- Cory, S. 1995. Regulation of lymphocyte survival by the BCL-2 gene family. *Annu. Rev. Immunol.* **12**:513.
- Ganz, T., and R. I. Lehrer. 1998. Antimicrobial peptides of vertebrates. *Curr. Opin. Immunol.* **10**:41.
- Liu, Y. J. 2001. Dendritic cell subsets and lineages, and their functions in innate and adaptive immunity. *Cell* **106**:259.
- Melchers, F., and A. Rolink. 1999. B-lymphocyte development and biology. In *Fundamental Immunology*, 4th ed., W. E. Paul, ed., p. 183. Lippincott-Raven, Philadelphia.
- Nathan, C., and M. U. Shiloh. 2000. Reactive oxygen and nitrogen intermediates in the relationship between mammalian hosts and microbial pathogens. *Proc. Natl. Acad. Sci.* **97**:8841.
- Pedersen, R. A. 1999. Embryonic stem cells for medicine. *Sci. Am.* **280**:68.
- Osborne, B. A. 1996. Apoptosis and the maintenance of homeostasis in the immune system. *Curr. Opin. Immunol.* **8**:245.
- Picker, L. J., and M. H. Siegelman. 1999. Lymphoid tissues and organs. In *Fundamental Immunology*, 4th ed., W. E. Paul, ed., p. 145. Lippincott-Raven, Philadelphia.
- Rothenberg, E. V. 2000. Stepwise specification of lymphocyte developmental lineages. *Current Opin. Gen. Dev.* **10**:370.
- Ward, A. C., D. M. Loeb, A. A. Soede-Bobok, I. P. Touw, and A. D. Friedman. 2000. Regulation of granulopoiesis by transcription factors and cytokine signals. *Leukemia* **14**:973.
- Weissman, I. L. 2000. Translating stem and progenitor cell biology to the clinic: barriers and opportunities. *Science* **287**:1442.



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This brilliantly illustrated atlas of normal and abnormal blood cells informatively displayed as stained cell smears has been assembled to help train medical students at the Harvard Medical School to recognize and remember cell morphology that is associated with many different pathologies, including leukemias, anemias, and even malarial infections.

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A well written and comprehensive presentation of stem cells and their biology is presented in an interesting and well-referenced monograph.

Study Questions

CLINICAL FOCUS QUESTION The T and B cells that differentiate from hematopoietic stem cells recognize as self the bodies in which they differentiate. Suppose a woman donates HSCs to a genetically unrelated man whose hematopoietic system was totally destroyed by a combination of radiation and chemotherapy. Suppose further that, although most of the donor HSCs differentiate into hematopoietic cells, some differentiate into cells of the pancreas, liver, and heart. Decide which of the following outcomes is likely and justify your choice.

- The T cells from the donor HSCs do not attack the pancreatic, heart, and liver cells that arose from donor cells, but mount a GVH response against all of the other host cells.
- The T cells from the donor HSCs mount a GVH response against all of the host cells.
- The T cells from the donor HSCs attack the pancreatic, heart, and liver cells that arose from donor cells, but fail to mount a GVH response against all of the other host cells.
- The T cells from the donor HSCs do not attack the pancreatic, heart, and liver cells that arose from donor cells and fail to mount a GVH response against all of the other host cells.

1. Explain why each of the following statements is false.

- All T_H cells express CD4 and recognize only antigen associated with class II MHC molecules.
- The pluripotent stem cell is one of the most abundant cell types in the bone marrow.
- Activation of macrophages increases their expression of class I MHC molecules, making the cells present antigen more effectively.



- d. Lymphoid follicles are present only in the spleen and lymph nodes.
 - e. Infection has no influence on the rate of hematopoiesis.
 - f. Follicular dendritic cells can process and present antigen to T lymphocytes.
 - g. All lymphoid cells have antigen-specific receptors on their membrane.
 - h. All vertebrates generate B lymphocytes in bone marrow.
 - i. All vertebrates produce B or T lymphocytes and most produce both.
2. For each of the following situations, indicate which type(s) of lymphocyte(s), if any, would be expected to proliferate rapidly in lymph nodes and where in the nodes they would do so.
 - a. Normal mouse immunized with a soluble protein antigen
 - b. Normal mouse with a viral infection
 - c. Neonatally thymectomized mouse immunized with a protein antigen
 - d. Neonatally thymectomized mouse immunized with the thymus-independent antigen bacterial lipopolysaccharide (LPS), which does not require the aid of T_H cells to activate B cells
 3. List the primary lymphoid organs and summarize their functions in the immune response.
 4. List the secondary lymphoid organs and summarize their functions in the immune response.
 5. What are the two primary characteristics that distinguish hematopoietic stem cells and progenitor cells?
 6. What are the two primary roles of the thymus?
 7. What do nude mice and humans with DiGeorge's syndrome have in common?
 8. At what age does the thymus reach its maximal size?
 - a. During the first year of life
 - b. Teenage years (puberty)
 - c. Between 40 and 50 years of age
 - d. After 70 years of age
 9. Preparations enriched in hematopoietic stem cells are useful for research and clinical practice. In Weissman's method for enriching hematopoietic stem cells, why is it necessary to use lethally irradiated mice to demonstrate enrichment?
 10. What effect does thymectomy have on a neonatal mouse? On an adult mouse? Explain why these effects differ.
 11. What effect would removal of the bursa of Fabricius (bursectomy) have on chickens?
 12. Some microorganisms (e.g., *Neisseria gonorrhoeae*, *Mycobacterium tuberculosis*, and *Candida albicans*) are classified as intracellular pathogens. Define this term and explain why the immune response to these pathogens differs from that to other pathogens such as *Staphylococcus aureus* and *Streptococcus pneumoniae*.
 13. Indicate whether each of the following statements about the spleen is true or false. If you think a statement is false, explain why.
 - a. It filters antigens out of the blood.
 - b. The marginal zone is rich in T cells, and the periarteriolar lymphoid sheath (PALS) is rich in B cells.
 - c. It contains germinal centers.
 - d. It functions to remove old and defective red blood cells.
 - e. Lymphatic vessels draining the tissue spaces enter the spleen.
 - f. Lymph node but not spleen function is affected by a knockout of the Ikaros gene
 14. For each type of cell indicated (a–p), select the most appropriate description (1–16) listed below. Each description may be used once, more than once, or not at all.

Cell Types

- a. _____ Common myeloid progenitor cells
- b. _____ Monocytes
- c. _____ Eosinophils
- d. _____ Dendritic cells
- e. _____ Natural killer (NK) cells
- f. _____ Kupffer cells
- g. _____ Lymphoid dendritic cell
- h. _____ Mast cells
- i. _____ Neutrophils
- j. _____ M cells
- k. _____ Bone-marrow stromal cells
- l. _____ Lymphocytes
- m. _____ NK1-T cell
- n. _____ Microglial cell
- o. _____ Myeloid dendritic cell
- p. _____ Hematopoietic stem cell

Descriptions

- (1) Major cell type presenting antigen to T_H cells
- (2) Phagocytic cell of the central nervous system
- (3) Phagocytic cells important in the body's defense against parasitic organisms
- (4) Macrophages found in the liver
- (5) Give rise to red blood cells
- (6) An antigen-presenting cell derived from monocytes that is not phagocytic
- (7) Generally first cells to arrive at site of inflammation
- (8) Secrete colony-stimulating factors (CSFs)
- (9) Give rise to thymocytes
- (10) Circulating blood cells that differentiate into macrophages in the tissues
- (11) An antigen-presenting cell that arises from the same precursor as a T cell but not the same as a macrophage
- (12) Cells that are important in sampling antigens of the intestinal lumen
- (13) Nonphagocytic granulocytic cells that release various pharmacologically active substances
- (14) White blood cells that migrate into the tissues and play an important role in the development of allergies
- (15) These cells sometimes recognize their targets with the aid of an antigen-specific cell-surface receptor and sometimes by mechanisms that resemble those of natural killer cells.
- (16) Members of this category of cells are not found in jawless fishes.

Antigens

chapter 3

SUBSTANCES THAT CAN BE RECOGNIZED BY THE immunoglobulin receptor of B cells, or by the T-cell receptor when complexed with MHC, are called **antigens**. The molecular properties of antigens and the way in which these properties ultimately contribute to immune activation are central to our understanding of the immune system. This chapter describes some of the molecular features of antigens recognized by B or T cells. The chapter also explores the contribution made to immunogenicity by the biological system of the host; ultimately the biological system determines whether a molecule that combines with a B or T cell's antigen-binding receptor can then induce an immune response. Fundamental differences in the way B and T lymphocytes recognize antigen determine which molecular features of an antigen are recognized by each branch of the immune system. These differences are also examined in this chapter.

Immunogenicity Versus Antigenicity

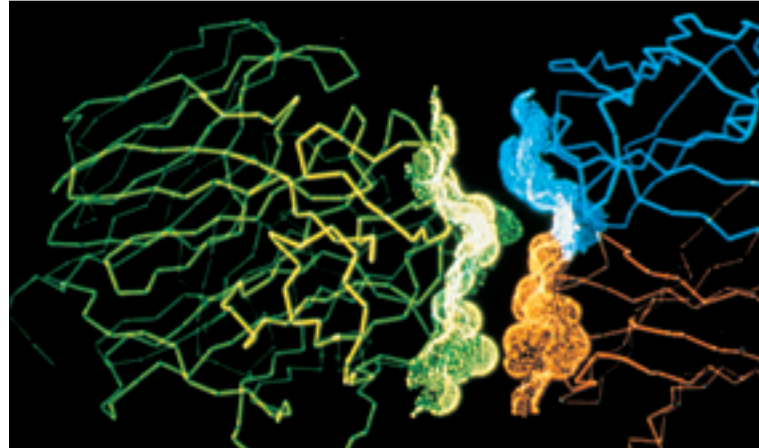
Immunogenicity and antigenicity are related but distinct immunologic properties that sometimes are confused. **Immunogenicity** is the ability to induce a humoral and/or cell-mediated immune response:

B cells + antigen → effector B cells + memory B cells
↓
(plasma cells)

T cells + antigen → effector T cells + memory T cells
↓
(e.g., CTLs, T_Hs)

Although a substance that induces a specific immune response is usually called an antigen, it is more appropriately called an **immunogen**.

Antigenicity is the ability to combine specifically with the final products of the above responses (i.e., antibodies and/or cell-surface receptors). Although all molecules that have the property of immunogenicity also have the property of antigenicity, the reverse is not true. Some small molecules, called *haptens*, are antigenic but incapable, by themselves, of inducing a specific immune response. In other words, they lack immunogenicity.



Complementarity of Interacting Surfaces of Antibody (left) and Antigen (right)

- Immunogenicity Versus Antigenicity
- Factors That Influence Immunogenicity
- Epitopes
- Haptens and the Study of Antigenicity
- Pattern-Recognition Receptors

Factors That Influence Immunogenicity

To protect against infectious disease, the immune system must be able to recognize bacteria, bacterial products, fungi, parasites, and viruses as immunogens. In fact, the immune system actually recognizes particular macromolecules of an infectious agent, generally either proteins or polysaccharides. Proteins are the most potent immunogens, with polysaccharides ranking second. In contrast, lipids and nucleic acids of an infectious agent generally do not serve as immunogens unless they are complexed with proteins or polysaccharides. Immunologists tend to use proteins or polysaccharides as immunogens in most experimental studies of humoral immunity (Table 3-1). For cell-mediated immunity, only proteins and some lipids and glycolipids serve as immunogens. These molecules are not recognized directly. Proteins must first be processed into small peptides and then presented together with MHC molecules on the membrane of a cell before they can be recognized as immunogens. Recent work shows that those lipids and glycolipids that can elicit cell-mediated immunity must also be combined with MHC-like membrane molecules called CD1 (see Chapter 8).

TABLE 3-1

Molecular weight of some common experimental antigens used in immunology

Antigen	Approximate molecular mass (Da)
Bovine gamma globulin (BGG)	150,000
Bovine serum albumin (BSA)	69,000
Flagellin (monomer)	40,000
Hen egg-white lysozyme (HEL)	15,000
Keyhole limpet hemocyanin (KLH)	>2,000,000
Ovalbumin (OVA)	44,000
Sperm whale myoglobin (SWM)	17,000
Tetanus toxoid (TT)	150,000

Immunogenicity is not an intrinsic property of an antigen but rather depends on a number of properties of the particular biological system that the antigen encounters. The next two sections describe the properties that most immunogens share and the contribution that the biological system makes to the expression of immunogenicity.

The Nature of the Immunogen Contributes to Immunogenicity

Immunogenicity is determined, in part, by four properties of the immunogen: its foreignness, molecular size, chemical composition and complexity, and ability to be processed and presented with an MHC molecule on the surface of an antigen-presenting cell or altered self-cell.

FOREIGNNESS

In order to elicit an immune response, a molecule must be recognized as nonself by the biological system. The capacity to recognize nonself is accompanied by tolerance of self, a specific unresponsiveness to self antigens. Much of the ability to tolerate self antigens arises during lymphocyte development, during which immature lymphocytes are exposed to self-components. Antigens that have not been exposed to immature lymphocytes during this critical period may be later recognized as nonself, or foreign, by the immune system. When an antigen is introduced into an organism, the degree of its immunogenicity depends on the degree of its foreignness. Generally, the greater the phylogenetic distance between two species, the greater the structural (and therefore the antigenic) disparity between them.

For example, the common experimental antigen bovine serum albumin (BSA) is not immunogenic when injected

into a cow but is strongly immunogenic when injected into a rabbit. Moreover, BSA would be expected to exhibit greater immunogenicity in a chicken than in a goat, which is more closely related to bovines. There are some exceptions to this rule. Some macromolecules (e.g., collagen and cytochrome c) have been highly conserved throughout evolution and therefore display very little immunogenicity across diverse species lines. Conversely, some self-components (e.g., corneal tissue and sperm) are effectively sequestered from the immune system, so that if these tissues are injected even into the animal from which they originated, they will function as immunogens.

MOLECULAR SIZE

There is a correlation between the size of a macromolecule and its immunogenicity. The most active immunogens tend to have a molecular mass of 100,000 daltons (Da). Generally, substances with a molecular mass less than 5000–10,000 Da are poor immunogens, although a few substances with a molecular mass less than 1000 Da have proven to be immunogenic.

CHEMICAL COMPOSITION AND HETEROGENEITY

Size and foreignness are not, by themselves, sufficient to make a molecule immunogenic; other properties are needed as well. For example, synthetic homopolymers (polymers composed of a single amino acid or sugar) tend to lack immunogenicity regardless of their size. Studies have shown that copolymers composed of different amino acids or sugars are usually more immunogenic than homopolymers of their constituents. These studies show that chemical complexity contributes to immunogenicity. In this regard it is notable that all four levels of protein organization—primary, secondary, tertiary, and quaternary—contribute to the structural complexity of a protein and hence affect its immunogenicity (Figure 3-1).

LIPIDS AS ANTIGENS

Appropriately presented lipoidal antigens can induce B- and T-cell responses. For the stimulation of B-cell responses, lipids are used as haptens and attached to suitable carrier molecules such as the proteins keyhole limpet hemocyanin (KLH) or bovine serum albumin (BSA). By immunizing with these lipid-protein conjugates it is possible to obtain antibodies that are highly specific for the target lipids. Using this approach, antibodies have been raised against a wide variety of lipid molecules including steroids, complex fatty-acid derivatives, and fat-soluble vitamins such as vitamin E. Such antibodies are of considerable practical importance since many clinical assays for the presence and amounts of medically important lipids are antibody-based. For example, a determination of the levels of a complex group of lipids known as leukotrienes can be useful in evaluating asthma patients. Prednisone, an immunosuppressive steroid, is often given as part of the effort to prevent the rejection of a trans-

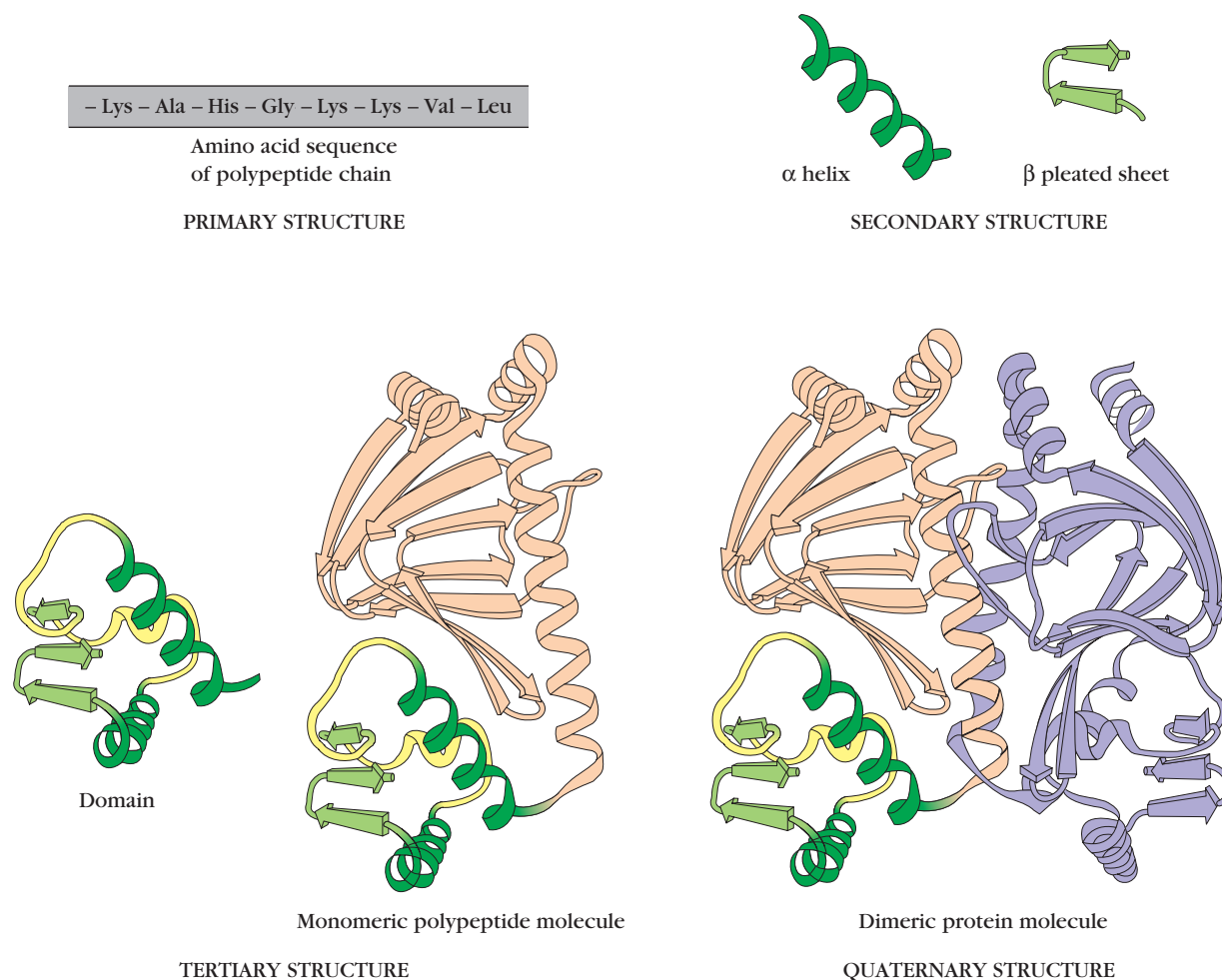


FIGURE 3-1 The four levels of protein organizational structure. The linear arrangement of amino acids constitutes the primary structure. Folding of parts of a polypeptide chain into regular structures (e.g., α helices and β pleated sheets) generates the secondary structure. Tertiary structure refers to the folding of regions between sec-

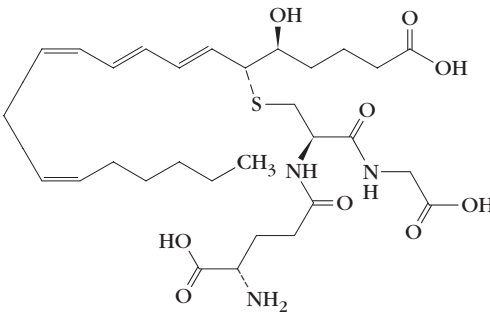
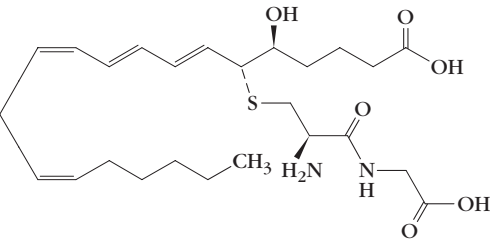
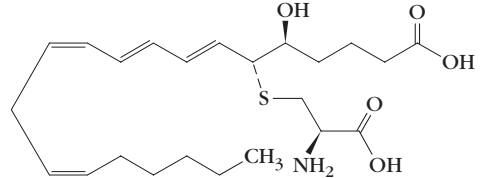
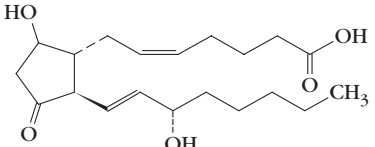
ondary features to give the overall shape of the molecule or parts of it (domains) with specific functional properties. Quaternary structure results from the association of two or more polypeptide chains into a single polymeric protein molecule.

planted organ. The achievement and maintenance of adequate blood levels of this and other immunosuppressive drugs is important to a successful outcome of transplantation, and antibody-based immunoassays are routinely used to make these evaluations. The extraordinary sensitivity and specificity of assays based on the use of anti-lipid antibodies is illustrated by Table 3-2, which shows the specificity of an antibody raised against leukotriene C_4 . This antibody allows the detection of as little as 16–32 picograms per ml of leukotriene C_4 . Because it has little or no reactivity with similar compounds, such as leukotriene D_4 or leukotriene E_4 , it can be used to assay leukotriene C_4 in samples that contain this compound and a variety of other structurally related lipids.

T cells recognize peptides derived from protein antigens when they are presented as peptide-MHC complexes. However, some lipids can also be recognized by T cells. Lipoidal

compounds such as glycolipids and some phospholipids can be recognized by T-cell receptors when presented as complexes with molecules that are very much like MHC molecules. These lipid-presenting molecules are members of the CD1 family (see Chapter 8) and are close structural relatives of class I MHC molecules. The lipid molecules recognized by the CD1–T-cell receptor system all appear to share the common feature of a hydrophobic portion and a hydrophilic head group. The hydrophobic portion is a long-chain fatty acid or alcohol and the hydrophilic head group is composed of highly polar groups that often contain carbohydrates. Recognition of lipids is a part of the immune response to some pathogens, and T cells that recognize lipids arising from *Mycobacterium tuberculosis* and *Mycobacterium leprae*, which respectively cause tuberculosis and leprosy, have been isolated from humans infected by these mycobacteria. More about the presentation of lipoidal antigens can be found in Chapter 8.

TABLE 3-2 Specificity of an antibody against a complex lipid

Lipid	Structure	Antibody reactivity* (on scale of 1 to 100)
Leukotriene C ₄		100.0
Leukotriene D ₄		5.0
Leukotriene E ₄		0.5
Prostaglandin D ₂		0.001

*The reactivity of the antibody with the immunizing antigen leukotriene C₄ is assigned a value of 100 in arbitrary units.

SUSCEPTIBILITY TO ANTIGEN PROCESSING AND PRESENTATION

The development of both humoral and cell-mediated immune responses requires interaction of T cells with antigen that has been processed and presented together with MHC molecules. Large, insoluble macromolecules generally are more immunogenic than small, soluble ones because the larger molecules are more readily phagocytosed and processed. Macromolecules that cannot be degraded and presented with MHC molecules are poor immunogens. This can be illustrated with polymers of D-amino acids, which are stereoisomers of the naturally occurring L-amino acids. Because the degradative enzymes within antigen-presenting cells can degrade only proteins containing L-amino acids, polymers of D-amino acids cannot be processed and thus are poor immunogens.

The Biological System Contributes to Immunogenicity

Even if a macromolecule has the properties that contribute to immunogenicity, its ability to induce an immune response will depend on certain properties of the biological system that the antigen encounters. These properties include the genotype of the recipient, the dose and route of antigen administration, and the administration of substances, called adjuvants, that increase immune responses.

GENOTYPE OF THE RECIPIENT ANIMAL

The genetic constitution (**genotype**) of an immunized animal influences the type of immune response the animal manifests, as well as the degree of the response. For example, Hugh McDevitt showed that two different inbred strains of

mice responded very differently to a synthetic polypeptide immunogen. After exposure to the immunogen, one strain produced high levels of serum antibody, whereas the other strain produced low levels. When the two strains were crossed, the F_1 generation showed an intermediate response to the immunogen. By backcross analysis, the gene controlling immune responsiveness was mapped to a subregion of the major histocompatibility complex (MHC). Numerous experiments with simple defined immunogens have demonstrated genetic control of immune responsiveness, largely confined to genes within the MHC. These data indicate that MHC gene products, which function to present processed antigen to T cells, play a central role in determining the degree to which an animal responds to an immunogen.

The response of an animal to an antigen is also influenced by the genes that encode B-cell and T-cell receptors and by genes that encode various proteins involved in immune regulatory mechanisms. Genetic variability in all of these genes affects the immunogenicity of a given macromolecule in different animals. These genetic contributions to immunogenicity will be described more fully in later chapters.

IMMUNOGEN DOSAGE AND ROUTE OF ADMINISTRATION

Each experimental immunogen exhibits a particular dose-response curve, which is determined by measuring the immune response to different doses and different administration routes. An antibody response is measured by determining the level of antibody present in the serum of immunized animals. Evaluating T-cell responses is less simple but may be determined by evaluating the increase in the number of T cells bearing TCRs that recognize the immunogen. Some combination of optimal dosage and route of administration will induce a peak immune response in a given animal.

An insufficient dose will not stimulate an immune response either because it fails to activate enough lymphocytes or because, in some cases, certain ranges of low doses can induce a state of immunologic unresponsiveness, or tolerance. The phenomenon of tolerance is discussed in chapters 10 and 21. Conversely, an excessively high dose can also induce tolerance. The immune response of mice to the purified pneumococcal capsular polysaccharide illustrates the importance of dose. A 0.5 mg dose of antigen fails to induce an immune response in mice, whereas a thousand-fold lower dose of the same antigen (5×10^{-4} mg) induces a humoral antibody response. A single dose of most experimental immunogens will not induce a strong response; rather, repeated administration over a period of weeks is usually required. Such repeated administrations, or **boosters**, increase the clonal proliferation of antigen-specific T cells or B cells and thus increase the lymphocyte populations specific for the immunogen.

Experimental immunogens are generally administered parenterally (*para*, around; *enteric*, gut)—that is, by routes other than the digestive tract. The following administration routes are common:

- Intravenous (iv): into a vein
- Intradermal (id): into the skin
- Subcutaneous (sc): beneath the skin
- Intramuscular (im): into a muscle
- Intraperitoneal (ip): into the peritoneal cavity

The administration route strongly influences which immune organs and cell populations will be involved in the response. Antigen administered intravenously is carried first to the spleen, whereas antigen administered subcutaneously moves first to local lymph nodes. Differences in the lymphoid cells that populate these organs may be reflected in the subsequent immune response.

ADJUVANTS

Adjuvants (from Latin *adjuvare*, to help) are substances that, when mixed with an antigen and injected with it, enhance the immunogenicity of that antigen. Adjuvants are often used to boost the immune response when an antigen has low immunogenicity or when only small amounts of an antigen are available. For example, the antibody response of mice to immunization with BSA can be increased fivefold or more if the BSA is administered with an adjuvant. Precisely how adjuvants augment the immune response is not entirely known, but they appear to exert one or more of the following effects (Table 3-3):

- Antigen persistence is prolonged.
- Co-stimulatory signals are enhanced.
- Local inflammation is increased.
- The nonspecific proliferation of lymphocytes is stimulated.

Aluminum potassium sulfate (alum) prolongs the persistence of antigen. When an antigen is mixed with alum, the salt precipitates the antigen. Injection of this alum precipitate results in a slower release of antigen from the injection site, so that the effective time of exposure to the antigen increases from a few days without adjuvant to several weeks with the adjuvant. The alum precipitate also increases the size of the antigen, thus increasing the likelihood of phagocytosis.

Water-in-oil adjuvants also prolong the persistence of antigen. A preparation known as **Freund's incomplete adjuvant** contains antigen in aqueous solution, mineral oil, and an emulsifying agent such as mannide monooleate, which disperses the oil into small droplets surrounding the antigen; the antigen is then released very slowly from the site of injection. This preparation is based on **Freund's complete adjuvant**, the first deliberately formulated highly effective adjuvant, developed by Jules Freund many years ago and containing heat-killed *Mycobacteria* as an additional ingredient. Muramyl dipeptide, a component of the mycobacterial cell wall, activates macrophages, making

TABLE 3-3 Postulated mode of action of some commonly used adjuvants

Adjuvant	POSTULATED MODE OF ACTION			
	Prolongs antigen persistence	Enhances co-stimulatory signal	Induces granuloma formation	Stimulates lymphocytes nonspecifically
Freund's incomplete adjuvant	+	+	+	—
Freund's complete adjuvant	+	++	++	—
Aluminum potassium sulfate (alum)	+	?	+	—
<i>Mycobacterium tuberculosis</i>	—	?	+	—
<i>Bordetella pertussis</i>	—	?	—	+
Bacterial lipopolysaccharide (LPS)	—	+	—	+
Synthetic polynucleotides (poly IC/poly AU)	—	?	—	+

Freund's complete adjuvant far more potent than the incomplete form. Activated macrophages are more phagocytic than unactivated macrophages and express higher levels of class II MHC molecules and the membrane molecules of the B7 family. The increased expression of class II MHC increases the ability of the antigen-presenting cell to present antigen to T_H cells. B7 molecules on the antigen-presenting cell bind to CD28, a cell-surface protein on T_H cells, triggering co-stimulation, an enhancement of the T-cell immune response. Thus, antigen presentation and the requisite co-stimulatory signal usually are increased in the presence of adjuvant.

Alum and Freund's adjuvants also stimulate a local, chronic inflammatory response that attracts both phagocytes and lymphocytes. This infiltration of cells at the site of the adjuvant injection often results in formation of a dense, macrophage-rich mass of cells called a **granuloma**. Because the macrophages in a granuloma are activated, this mechanism also enhances the activation of T_H cells.

Other adjuvants (e.g., synthetic polyribonucleotides and bacterial lipopolysaccharides) stimulate the nonspecific proliferation of lymphocytes and thus increase the likelihood of antigen-induced clonal selection of lymphocytes.

Epitopes

As mentioned in Chapter 1, immune cells do not interact with, or recognize, an entire immunogen molecule; instead, lymphocytes recognize discrete sites on the macromolecule called **epitopes**, or **antigenic determinants**. Epitopes are the immunologically active regions of an immunogen that bind to antigen-specific membrane receptors on lymphocytes or to secreted antibodies. Studies with small antigens have revealed that B and T cells recognize different epitopes on the same antigenic molecule. For example, when mice were immunized with glucagon, a small human hormone of 29 amino acids, antibody was elicited to epitopes in the amino-

terminal portion, whereas the T cells responded only to epitopes in the carboxyl-terminal portion.

Lymphocytes may interact with a complex antigen on several levels of antigen structure. An epitope on a protein antigen may involve elements of the primary, secondary, tertiary, and even quaternary structure of the protein (see Figure 3-1). In polysaccharides, branched chains are commonly present, and multiple branches may contribute to the conformation of epitopes.

The recognition of antigens by T cells and B cells is fundamentally different (Table 3-4). B cells recognize soluble antigen when it binds to their membrane-bound antibody. Because B cells bind antigen that is free in solution, the epitopes they recognize tend to be highly accessible sites on the exposed surface of the immunogen. As noted previously, most T cells recognize only peptides combined with MHC molecules on the surface of antigen-presenting cells and altered self-cells; T-cell epitopes, as a rule, cannot be considered apart from their associated MHC molecules.

Properties of B-Cell Epitopes Are Determined by the Nature of the Antigen-Binding Site

Several generalizations have emerged from studies in which the molecular features of the epitope recognized by B cells have been established.

The ability to function as a B-cell epitope is determined by the nature of the antigen-binding site on the antibody molecules displayed by B cells. Antibody binds to an epitope by weak noncovalent interactions, which operate only over short distances. For a strong bond, the antibody's binding site and the epitope must have complementary shapes that place the interacting groups near each other. This requirement poses some restriction on the properties of the epitope. The size of the epitope recognized by a B cell can be no larger than the size of the antibody's binding site. For any given antigen-antibody reaction, the shape of the epitope that can be recognized by the antibody is determined by the shape assumed by

TABLE 3-4 Comparison of antigen recognition by T cells and B cells

Characteristic	B cells	T cells
Interaction with antigen	Involves binary complex of membrane Ig and Ag	Involves ternary complex of T-cell receptor, Ag, and MHC molecule
Binding of soluble antigen	Yes	No
Involvement of MHC molecules	None required	Required to display processed antigen
Chemical nature of antigens	Protein, polysaccharide, lipid	Mostly proteins, but some lipids and glycolipids presented on MHC-like molecules
Epitope properties	Accessible, hydrophilic, mobile peptides containing sequential or nonsequential amino acids	Internal linear peptides produced by processing of antigen and bound to MHC molecules

the sequences of amino acids in the binding site and the chemical environment that they produce.

Smaller ligands such as carbohydrates, small oligonucleotides, peptides, and haptens often bind within a deep pocket of an antibody. For example, angiotensin II, a small octapeptide hormone, binds within a deep and narrow groove (725 Å²) of a monoclonal antibody specific for the hormone (Figure 3-2). Within this groove, the bound peptide hormone folds into a compact structure with two turns, which brings its amino (N-terminal) and carboxyl (C-terminal) termini close together. All eight amino acid residues of the octapeptide are in van der Waals contact with 14 residues of the antibody's groove.

A quite different picture of epitope structure emerges from x-ray crystallographic analyses of monoclonal antibodies bound to globular protein antigens such as hen egg-white lysozyme (HEL) and neuraminidase (an envelope glycoprotein of influenza virus). These antibodies make contact with the antigen across a large flat face (Figure 3-3). The interacting face between antibody and epitope is a flat or undulating surface in which protrusions on the epitope or antibody are matched by corresponding depressions on the antibody or epitope. These studies have revealed that 15–22 amino acids on the surface of the antigen make contact with a similar number of residues in the antibody's binding site; the surface area of this large complementary interface is between 650 Å² and 900 Å². For these globular protein antigens, then, the shape of the epitope is entirely determined by the tertiary conformation of the native protein.

Thus, globular protein antigens and small peptide antigens interact with antibody in different ways (Figure 3-4). Typically, larger areas of protein antigens are engaged by the antibody binding site. In contrast, a small peptide such as angiotensin II can fold into a compact structure that occupies less space and fits into a pocket or cleft of the binding site. This pattern is not unique to small peptides; it extends to the binding of low-molecular-weight antigens of various chemical types. However, these differences between the binding of small and large antigenic determinants do not reflect fundamental differences in the regions of the antibody molecule

that make up the binding site. Despite differences in the binding patterns of small haptens and large antigens, Chapter 4 will show that all antibody binding sites are assembled from the same regions of the antibody molecule—namely, parts of the variable regions of its polypeptide chains.

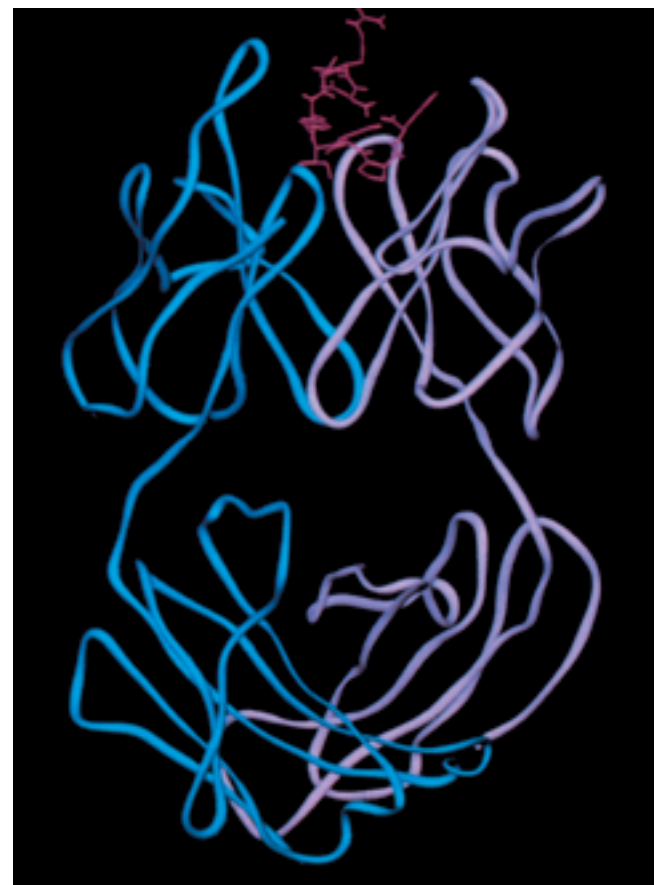


FIGURE 3-2 Three-dimensional structure of an octapeptide hormone (angiotensin II) complexed with a monoclonal antibody Fab fragment, the antigen-binding unit of the antibody molecule. The angiotensin II peptide is shown in red, the heavy chain in blue, and the light chain in purple. [From K. C. García et al., 1992, *Science* **257**:502.]

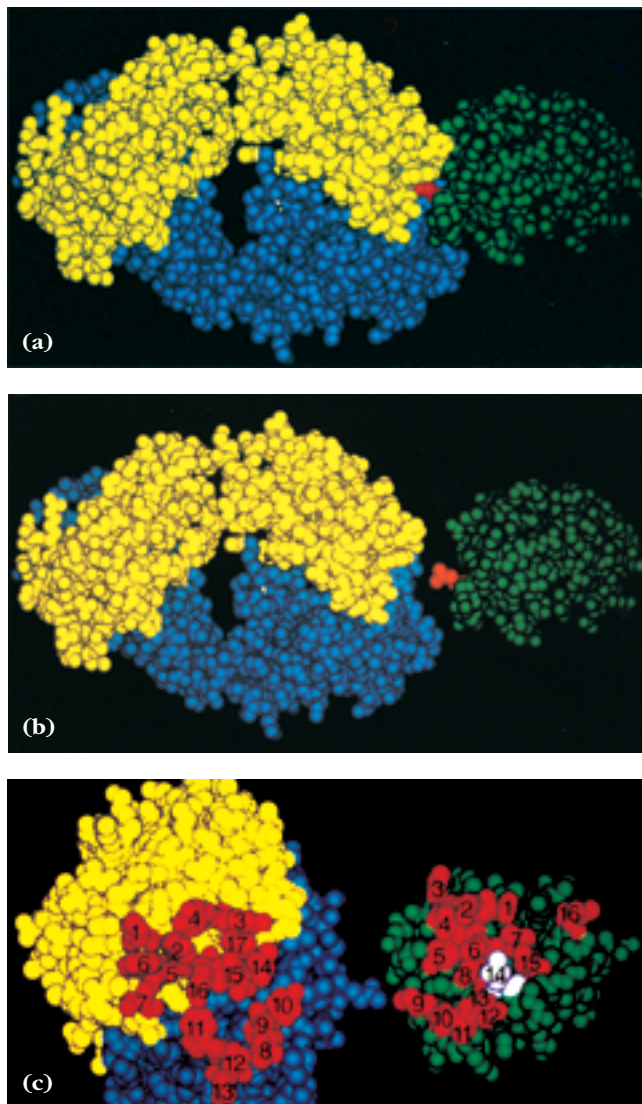


FIGURE 3-3 (a) Model of interaction between hen egg-white lysozyme (HEL) and Fab fragment of anti-HEL antibody based on x-ray diffraction analysis. HEL is shown in green, the Fab heavy chain in blue, and the Fab light chain in yellow. A glutamine residue of lysozyme (red) fits into a pocket in the Fab fragment. (b) Representation of HEL and the Fab fragment when pulled apart showing complementary surface features. (c) View of the interacting surfaces of the Fab fragment and HEL obtained by rotating each of the molecules. The contacting residues are numbered and shown in red with the protruding glutamine (#14) in HEL now shown in white. [From A. G. Amit *et al.*, 1986, *Science* **233**: 747.]

The B-cell epitopes on native proteins generally are composed of hydrophilic amino acids on the protein surface that are topographically accessible to membrane-bound or free antibody. A B-cell epitope must be accessible in order to be able to bind to an antibody; in general, protruding regions on the surface of the protein are the most likely to be recognized as epitopes, and these regions are usually composed of predominantly hydrophilic amino acids. Amino acid sequences that

are hidden within the interior of a protein often consist of predominantly hydrophobic amino acids, and cannot function as B-cell epitopes unless the protein is first denatured. In the crystallized antigen-antibody complexes analyzed to date, the interface between antibody and antigen shows numerous complementary protrusions and depressions (Figure 3-5). Between 15 and 22 amino acids on the antigen contact the antibody by 75–120 hydrogen bonds as well as by ionic and hydrophobic interactions.

B-cell epitopes can contain sequential or nonsequential amino acids. Epitopes may be composed of sequential contiguous residues along the polypeptide chain or nonsequential residues from segments of the chain brought together by the folded conformation of an antigen. Most antibodies elicited by globular proteins bind to the protein only when it is in its native conformation. Because denaturation of such antigens usually changes the structure of their epitopes, antibodies to the native protein do not bind to the denatured protein.

Five distinct **sequential epitopes**, each containing six to eight contiguous amino acids, have been found in sperm whale myoglobin. Each of these epitopes is on the surface of the molecule at bends between the α -helical regions (Figure 3-6a). Sperm whale myoglobin also contains several **nonsequential epitopes**, or **conformational determinants**. The residues that constitute these epitopes are far apart in the primary amino acid sequence but close together in the tertiary structure of the molecule. Such epitopes depend on the

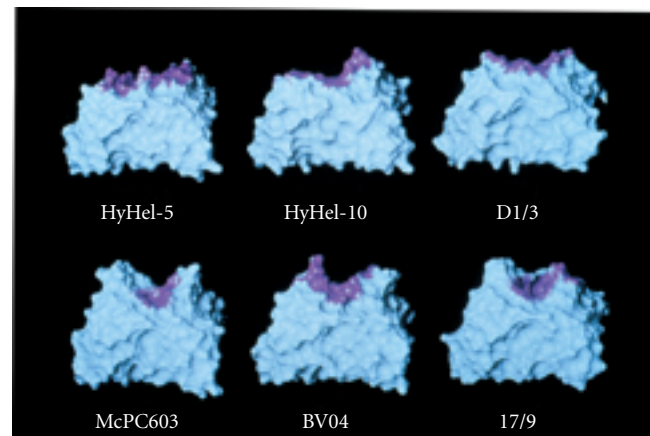


FIGURE 3-4 Models of the variable domains of six Fab fragments with their antigen-binding regions shown in purple. The top three antibodies are specific for lysozyme, a large globular protein. The lower three antibodies are specific for smaller molecules or very small segments of macromolecules: McPC603 for phosphocholine; BV04 for a small segment of a single-stranded DNA molecule; and 17/9 for a peptide from hemagglutinin, an envelope protein of influenza virus. In general, the binding sites for small molecules are deep pockets, whereas binding sites for large proteins are flatter, more undulating surfaces. [From I. A. Wilson and R. L. Stanfield, 1993, *Curr. Opin. Struc. Biol.* **3**:113.]

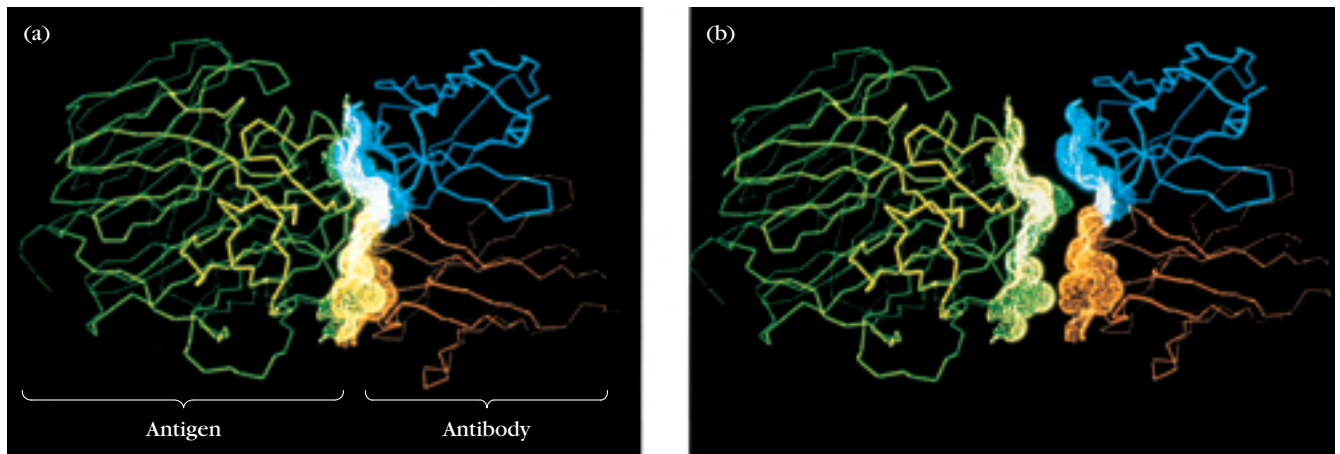


FIGURE 3-5 Computer simulation of an interaction between antibody and influenza virus antigen, a globular protein. (a) The antigen (yellow) is shown interacting with the antibody molecule; the variable region of the heavy chain is red, and the variable region of the light

chain is blue. (b) The complementarity of the two molecules is revealed by separating the antigen from the antibody by 8 Å. [Based on x-ray crystallography data collected by P. M. Colman and W. R. Tulip. From G. J. V. H. Nossal, 1993, *Sci. Am.* **269**(3):22.]



VISUALIZING CONCEPTS

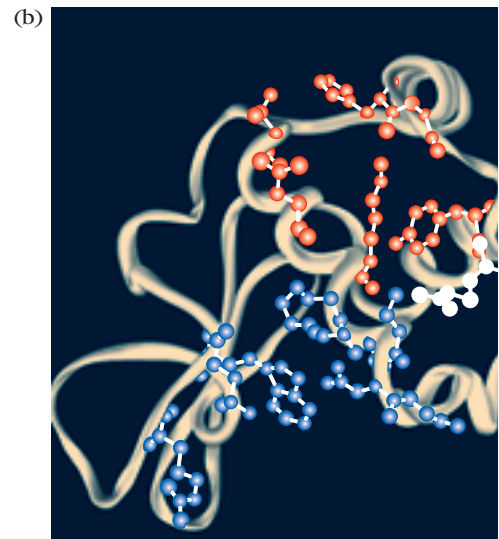
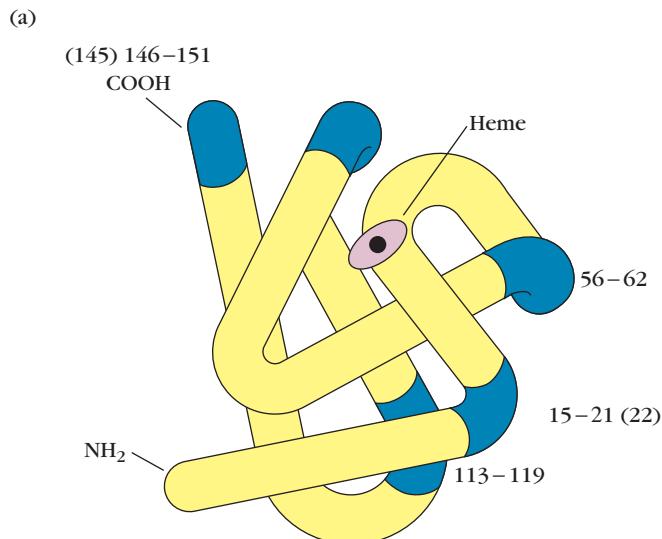


FIGURE 3-6 Protein antigens usually contain both sequential and nonsequential B-cell epitopes. (a) Diagram of sperm whale myoglobin showing locations of five sequential B-cell epitopes (blue). (b) Ribbon diagram of hen egg-white lysozyme showing residues that compose one nonsequential (conformational) epitope. Residues that contact antibody light chains, heavy chains, or

both are shown in red, blue, and white, respectively. These residues are widely spaced in the amino acid sequence but are brought into proximity by folding of the protein. [Part (a) adapted from M. Z. Atassi and A. L. Kazim. 1978, *Adv. Exp. Med. Biol.* **98**:9; part (b) from W. G. Laver et al., 1990, *Cell* **61**:554.]

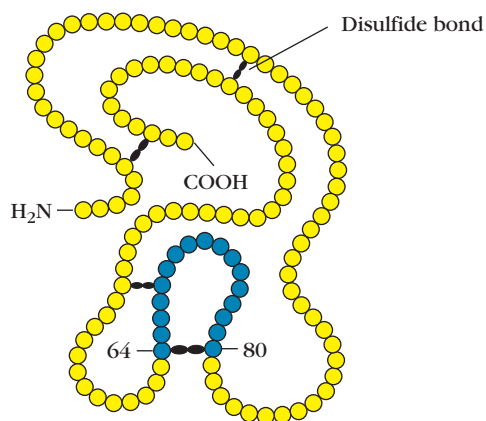
native protein conformation for their topographical structure. One well-characterized nonsequential epitope in hen egg-white lysozyme (HEL) is shown in Figure 3-6b. Although the amino acid residues that compose this epitope of HEL are far apart in the primary amino acid sequence, they are brought together by the tertiary folding of the protein.

Sequential and nonsequential epitopes generally behave differently when a protein is denatured, fragmented, or reduced. For example, appropriate fragmentation of sperm whale myoglobin can yield five fragments, each retaining one sequential epitope, as demonstrated by the observation that antibody can bind to each fragment. On the other hand, fragmentation of a protein or reduction of its disulfide bonds often destroys nonsequential epitopes. For example, HEL has four intrachain disulfide bonds, which determine the final protein conformation (Figure 3-7a). Many antibodies to HEL recognize several epitopes, and each of eight different epitopes have been recognized by a distinct antibody. Most of

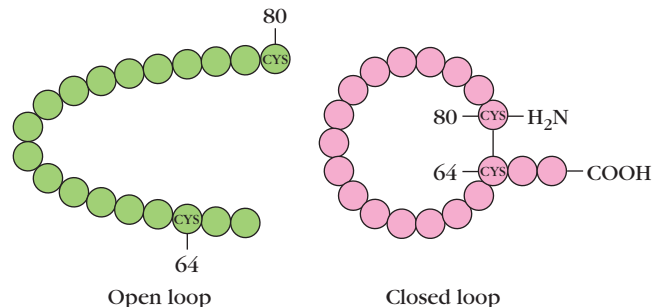
these epitopes are conformational determinants dependent on the overall structure of the protein. If the intrachain disulfide bonds of HEL are reduced with mercaptoethanol, the nonsequential epitopes are lost; for this reason, antibody to native HEL does not bind to reduced HEL.

The inhibition experiment shown in Figure 3-7 nicely demonstrates this point. An antibody to a conformational determinant, in this example a peptide loop present in native HEL, was able to bind the epitope only if the disulfide bond that maintains the structure of the loop was intact. Information about the structural requirements of the antibody combining site was obtained by examining the ability of structural relatives of the natural antigen to bind to that antibody. If a structural relative has the critical epitopes present in the natural antigen, it will bind to the antibody combining site, thereby blocking its occupation by the natural antigen. In this inhibition assay, the ability of the closed loop to inhibit binding showed that the closed loop was sufficiently

(a) Hen egg-white lysozyme



(b) Synthetic loop peptides



(c) Inhibition of reaction between HEL loop epitope and anti-loop antiserum

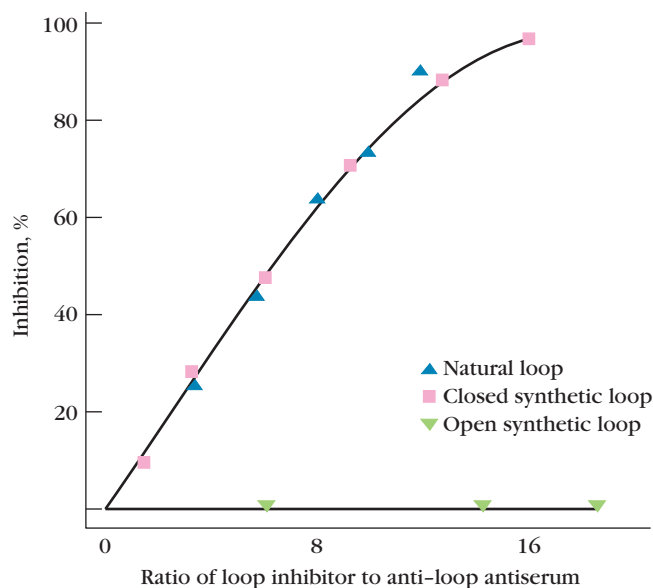


FIGURE 3-7 Experimental demonstration that binding of antibody to conformational determinants in hen egg-white lysozyme (HEL) depends on maintenance of the tertiary structure of the epitopes by intrachain disulfide bonds. (a) Diagram of HEL primary structure, in which circles represent amino acid residues. The loop (blue circles) formed by the disulfide bond between the cysteine residues at positions 64 and 80 constitutes one of the conformational determinants in HEL. (b) Synthetic open-loop and closed-loop peptides corresponding to the HEL loop epitope. (c) Inhibition of binding between HEL loop epitope and anti-loop antiserum. Anti-loop antiserum was first incubated with the natural loop sequence, the synthetic closed-loop peptide, or the synthetic open-loop peptide; the ability of the antiserum to bind the natural loop sequence then was measured. The absence of any inhibition by the open-loop peptide indicates that it does not bind to the anti-loop antiserum. [Adapted from D. Benjamin *et al.*, 1984, *Annu. Rev. Immunol.* **2**:67.]

similar to HEL to be recognized by antibody to native HEL. Even though the open loop had the same sequence of amino acids as the closed loop, it lacked the epitopes recognized by the antibody and therefore was unable to block binding of HEL.

B-cell epitopes tend to be located in flexible regions of an immunogen and display site mobility. John A. Tainer and his colleagues analyzed the epitopes on a number of protein antigens (myohemerytherin, insulin, cytochrome c, myoglobin, and hemoglobin) by comparing the positions of the known B-cell epitopes with the mobility of the same residues. Their analysis revealed that the major antigenic determinants in these proteins generally were located in the most mobile regions. These investigators proposed that site mobility of epitopes maximizes complementarity with the antibody's binding site, permitting an antibody to bind with an epitope that it might bind ineffectively if it were rigid. However, because of the loss of entropy due to binding to a flexible site, the binding of antibody to a flexible epitope is generally of lower affinity than the binding of antibody to a rigid epitope.

Complex proteins contain multiple overlapping B-cell epitopes, some of which are immunodominant. For many years, it was dogma in immunology that each globular protein had a small number of epitopes, each confined to a highly accessible region and determined by the overall conformation of the protein. However, it has been shown more recently that most of the surface of a globular protein is potentially antigenic. This has been demonstrated by comparing the antigen-binding profiles of different monoclonal antibodies to various globular proteins. For example, when 64 different monoclonal antibodies to BSA were compared for their ability to bind to a panel of 10 different mammalian albumins, 25 different overlapping antigen-binding profiles emerged, suggesting that these 64 different antibodies recognized a minimum of 25 different epitopes on BSA. Similar findings have emerged for other globular proteins, such as myoglobin and HEL.

The surface of a protein, then, presents a large number of potential antigenic sites. The subset of antigenic sites on a given protein that is recognized by the immune system of an animal is much smaller than the potential antigenic repertoire, and it varies from species to species and even among in-

dividual members of a given species. Within an animal, certain epitopes of an antigen are recognized as immunogenic, but others are not. Furthermore, some epitopes, called **immunodominant**, induce a more pronounced immune response than other epitopes in a particular animal. It is highly likely that the intrinsic topographical properties of the epitope as well as the animal's regulatory mechanisms influence the immunodominance of epitopes.

Antigen-Derived Peptides Are the Key Elements of T-Cell Epitopes

Studies by P. G. H. Gell and Baruj Benacerraf in 1959 suggested that there was a qualitative difference between the T-cell and the B-cell response to protein antigens. Gell and Benacerraf compared the humoral (B-cell) and cell-mediated (T-cell) responses to a series of native and denatured protein antigens (Table 3-5). They found that when primary immunization was with a native protein, only native protein, not denatured protein, could elicit a secondary antibody (humoral) response. In contrast, both native and denatured protein could elicit a secondary cell-mediated response. The finding that a secondary response mediated by T cells was induced by denatured protein, even when the primary immunization had been with native protein, initially puzzled immunologists. In the 1980s, however, it became clear that T cells do not recognize soluble native antigen but rather recognize antigen that has been processed into **antigenic peptides**, which are presented in combination with MHC molecules. For this reason, destruction of the conformation of a protein by denaturation does not affect its T-cell epitopes.

Because the T-cell receptor does not bind free peptides, experimental systems for studying T-cell epitopes must include antigen-presenting cells or target cells that can display the peptides bound to an MHC molecule.

Antigenic peptides recognized by T cells form trimolecular complexes with a T-cell receptor and an MHC molecule (Figure 3-8). The structures of TCR-peptide-MHC trimolecular complexes have been determined by x-ray crystallography and are described in Chapter 9. These structural studies of class I or class II MHC molecules crystallized with known T-cell antigenic peptides has shown that the peptide binds to a

TABLE 3-5 Antigen recognition by T and B lymphocytes reveals qualitative differences

Primary immunization	Secondary immunization	SECONDARY IMMUNE RESPONSE	
		Antibody production	Cell-mediated T _{DTH} response*
Native protein	Native protein	+	+
Native protein	Denatured protein	—	+

*T_{DTH} is a subset of CD4⁺ T_H cells that mediate a cell-mediated response called delayed-type hypersensitivity (see Chapter 14).

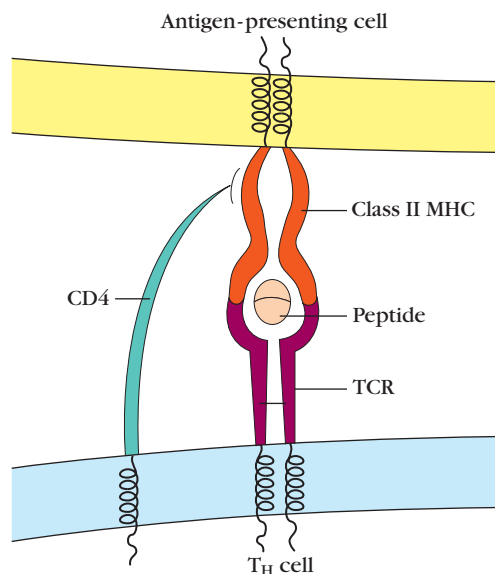


FIGURE 3-8 Schematic diagram of the ternary complex formed between a T-cell receptor (TCR) on a T_H cell, an antigen, and a class II MHC molecule. Antigens that are recognized by T cells yield peptides that interact with MHC molecules to form a peptide-MHC complex that is recognized by the T-cell receptor. As described in later chapters, the coreceptor, CD4, on T_H cells also interacts with MHC molecules. T_C cells form similar ternary complexes with class I MHC molecules on target cells however, these cells bear MHC-interacting CD8 coreceptors.

cleft in the MHC molecule (see Figure 7-8). Unlike B-cell epitopes, which can be viewed strictly in terms of their ability to interact with antibody, T-cell epitopes must be viewed in terms of their ability to interact with both a T-cell receptor and an MHC molecule.

The binding of an MHC molecule to an antigenic peptide does not have the fine specificity of the interaction between an antibody and its epitope. Instead, a given MHC molecule can selectively bind a variety of different peptides. For example, the class II MHC molecule designated IA^d can bind peptides from ovalbumin (residues 323–339), hemagglutinin (residues 130–142), and lambda repressor (residues 12–26). Studies revealing structural features, or motifs, common to different peptides that bind to a single MHC molecule are described in Chapter 7.

Antigen processing is required to generate peptides that interact specifically with MHC molecules. As mentioned in Chapter 1, endogenous and exogenous antigens are usually processed by different intracellular pathways (see Figure 1-9). Endogenous antigens are processed into peptides within the cytoplasm, while exogenous antigens are processed by the endocytic pathway. The details of antigen processing and presentation are described in Chapter 8.

Epitopes recognized by T cells are often internal. T cells tend to recognize internal peptides that are exposed by processing within antigen-presenting cells or altered self-cells. J. Rothbard analyzed the tertiary conformation of hen egg-white lysozyme and sperm whale myoglobin to determine which amino acids protruded from the natural molecule. He then mapped the major T-cell epitopes for both proteins and found that, in each case, the T-cell epitopes tended to be on the “inside” of the protein molecule (Figure 3-9).

Haptens and the Study of Antigenicity

The pioneering work of Karl Landsteiner in the 1920s and 1930s created a simple, chemically defined system for studying the binding of an individual antibody to a unique epitope

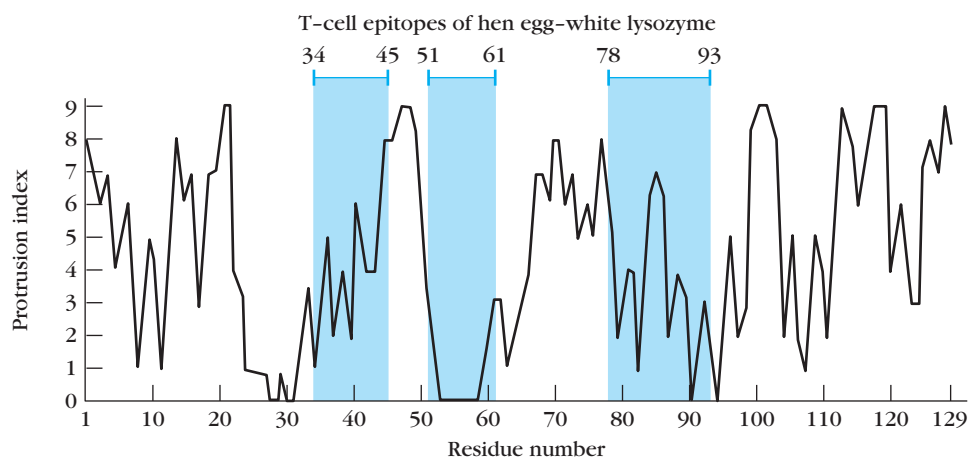


FIGURE 3-9 Experimental evidence that T_H cells tend to recognize internal peptides of antigens. This plot shows the relative protrusion of amino acid residues in the tertiary conformation of hen egg-white lysozyme. The known T-cell epitopes in HEL are indicated by the blue bars at the top. Notice that, in general, the amino acid residues that

correspond to the T-cell epitopes exhibit less overall protrusion. In contrast, note that the B-cell epitope consisting of residues 64–80, which form a conformational determinant in native HEL that is recognized by antibody (see Figure 3-7), exhibit greater overall protrusion. [From J. Rothbard et al., 1987, *Mod. Trends Hum. Leuk.*, vol. 7.]

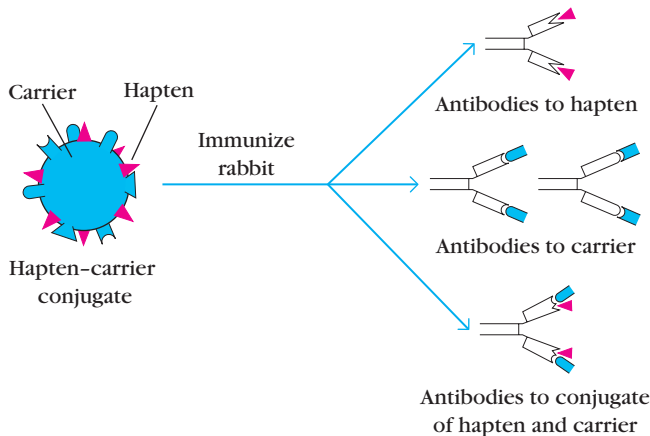
on a complex protein antigen. Landsteiner employed various **haptens**, small organic molecules that are antigenic but not immunogenic. Chemical coupling of a hapten to a large protein, called a **carrier**, yields an immunogenic **hapten-carrier conjugate**. Animals immunized with such a conjugate produce antibodies specific for (1) the hapten determinant, (2) unaltered epitopes on the carrier protein, and (3) new epitopes formed by combined parts of both the hapten and carrier (Figure 3-10). By itself, a hapten cannot function as an immunogenic epitope. But when multiple molecules of a single hapten are coupled to a carrier protein (or nonimmunogenic homopolymer), the hapten becomes accessible to the immune system and can function as an immunogen.

The beauty of the hapten-carrier system is that it provides immunologists with a chemically defined determinant that can be subtly modified by chemical means to determine the effect of various chemical structures on immune specificity. In his studies, Landsteiner immunized rabbits with a hapten-carrier conjugate and then tested the reactivity of the rabbit's immune sera with that hapten and with closely related haptens coupled to a different carrier protein. He was thus able to measure, specifically, the reaction of the anti-hapten antibodies in the immune serum and not that of antibodies to the

original carrier epitopes. Landsteiner tested whether an anti-hapten antibody could bind to other haptens having a slightly different chemical structure. If a reaction occurred, it was called a **cross-reaction**. By observing which hapten modifications prevented or permitted cross-reactions, Landsteiner was able to gain insight into the specificity of antigen-antibody interactions.

Using various derivatives of aminobenzene as haptens, Landsteiner found that the overall configuration of a hapten plays a major role in determining whether it can react with a given antibody. For example, antiserum from rabbits immunized with aminobenzene or one of its carboxyl derivatives (*o*-aminobenzoic acid, *m*-aminobenzoic acid, or *p*-aminobenzoic acid) coupled to a carrier protein reacted only with the original immunizing hapten and did not cross-react with any of the other haptens (Table 3-6). In contrast, if the overall configuration of the hapten was kept the same and the hapten was modified in the para position with various nonionic derivatives, then the antisera showed various degrees of cross-reactivity. Landsteiner's work not only demonstrated the specificity of the immune system, but also demonstrated the enormous diversity of epitopes that the immune system is capable of recognizing.

Many biologically important substances, including drugs, peptide hormones, and steroid hormones, can function as haptens. Conjugates of these haptens with large protein carriers can be used to produce hapten-specific antibodies. These antibodies are useful for measuring the presence of various substances in the body. For instance, the original home pregnancy test kit employed anti-hapten antibodies to determine whether a woman's urine contained human chorionic gonadotropin (HCG), which is a sign of pregnancy. However, as shown in the Clinical Focus, the formation of drug-protein conjugates in the body can produce drug allergies that may be life-threatening.



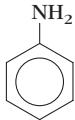
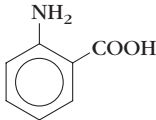
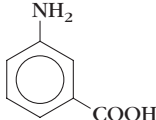
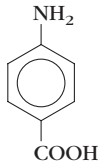
Injection with:	Antibodies formed:
Hapten (DNP)	None
Protein carrier (BSA)	Anti-BSA
Hapten-carrier conjugate (DNP-BSA)	Anti-DNP (major) Anti-BSA (minor) Anti-DNP/BSA (minor)

FIGURE 3-10 A hapten-carrier conjugate contains multiple copies of the hapten—a small nonimmunogenic organic compound such as dinitrophenol (DNP)—chemically linked to a large protein carrier such as bovine serum albumin (BSA). Immunization with DNP alone elicits no anti-DNP antibodies, but immunization with DNP-BSA elicits three types of antibodies. Of these, anti-DNP antibody is predominant, indicating that in this case the hapten is the immunodominant epitope in a hapten-carrier conjugate, as it often is in such conjugates.

Pattern-Recognition Receptors

The receptors of adaptive and innate immunity differ. Antibodies and T-cell receptors, the receptors of adaptive immunity, recognize details of molecular structure and can discriminate with exquisite specificity between antigens featuring only slight structural differences. The receptors of innate immunity recognize broad structural motifs that are highly conserved within microbial species but are generally absent from the host. Because they recognize particular overall molecular patterns, such receptors are called **pattern-recognition receptors (PRRs)**. Patterns recognized by this type of receptor include combinations of sugars, certain proteins, particular lipid-bearing molecules, and some nucleic acid motifs. Typically, the ability of pattern-recognition receptors to distinguish between self and nonself is perfect because the molecular pattern targeted by the receptor is produced only by the pathogen and never by the host. This contrasts sharply with the occasional recognition of self

TABLE 3-6 Reactivity of antisera with various haptens

Antiserum against	REACTIVITY WITH			
				
	Aminobenzene (aniline)	<i>o</i> -Aminobenzoic acid	<i>m</i> -Aminobenzoic acid	<i>p</i> -Aminobenzoic acid
Aminobenzene	+	0	0	0
<i>o</i> -Aminobenzoic acid	0	+	0	0
<i>m</i> -Aminobenzoic acid	0	0	+	0
<i>p</i> -Aminobenzoic acid	0	0	0	+

KEY: 0 = no reactivity; + = strong reactivity

SOURCE: Based on K. Landsteiner, 1962, *The Specificity of Serologic Reactions*, Dover Press. Modified by J. Klein, 1982, *Immunology: The Science of Self-Nonself Discrimination*, John Wiley.

antigens by receptors of adaptive immunity, which can lead to autoimmune disorders. Like antibodies and T-cell receptors, pattern-recognition receptors are proteins. However, the genes that encode PRRs are present in the germline of the organism. In contrast, the genes that encode the enormous diversity of antibodies and TCRs are not present in the germline. They are generated by an extraordinary process of genetic recombination that is discussed in Chapter 5.

Many different pattern-recognition receptors have been identified and several examples appear in Table 3-7. Some are present in the bloodstream and tissue fluids as soluble circulating proteins and others are on the membrane of cells such as macrophages, neutrophils, and dendritic cells. Mannose-binding lectin (MBL) and C-reactive protein (CRP) are soluble pattern receptors that bind to microbial surfaces and promote their opsonization. Both of these receptors also have the ability to activate the complement system when they are bound to the surface of microbes, thereby making the invader a likely target of complement-mediated lysis. Yet another soluble receptor of the innate immune system, lipopolysaccharide-binding protein, is an important part of the system that recognizes and signals a response to lipopolysaccharide, a component of the outer cell wall of gram-negative bacteria.

Pattern-recognition receptors found on the cell membrane include scavenger receptors and the toll-like receptors. Scavenger receptors (SRs) are present on macrophages and many types of dendritic cells, and are involved in the binding and internalization of gram-positive and gram-negative bacteria, as well as the phagocytosis of apoptotic host cells. The exact roles and mechanisms of action of the many types of scavenger receptors known to date are under active investigation. The toll-like receptors (TLRs) are important in recognizing many microbial patterns. This family of proteins is

ancient—toll-like receptors mediate the recognition and generation of defensive responses to pathogens in organisms as widely separated in evolutionary history as humans and flies. Typically, signals transduced through the TLRs cause transcriptional activation and the synthesis and secretion of cytokines, which promote inflammatory responses that bring macrophages and neutrophils to sites of inflammation.

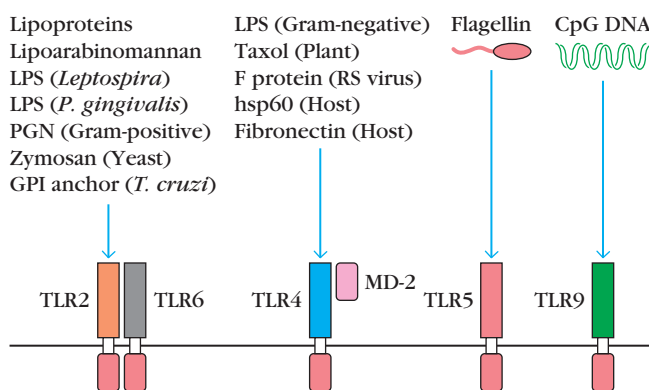


FIGURE 3-11 Location and targets of some pattern-recognition receptors. Many pattern-recognition receptors are extracellular and target microbes or microbial components in the bloodstream and tissue fluids, causing their lysis or marking them for removal by phagocytes. Other pattern-recognition receptors are present on the cell membrane and bind to a broad variety of microbes or microbial products. Engagement of these receptors triggers signaling pathways that promote inflammation or, in the case of the scavenger receptors, phagocytosis or endocytosis. dsRNA = double stranded RNA; LPS = lipopolysaccharide. [S. Akira *et al.*, 2001, *Nature Immunology* 2:675.]

TABLE 3-7 Receptors of innate and adaptive immunity

Characteristic	Innate immunity	Adaptive immunity
Specificity	Specific for conserved molecular patterns or types	Specific for details of antigen structure
Self/nonself discrimination	Perfect: evolutionarily selected to distinguish phylogenetic differences. Never recognizes self.	Excellent: but imperfect. Occasional reaction with self antigens
RECEPTORS OF THE ADAPTIVE IMMUNE SYSTEM		
Receptor (location)	Target (source)	Effect of recognition
Antibody (B-cell membrane, blood, tissue fluids)	Specific components of pathogen	Labeling of pathogen for destruction and removal
T-cell receptor (T-cell membrane)	Proteins or certain lipids of pathogen	Induction of pathogen-specific humoral and cell-mediated immunity
RECEPTORS OF THE INNATE IMMUNE SYSTEM		
Complement (bloodstream, tissue fluids)	Microbial cell-wall components	Complement activation, opsonization
Mannose-binding lectin (MBL) (bloodstream, tissue fluids)	Mannose-containing microbial carbohydrates (cell walls)	Complement activation, opsonization
C-reactive protein (CRP) (bloodstream, tissue fluids)	Phosphatidylcholine (microbial membranes)	Complement activation, opsonization
LPS-binding protein (LBP) (bloodstream, tissue fluids)	Bacterial lipopolysaccharide (LPS)	Delivery to cell-membrane LPS receptor (TLR-CD14-MD-2 complex*)
TLR2 (cell membrane)	Cell-wall components of gram-positive bacteria, LPS*. Yeast cell-wall component (zymosan)	Attracts phagocytes, activates macrophages, dendritic cells. Induces secretion of several cytokines
TLR3 (cell membrane)	Double-stranded RNA (dsRNA) (replication of many RNA viruses)	Induces production of interferon, an antiviral cytokine
TLR4 (cell membrane)	LPS*	Attracts phagocytes, activates macrophages, dendritic cells. Induces secretion of several cytokines
TLR5 (cell membrane)	Flagellin (flagella of gram-positive and gram-negative bacteria)	Attracts phagocytes, activates macrophages, dendritic cells. Induces secretion of several cytokines
TLR9 (cell membrane)	CpG	Attracts phagocytes, macrophages, dendritic cells. Induces secretion of several cytokines
Scavenger receptors (many) (cell membrane)	Many targets; gram-positive and gram-negative bacteria, apoptotic host cells	Induces phagocytosis or endocytosis

*LPS is bound at the cell membrane by a complex of proteins that includes CD14, MD-2, and a TLR (usually TLR4).



CLINICAL FOCUS

Drug Allergies—When Medicines Become Immunogens

Since World War II,

penicillin has been used to successfully treat a wide variety of bacterial infections. However, the penicillin family of antibiotics is not without drawbacks. One is the role of penicillins and other antibiotics in the evolution of antibiotic-resistant bacterial strains. Another is their capacity to induce allergic reactions in some patients. Penicillin and its relatives are responsible for most of the recorded allergic reactions to drugs and 97% of the deaths caused each year by drug allergies.

Allergies to penicillin and other drugs can be induced by small doses and are not consequences of the pharmacological or physiological effects of the drugs. An allergic response usually occurs about a week or so after the patient's first exposure to the agent, with typically mild symptoms often including hives, fever, swelling of lymph nodes, and occasion-

ally an arthritis-like discomfort. Subsequent treatments with the drug usually cause much more rapid and often more severe reactions. Within minutes the throat and eyelids may swell. Grave danger arises if these symptoms progress to anaphylaxis, a physiological collapse that often involves the respiratory, circulatory, and digestive systems. Hives, vomiting, abdominal pain, and diarrhea may be a preamble to respiratory and circulatory problems that are life threatening. Wheezing and shortness of breath may be accompanied by swelling of the larynx and epiglottis that can block airflow, and a profound drop in blood pressure causes shock, frequently accompanied by weakened heart contractions.

The treatment of choice for anaphylaxis is injection of the drug epinephrine (adrenaline), which can reverse the body's slide into deep anaphylaxis by raising blood pressure, easing constriction of the air passages, and inhibiting

the release from mast cells and basophils of the agents that induce anaphylaxis. Other drugs may be used to raise the low blood pressure, strengthen heart contractions, and expand the blocked airways. After a case of drug-induced anaphylaxis, affected individuals are advised to carry a notice warning future healthcare providers of the drug allergy.

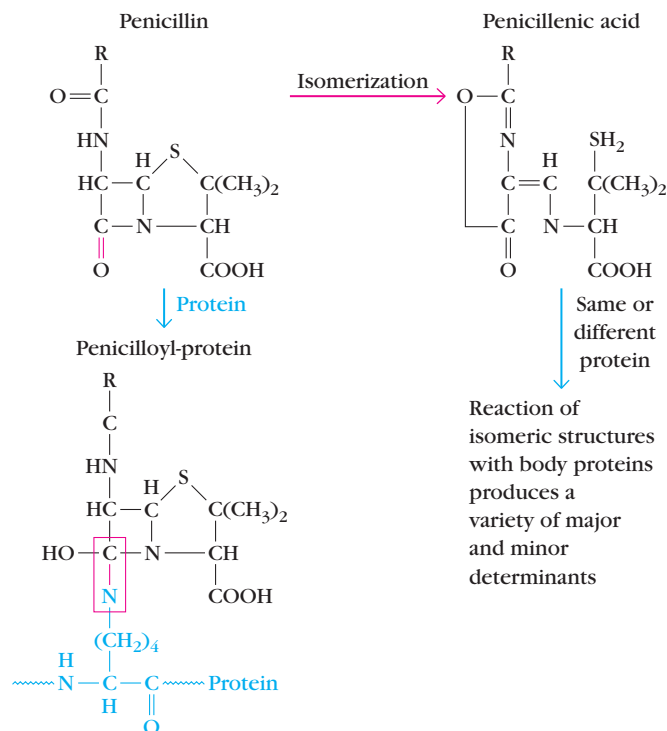
Most drugs, including penicillin, are low-molecular-weight compounds that cannot induce immune responses unless they are conjugated with a larger molecule. Intensive investigation of allergy to penicillin has provided critical insight into the basis of allergic reactions to this and other drugs. As shown in the accompanying figure, penicillin can react with proteins to form a penicilloyl-protein derivative. The penicilloyl-protein behaves as a hapten-carrier conjugate, with the penicilloyl group acting as a haptenic epitope. This epitope is readily recognized by the immune system, and antibodies are produced against it. Some individuals respond to penicillin by producing significant amounts of a type of antibody known as immunoglobulin E (IgE). Once generated, these IgE antibodies are dispersed throughout the body and are bound by IgE receptors on the surfaces of mast cells and basophils,

TLR signaling can also result in the recruitment and activation of macrophages, NK cells, and dendritic cells, key agents in the presentation of antigen to T cells. The links to T cells and cytokine release shows the intimate relationship between innate and adaptive responses.

A search of the human genome has uncovered 10 TLRs, and the functions of six members of this PRR family have been determined. TLR2, often with the collaboration of TLR6, binds a wide variety of molecular classes found in microbes, including peptidoglycans, zymosans, and bacterial lipopeptides. TLR4 is the key receptor for most bacterial lipopolysaccharides, although TLR2 also binds some varieties of LPS. The binding of LPS by either of these TLRs is complex and involves the participation of three additional proteins, one of which is the lipopolysaccharide-binding protein mentioned above, abbreviated LBP. The first step in the process is the binding of LPS by circulating LBP, which

then delivers it to a complex of TLR4 (or TLR2) with two additional proteins, CD14 and MD2. The engagement of LPS by this complex causes its TLR component to initiate a signal-transduction process that can produce a cellular response. Another family member, TLR5, recognizes flagellin, the major structural component of bacterial flagella. TLR3 recognizes the double-stranded RNA (dsRNA) that appears after infection by RNA viruses. As shown in Table 3-7, dsRNA is also recognized by dsRNA-activated kinase. Finally, TLR9 recognizes and initiates a response to CpG (unmethylated cytosine linked to guanine) sequences. These sequences are represented in abundance in microbial sequences but are much less common in mammalian sequences. Table 3-7 summarizes the receptors of adaptive immunity and lists many pattern-recognition receptors of innate immunity. The microbial targets and physiological sites of many PRRs are shown in Figure 3-11.

When nucleophiles such as amino groups or hydroxyl groups are present on soluble proteins or on the membrane of cells, they can react with penicillin and its relatives to form covalent linkages between host macromolecular structures and the drug. This is illustrated by the reaction of the free amino group of a lysine residue with penicillin (or with its spontaneously forming isomeric compounds, such as penicillenic acid). Such adducts are the major immunogenic species that elicit immune responses to this antibiotic. However, as indicated, other hapten-carrier conjugates of somewhat different structure are also formed and, because of their structural similarity, can also induce immune responses to penicillin. [Adapted from N. F. Adkinson, 1995, in *Manual of Clinical Laboratory Immunology*, N. Rose et al., eds., American Society for Microbiology, Washington, D.C.]



where they can remain for a long time. If a person with penicillin-specific IgE antibody bound to mast cells is subsequently treated with penicillin, there may be an allergic reaction. In fact, between 1 and 5 percent of people treated with penicillin develop some degree of allergy to it.

Penicillin is not the only drug against which patients can develop allergies. Others include streptomycin, aspirin, the so-called “sulfa-drugs” such as the sulfonamides, some anesthetics (e.g., succinyl choline), and some opiates. All of these small molecules first react with proteins to form drug-protein deriva-

tives. When this happens, there is a possibility that the immune system will produce an anti-hapten response to the drug, just as with penicillin. Drugs (and their metabolites) that are incapable of forming drug-protein conjugates rarely elicit allergic reactions.

SUMMARY

- All immunogens are antigens but not all antigens are immunogens.
- Immunogenicity is determined by many factors including foreignness, molecular size, chemical composition, complexity, dose, susceptibility to antigen processing and presentation, the genotype of the recipient animal (in particular, its MHC genes), route of administration, and adjuvants.
- The sizes of B-cell epitopes range widely. Some are quite small (e.g., small peptides or small organic molecules), and are often bound in narrow grooves or deep pockets of the antibody. Protein B-cell epitopes are much larger and interact with a larger, flatter complementary surface on the antibody molecule.

- T-cell epitopes are generated by antigen processing, which fragments protein into small peptides that combine with class I or class II MHC molecules to form peptide-MHC complexes that are displayed on the surface of cells. T-cell activation requires the formation of a ternary complex between a T cell's TCR and peptide-MHC on antigen-presenting or altered self cells.
- Haptens are small molecules that can bind to antibodies but cannot by themselves induce an immune response. However, the conjugate formed by coupling a hapten to a large carrier protein is immunogenic and elicits production of anti-hapten antibodies when injected into an animal. Such injections also produce anti-carrier and anti-hapten/carrier antibodies as well.
- In the body, the formation of hapten-carrier conjugates is the basis of allergic responses to drugs such as penicillin.

- The innate immune system uses pattern-recognition receptors to recognize and respond to broad structural motifs that are highly conserved within microbial species but are generally absent from the host.

References

- Berzofsky, J. A., and J. J. Berkower. 1999. Immunogenicity and antigen structure. In *Fundamental Immunology*, 4th ed., W. E. Paul, ed., Lippincott-Raven, Philadelphia.
- Dale, D., and D. Federman, eds. 1997. Drug allergy. In *Scientific American Medicine*. Chapter VIII, Hypersensitivity and allergy, p. 27.
- Demotz, S., H. M. Grey, E. Appella, and A. Sette. 1989. Characterization of a naturally processed MHC class II-restricted T-cell determinant of hen egg lysozyme. *Nature* **342**:682.
- Grey, H. M., A. Sette, and S. Buus. 1989. How T cells see antigen. *Sci. Am.* **261**(5):56.
- Landsteiner, K. 1945. *The Specificity of Serological Reactions*. Harvard University Press, Cambridge, Massachusetts.
- Laver, W. G., G. M. Air, R. G. Webster, and S. J. Smith-Gill. 1990. Epitopes on protein antigens: misconceptions and realities. *Cell* **61**:553.
- Peiser, L., S. Mukhopadhyay, and S. Gordon. 2002. Scavenger receptors in innate immunity. *Curr. Opin. Immunol.* **14**:123.
- Stanfield, R. L., and I. A. Wilson. 1995. Protein-peptide interactions. *Curr. Opin. Struc. Biol.* **5**:103.
- Tainer, J. A., et al. 1985. The atomic mobility component of protein antigenicity. *Annu. Rev. Immunol.* **3**:501.
- Underhill, D. M., and A. Ozinsky. 2002. Toll-like receptors: key mediators of microbe detection. *Curr. Opin. Immunol.* **14**:103.



USEFUL WEB SITES

<http://www.umass.edu/microbio/rasmol/>

RASMOL is free software for visualizing molecular structures that can be run on Windows-based, Macintosh, or Unix PCs. With it one can view three-dimensional structures of many types of molecules, including proteins and nucleic acids.

<http://www.expasy.ch/>

This is the excellent and comprehensive Swiss Institute of Bioinformatics (SIB) Web site, which contains extensive information on protein structure. From it one can obtain protein sequences and three-dimensional structures of proteins, as well as the versatile Swiss-PdbViewer software, which has several advanced capabilities not found in RASMOL.

Study Questions

CLINICAL FOCUS QUESTION Consider the following situations and provide a likely diagnosis or appropriate response.

- a. Six hours after receiving a dose of penicillin, a young child who has never been treated with penicillin develops a case of hives and diarrhea. The parents report the illness and ask if it might be an allergic reaction to penicillin.
 - b. A patient who has never taken sulfonamides but is known to be highly allergic to penicillin develops a bladder infection that is best treated with a “sulfa” drug. The patient wonders if “sulfa” drugs should be avoided.
 - c. A student who is unaware that he had developed a significant allergy to penicillin received an injection of the antibiotic and within minutes experienced severe respiratory distress and a drop in blood pressure. An alert intern administered epinephrine and the patient’s condition improved quickly. Frightened but impressed by the effectiveness of the treatment, he asked the intern why the shot of adrenaline made him feel better.
 - d. A pet owner asks whether the same mechanism that causes his allergy to penicillin could also be responsible for his dog’s development of a similar allergy to the drug. (Please go beyond yes or no.)
1. Indicate whether each of the following statements is true or false. If you think a statement is false, explain why.
 - a. Most antigens induce a response from more than one clone.
 - b. A large protein antigen generally can combine with many different antibody molecules.
 - c. A hapten can stimulate antibody formation but cannot combine with antibody molecules.
 - d. MHC genes play a major role in determining the degree of immune responsiveness to an antigen.
 - e. T-cell epitopes tend to be accessible amino acid residues that can combine with the T-cell receptor.
 - f. Many B-cell epitopes are nonsequential amino acids brought together by the tertiary conformation of a protein antigen.
 - g. Both T_H and T_C cells recognize antigen that has been processed and presented with an MHC molecule.
 - h. Each MHC molecule binds a unique peptide.
 - i. All antigens are also immunogens.
 - j. Antibodies can bind hydrophilic or hydrophobic compounds, but T-cell receptors can only bind peptide-MHC complexes.
 2. What would be the likely outcome of each of the developments indicated below. Please be as specific as you can.
 - a. An individual is born with a mutation in C-reactive protein that enables it to recognize phospholipids in both bacterial and mammalian cell membranes.
 - b. A group of mice in which the CD1 family has been “knocked out” are immunized with *Mycobacterium tuberculosis*. Spleen cells from these mice are isolated and divided into two batches. One batch is treated with a lipid extract of the bacteria and a second batch is treated with a protein derived from the bacteria known as purified protein derivative (PPD).
 3. Two vaccines are described below. Would you expect either or both of them to activate T_C cells? Explain your answer.
 - a. A UV-inactivated (“killed”) viral preparation that has retained its antigenic properties but cannot replicate.

- b. An attenuated viral preparation that has low virulence but can still replicate within host cells.
4. For each pair of antigens listed below, indicate which is likely to be more immunogenic. Explain your answer.
- a. Native bovine serum albumin (BSA)
Heat-denatured BSA
 - b. Hen egg-white lysozyme (HEL)
Hen collagen
 - c. A protein with a molecular weight of 30,000
A protein with a molecular weight of 150,000
 - d. BSA in Freund's complete adjuvant
BSA in Freund's incomplete adjuvant
5. Indicate which of the following statements regarding haptens and carriers are true.
- a. Haptens are large protein molecules such as BSA.
 - b. When a hapten-carrier complex containing multiple hapten molecules is injected into an animal, most of the induced antibodies are specific for the hapten.
 - c. Carriers are needed only if one wants to elicit a cell-mediated response.
 - d. It is necessary to immunize with a hapten-carrier complex in order to obtain antibodies directed against the hapten.
 - e. Carriers include small molecules such as dinitrophenol and penicillenic acid (derived from penicillin).
6. For each of the following statements, indicate whether it is true only of B-cell epitopes (B), only of T-cell epitopes (T), or both types of epitopes (BT) within a large antigen.
- a. They almost always consist of a linear sequence of amino acid residues.
 - b. They generally are located in the interior of a protein antigen.
 - c. They generally are located on the surface of a protein antigen.
 - d. They lose their immunogenicity when a protein antigen is denatured by heat.
 - e. Immunodominant epitopes are determined in part by the MHC molecules expressed by an individual.
 - f. They generally arise from proteins.
 - g. Multiple different epitopes may occur in the same antigen.
 - h. Their immunogenicity may depend on the three-dimensional structure of the antigen.
 - i. The immune response to them may be enhanced by co-administration of Freund's complete adjuvant.

Antibodies:

Structure and Function

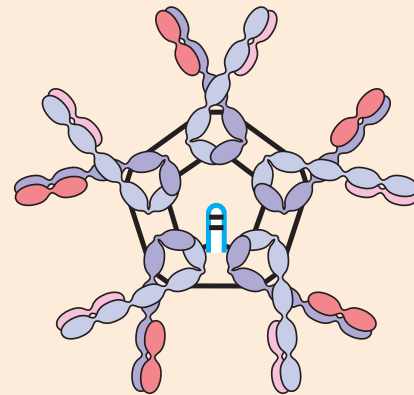
ANTIBODIES ARE THE ANTIGEN-BINDING PROTEINS present on the B-cell membrane and secreted by plasma cells. Membrane-bound antibody confers antigenic specificity on B cells; antigen-specific proliferation of B-cell clones is elicited by the interaction of membrane antibody with antigen. Secreted antibodies circulate in the blood, where they serve as the effectors of humoral immunity by searching out and neutralizing antigens or marking them for elimination. All antibodies share structural features, bind to antigen, and participate in a limited number of effector functions.

The antibodies produced in response to a particular antigen are heterogeneous. Most antigens are complex and contain many different antigenic determinants, and the immune system usually responds by producing antibodies to several epitopes on the antigen. This response requires the recruitment of several clones of B cells. Their outputs are monoclonal antibodies, each of which specifically binds a single antigenic determinant. Together, these monoclonal antibodies make up the polyclonal and heterogeneous serum antibody response to an immunizing antigen.

Basic Structure of Antibodies

Blood can be separated in a centrifuge into a fluid and a cellular fraction. The fluid fraction is the **plasma** and the cellular fraction contains red blood cells, leukocytes, and platelets. Plasma contains all of the soluble small molecules and macromolecules of blood, including fibrin and other proteins required for the formation of blood clots. If the blood or plasma is allowed to clot, the fluid phase that remains is called **serum**. It has been known since the turn of the century that antibodies reside in the serum. The first evidence that antibodies were contained in particular serum protein fractions came from a classic experiment by A. Tiselius and E. A. Kabat, in 1939. They immunized rabbits with the protein ovalbumin (the albumin of egg whites) and then divided the immunized rabbits' serum into two aliquots. Electrophoresis of one serum aliquot revealed four peaks corresponding to albumin and the alpha (α), beta (β), and gamma (γ) globulins. The other serum aliquot was reacted with ovalbumin, and the precipitate that formed was removed; the remaining serum proteins, which did not react with the antigen, were then electrophoresed. A comparison of the electrophoretic profiles of these two serum aliquots revealed that there was a significant drop in the γ -globulin

chapter 4



IgM, the First Responder

- Basic Structure of Antibodies
- Obstacles to Antibody Sequencing
- Immunoglobulin Fine Structure
- Antibody-Mediated Effector Functions
- Antibody Classes and Biological Activities
- Antigenic Determinants on Immunoglobulins
- The B-Cell Receptor
- The Immunoglobulin Superfamily
- Monoclonal Antibodies

peak in the aliquot that had been reacted with antigen (Figure 4-1). Thus, the **γ -globulin fraction** was identified as containing serum antibodies, which were called **immunoglobulins**, to distinguish them from any other proteins that might be contained in the γ -globulin fraction. The early experiments of Kabat and Tiselius resolved serum proteins into three major nonalbumin peaks— α , β and γ . We now know that although immunoglobulin G (IgG), the main class of antibody molecules, is indeed mostly found in the γ -globulin fraction, significant amounts of it and other important classes of antibody molecules are found in the α and the β fractions of serum.

Antibodies Are Heterodimers

Antibody molecules have a common structure of four peptide chains (Figure 4-2). This structure consists of two identical **light (L) chains**, polypeptides of about 25,000 molecular weight, and two identical **heavy (H) chains**, larger

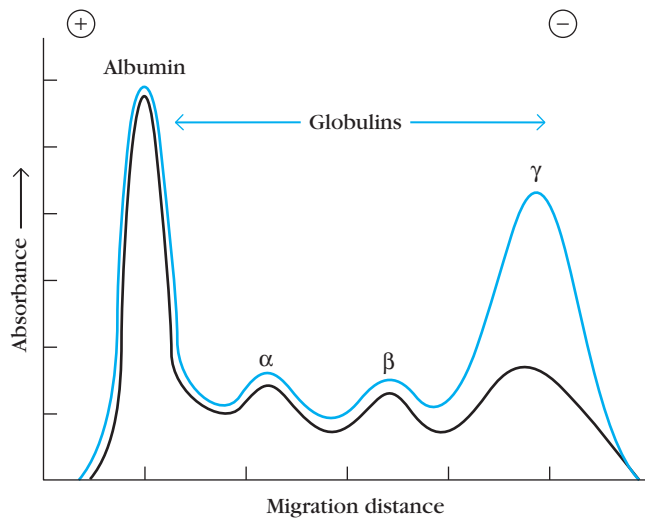


FIGURE 4-1 Experimental demonstration that most antibodies are in the γ -globulin fraction of serum proteins. After rabbits were immunized with ovalbumin (OVA), their antisera were pooled and electrophoresed, which separated the serum proteins according to their electric charge and mass. The blue line shows the electrophoretic pattern of untreated antiserum. The black line shows the pattern of antiserum that was incubated with OVA to remove anti-OVA antibody and then electrophoresed. [Adapted from A. Tiselius and E. A. Kabat, 1939, *J. Exp. Med.* **69**:119, with copyright permission of the Rockefeller University Press.]

polypeptides of molecular weight 50,000 or more. Like the antibody molecules they constitute, H and L chains are also called immunoglobulins. Each light chain is bound to a heavy chain by a disulfide bond, and by such noncovalent interactions as salt linkages, hydrogen bonds, and hydrophobic bonds, to form a heterodimer (H-L). Similar noncovalent interactions and disulfide bridges link the two identical heavy and light (H-L) chain combinations to each other to form the basic four-chain (H-L)₂ antibody structure, a dimer of dimers. As we shall see, the exact number and precise positions of these interchain disulfide bonds differs among antibody classes and subclasses.

The first 110 or so amino acids of the amino-terminal region of a light or heavy chain varies greatly among antibodies of different specificity. These segments of highly variable sequence are called *V regions*: V_L in light chains and V_H in heavy. All of the differences in specificity displayed by different antibodies can be traced to differences in the amino acid sequences of V regions. In fact, most of the differences among antibodies fall within areas of the V regions called *complementarity-determining regions (CDRs)*, and it is these CDRs, on both light and heavy chains, that constitute the antigen-binding site of the antibody molecule. By contrast, within the same antibody class, far fewer differences are seen when one compares sequences throughout the rest of the molecule. The regions of relatively constant sequence beyond the variable regions have been dubbed C regions, C_L on the light chain and

C_H on the heavy chain. Antibodies are glycoproteins; with few exceptions, the sites of attachment for carbohydrates are restricted to the constant region. We do not completely understand the role played by glycosylation of antibodies, but it probably increases the solubility of the molecules. Inappropriate glycosylation, or its absence, affects the rate at which antibodies are cleared from the serum, and decreases the efficiency of interaction between antibody and the complement system and between antibodies and Fc receptors.

Chemical and Enzymatic Methods Revealed Basic Antibody Structure

Our knowledge of basic antibody structure was derived from a variety of experimental observations. When the γ -globulin fraction of serum is separated into high- and low-molecular-weight fractions, antibodies of around 150,000-MW, designated as immunoglobulin G (IgG) are found in the low-molecular-weight fraction. In a key experiment, brief digestion of IgG with the enzyme papain produced three fragments, two of which were identical fragments and a third that was quite different (Figure 4-3). The two identical fragments

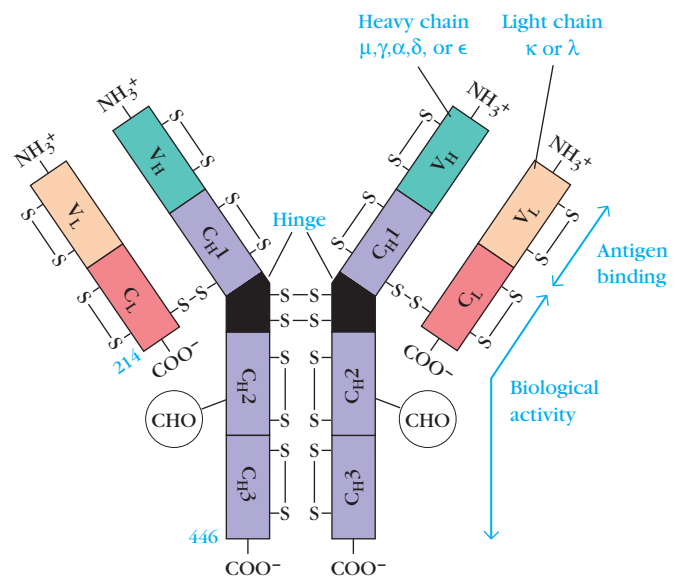


FIGURE 4-2 Schematic diagram of structure of immunoglobulins derived from amino acid sequencing studies. Each heavy and light chain in an immunoglobulin molecule contains an amino-terminal variable (V) region (aqua and tan, respectively) that consists of 100–110 amino acids and differs from one antibody to the next. The remainder of each chain in the molecule—the constant (C) regions (purple and red)—exhibits limited variation that defines the two light-chain subtypes and the five heavy-chain subclasses. Some heavy chains (γ , δ , and α) also contain a proline-rich hinge region (black). The amino-terminal portions, corresponding to the V regions, bind to antigen; effector functions are mediated by the other domains. The μ and ϵ heavy chains, which lack a hinge region, contain an additional domain in the middle of the molecule.

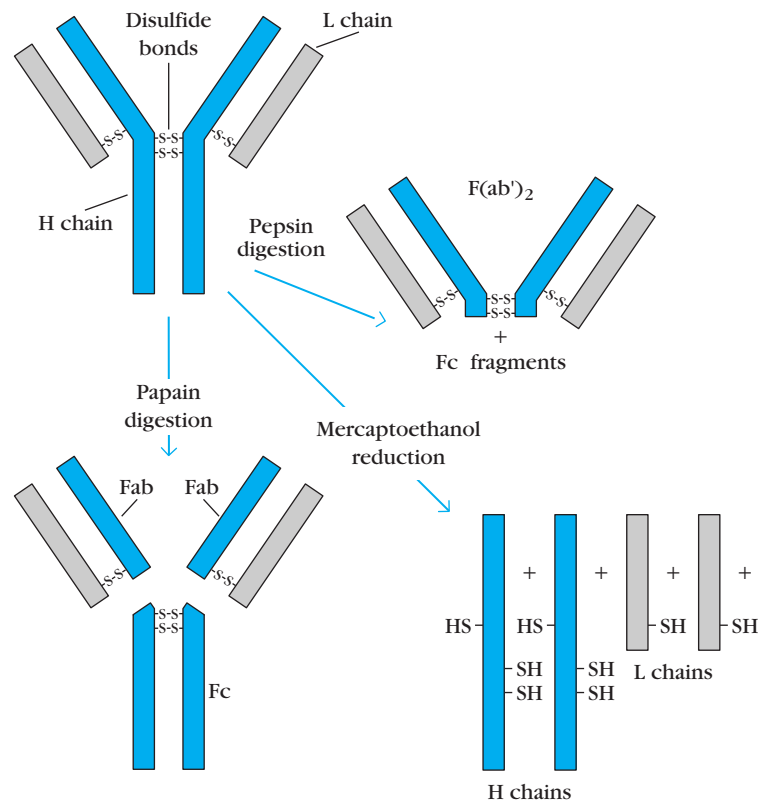


FIGURE 4-3 Prototype structure of IgG, showing chain structure and interchain disulfide bonds. The fragments produced by various

treatments are also indicated. Light (L) chains are in gray and heavy (H) chains in blue.

(each with a MW of 45,000), had antigen-binding activity and were called **Fab fragments** (“fragment, antigen binding”). The other fragment (MW of 50,000) had no antigen-binding activity at all. Because it was found to crystallize during cold storage, it was called the **Fc fragment** (“fragment, crystallizable”). Digestion with pepsin, a different proteolytic enzyme, also demonstrated that the antigen-binding properties of an antibody can be separated from the rest of the molecule. Pepsin digestion generated a single 100,000-MW fragment composed of two Fab-like fragments designated the **$F(ab')_2$ fragment**, which binds antigen. The Fc fragment was not recovered from pepsin digestion because it had been digested into multiple fragments.

A key observation in deducing the multichain structure of IgG was made when the molecule was subjected to mercaptoethanol reduction and alkylation, a chemical treatment that irreversibly cleaves disulfide bonds. If the sample is chromatographed on a column that separates molecules by size following cleavage of disulfide bonds, it is clear that the intact 150,000-MW IgG molecule is, in fact, composed of subunits. Each IgG molecule contains two 50,000-MW polypeptide chains, designated as heavy (H) chains, and two 25,000-MW chains, designated as light (L) chains (see Figure 4-3).

Antibodies themselves were used to determine how the enzyme digestion products—Fab, $F(ab')_2$, and Fc—were related to the heavy-chain and light-chain reduction products.

This question was answered by using antisera from goats that had been immunized with either the Fab fragments or the Fc fragments of rabbit IgG. The antibody to the Fab fragment could react with both the H and the L chains, whereas antibody to the Fc fragment reacted only with the H chain. These observations led to the conclusion that the Fab fragment consists of portions of a heavy and a light chain and that Fc contains only heavy-chain components. From these results, and those mentioned above, the structure of IgG shown in Figure 4-3 was deduced. According to this model, the IgG molecule consists of two identical H chains and two identical L chains, which are linked by disulfide bridges. The enzyme papain cleaves just above the interchain disulfide bonds linking the heavy chains, whereas the enzyme pepsin cleaves just below these bonds, so that the two proteolytic enzymes generate different digestion products. Mercaptoethanol reduction and alkylation allow separation of the individual heavy and light chains.

Obstacles to Antibody Sequencing

Initial attempts to determine the amino acid sequence of the heavy and light chains of antibody were hindered because insufficient amounts of homogeneous protein were available. Although the basic structure and chemical properties of differ-

ent antibodies are similar, their antigen-binding specificities, and therefore their exact amino acid sequences, are very different. The population of antibodies in the serum γ -globulin fraction consists of a heterogeneous spectrum of antibodies. Even if immunization is done with a hapten-carrier conjugate, the antibodies formed just to the hapten alone are heterogeneous: they recognize different epitopes of the hapten and have different binding affinities. This heterogeneity of serum antibodies made them unsuitable for sequencing studies.

Pure Immunoglobulin Obtained from Multiple Myeloma Patients Made Sequencing Possible

Sequencing analysis finally became feasible with the discovery of **multiple myeloma**, a cancer of antibody-producing plasma cells. The plasma cells in a normal individual are end-stage cells that secrete a single molecular species of antibody for a limited period of time and then die. In contrast, a clone of plasma cells in an individual with multiple myeloma has escaped normal controls on their life span and proliferation and are not end-stage cells; rather, they divide over and over in an unregulated way without requiring any activation by antigen to induce proliferation. Although such a cancerous plasma cell, called a **myeloma cell**, has been transformed, its protein-synthesizing machinery and secretory functions are not altered; thus, the cell continues to secrete molecularly homogeneous antibody. This antibody is indistinguishable from normal antibody molecules but is called **myeloma protein** to denote its source. In a patient afflicted with multiple myeloma, myeloma protein can account for 95% of the serum immunoglobulins. In most patients, the myeloma cells also secrete excessive amounts of light chains. These excess light chains were first discovered in the urine of myeloma patients and were named **Bence-Jones proteins**, for their discoverer.

Multiple myeloma also occurs in other animals. In mice it can arise spontaneously, as it does in humans, or conditions favoring myeloma induction can be created by injecting mineral oil into the peritoneal cavity. The clones of malignant plasma cells that develop are called **plasmacytomas**, and many of these are designated MOPCs, denoting the mineral-oil induction of plasmacytoma cells. A large number of mouse MOPC lines secreting different immunoglobulin classes are presently carried by the American Type-Culture Collection, a nonprofit repository of cell lines commonly used in research.

Light-Chain Sequencing Revealed That Immunoglobulins Have Constant and Variable Regions

When the amino acid sequences of several Bence-Jones proteins (light chains) from different individuals were compared, a striking pattern emerged. The amino-terminal half of the chain, consisting of 100–110 amino acids, was found to vary among different Bence-Jones proteins. This region

was called the **variable (V) region**. The carboxyl-terminal half of the molecule, called the **constant (C) region**, had two basic amino acid sequences. This led to the recognition that there were two light chain types, **kappa (κ)** and **lambda (λ)**. In humans, 60% of the light chains are kappa and 40% are lambda, whereas in mice, 95% of the light chains are kappa and only 5% are lambda. A single antibody molecule contains only one light chain type, either κ or λ , never both.

The amino acid sequences of λ light chains show minor differences that are used to classify λ light chains into subtypes. In mice, there are three subtypes ($\lambda 1$, $\lambda 2$, and $\lambda 3$); in humans, there are four subtypes. Amino acid substitutions at only a few positions are responsible for the subtype differences.

Heavy-Chain Sequencing Revealed Five Basic Varieties of Heavy Chains

For heavy-chain sequencing studies, myeloma proteins were reduced with mercaptoethanol and alkylated, and the heavy chains were separated by gel filtration in a denaturing solvent. When the amino acid sequences of several myeloma protein heavy chains were compared, a pattern similar to that of the light chains emerged. The amino-terminal part of the chain, consisting of 100–110 amino acids, showed great sequence variation among myeloma heavy chains and was therefore called the variable (V) region. The remaining part of the protein revealed five basic sequence patterns, corresponding to five different heavy-chain constant (C) regions (μ , δ , γ , ϵ and α). Each of these five different heavy chains is called an **isotype**. The length of the constant regions is approximately 330 amino acids for δ , γ , and α , and 440 amino acids for μ and ϵ . The heavy chains of a given antibody molecule determine the class of that antibody: IgM(μ), IgG(γ), IgA(α), IgD(δ), or IgE(ϵ). Each class can have either κ or λ light chains. A single antibody molecule has two identical heavy chains and two identical light chains, H_2L_2 , or a multiple $(H_2L_2)_n$ of this basic four-chain structure (Table 4-1).

Minor differences in the amino acid sequences of the α and γ heavy chains led to further classification of the heavy chains into subisotypes that determine the subclass of antibody molecules they constitute. In humans, there are two subisotypes of α heavy chains— $\alpha 1$ and $\alpha 2$ —(and thus two subclasses, IgA1 and IgA2)—and four subisotypes of γ heavy chains: $\gamma 1$, $\gamma 2$, $\gamma 3$, and $\gamma 4$ (therefore four subclasses, IgG1, IgG2, IgG3, and IgG4). In mice, there are four subisotypes, $\gamma 1$, $\gamma 2a$, $\gamma 2b$, and $\gamma 3$, and the corresponding subclasses.

Immunoglobulin Fine Structure

The structure of the immunoglobulin molecule is determined by the primary, secondary, tertiary, and quaternary organization of the protein. The primary structure, the amino acid sequence, accounts for the variable and constant regions of the heavy and light chains. The secondary structure is formed by folding of the extended polypeptide chain



TABLE 4-1 Chain composition of the five immunoglobulin classes in humans

Class	Heavy chain	Subclasses	Light chain	Molecular formula
IgG	γ	γ1, γ2, γ3, γ4	κ or λ	γ ₂ κ ₂ γ ₂ λ ₂
IgM	μ	None	κ or λ	(μ ₂ κ ₂) _n (μ ₂ λ ₂) _n n = 1 or 5
IgA	α	α1, α2	κ or λ	(α ₂ κ ₂) _n (α ₂ λ ₂) _n n = 1, 2, 3, or 4
IgE	ε	None	κ or λ	ε ₂ κ ₂ ε ₂ λ ₂
IgD	δ	None	κ or λ	δ ₂ κ ₂ δ ₂ λ ₂

back and forth upon itself into an antiparallel β pleated sheet (Figure 4-4). The chains are then folded into a tertiary structure of compact globular domains, which are connected to neighboring domains by continuations of the polypeptide chain that lie outside the β pleated sheets. Finally, the globular domains of adjacent heavy and light polypeptide chains interact in the quaternary structure (Figure 4-5), forming functional domains that enable the molecule to specifically bind antigen and, at the same time, perform a number of biological effector functions.

Immunoglobulins Possess Multiple Domains Based on the Immunoglobulin Fold

Careful analysis of the amino acid sequences of immunoglobulin heavy and light chains showed that both chains contain

several homologous units of about 110 amino acid residues. Within each unit, termed a domain, an intrachain disulfide bond forms a loop of about 60 amino acids. Light chains contain one variable domain (V_L), and one constant domain (C_L); heavy chains contain one variable domain (V_H), and either three or four constant domains (C_H1, C_H2, C_H3, and C_H4), depending on the antibody class (Figure 4-6).

X-ray crystallographic analysis revealed that immunoglobulin domains are folded into a characteristic compact structure called the **immunoglobulin fold**. This structure consists of a “sandwich” of two β pleated sheets, each containing antiparallel β strands of amino acids, which are connected by loops of various lengths (Figure 4-7). The β strands within a sheet are stabilized by hydrogen bonds that connect the –NH groups in one strand with carbonyl groups of an adjacent strand (see Figure 4-4). The β strands are characterized by alternating hydrophobic and hydrophilic amino acids whose side chains are arranged perpendicular to the plane of the sheet; the hydrophobic amino acids are oriented toward the interior of the sandwich, and the hydrophilic amino acids face outward.

The two β sheets within an immunoglobulin fold are stabilized by the hydrophobic interactions between them and by the conserved disulfide bond. An analogy has been made to two pieces of bread, the butter between them, and a toothpick holding the slices together. The bread slices represent the two β pleated sheets; the butter represents the hydrophobic interactions between them; and the toothpick represents the intrachain disulfide bond. Although variable and constant domains have a similar structure, there are subtle differences between them. The V domain is slightly longer than the C domain and contains an extra pair of β strands within the β-sheet structure, as well as the extra loop sequence connecting this pair of β strands (see Figure 4-7).

The basic structure of the immunoglobulin fold contributes to the quaternary structure of immunoglobulins by facilitating noncovalent interactions between domains

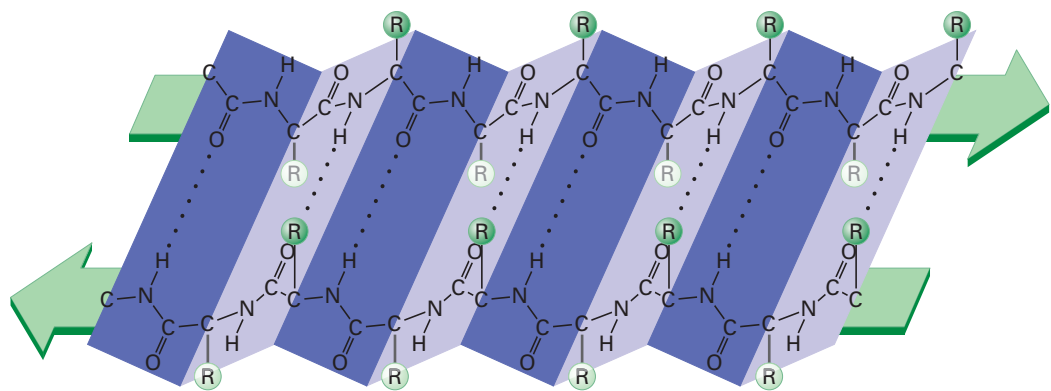


FIGURE 4-4 Structural formula of a β pleated sheet containing two antiparallel β strands. The structure is held together by hydrogen bonds between peptide bonds of neighboring stretches of polypeptide chains. The amino acid side groups (R) are arranged perpendicular to the plane of the sheet. [Adapted from H. Lodish et al., 1995, Molecular Cell Biology, 4th ed., Scientific American Books, New York; reprinted by permission of W. H. Freeman and Company.]

ular to the plane of the sheet. [Adapted from H. Lodish et al., 1995, Molecular Cell Biology, 4th ed., Scientific American Books, New York; reprinted by permission of W. H. Freeman and Company.]

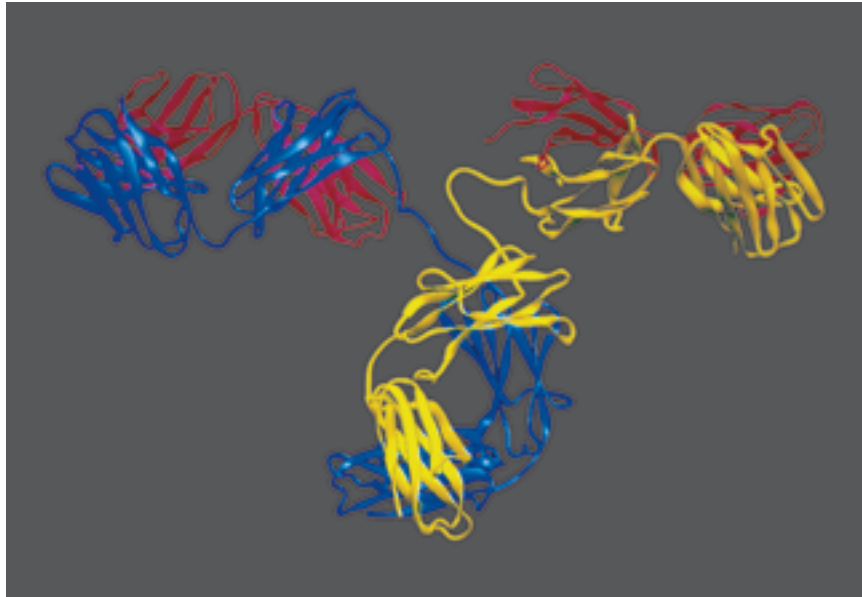


FIGURE 4-5 Ribbon representation of an intact monoclonal antibody depicting the heavy chains (yellow and blue) and light chains (red). The domains of the molecule composed of β pleated sheets are readily visible as is the extended conformation of the hinge re-

gion. [The laboratory of A. McPherson provided this image, which is based on x-ray crystallography data determined by L. J. Harris et al., 1992, *Nature* **360**:369. The image was generated using the computer program RIBBONS.]

across the faces of the β sheets (Figure 4-8). Interactions form links between identical domains (e.g., C_{H2}/C_{H2} , C_{H3}/C_{H3} , and C_{H4}/C_{H4}) and between nonidentical domains (e.g., V_H/V_L and C_{H1}/C_L). The structure of the immunoglobulin fold also allows for variable lengths and

sequences of amino acids that form the loops connecting the β strands. As the next section explains, some of the loop sequences of the V_H and V_L domains contain variable amino acids and constitute the antigen-binding site of the molecule.

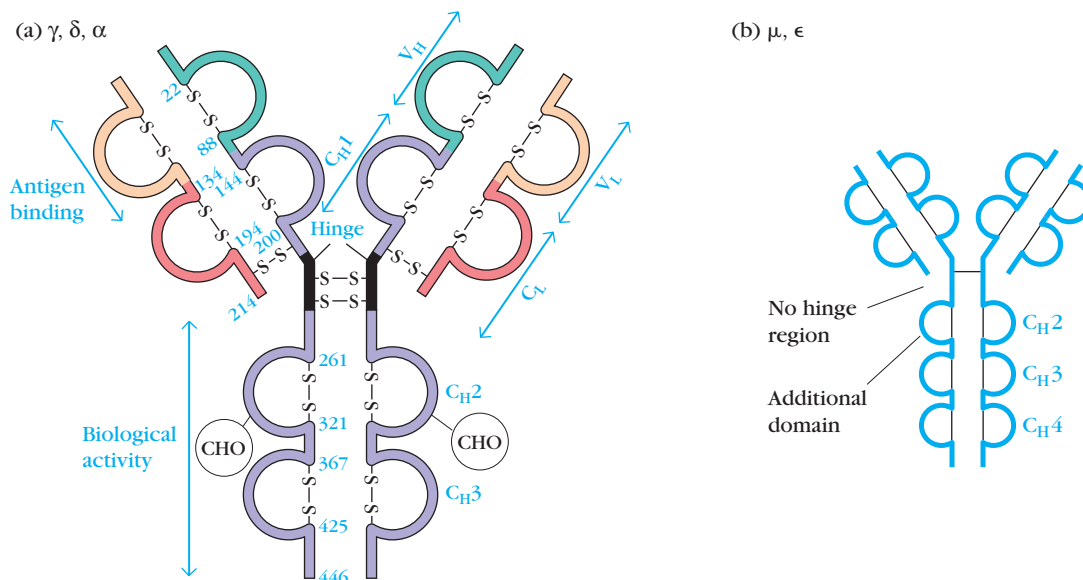


FIGURE 4-6 (a) Heavy and light chains are folded into domains, each containing about 110 amino acid residues and an intrachain disulfide bond that forms a loop of 60 amino acids. The amino-terminal domains, corresponding to the V regions, bind to antigen;

effector functions are mediated by the other domains. (b) The μ and ϵ heavy chains contain an additional domain that replaces the hinge region.

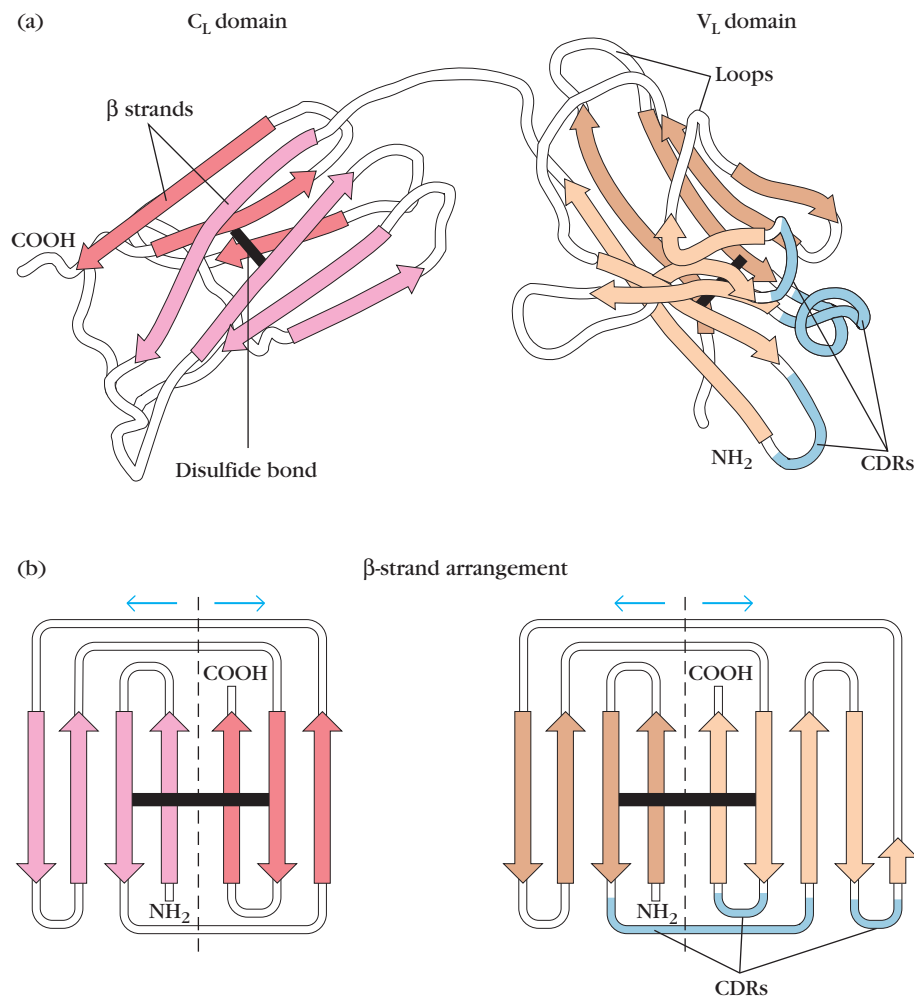


FIGURE 4-7 (a) Diagram of an immunoglobulin light chain depicting the immunoglobulin-fold structure of its variable and constant domains. The two β pleated sheets in each domain are held together by hydrophobic interactions and the conserved disulfide bond. The β strands that compose each sheet are shown in different colors. The amino acid sequences in three loops of each variable domain show considerable variation; these hypervariable regions (blue) make up the antigen-binding site. Hypervariable regions are usually called

CDRs (complementarity-determining regions). Heavy-chain domains have the same characteristic structure. (b) The β pleated sheets are opened out to reveal the relationship of the individual β strands and joining loops. Note that the variable domain contains two more β strands than the constant domain. [Part (a) adapted from M. Schiffer et al., 1973, *Biochemistry* **12**:4620; reprinted with permission; part (b) adapted from Williams and Barclay, 1988, *Annu. Rev. Immunol.* **6**:381.]

Diversity in the Variable-Region Domain Is Concentrated in CDRs

Detailed comparisons of the amino acid sequences of a large number of V_L and V_H domains revealed that the sequence variation is concentrated in a few discrete regions of these domains. The pattern of this variation is best summarized by a quantitative plot of the variability at each position of the polypeptide chain. The **variability** is defined as:

$$\text{Variability} = \frac{\text{\# of different amino acids at a given position}}{\text{Frequency of the most common amino acid at given position}}$$

Thus if a comparison of the sequences of 100 heavy chains revealed that a serine was found in position 7 in 51 of the sequences (frequency 0.51), it would be the most common amino acid. If examination of the other 49 sequences showed that position 7 was occupied by either glutamine, histidine, proline, or tryptophan, the variability at that position would be 9.8 (5/0.51). Variability plots of V_L and V_H domains of human antibodies show that maximum variation is seen in those portions of the sequence that correspond to the loops that join the β strands (Figure 4-9). These regions were originally called **hypervariable regions** in recognition of their high variability. Hypervariable regions form the antigen-binding site of the antibody molecule. Because the antigen binding site is complementary to the structure of the epitope,

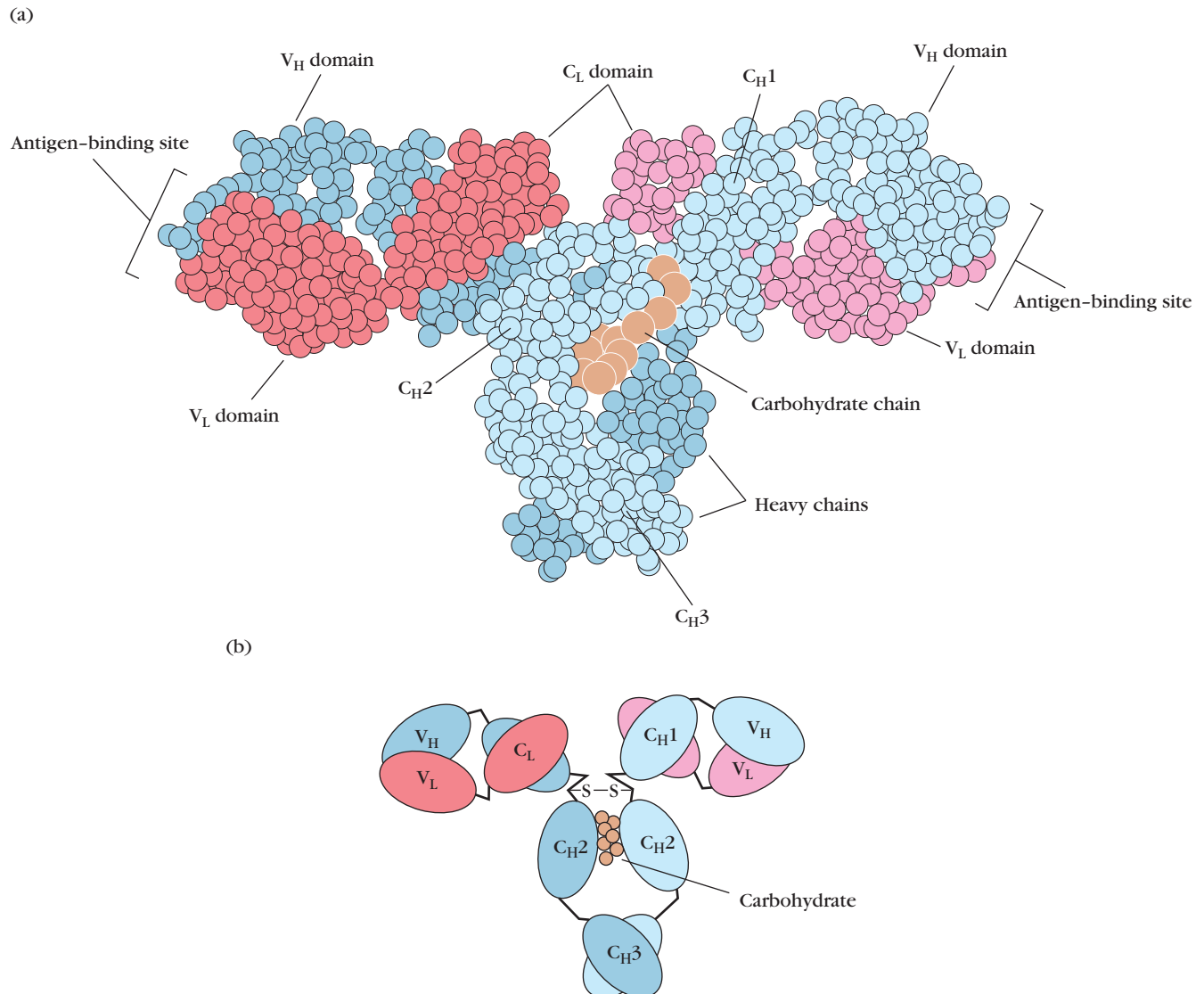


FIGURE 4-8 Interactions between domains in the separate chains of an immunoglobulin molecule are critical to its quaternary structure. (a) Model of IgG molecule, based on x-ray crystallographic analysis, showing associations between domains. Each solid ball represents an amino acid residue; the larger tan balls are carbohydrate. The two light chains are shown in shades of red; the two heavy chains, in shades of blue. (b) A schematic diagram showing the in-

teracting heavy- and light-chain domains. Note that the C_H2/C_H2 domains protrude because of the presence of carbohydrate (tan) in the interior. The protrusion makes this domain more accessible, enabling it to interact with molecules such as certain complement components. [Part (a) from E. W. Silverton *et al.*, 1977, *Proc. Nat. Acad. Sci. U.S.A.* **74**:5140.]

these areas are now more widely called **complementarity determining regions (CDRs)**. The three heavy-chain and three light-chain CDR regions are located on the loops that connect the β strands of the V_H and V_L domains. The remainder of the V_L and V_H domains exhibit far less variation; these stretches are called the **framework regions (FRs)**. The wide range of specificities exhibited by antibodies is due to variations in the length and amino acid sequence of the six CDRs in each Fab fragment. The framework region acts as a scaffold that supports these six loops. The three-dimensional structure of the framework regions of virtually all antibodies

analyzed to date can be superimposed on one another; in contrast, the hypervariable loops (i.e., the CDRs) have different orientations in different antibodies.

CDRs Bind Antigen

The finding that CDRs are the antigen-binding regions of antibodies has been confirmed directly by high-resolution x-ray crystallography of antigen-antibody complexes. Crystallographic analysis has been completed for many Fab fragments of monoclonal antibodies complexed either with



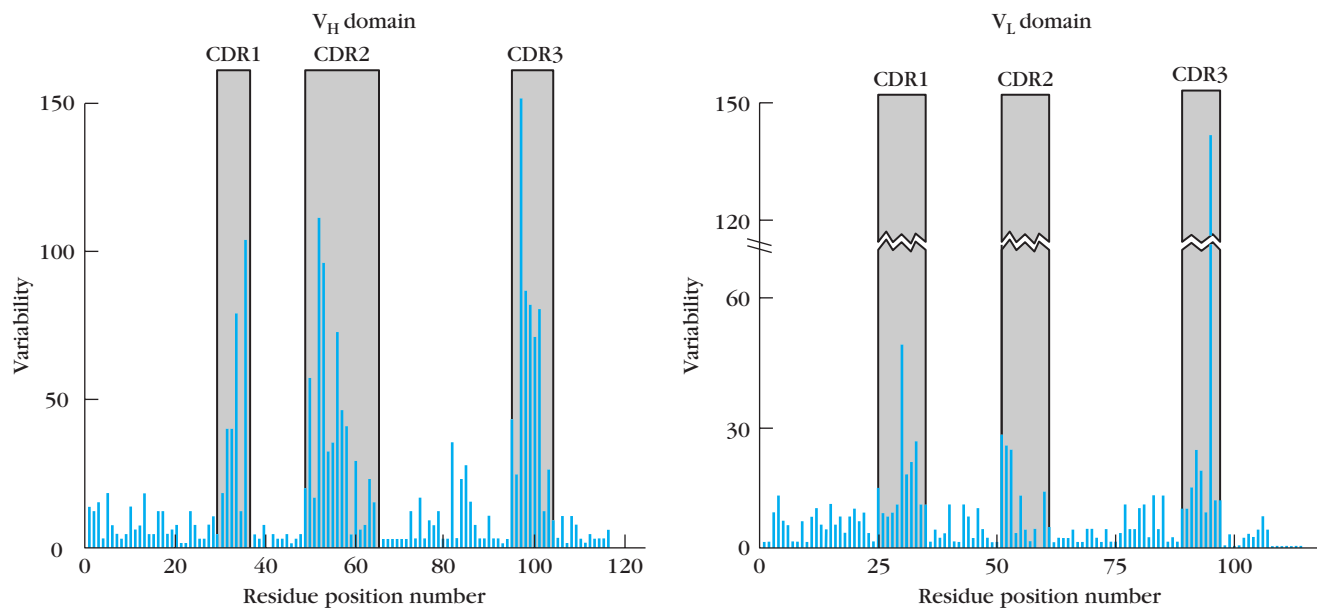


FIGURE 4-9 Variability of amino acid residues in the V_L and V_H domains of human antibodies with different specificities. Three hyper-variable (HV) regions, also called complementarity-determining regions (CDRs), are present in both heavy- and light-chain V domains. As shown in Figure 4-7 (right), the three HV regions in the

light-chain V domain are brought into proximity in the folded structure. The same is true of the heavy-chain V domain. [Based on E. A. Kabat *et al.*, 1977, *Sequence of Immunoglobulin Chains*, U.S. Dept. of Health, Education, and Welfare.]

large globular protein antigens or with a number of smaller antigens including carbohydrates, nucleic acids, peptides, and small haptens. In addition, complete structures have been obtained for several intact monoclonal antibodies. X-ray diffraction analysis of antibody-antigen complexes has

shown that several CDRs may make contact with the antigen, and a number of complexes have been observed in which all six CDRs contact the antigen. In general, more residues in the heavy-chain CDRs appear to contact antigen than in the light-chain CDRs. Thus the V_H domain often contributes

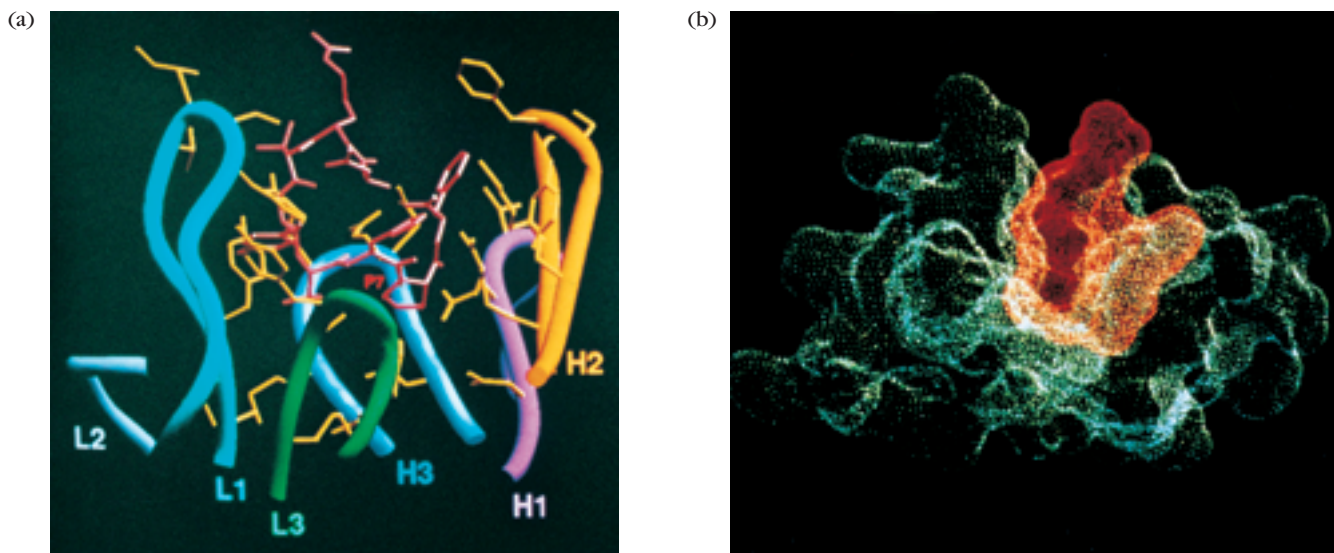


FIGURE 4-10 (a) Side view of the three-dimensional structure of the combining site of an angiotensin II–Fab complex. The peptide is in red. The three heavy-chain CDRs (H1, H2, H3) and three light-chain CDRs (L1, L2, L3) are each shown in a different color. All six CDRs contain side chains, shown in yellow, that are within van der

Waals contact of the angiotensin peptide. (b) Side view of the van der Waals surface of contact between angiotensin II and Fab fragment. [From K. C. Garcia *et al.*, 1992, *Science* **257**:502; courtesy of M. Amzel, Johns Hopkins University.]

more to antigen binding than the V_L domain. The dominant role of the heavy chain in antigen binding was demonstrated in a study in which a single heavy chain specific for a glycoprotein antigen of the human immunodeficiency virus (HIV) was combined with various light chains of different antigenic specificity. All of the hybrid antibodies bound to the HIV glycoprotein antigen, indicating that the heavy chain alone was sufficient to confer specificity. However, one should not conclude that the light chain is largely irrelevant; in some antibody-antigen reactions, the light chain makes the more important contribution.

The actual shape of the antigen binding site formed by whatever combination of CDRs are used in a particular antibody has been shown to vary dramatically. As pointed out in Chapter 3, contacts between a large globular protein antigen and antibody occur over a broad, often rather flat, undulating face. In the area of contact, protrusions or depressions on the antigen are likely to match complementary depressions or protrusions on the antibody. In the case of the well studied lysozyme/anti-lysozyme system, crystallographic studies have shown that the surface areas of interaction are quite large, ranging from about $650 \text{ \AA}^2$ to more than $900 \text{ \AA}^2$. Within this area, some 15–22 amino acids in the antibody contact the same number of residues in the protein antigen. In contrast, antibodies bind smaller antigens, such as small haptens, in smaller, recessed pockets in which the ligand is buried. This is nicely illustrated by the interaction of the

small octapeptide hormone angiotensin II with the binding site of an anti-angiotensin antibody (Figure 4-10).

Conformational Changes May Be Induced by Antigen Binding

As more x-ray crystallographic analyses of Fab fragments were completed, it became clear that in some cases binding of antigen induces conformational changes in the antibody, antigen, or both. Formation of the complex between neuraminidase and anti-neuraminidase is accompanied by a change in the orientation of side chains of both the epitope and the antigen-binding site. This conformational change results in a closer fit between the epitope and the antibody's binding site.

In another example, comparison of an anti-hemagglutinin Fab fragment before and after binding to a hemagglutinin peptide antigen has revealed a visible conformational change in the heavy-chain CDR3 loop and in the accessible surface of the binding site. Another striking example of conformational change has been seen in the complex between an Fab fragment derived from a monoclonal antibody against the HIV protease and the peptide epitope of the protease. As shown in Figure 4-11, there are significant changes in the Fab upon binding. In fact, upon antigen binding, the CDR1 region of the light chain moves as much as $1 \text{ \AA}$ and the heavy chain CDR3 moves $2.7 \text{ \AA}$. Thus, in addition to variability in the

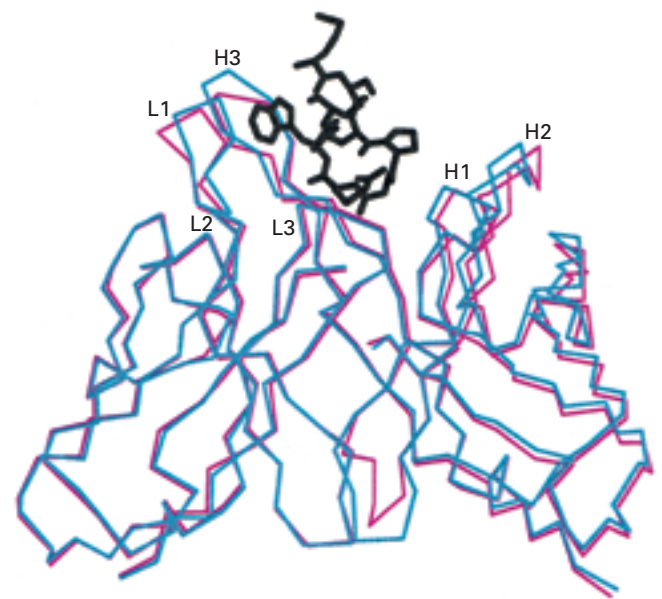
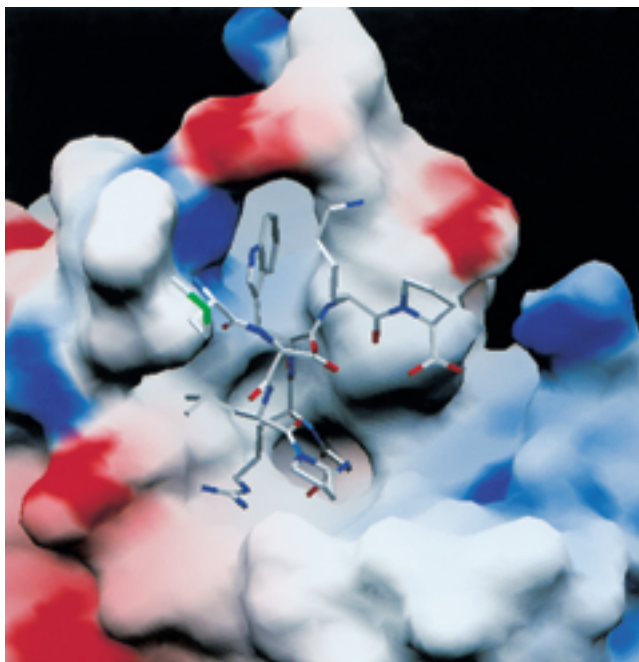


FIGURE 4-11 Structure of a complex between a peptide derived from HIV protease and an Fab fragment from an anti-protease antibody (*left*) and comparison of the Fab structure before and after peptide binding (*right*). In the right panel, the red line shows the structure of the Fab fragment before it binds the peptide and the blue

line shows its structure when bound. There are significant conformational changes in the CDRs of the Fab on binding the antigen. These are especially pronounced in the light chain CDR1 (L1) and the heavy chain CDR3 (H3). [From J. Lescar et al., 1997, *J. Mol. Biol.* **267**:1207; courtesy of G. Bentley, *Institute Pasteur*.]

length and amino acid composition of the CDR loops, the ability of these loops to significantly change conformation upon antigen binding enables antibodies to assume a shape more effectively complementary to that of their epitopes.

As already indicated, conformational changes following antigen binding need not be limited to the antibody. Although it is not shown in Figure 4-11, the conformation of the protease peptide bound to the Fab shows no structural similarity to the corresponding epitope in the native HIV protease. It has been suggested that the inhibition of protease activity by this anti-HIV protease antibody is a result of its distortion of the enzyme's native conformation.

Constant-Region Domains

The immunoglobulin constant-region domains take part in various biological functions that are determined by the amino acid sequence of each domain.

C_{H1} AND C_L DOMAINS

The C_{H1} and C_L domains serve to extend the Fab arms of the antibody molecule, thereby facilitating interaction with antigen and increasing the maximum rotation of the Fab arms. In addition, these constant-region domains help to hold the V_H and V_L domains together by virtue of the interchain disulfide bond between them (see Figure 4-6). Also, the C_{H1} and C_L domains may contribute to antibody diversity by allowing more random associations between V_H and V_L domains than would occur if this association were driven by the

V_H/V_L interaction alone. These considerations have important implications for building a diverse repertoire of antibodies. As Chapter 5 will show, random rearrangements of the immunoglobulin genes generate unique V_H and V_L sequences for the heavy and light chains expressed by each B lymphocyte; association of the V_H and V_L sequences then generates a unique antigen-binding site. The presence of C_{H1} and C_L domains appears to increase the number of stable V_H and V_L interactions that are possible, thus contributing to the overall diversity of antibody molecules that can be expressed by an animal.

HINGE REGION

The γ , δ , and α heavy chains contain an extended peptide sequence between the C_{H1} and C_{H2} domains that has no homology with the other domains (see Figure 4-8). This region, called the **hinge region**, is rich in proline residues and is flexible, giving IgG, IgD, and IgA segmental flexibility. As a result, the two Fab arms can assume various angles to each other when antigen is bound. This flexibility of the hinge region can be visualized in electron micrographs of antigen-antibody complexes. For example, when a molecule containing two dinitrophenol (DNP) groups reacts with anti-DNP antibody and the complex is captured on a grid, negatively stained, and observed by electron microscopy, large complexes (e.g., dimers, trimers, tetramers) are seen. The angle between the arms of the Y-shaped antibody molecules differs in the different complexes, reflecting the flexibility of the hinge region (Figure 4-12).

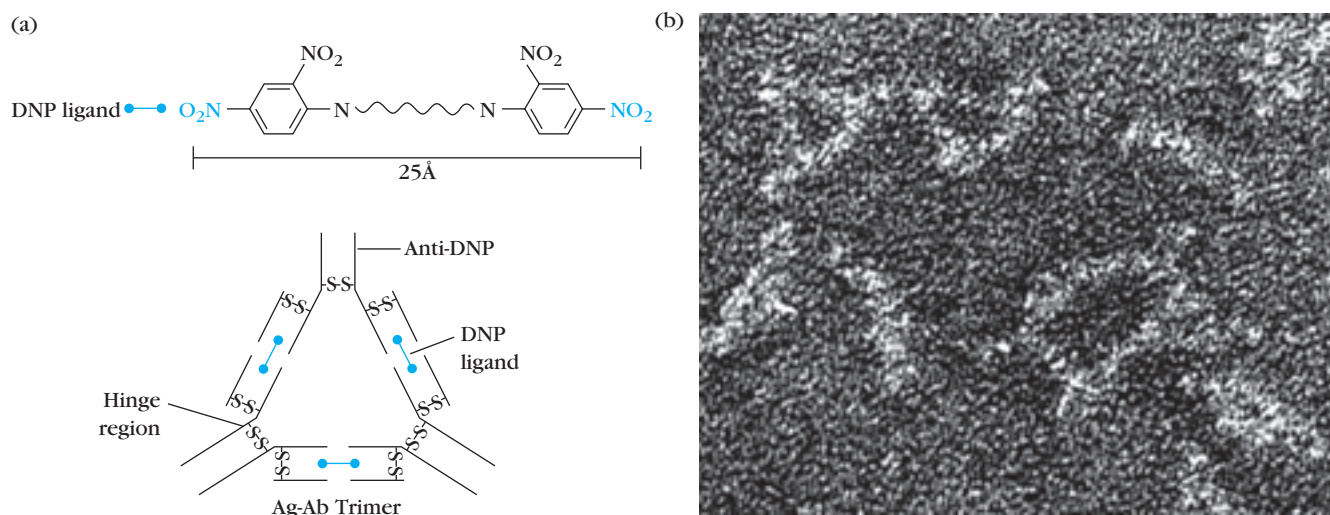


FIGURE 4-12 Experimental demonstration of the flexibility of the hinge region in antibody molecules. (a) A hapten in which two dinitrophenyl (DNP) groups are tethered by a short connecting spacer group reacts with anti-DNP antibodies to form trimers, tetramers, and other larger antigen-antibody complexes. A trimer is shown schematically. (b) In an electron micrograph of a negatively stained preparation of these complexes, two triangular trimeric structures

are clearly visible. The antibody protein stands out as a light structure against the electron-dense background. Because of the flexibility of the hinge region, the angle between the arms of the antibody molecules varies. [Photograph from R. C. Valentine and N. M. Green, 1967, *J. Mol. Biol.* **27**:615; reprinted by permission of Academic Press Inc. (London) Ltd.]

Two prominent amino acids in the hinge region are proline and cysteine. The large number of proline residues in the hinge region gives it an extended polypeptide conformation, making it particularly vulnerable to cleavage by proteolytic enzymes; it is this region that is cleaved with papain or pepsin (see Figure 4-3). The cysteine residues form interchain disulfide bonds that hold the two heavy chains together. The number of interchain disulfide bonds in the hinge region varies considerably among different classes of antibodies and between species. Although μ and ϵ chains lack a hinge region, they have an additional domain of 110 amino acids (C_{H2}/C_{H2}) that has hingelike features.

OTHER CONSTANT-REGION DOMAINS

As noted already, the heavy chains in IgA, IgD, and IgG contain three constant-region domains and a hinge region, whereas the heavy chains in IgE and IgM contain four constant-region domains and no hinge region. The corresponding domains of the two groups are as follows:

IgA, IgD, IgG	IgE, IgM
C_{H1}/C_{H1}	C_{H1}/C_{H1}
Hinge region	C_{H2}/C_{H2}
C_{H2}/C_{H2}	C_{H3}/C_{H3}
C_{H3}/C_{H3}	C_{H4}/C_{H4}

Although the C_{H2}/C_{H2} domains in IgE and IgM occupy the same position in the polypeptide chains as the hinge region in the other classes of immunoglobulin, a function for this extra domain has not yet been determined.

X-ray crystallographic analyses have revealed that the two C_{H2} domains of IgA, IgD, and IgG (and the C_{H3} domains of IgE and IgM) are separated by oligosaccharide side chains; as a result, these two globular domains are much more accessible than the others to the aqueous environment (see Figure 4-8b). This accessibility is one of the elements that contributes to the biological activity of these domains in the activation of complement components by IgG and IgM.

The carboxyl-terminal domain is designated C_{H3}/C_{H3} in IgA, IgD, and IgG and C_{H4}/C_{H4} in IgE and IgM. The five classes of antibody and their subclasses can be expressed either as **secreted immunoglobulin (sIg)** or as **membrane-bound immunoglobulin (mIg)**. The carboxyl-terminal domain in secreted immunoglobulin differs in both structure and function from the corresponding domain in membrane-bound immunoglobulin. Secreted immunoglobulin has a hydrophilic amino acid sequence of various lengths at the carboxyl-terminal end. The functions of this domain in the various classes of secreted antibody will be described later. In membrane-bound immunoglobulin, the carboxyl-terminal domain contains three regions:

- An extracellular hydrophilic “spacer” sequence composed of 26 amino acid residues
- A hydrophobic transmembrane sequence
- A short cytoplasmic tail

The length of the transmembrane sequence is constant among all immunoglobulin isotypes, whereas the lengths of the extracellular spacer sequence and the cytoplasmic tail vary.

B cells express different classes of mIg at different developmental stages. The immature B cell, called a pre-B cell, expresses only mIgM; later in maturation, mIgD appears and is coexpressed with IgM on the surface of mature B cells before they have been activated by antigen. A memory B cell can express mIgM, mIgG, mIgA, or mIgE. Even when different classes are expressed sequentially on a single cell, the antigenic specificity of all the membrane antibody molecules expressed by a single cell is identical, so that each antibody molecule binds to the same epitope. The genetic mechanism that allows a single B cell to express multiple immunoglobulin isotypes all with the same antigenic specificity is described in Chapter 5.

Antibody-Mediated Effector Functions

In addition to binding antigen, antibodies participate in a broad range of other biological activities. When considering the role of antibody in defending against disease, it is important to remember that antibodies generally do not kill or remove pathogens solely by binding to them. In order to be effective against pathogens, antibodies must not only recognize antigen, but also invoke responses—effector functions—that will result in removal of the antigen and death of the pathogen. While variable regions of antibody are the sole agents of binding to antigen, the heavy-chain constant region (C_H) is responsible for a variety of collaborative interactions with other proteins, cells, and tissues that result in the effector functions of the humoral response.

Because these effector functions result from interactions between heavy-chain constant regions and other serum proteins or cell-membrane receptors, not all classes of immunoglobulin have the same functional properties. An overview of four major effector functions mediated by domains of the constant region is presented here. A fifth function unique to IgE, the activation of mast cells, eosinophils, and basophils, will be described later.

Opsonization Is Promoted by Antibody

Opsonization, the promotion of phagocytosis of antigens by macrophages and neutrophils, is an important factor in antibacterial defenses. Protein molecules called *Fc receptors* (*FcR*), which can bind the constant region of Ig molecules, are present on the surfaces of macrophages and neutrophils.

The binding of phagocyte Fc receptors with several antibody molecules complexed with the same target, such as a bacterial cell, produces an interaction that results in the binding of the pathogen to the phagocyte membrane. This crosslinking of the FcR by binding to an array of antibody Fc regions initiates a signal-transduction pathway that results in the phagocytosis of the antigen-antibody complex. Inside the phagocyte, the pathogen becomes the target of various destructive processes that include enzymatic digestion, oxidative damage, and the membrane-disrupting effects of antibacterial peptides.

Antibodies Activate Complement

IgM and, in humans, most IgG subclasses can activate a collection of serum glycoproteins called the **complement system**. Complement includes a collection of proteins that can perforate cell membranes. An important byproduct of the complement activation pathway is a protein fragment called C3b, which binds nonspecifically to cell- and antigen-antibody complexes near the site at which complement was activated. Many cell types—for example, red blood cells and macrophages—have receptors for C3b and so bind cells or complexes to which C3b has adhered. Binding of adherent C3b by macrophages leads to phagocytosis of the cells or molecular complexes attached to C3b. Binding of antigen-antibody complexes by the C3b receptors of a red blood cell allows the erythrocyte to deliver the complexes to liver or spleen, where resident macrophages remove them without destroying the red cell. The collaboration between antibody and the complement system is important for the inactivation and removal of antigens and the killing of pathogens. The process of complement activation is described in detail in Chapter 13.

Antibody-Dependent Cell-Mediated Cytotoxicity (ADCC) Kills Cells

The linking of antibody bound to target cells (virus infected cells of the host) with the Fc receptors of a number of cell types, particularly natural killer (NK) cells, can direct the cytotoxic activities of the effector cell against the target cell. In this process, called **antibody-dependent cell-mediated cytotoxicity (ADCC)**, the antibody acts as a newly acquired receptor enabling the attacking cell to recognize and kill the target cell. The phenomenon of ADCC is discussed in Chapter 14.

Some Antibodies Can Cross Epithelial Layers by Transcytosis

The delivery of antibody to the mucosal surfaces of the respiratory, gastrointestinal, and urogenital tracts, as well as its export to breast milk, requires the movement of immunoglobulin across epithelial layers, a process called **transcytosis**. The capacity to be transported depends on properties of the

constant region. In humans and mice, IgA is the major antibody species that undergoes such transcytosis, although IgM can also be transported to mucosal surfaces. Some mammalian species, such as humans and mice, also transfer significant amounts of most subclasses of IgG from mother to fetus. Since maternal and fetal circulatory systems are separate, antibody must be transported across the placental tissue that separates mother and fetus. In humans, this transfer takes place during the third trimester of gestation. The important consequence is that the developing fetus receives a sample of the mother's repertoire of antibody as a protective endowment against pathogens. As with the other effector functions described here, the capacity to undergo transplacental transport depends upon properties of the constant region of the antibody molecule.

The transfer of IgG from mother to fetus is a form of **passive immunization**, which is the acquisition of immunity by receipt of preformed antibodies rather than by active production of antibodies after exposure to antigen. The ability to transfer immunity from one individual to another by the transfer of antibodies is the basis of passive antibody therapy, an important and widely practiced medical procedure (see Clinical Focus).

Antibody Classes and Biological Activities

The various immunoglobulin isotypes and classes have been mentioned briefly already. Each class is distinguished by unique amino acid sequences in the heavy-chain constant region that confer class-specific structural and functional properties. In this section, the structure and effector functions of each class are described in more detail. The molecular properties and biological activities of the immunoglobulin classes are summarized in Table 4-2 (page 90). The structures of the five major classes are diagrammed in Figure 4-13 (page 91).

Immunoglobulin G (IgG)

IgG, the most abundant class in serum, constitutes about 80% of the total serum immunoglobulin. The IgG molecule consists of two γ heavy chains and two κ or two λ light chains (see Figure 4-13a). There are four human IgG subclasses, distinguished by differences in γ -chain sequence and numbered according to their decreasing average serum concentrations: IgG1, IgG2, IgG3, and IgG4 (see Table 4-2).

The amino acid sequences that distinguish the four IgG subclasses are encoded by different germ-line C_H genes, whose DNA sequences are 90%–95% homologous. The structural characteristics that distinguish these subclasses from one another are the size of the hinge region and the number and position of the interchain disulfide bonds between the heavy chains (Figure 4-14, page 92). The subtle



CLINICAL FOCUS

Passive Antibody Therapy

In 1890, Emil Behring and Shibasaburo Kitasato reported an extraordinary experiment. They immunized rabbits with tetanus and then collected serum from these animals. Subsequently, they injected 0.2 ml of the immune serum into the abdominal cavity of six mice. After 24 hours, they infected the treated animals and untreated controls with live, virulent tetanus bacteria. All of the control mice died within 48 hours of infection, whereas the treated mice not only survived but showed no effects of infection. This landmark experiment demonstrated two important points. One, it showed that following immunization, substances appeared in serum that could protect an animal against pathogens. Two, this work demonstrated that immunity could be passively acquired. Immunity could be transferred from one animal to another by taking serum from an immune animal and injecting it into a nonimmune one. These and subsequent experiments did not go unnoticed. Both men eventually received titles (Behring became von Behring and Kitasato became Baron Kitasato). A few years later, in 1901, von Behring was awarded the first Nobel prize in Medicine.

These early observations and others paved the way for the introduction of passive immunization into clinical prac-

tice. During the 1930s and 1940s, passive immunotherapy, the endowment of resistance to pathogens by transfer of the agent of immunity from an immunized donor to an unimmunized recipient, was used to prevent or modify the course of measles and hepatitis A. During subsequent years, clinical experience and advances in the technology of preparation of immunoglobulin for passive immunization have made this approach a standard medical practice. Passive immunization based on the transfer of antibodies is widely used in the treatment of immunodeficiency diseases and as a protection against anticipated exposure to infectious agents against which one does not have immunity.

Immunoglobulin for passive immunization is prepared from the pooled plasma of thousands of donors. In effect, recipients of these antibody preparations are receiving a sample of the antibodies produced by many people to a broad diversity of different pathogens. In fact a gram of intravenous immune globulin (IVIG) contains about 10^{18} molecules of antibody (mostly IgG) and may incorporate more than 10^7 different antibody specificities. During the course of therapy, patients receive significant amounts of IVIG, usually 200–400 mg per kilogram of body weight. This means that an immunodeficient patient weighing

70 kilograms would receive 14 to 28 grams of IVIG every 3 to 4 weeks. A product derived from the blood of such a large number of donors carries a risk of harboring pathogenic agents, particularly viruses. The risk is minimized by the processes used to produce intravenous immune globulin. The manufacture of IVIG involves treatment with solvents, such as ethanol, and the use of detergents that are highly effective in inactivating viruses such as HIV and hepatitis. In addition to removing or inactivating infectious agents, the production process must also eliminate aggregated immunoglobulin, because antibody aggregates can trigger massive activation of the complement pathway, leading to severe, even fatal, anaphylaxis.

Passively administered antibody exerts its protective action in a number of ways. One of the most important is the recruitment of the complement pathway to the destruction or removal of a pathogen. In bacterial infections, antibody binding to bacterial surfaces promotes opsonization, the phagocytosis and killing of the invader by macrophages and neutrophils. Toxins and viruses can be bound and neutralized by antibody, even as the antibody marks the pathogen for removal from the body by phagocytes and by organs such as liver and kidneys. By the initiation of antibody-dependent cell-mediated cytotoxicity (ADCC), antibodies can also mediate the killing of target cells by cytotoxic cell populations such as natural killer cells.

amino acid differences between subclasses of IgG affect the biological activity of the molecule:

- IgG1, IgG3, and IgG4 readily cross the placenta and play an important role in protecting the developing fetus.
- IgG3 is the most effective complement activator, followed by IgG1; IgG2 is less efficient, and IgG4 is not able to activate complement at all.
- IgG1 and IgG3 bind with high affinity to Fc receptors on phagocytic cells and thus mediate opsonization. IgG4 has an intermediate affinity for Fc receptors, and IgG2 has an extremely low affinity.

Immunoglobulin M (IgM)

IgM accounts for 5%–10% of the total serum immunoglobulin, with an average serum concentration of 1.5 mg/ml. Monomeric IgM, with a molecular weight of 180,000, is expressed as membrane-bound antibody on B cells. IgM is secreted by plasma cells as a pentamer in which five monomer units are held together by disulfide bonds that link their carboxyl-terminal heavy chain domains ($C_{\mu}4/C_{\mu}4$) and their $C_{\mu}3/C_{\mu}3$ domains (see Figure 4-13e). The five monomer subunits are arranged with their Fc regions in the center of the pentamer and the ten antigen-binding sites on the periphery of the molecule. Each pentamer contains an

TABLE 4-2 Properties and biological activities* of classes and subclasses of human serum immunoglobulins

Property/Activity	IgG1	IgG2	IgG3	IgG4	IgA1	IgA2	IgM [‡]	IgE	IgD
Molecular weight [†]	150,000	150,000	150,000	150,000	150,000–600,000	150,000–600,000	900,000	190,000	150,000
Heavy-chain component	γ1	γ2	γ3	γ4	α1	α2	μ	ε	δ
Normal serum level (mg/ml)	9	3	1	0.5	3.0	0.5	1.5	0.0003	0.03
In vivo serum half life (days)	23	23	8	23	6	6	5	2.5	3
Activates classical complement pathway	+	+/-	++	–	–	–	+++	–	–
Crosses placenta	+	+/-	+	+	–	–	–	–	–
Present on membrane of mature B cells	–	–	–	–	–	–	+	–	+
Binds to Fc receptors of phagocytes	++	+/-	++	+	–	–	?	–	–
Mucosal transport	–	–	–	–	++	++	+	–	–
Induces mast-cell degranulation	–	–	–	–	–	–	–	+	–

*Activity levels indicated as follows: ++ = high; + = moderate; +/- = minimal; – = none; ? = questionable.

[†]IgG, IgE, and IgD always exist as monomers; IgA can exist as a monomer, dimer, trimer, or tetramer. Membrane-bound IgM is a monomer, but secreted IgM in serum is a pentamer.

[‡]IgM is the first isotype produced by the neonate and during a primary immune response.

additional Fc-linked polypeptide called the **J (joining) chain**, which is disulfide-bonded to the carboxyl-terminal cysteine residue of two of the ten μ chains. The J chain appears to be required for polymerization of the monomers to form pentameric IgM; it is added just before secretion of the pentamer.

IgM is the first immunoglobulin class produced in a primary response to an antigen, and it is also the first immunoglobulin to be synthesized by the neonate. Because of its pentameric structure with 10 antigen-binding sites, serum IgM has a higher valency than the other isotypes. An IgM molecule can bind 10 small hapten molecules; however, because of steric hindrance, only 5 or fewer molecules of larger antigens can be bound simultaneously. Because of its high valency, pentameric IgM is more efficient than other isotypes in binding antigens with many repeating epitopes such as viral particles and red blood cells (RBCs). For example, when RBCs are incubated with specific antibody, they clump together into large aggregates in a process called agglutination. It takes 100 to 1000 times more molecules of IgG than of IgM to achieve the same level of agglutination. A similar phenomenon occurs with viral particles: less IgM than IgG is required

to neutralize viral infectivity. IgM is also more efficient than IgG at activating complement. Complement activation requires two Fc regions in close proximity, and the pentameric structure of a single molecule of IgM fulfills this requirement.

Because of its large size, IgM does not diffuse well and therefore is found in very low concentrations in the intercellular tissue fluids. The presence of the J chain allows IgM to bind to receptors on secretory cells, which transport it across epithelial linings to enter the external secretions that bathe mucosal surfaces. Although IgA is the major isotype found in these secretions, IgM plays an important accessory role as a secretory immunoglobulin.

Immunoglobulin A (IgA)

Although IgA constitutes only 10%–15% of the total immunoglobulin in serum, it is the predominant immunoglobulin class in external secretions such as breast milk, saliva, tears, and mucus of the bronchial, genitourinary, and digestive tracts. In serum, IgA exists primarily as a monomer, but polymeric forms (dimers, trimers, and some tetramers) are sometimes seen, all containing a J-chain

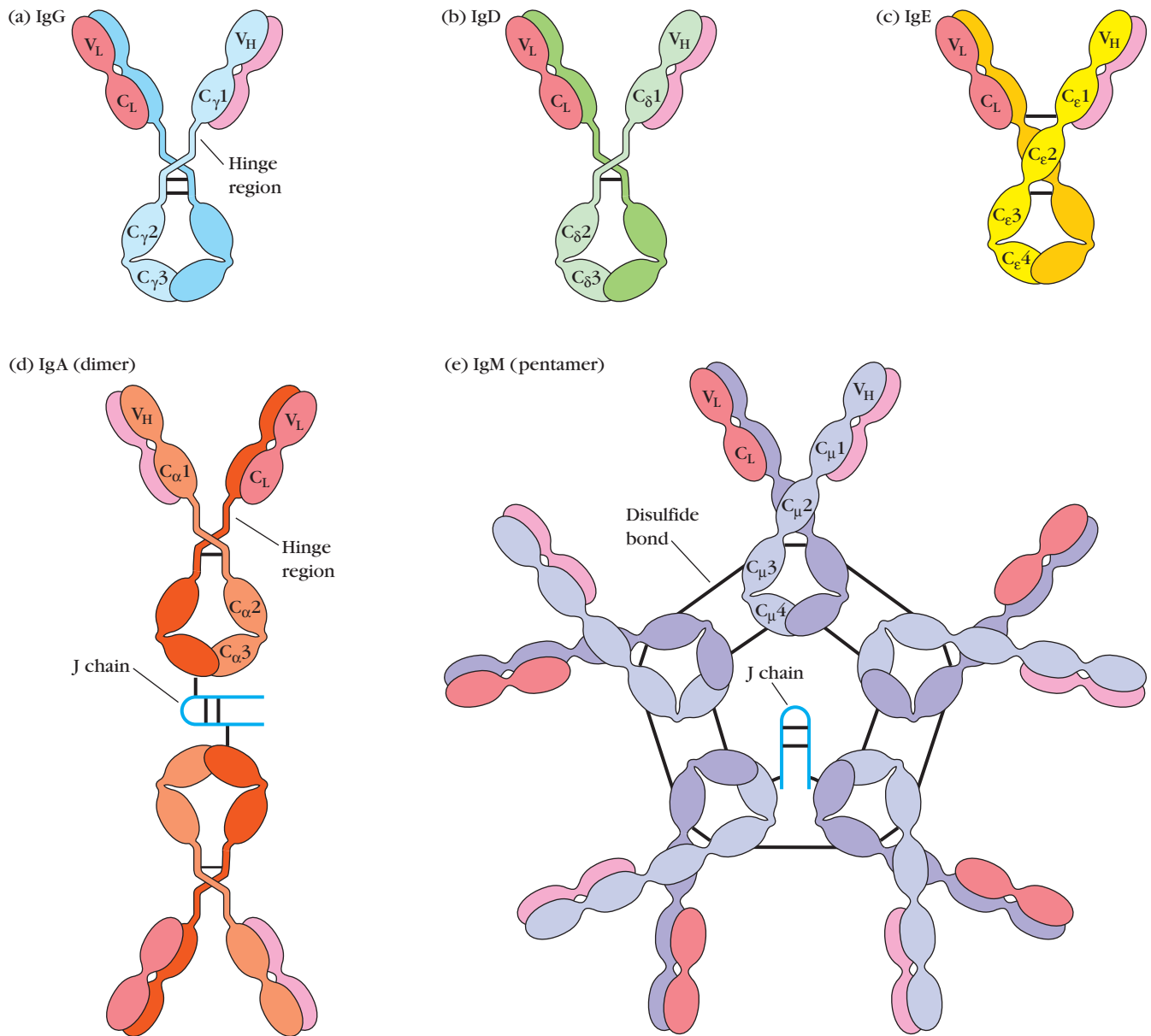


FIGURE 4-13 General structures of the five major classes of secreted antibody. Light chains are shown in shades of pink, disulfide bonds are indicated by thick black lines. Note that the IgG, IgA, and IgD heavy chains (blue, orange, and green, respectively) contain four domains and a hinge region, whereas the IgM and IgE heavy chains (purple and yellow, respectively) contain five domains but no hinge region. The polymeric forms of IgM and IgA contain a polypeptide,

called the J chain, that is linked by two disulfide bonds to the Fc region in two different monomers. Serum IgM is always a pentamer; most serum IgA exists as a monomer, although dimers, trimers, and even tetramers are sometimes present. Not shown in these figures are intrachain disulfide bonds and disulfide bonds linking light and heavy chains (see Figure 4-2).

polypeptide (see Figure 4-13d). The IgA of external secretions, called **secretory IgA**, consists of a dimer or tetramer, a J-chain polypeptide, and a polypeptide chain called **secretory component** (Figure 4-15a, page 93). As is explained below, secretory component is derived from the receptor that is responsible for transporting polymeric IgA across cell membranes. The J-chain polypeptide in IgA is identical to that found in pentameric IgM and serves a similar function in fa-

cilitating the polymerization of both serum IgA and secretory IgA. The secretory component is a 70,000-MW polypeptide produced by epithelial cells of mucous membranes. It consists of five immunoglobulin-like domains that bind to the Fc region domains of the IgA dimer. This interaction is stabilized by a disulfide bond between the fifth domain of the secretory component and one of the chains of the dimeric IgA.

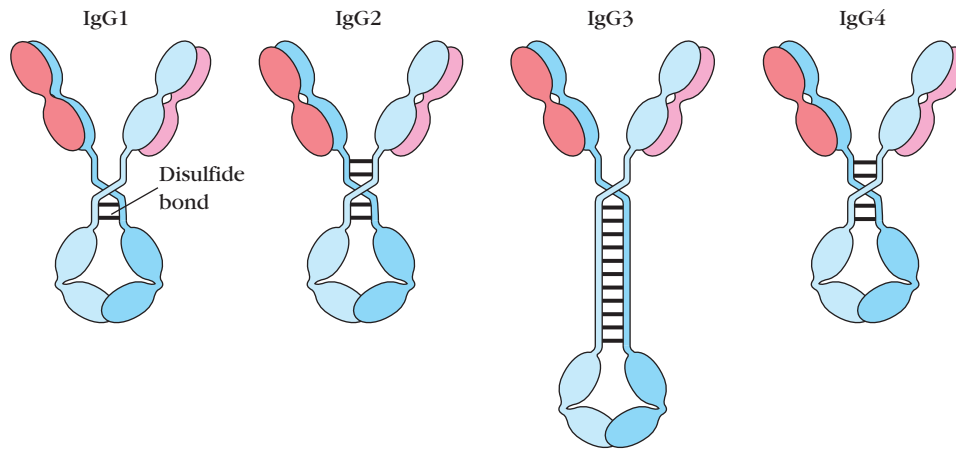


FIGURE 4-14 General structure of the four subclasses of human IgG, which differ in the number and arrangement of the interchain

disulfide bonds (thick black lines) linking the heavy chains. A notable feature of human IgG3 is its 11 interchain disulfide bonds.

The daily production of secretory IgA is greater than that of any other immunoglobulin class. IgA-secreting plasma cells are concentrated along mucous membrane surfaces. Along the jejunum of the small intestine, for example, there are more than 2.5×10^{10} IgA-secreting plasma cells—a number that surpasses the total plasma cell population of the bone marrow, lymph, and spleen combined! Every day, a human secretes from 5 g to 15 g of secretory IgA into mucous secretions.

The plasma cells that produce IgA preferentially migrate (home) to subepithelial tissue, where the secreted IgA binds tightly to a receptor for polymeric immunoglobulin molecules (Figure 4-15b). This **poly-Ig receptor** is expressed on the basolateral surface of most mucosal epithelia (e.g., the lining of the digestive, respiratory, and genital tracts) and on glandular epithelia in the mammary, salivary, and lacrimal glands. After polymeric IgA binds to the poly-Ig receptor, the receptor-IgA complex is transported across the epithelial barrier to the lumen. Transport of the receptor-IgA complex involves receptor-mediated endocytosis into coated pits and directed transport of the vesicle across the epithelial cell to the luminal membrane, where the vesicle fuses with the plasma membrane. The poly-Ig receptor is then cleaved enzymatically from the membrane and becomes the secretory component, which is bound to and released together with polymeric IgA into the mucous secretions. The secretory component masks sites susceptible to protease cleavage in the hinge region of secretory IgA, allowing the polymeric molecule to exist longer in the protease-rich mucosal environment than would be possible otherwise. Pentameric IgM is also transported into mucous secretions by this mechanism, although it accounts for a much lower percentage of antibody in the mucous secretions than does IgA. The poly-Ig receptor interacts with the J chain of both polymeric IgA and IgM antibodies.

Secretory IgA serves an important effector function at mucous membrane surfaces, which are the main entry sites

for most pathogenic organisms. Because it is polymeric, secretory IgA can cross-link large antigens with multiple epitopes. Binding of secretory IgA to bacterial and viral surface antigens prevents attachment of the pathogens to the mucosal cells, thus inhibiting viral infection and bacterial colonization. Complexes of secretory IgA and antigen are easily entrapped in mucus and then eliminated by the ciliated epithelial cells of the respiratory tract or by peristalsis of the gut. Secretory IgA has been shown to provide an important line of defense against bacteria such as *Salmonella*, *Vibrio cholerae*, and *Neisseria gonorrhoeae* and viruses such as polio, influenza, and reovirus.

Breast milk contains secretory IgA and many other molecules that help protect the newborn against infection during the first month of life (Table 4-3). Because the immune system of infants is not fully functional, breast-feeding plays an important role in maintaining the health of newborns.

Immunoglobulin E (IgE)

The potent biological activity of IgE allowed it to be identified in serum despite its extremely low average serum concentration ($0.3 \mu\text{g/ml}$). IgE antibodies mediate the immediate hypersensitivity reactions that are responsible for the symptoms of hay fever, asthma, hives, and anaphylactic shock. The presence of a serum component responsible for allergic reactions was first demonstrated in 1921 by K. Prausnitz and H. Kustner, who injected serum from an allergic person intra-dermally into a nonallergic individual. When the appropriate antigen was later injected at the same site, a wheal and flare reaction (analogous to hives) developed there. This reaction, called the **P-K reaction** (named for its originators, Prausnitz and Kustner), was the basis for the first biological assay for IgE activity.

Actual identification of IgE was accomplished by K. and T. Ishizaka in 1966. They obtained serum from an allergic in-

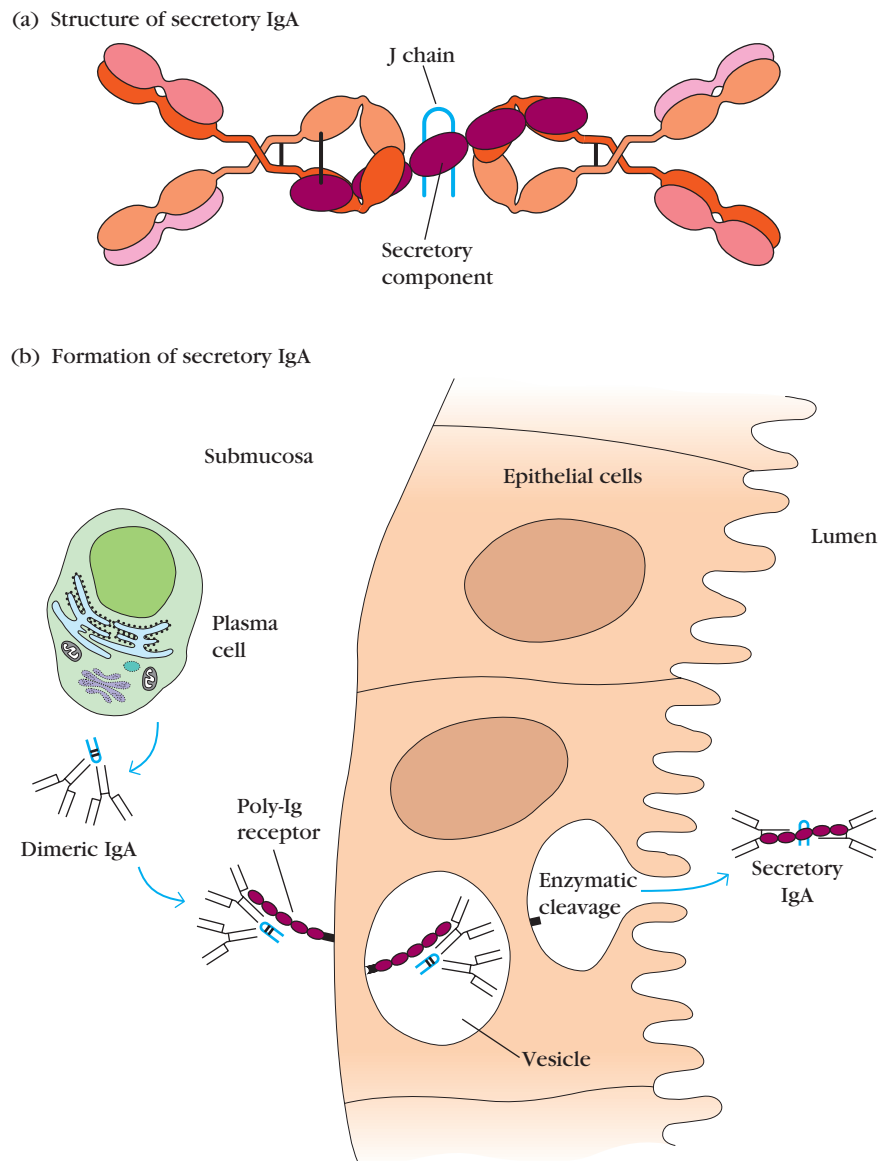


FIGURE 4-15 Structure and formation of secretory IgA. (a) Secretory IgA consists of at least two IgA molecules, which are covalently linked to each other through a J chain and are also covalently linked with the secretory component. The secretory component contains five Ig-like domains and is linked to dimeric IgA by a disulfide bond between its fifth domain and one of the IgA heavy chains. (b) Secre-

tory IgA is formed during transport through mucous membrane epithelial cells. Dimeric IgA binds to a poly-Ig receptor on the basolateral membrane of an epithelial cell and is internalized by receptor-mediated endocytosis. After transport of the receptor-IgA complex to the luminal surface, the poly-Ig receptor is enzymatically cleaved, releasing the secretory component bound to the dimeric IgA.

dividual and immunized rabbits with it to prepare anti-isotype antiserum. The rabbit antiserum was then allowed to react with each class of human antibody known at that time (i.e., IgG, IgA, IgM, and IgD). In this way, each of the known anti-isotype antibodies was precipitated and removed from the rabbit anti-serum. What remained was an anti-isotype antibody specific for an unidentified class of antibody. This antibody turned out to completely block the P-K reaction. The new antibody was called IgE (in reference to the E antigen of ragweed pollen, which is a potent inducer of this class of antibody).

IgE binds to Fc receptors on the membranes of blood basophils and tissue mast cells. Cross-linkage of receptor-bound IgE molecules by antigen (allergen) induces basophils and mast cells to translocate their granules to the plasma membrane and release their contents to the extracellular environment, a process known as degranulation. As a result, a variety of pharmacologically active mediators are released and give rise to allergic manifestations (Figure 4-16). Localized mast-cell degranulation induced by IgE also may release mediators that facilitate a buildup of various cells necessary for antiparasitic defense (see Chapter 15).

TABLE 4-3 Immune benefits of breast milk

Antibodies of secretory IgA class	Bind to microbes in baby's digestive tract and thereby prevent their attachment to the walls of the gut and their subsequent passage into the body's tissues.
B ₁₂ binding protein	Reduces amount of vitamin B ₁₂ , which bacteria need in order to grow.
Bifidus factor	Promotes growth of <i>Lactobacillus bifidus</i> , a harmless bacterium, in baby's gut. Growth of such nonpathogenic bacteria helps to crowd out dangerous varieties.
Fatty acids	Disrupt membranes surrounding certain viruses and destroy them.
Fibronectin	Increases antimicrobial activity of macrophages; helps to repair tissues that have been damaged by immune reactions in baby's gut.
Hormones and growth factors	Stimulate baby's digestive tract to mature more quickly. Once the initially "leaky" membranes lining the gut mature, infants become less vulnerable to microorganisms.
Interferon (IFN- γ)	Enhances antimicrobial activity of immune cells.
Lactoferrin	Binds to iron, a mineral many bacteria need to survive. By reducing the available amount of iron, lactoferrin thwarts growth of pathogenic bacteria.
Lysozyme	Kills bacteria by disrupting their cell walls.
Mucins	Adhere to bacteria and viruses, thus keeping such microorganisms from attaching to mucosal surfaces.
Oligosaccharides	Bind to microorganisms and bar them from attaching to mucosal surfaces.

SOURCE: Adapted from J. Newman, 1995, How breast milk protects newborns, *Sci. Am.* 273(6):76.

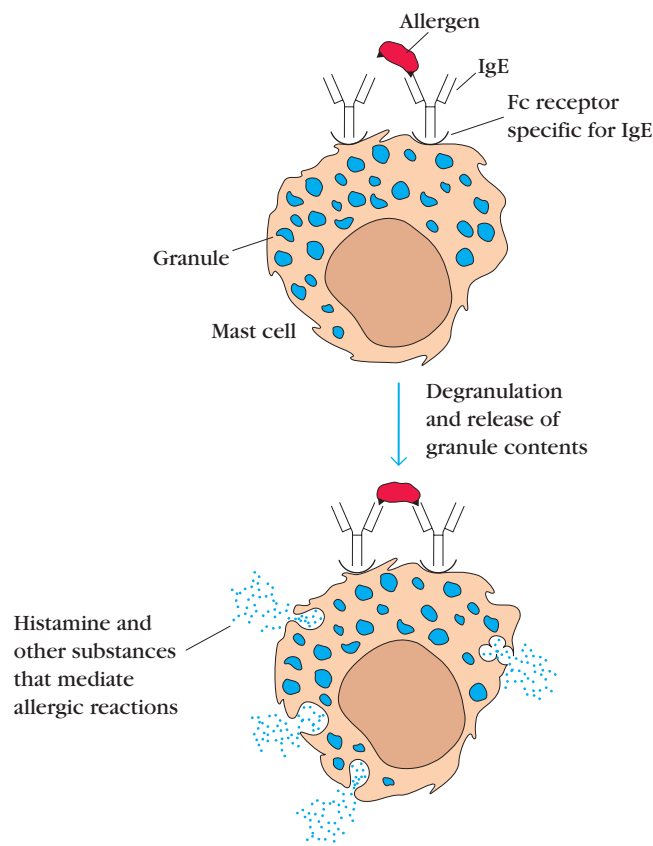


FIGURE 4-16 Allergen cross-linkage of receptor-bound IgE on mast cells induces degranulation, causing release of substances (blue dots) that mediate allergic manifestations.

Immunoglobulin D (IgD)

IgD was first discovered when a patient developed a multiple myeloma whose myeloma protein failed to react with anti-isotype antisera against the then-known isotypes: IgA, IgM, and IgG. When rabbits were immunized with this myeloma protein, the resulting antisera were used to identify the same class of antibody at low levels in normal human serum. The new class, called IgD, has a serum concentration of 30 $\mu\text{g/ml}$ and constitutes about 0.2% of the total immunoglobulin in serum. IgD, together with IgM, is the major membrane-bound immunoglobulin expressed by mature B cells, and its role in the physiology of B cells is under investigation. No biological effector function has been identified for IgD.

Antigenic Determinants on Immunoglobulins

Since antibodies are glycoproteins, they can themselves function as potent immunogens to induce an antibody response. Such anti-Ig antibodies are powerful tools for the study of B-cell development and humoral immune responses. The antigenic determinants, or epitopes, on immunoglobulin molecules fall into three major categories: isotypic, allotypic, and idiotypic determinants, which are located in characteristic portions of the molecule (Figure 4-17).

Isotype

Isotypic determinants are constant-region determinants that collectively define each heavy-chain class and subclass and

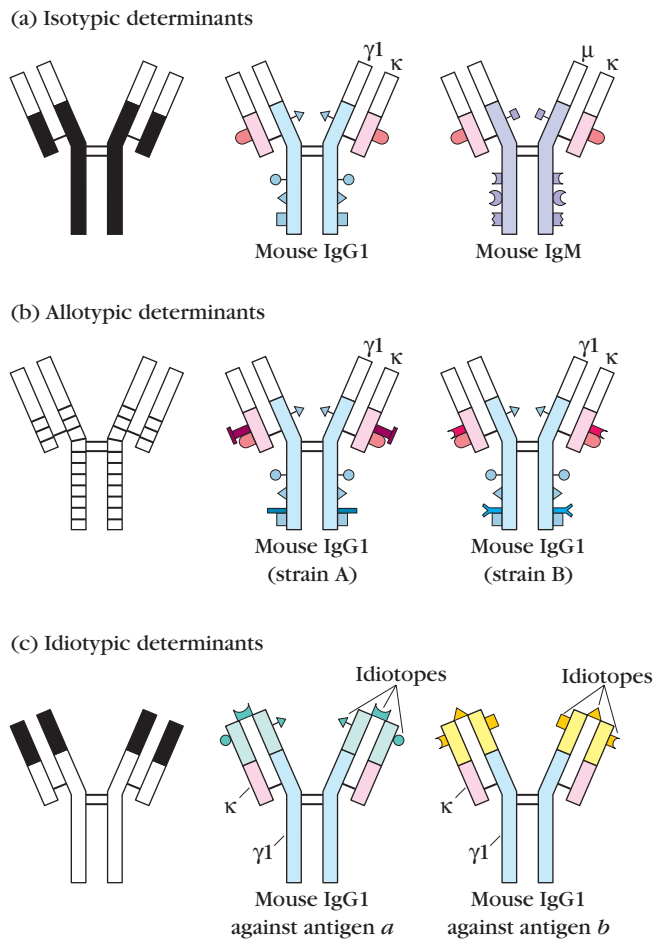


FIGURE 4-17 Antigenic determinants of immunoglobulins. For each type of determinant, the general location of determinants within the antibody molecule is shown (*left*) and two examples are illustrated (*center* and *right*). (a) Isotypic determinants are constant-region determinants that distinguish each Ig class and subclass within a species. (b) Allotypic determinants are subtle amino acid differences encoded by different alleles of isotype genes. Allotypic differences can be detected by comparing the same antibody class among different inbred strains. (c) Idiotypic determinants are generated by the conformation of the amino acid sequences of the heavy- and light-chain variable regions specific for each antigen. Each individual determinant is called an **idiotope**, and the sum of the individual idiotopes is the **idiotypic**.

each light-chain type and subtype within a species (see Figure 4-17a). Each isotype is encoded by a separate constant-region gene, and all members of a species carry the same constant-region genes (which may include multiple alleles). Within a species, each normal individual will express all isotypes in the serum. Different species inherit different constant-region genes and therefore express different isotypes. Therefore, when an antibody from one species is injected into another species, the isotypic determinants will be recognized as foreign, inducing an antibody response to the isotypic determinants on the foreign antibody. Anti-isotype

antibody is routinely used for research purposes to determine the class or subclass of serum antibody produced during an immune response or to characterize the class of membrane-bound antibody present on B cells.

Allotype

Although all members of a species inherit the same set of isotype genes, multiple alleles exist for some of the genes (see Figure 4-17b). These alleles encode subtle amino acid differences, called allotypic determinants, that occur in some, but not all, members of a species. The sum of the individual allotypic determinants displayed by an antibody determines its **allotype**. In humans, allotypes have been characterized for all four IgG subclasses, for one IgA subclass, and for the κ light chain. The γ -chain allotypes are referred to as Gm markers. At least 25 different Gm allotypes have been identified; they are designated by the class and subclass followed by the allele number, for example, G1m(1), G2m(23), G3m(11), G4m(4a). Of the two IgA subclasses, only the IgA2 subclass has allotypes, as A2m(1) and A2m(2). The κ light chain has three allotypes, designated κ m(1), κ m(2), and κ m(3). Each of these allotypic determinants represents differences in one to four amino acids that are encoded by different alleles.

Antibody to allotypic determinants can be produced by injecting antibodies from one member of a species into another member of the same species who carries different allotypic determinants. Antibody to allotypic determinants sometimes is produced by a mother during pregnancy in response to paternal allotypic determinants on the fetal immunoglobulins. Antibodies to allotypic determinants can also arise from a blood transfusion.

Idiotypic

The unique amino acid sequence of the V_H and V_L domains of a given antibody can function not only as an antigen-binding site but also as a set of antigenic determinants. The idiotypic determinants arise from the sequence of the heavy- and light-chain variable regions. Each individual antigenic determinant of the variable region is referred to as an **idiotope** (see Figure 4-17c). In some cases an idiotope may be the actual antigen-binding site, and in some cases an idiotope may comprise variable-region sequences outside of the antigen-binding site. Each antibody will present multiple idiotopes; the sum of the individual idiotopes is called the **idiotypic** of the antibody.

Because the antibodies produced by individual B cells derived from the same clone have identical variable-region sequences, they all have the same idiotypic. Anti-idiotypic antibody is produced by injecting antibodies that have minimal variation in their isotypes and allotypes, so that the idiotypic difference can be recognized. Often a homogeneous antibody such as myeloma protein or monoclonal antibody is used. Injection of such an antibody into a recipient who is

genetically identical to the donor will result in the formation of anti-idiotypic antibody to the idiotypic determinants.

The B-Cell Receptor

Immunologists have long been puzzled about how mIg mediates an activating signal after contact with an antigen. The dilemma is that all isotypes of mIg have very short cytoplasmic tails: the mIgM and mIgD cytoplasmic tails contain only 3 amino acids; the mIgA tail, 14 amino acids; and the mIgG and mIgE tails, 28 amino acids. In each case, the cytoplasmic tail is too short to be able to associate with intracellular signaling molecules (e.g., tyrosine kinases and G proteins).

The answer to this puzzle is that mIg does not constitute the entire antigen-binding receptor on B cells. Rather, the **B-cell receptor (BCR)** is a transmembrane protein complex composed of mIg and disulfide-linked heterodimers called Ig- α /Ig- β . Molecules of this heterodimer associate with an mIg molecule to form a BCR (Figure 4-18). The Ig- α chain

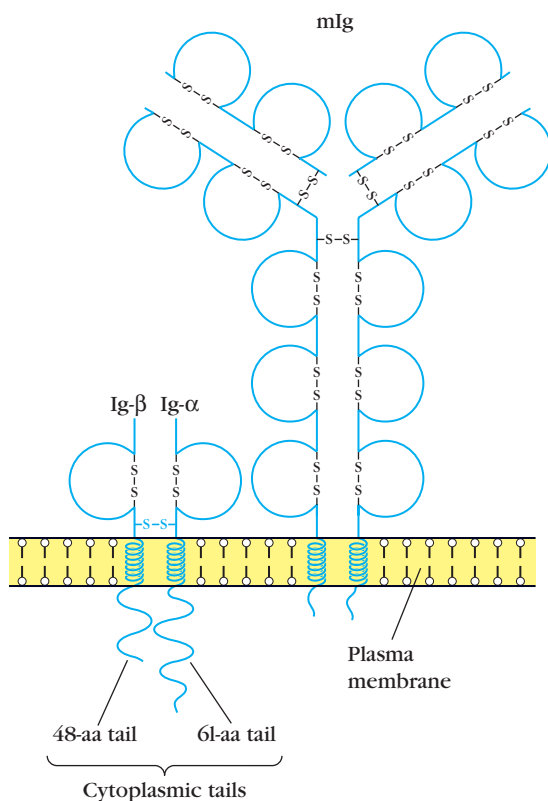


FIGURE 4-18 General structure of the B-cell receptor (BCR). This antigen-binding receptor is composed of membrane-bound immunoglobulin (mIg) and disulfide-linked heterodimers called Ig- α /Ig- β . Each heterodimer contains the immunoglobulin-fold structure and cytoplasmic tails much longer than those of mIg. As depicted, an mIg molecule is associated with one Ig- α /Ig- β heterodimer. [Adapted from A. D. Keegan and W. E. Paul, 1992, *Immunol. Today* **13**:63, and M. Reth, 1992, *Annu. Rev. Immunol.* **10**:97.]

has a long cytoplasmic tail containing 61 amino acids; the tail of the Ig- β chain contains 48 amino acids. The tails in both Ig- α and Ig- β are long enough to interact with intracellular signaling molecules. Discovery of the Ig- α /Ig- β heterodimer by Michael Reth and his colleagues in the early 1990s has substantially furthered understanding of B-cell activation, which is discussed in detail in Chapter 11.

Fc Receptors Bond to Fc Regions of Antibodies

Many cells feature membrane glycoproteins called **Fc receptors (FcR)** that have an affinity for the Fc portion of the antibody molecule. These receptors are essential for many of the biological functions of antibodies. Fc receptors are responsible for the movement of antibodies across cell membranes and the transfer of IgG from mother to fetus across the placenta. These receptors also allow passive acquisition of antibody by many cell types, including B and T lymphocytes, neutrophils, mast cells, eosinophils, macrophages, and natural killer cells. Consequently, Fc receptors provide a means by which antibodies—the products of the adaptive immune system—can recruit such key cellular elements of innate immunity as macrophages and natural killer cells. Engagement of antibody-bound antigens by the Fc receptors of macrophages or neutrophils provides an effective signal for the efficient phagocytosis (opsonization) of antigen-antibody complexes. In addition to triggering such effector functions as opsonization or ADCC, crosslinking of Fc receptors by antigen-mediated crosslinking of FcR-bound antibodies can generate immunoregulatory signals that affect cell activation, induce differentiation and, in some cases, downregulate cellular responses.

There are many different Fc receptors (Figure 4-19). The poly Ig receptor is essential for the transport of polymeric immunoglobulins (polymeric IgA and to some extent, pentameric IgM) across epithelial surfaces. In humans, the **neonatal Fc receptor (FcR_N)** transfers IgGs from mother to fetus during gestation and also plays a role in the regulation of IgG serum levels. Fc receptors have been discovered for all of the Ig classes. Thus there is an Fc α R receptor that binds IgA, an Fc ϵ R that binds IgE (see Figure 4-16 also), an Fc δ R that binds IgD, IgM is bound by an Fc μ R, and several varieties of Fc γ R receptors capable of binding IgG and its subclasses are found in humans. In many cases, the crosslinking of these receptors by binding of antigen-antibody complexes results in the initiation of signal-transduction cascades that result in such behaviors as phagocytosis or ADCC. The Fc receptor is often part of a signal-transducing complex that involves the participation of other accessory polypeptide chains. As shown in Figure 4-19, this may involve a pair of γ chains or, in the case of the IgE receptor, a more complex assemblage of two γ chains and a β chain. The association of an extracellular receptor with an intracellular signal-transducing unit was seen in the B cell receptor (Figure 4-18) and is a central feature of the T-cell-receptor complex (Chapter 9).

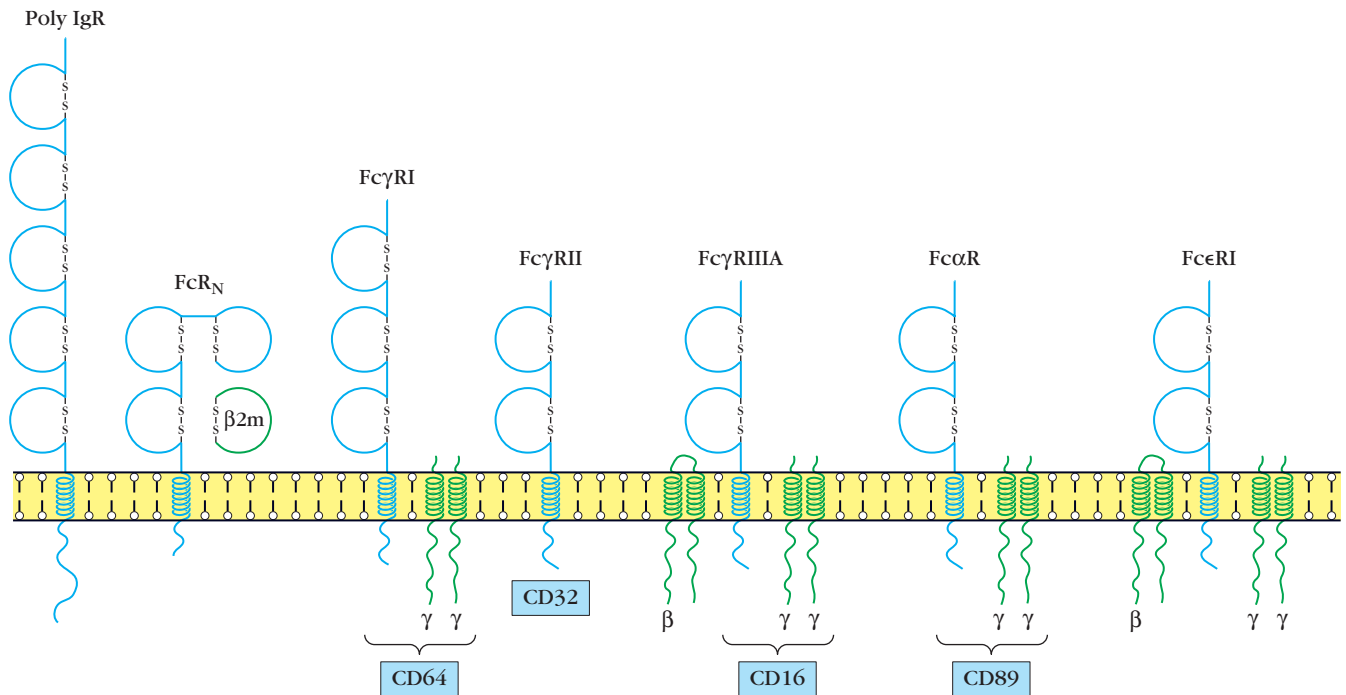


FIGURE 4-19 The structure of a number of human Fc-receptors. The Fc-binding polypeptides are shown in blue and, where present, accessory signal-transducing polypeptides are shown in green. The loops in these structures represent portions of the molecule with the characteristic immunoglobulin-fold structure. These molecules

appear on the plasma membrane as cell-surface antigens and, as indicated in the figure, many have been assigned CD designations (for clusters of differentiation; see Appendix). [Adapted from M. Daeron, 1999, in *The Antibodies*, vol. 5, p. 53. Edited by M. Zanetti and J. D. Capra.]

The Immunoglobulin Superfamily

The structures of the various immunoglobulin heavy and light chains described earlier share several features, suggesting that they have a common evolutionary ancestry. In particular, all heavy- and light-chain classes have the immunoglobulin-fold domain structure (see Figure 4-7). The presence of this characteristic structure in all immunoglobulin heavy and light chains suggests that the genes encoding them arose from a common primordial gene encoding a polypeptide of about 110 amino acids. Gene duplication and later divergence could then have generated the various heavy- and light-chain genes.

Large numbers of membrane proteins have been shown to possess one or more regions homologous to an immunoglobulin domain. Each of these membrane proteins is classified as a member of the **immunoglobulin superfamily**. The term *superfamily* is used to denote proteins whose corresponding genes derived from a common primordial gene encoding the basic domain structure. These genes have evolved independently and do not share genetic linkage or function. The following proteins, in addition to the immunoglobulins themselves, are representative members of the immunoglobulin superfamily (Figure 4-20):

- Ig-α/Ig-β heterodimer, part of the B-cell receptor
- Poly-Ig receptor, which contributes the secretory component to secretory IgA and IgM
- T-cell receptor
- T-cell accessory proteins, including CD2, CD4, CD8, CD28, and the γ, δ, and ε chains of CD3
- Class I and class II MHC molecules
- β₂-microglobulin, an invariant protein associated with class I MHC molecules
- Various cell-adhesion molecules, including VCAM-1, ICAM-1, ICAM-2, and LFA-3
- Platelet-derived growth factor

Numerous other proteins, some of them discussed in other chapters, also belong to the immunoglobulin superfamily.

X-ray crystallographic analysis has not been accomplished for all members of the immunoglobulin superfamily. Nevertheless, the primary amino acid sequence of these proteins suggests that they all contain the typical immunoglobulin-fold domain. Specifically, all members of the immunoglobulin superfamily contain at least one or more stretches of about 110 amino acids, capable of arrangement

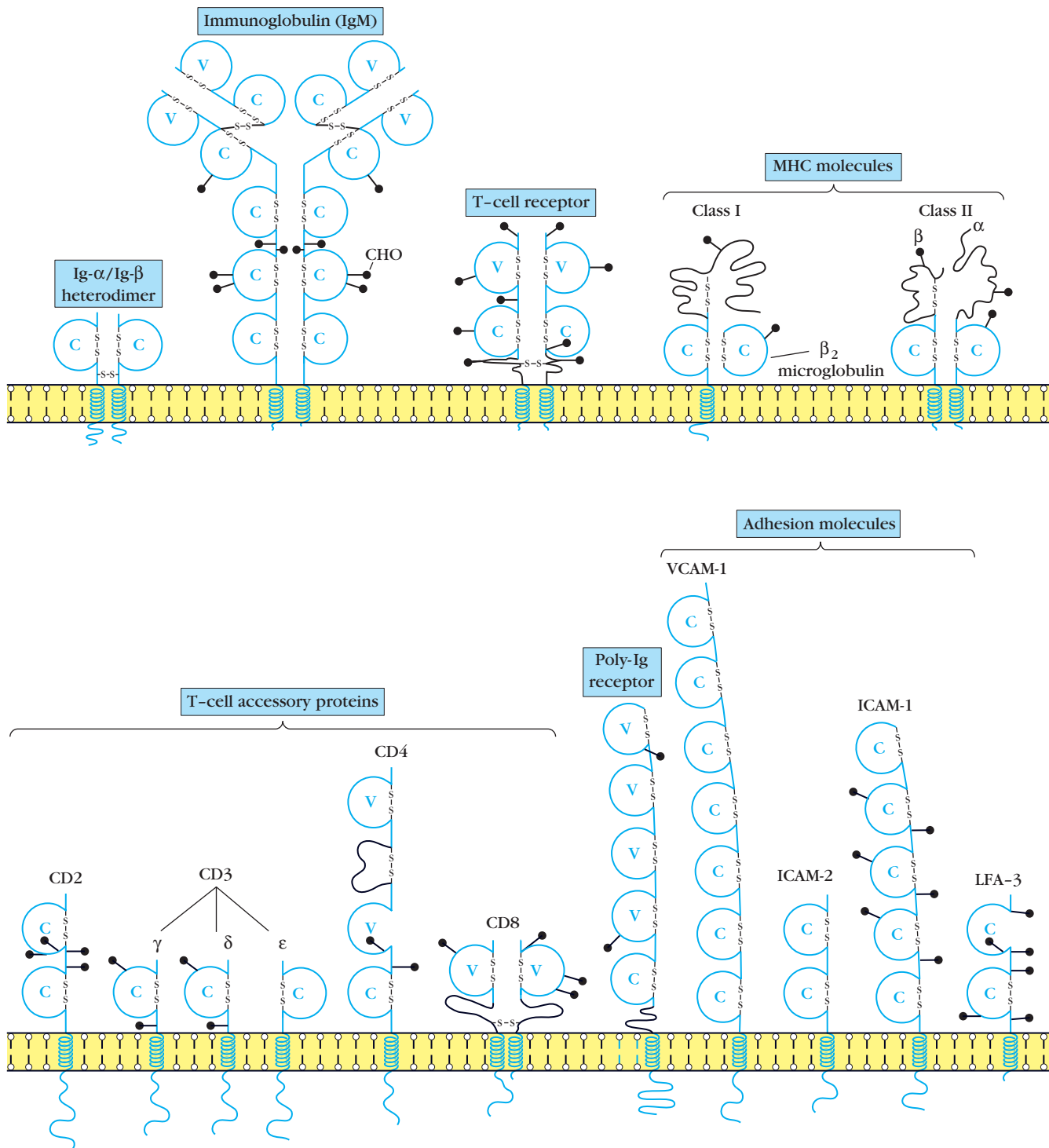


FIGURE 4-20 Some members of the immunoglobulin superfamily, a group of structurally related, usually membrane-bound glycopro-

teins. In all cases shown here except for β_2 -microglobulin, the carboxyl-terminal end of the molecule is anchored in the membrane.

into pleated sheets of antiparallel β strands, usually with an invariant intrachain disulfide bond that closes a loop spanning 50–70 residues.

Most members of the immunoglobulin superfamily cannot bind antigen. Thus, the characteristic Ig-fold structure

found in so many membrane proteins must have some function other than antigen binding. One possibility is that the immunoglobulin fold may facilitate interactions between membrane proteins. As described earlier, interactions can occur between the faces of β pleated sheets both of homo-

gous immunoglobulin domains (e.g., C_{H2}/C_{H2} interaction) and of nonhomologous domains (e.g., V_H/V_L and C_{H1}/C_L interactions).

Monoclonal Antibodies

As noted in Chapter 3, most antigens offer multiple epitopes and therefore induce proliferation and differentiation of a variety of B-cell clones, each derived from a B cell that recognizes a particular epitope. The resulting serum antibodies are heterogeneous, comprising a mixture of antibodies, each specific for one epitope (Figure 4-21). Such a **polyclonal antibody** response facilitates the localization, phagocytosis, and complement-mediated lysis of antigen; it thus has clear ad-

vantages for the organism in vivo. Unfortunately, the antibody heterogeneity that increases immune protection in vivo often reduces the efficacy of an antiserum for various in vitro uses. For most research, diagnostic, and therapeutic purposes, **monoclonal antibodies**, derived from a single clone and thus specific for a single epitope, are preferable.

Direct biochemical purification of a monoclonal antibody from a polyclonal antibody preparation is not feasible. In 1975, Georges Köhler and Cesar Milstein devised a method for preparing monoclonal antibody, which quickly became one of immunology's key technologies. By fusing a normal activated, antibody-producing B cell with a myeloma cell (a cancerous plasma cell), they were able to generate a hybrid cell, called a **hybridoma**, that possessed the immortal-growth properties of the myeloma cell and secreted the

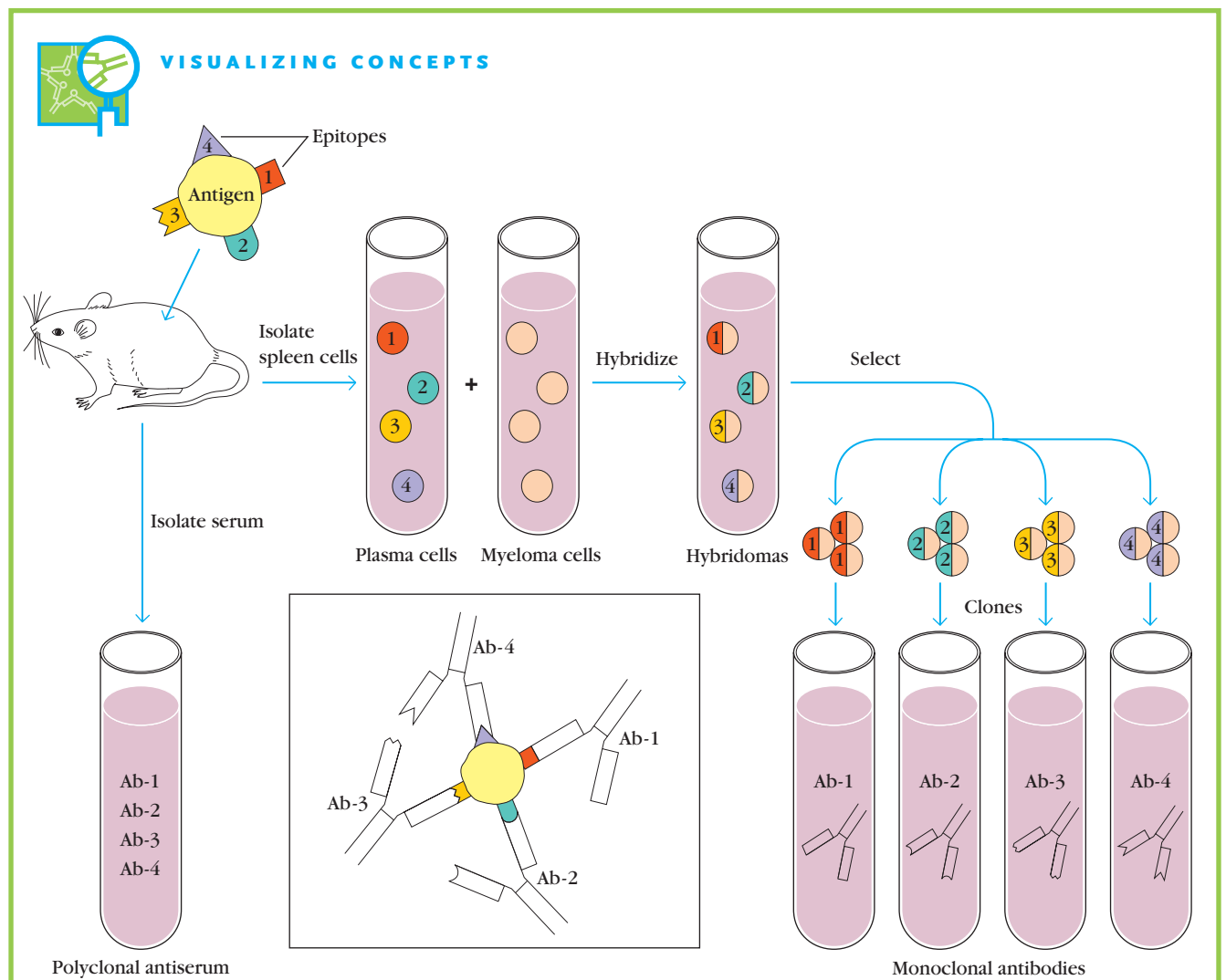


FIGURE 4-21 The conventional polyclonal antiserum produced in response to a complex antigen contains a mixture of monoclonal antibodies, each specific for one of the four epitopes shown on the antigen (inset). In contrast, a monoclonal antibody,

which is derived from a single plasma cell, is specific for one epitope on a complex antigen. The outline of the basic method for obtaining a monoclonal antibody is illustrated here.

antibody produced by the B cell (see Figure 4-21). The resulting clones of hybridoma cells, which secrete large quantities of monoclonal antibody, can be cultured indefinitely. The development of techniques for producing monoclonal antibodies, the details of which are discussed in Chapter 23, gave immunologists a powerful and versatile research tool. The significance of the work by Köhler and Milstein was acknowledged when each was awarded a Nobel Prize.

Monoclonal Antibodies Have Important Clinical Uses

Monoclonal antibodies are proving to be very useful as diagnostic, imaging, and therapeutic reagents in clinical medi-

cine. Initially, monoclonal antibodies were used primarily as in vitro diagnostic reagents. Among the many monoclonal antibody diagnostic reagents now available are products for detecting pregnancy, diagnosing numerous pathogenic microorganisms, measuring the blood levels of various drugs, matching histocompatibility antigens, and detecting antigens shed by certain tumors.

Radiolabeled monoclonal antibodies can also be used in vivo for detecting or locating tumor antigens, permitting earlier diagnosis of some primary or metastatic tumors in patients. For example, monoclonal antibody to breast-cancer cells is labeled with iodine-131 and introduced into the blood to detect the spread of a tumor to regional lymph nodes. This monoclonal imaging technique can reveal breast-cancer metastases that would be undetected by other, less sensitive scanning techniques.

Immunotoxins composed of tumor-specific monoclonal antibodies coupled to lethal toxins are potentially valuable therapeutic reagents. The toxins used in preparing immunotoxins include ricin, *Shigella* toxin, and diphtheria toxin, all of which inhibit protein synthesis. These toxins are so potent that a single molecule has been shown to kill a cell. Each of these toxins consists of two types of functionally distinct polypeptide components, an inhibitory (toxin) chain and one or more binding chains, which interact with receptors on cell surfaces; without the binding polypeptide(s) the toxin cannot get into cells and therefore is harmless. An immunotoxin is prepared by replacing the binding polypeptide(s) with a monoclonal antibody that is specific for a particular tumor cell (Figure 4-22a). In theory, the attached monoclonal antibody will deliver the toxin chain specifically to tumor cells, where it will cause death by inhibiting protein synthesis (Figure 4-22b). The initial clinical responses to such immunotoxins in patients with leukemia, lymphoma, and some other types of cancer have shown promise, and research to develop and demonstrate their safety and effectiveness is underway.

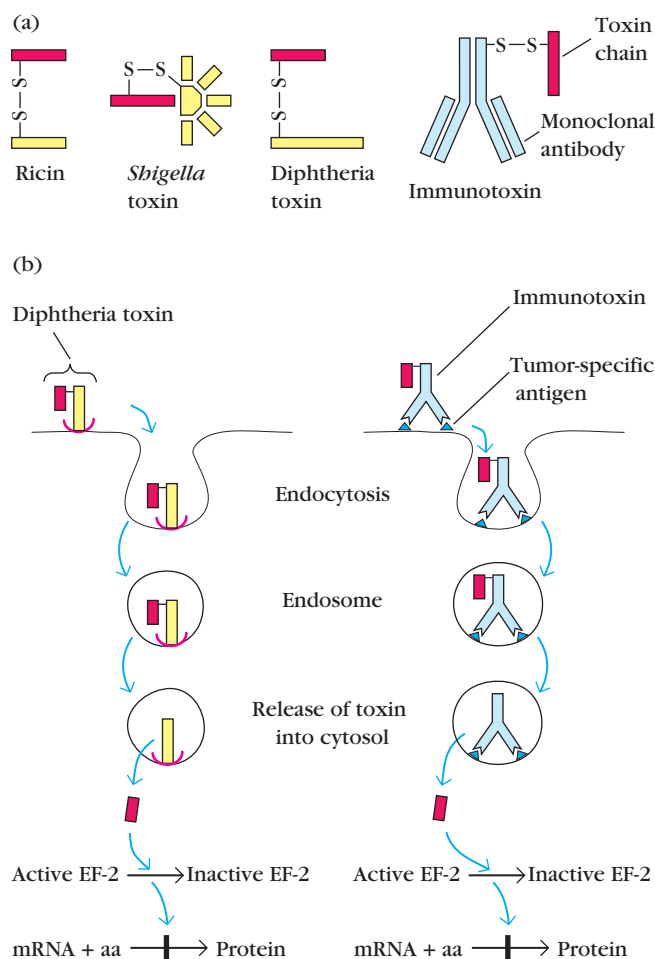


FIGURE 4-22 (a) Toxins used to prepare immunotoxins include ricin, *Shigella* toxin, and diphtheria toxin. Each toxin contains an inhibitory toxin chain (red) and a binding component (yellow). To make an immunotoxin, the binding component of the toxin is replaced with a monoclonal antibody (blue). (b) Diphtheria toxin binds to a cell-membrane receptor (left) and a diphtheria-immunotoxin binds to a tumor-associated antigen (right). In either case, the toxin is internalized in an endosome. The toxin chain is then released into the cytoplasm, where it inhibits protein synthesis by catalyzing the inactivation of elongation factor 2 (EF-2).

Abzymes Are Monoclonal Antibodies That Catalyze Reactions

The binding of an antibody to its antigen is similar in many ways to the binding of an enzyme to its substrate. In both cases the binding involves weak, noncovalent interactions and exhibits high specificity and often high affinity. What distinguishes an antibody-antigen interaction from an enzyme-substrate interaction is that the antibody does not alter the antigen, whereas the enzyme catalyzes a chemical change in its substrate. However, like enzymes, antibodies of appropriate specificity can stabilize the transition state of a bound substrate, thus reducing the activation energy for chemical modification of the substrate.

The similarities between antigen-antibody interactions and enzyme-substrate interactions raised the question of whether some antibodies could behave like enzymes and catalyze chemical reactions. To investigate this possibility, a

hapten-carrier complex was synthesized in which the hapten structurally resembled the transition state of an ester undergoing hydrolysis. Spleen cells from mice immunized with this transition state analogue were fused with myeloma cells to generate monoclonal anti-hapten monoclonal antibodies. When these monoclonal antibodies were incubated with an ester substrate, some of them accelerated hydrolysis by about 1000-fold; that is, they acted like the enzyme that normally catalyzes the substrate's hydrolysis. The catalytic activity of these antibodies was highly specific; that is, they hydrolyzed only esters whose transition-state structure closely resembled the transition state analogue used as a hapten in the immunizing conjugate. These catalytic antibodies have been called **abzymes** in reference to their dual role as antibody and enzyme.

A central goal of catalytic antibody research is the derivation of a battery of abzymes that cut peptide bonds at specific amino acid residues, much as restriction enzymes cut DNA at specific sites. Such abzymes would be invaluable tools in the structural and functional analysis of proteins. Additionally, it may be possible to generate abzymes with the ability to dissolve blood clots or to cleave viral glycoproteins at specific sites, thus blocking viral infectivity. Unfortunately, catalytic antibodies that cleave the peptide bonds of proteins have been exceedingly difficult to derive. Much of the research currently being pursued in this field is devoted to the solution of this important but difficult problem.

SUMMARY

- An antibody molecule consists of two identical light chains and two identical heavy chains, which are linked by disulfide bonds. Each heavy chain has an amino-terminal variable region followed by a constant region.
- In any given antibody molecule, the constant region contains one of five basic heavy-chain sequences (μ , γ , δ , α , or ϵ) called isotypes and one of two basic light-chain sequences (κ or λ) called types.
- The heavy-chain isotype determines the class of an antibody (μ , IgM; γ , IgG; δ , IgD; α , IgA; and ϵ , IgE).
- The five antibody classes have different effector functions, average serum concentrations, and half-lives.
- Each of the domains in the immunoglobulin molecule has a characteristic tertiary structure called the immunoglobulin fold. The presence of an immunoglobulin fold domain also identifies many other nonantibody proteins as members of the immunoglobulin superfamily.
- Within the amino-terminal variable domain of each heavy and light chain are three complementarity-determining regions (CDRs). These polypeptide regions contribute the antigen-binding site of an antibody, determining its specificity.
- Immunoglobulins are expressed in two forms: secreted antibody that is produced by plasma cells, and membrane-bound antibody that associates with Ig- α /Ig- β heterodimers to form the B-cell antigen receptor present on the surface of B cells.
- The three major effector functions that enable antibodies to remove antigens and kill pathogens are: opsonization, which promotes antigen phagocytosis by macrophages and neutrophils; complement activation, which activates a pathway that leads to the generation of a collection of proteins that can perforate cell membranes; and antibody-dependent cell-mediated cytotoxicity (ADCC), which can kill antibody-bound target cells.
- Unlike polyclonal antibodies that arise from many B cell clones and have a heterogeneous collection of binding sites, a monoclonal antibody is derived from a single B cell clone and is a homogeneous collection of binding sites.

References

- Frazer, J. K., and J. D. Capra. 1999. Immunoglobulins: structure and function. In *Fundamental Immunology*, 4th ed. W. E. Paul, ed. Philadelphia, Lippincott-Raven.
- Kohler, G., and C. Milstein. 1975. Continuous cultures of fused cells secreting antibody of predefined specificity. *Nature* **256**:495.
- Kraehenbuhl, J. P., and M. R. Neutra. 1992. Transepithelial transport and mucosal defence II: secretion of IgA. *Trends Cell Biol.* **2**:134.
- Immunology Today, The Immune Receptor Supplement*, 2nd ed. 1997. Elsevier Trends Journals, Cambridge, UK (ISSN 1365-1218).
- Newman, J. 1995. How breast milk protects newborns. *Sci. Am.* **273**(6):76.
- Reth, M. 1995. The B-cell antigen receptor complex and coreceptor. *Immunol. Today* **16**:310.
- Stanfield, R. L., and I. A. Wilson. 1995. Protein-peptide interactions. *Curr. Opin. Struc. Biol.* **5**:103.
- Wedemayer, G. J., P. A. Patten, L. H. Wang, P. G. Schultz, and R. C. Stevens. 1997. Structural insights into the evolution of an antibody combining site. *Science*, **276**:1665.
- Wentworth, P., and Janda, K. 1998. Catalytic Antibodies. *Curr. Opin. Chem. Biol.* **8**:138.
- Wilson, I. A., and R. L. Stanfield. 1994. Antibody-antibody interactions: new structures and new conformational changes. *Curr. Opin. Struc. Biol.* **4**:857.



USEFUL WEB SITES

<http://immuno.bme.nwu.edu/>

The Kabat Database of Sequences of Proteins of Immunological Interest: This site has the amino acid and DNA sequences of many antibodies and other proteins that play important roles in immunology.

<http://www.biochem.ucl.ac.uk/~martin/abs>

Antibodies—Structure and Sequence: This Web site summarizes useful information on antibody structure and sequence. It provides general information on antibodies and crystal structures and links to other antibody-related information.

<http://www.ncbi.nlm.nih.gov>

National Center for Biotechnology Information (NCBI): A unique and comprehensive resource of computerized databases of bibliographic information, nucleic acid sequences, protein sequences, and sequence analysis tools created and maintained by the National Library of Medicine.

<http://www.ncbi.nlm.nih.gov/Structure/>

The Molecular Modeling Database (MMDB) contains 3-dimensional structures determined by x-ray crystallography and NMR spectroscopy. The data for MMDB are obtained from the Protein Data Bank (PDB). The National Center for Biotechnology Information (NCBI) has structural data crosslinked to bibliographic information, to databases of protein and nucleic acid sequences, and to the NCBI animal taxonomy database. The NCBI has developed a 3D structure viewer, Cn3D, for easy interactive visualization of molecular structures.

<http://www.umass.edu/microbio/chime/explorer/>

Protein Explorer is a molecular visualization program created by Eric Martz with the support of the National Science Foundation to make it easier for students, educators, and scientists to use interactive and dynamic molecular visualization techniques. Many will find it easier to use than Chime and Rasmol.

<http://imgt.cines.fr>

IMGT, the international ImMunoGeneTics database created by Marie-Paule Lefranc, is a well organized, powerful, and comprehensive information system that specializes in immunoglobulins, T-cell receptors and major histocompatibility complex (MHC) molecules of all vertebrate species.

Study Questions

CLINICAL FOCUS QUESTION Two pharmaceutical companies make IVIG. Company A produces their product from pools of 100,000 donors drawn exclusively from the population of the United States. Company B makes their IVIG from pools of 60,000 donors drawn in equal numbers from North America, Europe, Brazil, and Japan.

- Which product would you expect to have the broadest spectrum of pathogen reactivities? Why?
 - Assume the patients receiving the antibody will (1) never leave the USA, or (2) travel extensively in many parts of the world. Which company's product would you choose for each of these patient groups? Justify your choices.
- Indicate whether each of the following statements is true or false. If you think a statement is false, explain why.
 - A rabbit immunized with human IgG3 will produce antibody that reacts with all subclasses of IgG in humans.

- All immunoglobulin molecules on the surface of a given B cell have the same idiotype.
- All immunoglobulin molecules on the surface of a given B cell have the same isotype.
- All myeloma protein molecules derived from a single myeloma clone have the same idiotype and allotype.
- Although IgA is the major antibody species that undergoes transcytosis, polymeric IgM, but not monomeric IgA, can also undergo transcytosis.
- The hypervariable regions make significant contact with the epitope.
- IgG functions more effectively than IgM in bacterial agglutination.
- Although monoclonal antibodies are often preferred for research and diagnostic purposes, both monoclonal and polyclonal antibodies can be highly specific.
- All isotypes are normally found in each individual of a species.
- The heavy-chain variable region (V_H) is twice as long as the light-chain variable region (V_L).

- You are an energetic immunology student who has isolated protein X, which you believe is a new isotype of human immunoglobulin.
 - What structural features would protein X have to have in order to be classified as an immunoglobulin?
 - You prepare rabbit antisera to whole human IgG, human κ chain, and human γ chain. Assuming protein X is, in fact, a new immunoglobulin isotype, to which of these antisera would it bind? Why?
 - Devise an experimental procedure for preparing an antiserum that is specific for protein X.
- According to the clonal selection theory, all the immunoglobulin molecules on a single B cell have the same antigenic specificity. Explain why the presence of both IgM and IgD on the same B cell does not violate the unispecificity implied by clonal selection.
- IgG, which contains γ heavy chains, developed much more recently during evolution than IgM, which contains μ heavy chains. Describe two advantages and two disadvantages that IgG has in comparison with IgM.
- Although the five immunoglobulin isotypes share many common structural features, the differences in their structures affect their biological activities.
 - Draw a schematic diagram of a typical IgG molecule and label each of the following parts: H chains, L chains, interchain disulfide bonds, intrachain disulfide bonds, hinge, Fab, Fc, and all the domains. Indicate which domains are involved in antigen binding.
 - How would you have to modify the diagram of IgG to depict an IgA molecule isolated from saliva?
 - How would you have to modify the diagram of IgG to depict serum IgM?
- Fill out the accompanying table relating to the properties of IgG molecules and their various parts. Insert a (+) if the molecule or part exhibits the property; a (−) if it does not; and a (+/−) if it does so only weakly.



Property	Whole IgG	H chain	L chain	Fab	F(ab') ₂	Fc
Binds antigen						
Bivalent antigen binding						
Binds to Fc receptors						
Fixed complement in presence of antigen						
Has V domains						
Has C domains						

7. Because immunoglobulin molecules possess antigenic determinants, they themselves can function as immunogens, inducing formation of antibody. For each of the following immunization scenarios, indicate whether anti-immunoglobulin antibodies would be formed to isotypic (IS), allotypic (AL), or idiotypic (ID) determinants:

- Anti-DNP antibodies produced in a BALB/c mouse are injected into a C57BL/6 mouse.
- Anti-BGG monoclonal antibodies from a BALB/c mouse are injected into another BALB/c mouse.
- Anti-BGG antibodies produced in a BALB/c mouse are injected into a rabbit.
- Anti-DNP antibodies produced in a BALB/c mouse are injected into an outbred mouse.
- Anti-BGG antibodies produced in a BALB/c mouse are injected into the same mouse.

8. Write YES or NO in the accompanying table to indicate whether the rabbit antisera listed at the top react with the mouse antibody components listed at the left.

	γ chain	κ chain	IgG Fab fragment	IgG Fc fragment	J chain
Mouse γ chain					
Mouse κ chain					
Mouse IgM whole					
Mouse IgM Fc fragment					

9. The characteristic structure of immunoglobulin domains, termed the immunoglobulin fold, also occurs in the numerous membrane proteins belonging to the immunoglobulin superfamily.

- Describe the typical features that define the immunoglobulin-fold domain structure.
- Consider proteins that belong to the immunoglobulin superfamily. What do all of these proteins have in common? Describe two different Ig superfamily members that bind antigen. Identify four different Ig superfamily members that do not bind antigen.

10. Where are the CDR regions located on an antibody molecule and what are their functions?

11. The variation in amino acid sequence at each position in a polypeptide chain can be expressed by a quantity termed the variability. What are the largest and smallest values of variability possible?

12. You prepare an immunotoxin by conjugating diphtheria toxin with a monoclonal antibody specific for a tumor antigen.

- If this immunotoxin is injected into an animal, will any normal cells be killed? Explain.
- If the antibody part of the immunotoxin is degraded so that the toxin is released, will normal cells be killed? Explain.

13. An investigator wanted to make a rabbit antiserum specific for mouse IgG. She injected a rabbit with purified mouse IgG and obtained an antiserum that reacted strongly with mouse IgG. To her dismay, however, the antiserum also reacted with each of the other mouse isotypes. Explain why she got this result. How could she make the rabbit antiserum specific for mouse IgG?

14. You fuse spleen cells having a normal genotype for immunoglobulin heavy chains (H) and light chains (L) with three myeloma-cell preparations differing in their immunoglobulin genotype as follows: (a) H^+, L^+ ; (b) H^-, L^+ ; and (c) H^-, L^- . For each hybridoma, predict how many unique antigen-binding sites, composed of one H and one L chain, theoretically could be produced and show the chain structure of the possible antibody molecules. For each possible antibody molecule indicate whether the chains would originate from the spleen (S) or from the myeloma (M) fusion partner (e.g., $H_S L_S / H_M L_M$).

15. For each immunoglobulin isotype (a–e) select the description(s) listed below (1–12) that describe that isotype. Each description may be used once, more than once, or not at all; more than one description may apply to some isotypes.

Isotypes

- _____ IgA
- _____ IgD
- _____ IgE
- _____ IgG
- _____ IgM

Descriptions

- Secreted form is a pentamer of the basic H_2L_2 unit
- Binds to Fc receptors on mast cells

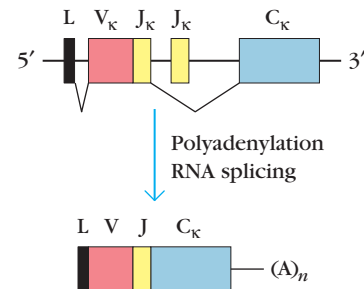
- (3) Multimeric forms have a J chain
 - (4) Present on the surface of mature, unprimed B cells
 - (5) The most abundant isotype in serum
 - (6) Major antibody in secretions such as saliva, tears, and breast milk
 - (7) Present on the surface of immature B cells
 - (8) The first serum antibody made in a primary immune response
 - (9) Plays an important role in immediate hypersensitivity
 - (10) Plays primary role in protecting against pathogens that invade through the gut or respiratory mucosa
 - (11) Multimeric forms may contain a secretory component
 - (12) Least abundant isotype in serum
16. Describe four distinct roles played by Fc receptors. In what ways is signal transduction from Fc receptors similar to signal transduction from the B-cell receptor?
17. What is IVIG and what are some of the mechanisms by which it might protect the body against infection? Suppose one had the option of collecting blood for the manufacture of IVIG from the following groups of healthy individuals: 35-year-old men who had lived all of their lives in isolated villages in the mountains of Switzerland, or 45–55-year-old men who had been international airline pilots for 20 years. Which group would provide the better pool of blood? Justify your answer.

chapter 5

Organization and Expression of Immunoglobulin Genes

ONE OF THE MOST REMARKABLE FEATURES OF the vertebrate immune system is its ability to respond to an apparently limitless array of foreign antigens. As immunoglobulin (Ig) sequence data accumulated, virtually every antibody molecule studied was found to contain a unique amino acid sequence in its variable region but only one of a limited number of invariant sequences in its constant region. The genetic basis for this combination of constancy and tremendous variation in a single protein molecule lies in the organization of the immunoglobulin genes.

In germ-line DNA, multiple gene segments encode portions of a single immunoglobulin heavy or light chain. These gene segments are carried in the germ cells but cannot be transcribed and translated into complete chains until they are rearranged into functional genes. During B-cell maturation in the bone marrow, certain of these gene segments are randomly shuffled by a dynamic genetic system capable of generating more than 10^6 combinations. Subsequent processes increase the diversity of the repertoire of antibody binding sites to a very large number that exceeds 10^6 by at least two or three orders of magnitude. The processes of B-cell development are carefully regulated: the maturation of a progenitor B cell progresses through an ordered sequence of Ig-gene rearrangements, coupled with modifications to the gene that contribute to the diversity of the final product. By the end of this process, a mature, immunocompetent B cell will contain coding sequences for one functional heavy-chain variable-region and one light-chain variable-region. The individual B cell is thus antigenically committed to a specific epitope. After antigenic stimulation of a mature B cell in peripheral lymphoid organs, further rearrangement of constant-region gene segments can generate changes in the isotype expressed, which produce changes in the biological effector functions of the immunoglobulin molecule without changing its specificity. Thus, mature B cells contain chromosomal DNA that is no longer identical to germ-line



Kappa Light-Chain Gene Rearrangement

- Genetic Model Compatible with Ig Structure
- Multigene Organization of Ig Genes
- Variable-Region Gene Rearrangements
- Mechanism of Variable-Region DNA Rearrangements
- Generation of Antibody Diversity
- Class Switching among Constant-Region Genes
- Expression of Ig Genes
- Synthesis, Assembly, and Secretion of Immunoglobulins
- Regulation of Ig-Gene Transcription
- Antibody Genes and Antibody Engineering

DNA. While we think of genomic DNA as a stable genetic blueprint, the lymphocyte cell lineage does not retain an intact copy of this blueprint. Genomic rearrangement is an essential feature of lymphocyte differentiation, and no other vertebrate cell type has been shown to undergo this process.

This chapter first describes the detailed organization of the immunoglobulin genes, the process of Ig-gene rearrangement, and the mechanisms by which the dynamic immunoglobulin genetic system generates more than 10^8 different antigenic specificities. Then it describes the mechanism of class switching, the role of differential RNA processing in the expression of immunoglobulin genes, and the regulation of Ig-gene transcription. The chapter concludes with the application of our knowledge of the molecular



VISUALIZING CONCEPTS

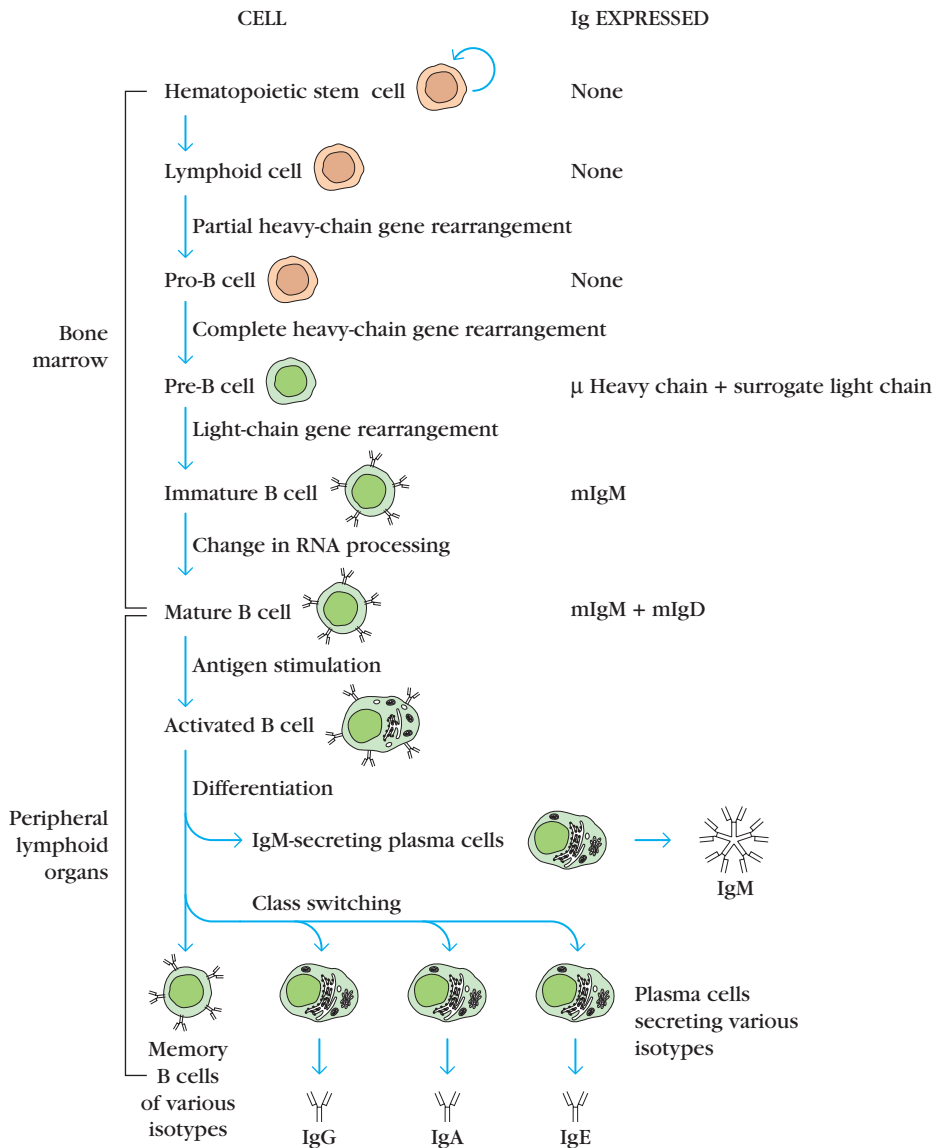


FIGURE 5-1 Overview of B-cell development. The events that occur during maturation in the bone marrow do not require antigen, whereas activation and differentiation of mature B cells in pe-

ripheral lymphoid organs require antigen. The labels mIgM and mIgD refer to membrane-associated Igs. IgG, IgA, and IgE are secreted immunoglobulins.

biology of immunoglobulin genes to the engineering of antibody molecules for therapeutic and research applications. Chapter 11 covers in detail the entire process of B-cell development from the first gene rearrangements in progenitor B cells to final differentiation into memory B cells and antibody-secreting plasma cells. Figure 5-1 outlines the sequential stages in B-cell development, many of which result from critical rearrangements.

Genetic Model Compatible with Ig Structure

The results of the immunoglobulin-sequencing studies described in Chapter 4 revealed a number of features of immunoglobulin structure that were difficult to reconcile with classic genetic models. Any viable model of the

immunoglobulin genes had to account for the following properties of antibodies:

- The vast diversity of antibody specificities
- The presence in Ig heavy and light chains of a variable region at the amino-terminal end and a constant region at the carboxyl-terminal end
- The existence of isotypes with the same antigenic specificity, which result from the association of a given variable region with different heavy-chain constant regions

Germ-Line and Somatic-Variation Models Contended To Explain Antibody Diversity

For several decades, immunologists sought to imagine a genetic mechanism that could explain the tremendous diversity of antibody structure. Two different sets of theories emerged. The **germ-line theories** maintained that the genome contributed by the germ cells, egg and sperm, contains a large repertoire of immunoglobulin genes; thus, these theories invoked no special genetic mechanisms to account for antibody diversity. They argued that the immense survival value of the immune system justified the dedication of a significant fraction of the genome to the coding of antibodies. In contrast, the **somatic-variation theories** maintained that the genome contains a relatively small number of immunoglobulin genes, from which a large number of antibody specificities are generated in the somatic cells by mutation or recombination.

As the amino acid sequences of more and more immunoglobulins were determined, it became clear that there must be mechanisms not only for generating antibody diversity but also for maintaining constancy. Whether diversity was generated by germ-line or by somatic mechanisms, a paradox remained: How could stability be maintained in the constant (C) region while some kind of diversifying mechanism generated the variable (V) region?

Neither the germ-line nor the somatic-variation proponents could offer a reasonable explanation for this central feature of immunoglobulin structure. Germ-line proponents found it difficult to account for an evolutionary mechanism that could generate diversity in the variable part of the many heavy- and light-chain genes while preserving the constant region of each unchanged. Somatic-variation proponents found it difficult to conceive of a mechanism that could diversify the variable region of a single heavy- or light-chain gene in the somatic cells without allowing alteration in the amino acid sequence encoded by the constant region.

A third structural feature requiring an explanation emerged when amino acid sequencing of the human myeloma protein called Ti1 revealed that identical variable-region sequences were associated with both γ and μ heavy-chain constant regions. A similar phenomenon was observed

in rabbits by C. Todd, who found that a particular allotypic marker in the heavy-chain variable region could be associated with α , γ , and μ heavy-chain constant regions. Considerable additional evidence has confirmed that a single variable-region sequence, defining a particular antigenic specificity, can be associated with multiple heavy-chain constant-region sequences; in other words, different classes, or isotypes, of antibody (e.g., IgG, IgM) can be expressed with identical variable-region sequences.

Dreyer and Bennett Proposed the Two-Gene Model

In an attempt to develop a genetic model consistent with the known findings about the structure of immunoglobulins, W. Dreyer and J. Bennett suggested, in their classic theoretical paper of 1965, that two separate genes encode a single immunoglobulin heavy or light chain, one gene for the V region (variable region) and the other for the C region (constant region). They suggested that these two genes must somehow come together at the DNA level to form a continuous message that can be transcribed and translated into a single Ig heavy or light chain. Moreover, they proposed that hundreds or thousands of V-region genes were carried in the germ line, whereas only single copies of C-region class and subclass genes need exist.

The strength of this type of recombinational model (which combined elements of the germ-line and somatic-variation theories) was that it could account for those immunoglobulins in which a single V region was combined with various C regions. By postulating a single constant-region gene for each immunoglobulin class and subclass, the model also could account for the conservation of necessary biological effector functions while allowing for evolutionary diversification of variable-region genes.

At first, support for the Dreyer and Bennett hypothesis was indirect. Early studies of DNA hybridization kinetics using a radioactive constant-region DNA probe indicated that the probe hybridized with only one or two genes, confirming the model's prediction that only one or two copies of each constant-region class and subclass gene existed. However, indirect evidence was not enough to overcome stubborn resistance in the scientific community to the hypothesis of Dreyer and Bennett. The suggestion that two genes encoded a single polypeptide contradicted the existing one gene—one polypeptide principle and was without precedent in any known biological system.

As so often is the case in science, theoretical and intellectual understanding of Ig-gene organization progressed ahead of the available methodology. Although the Dreyer and Bennett model provided a theoretical framework for reconciling the dilemma between Ig-sequence data and gene organization, actual validation of their hypothesis had to wait for several major technological advances in the field of molecular biology.

Tonegawa's Bombshell—Immunoglobulin Genes Rearrange

In 1976, S. Tonegawa and N. Hozumi found the first direct evidence that separate genes encode the V and C regions of immunoglobulins and that the genes are rearranged in the course of B-cell differentiation. This work changed the field of immunology. In 1987, Tonegawa was awarded the Nobel Prize for this work.

Selecting DNA from embryonic cells and adult myeloma cells—cells at widely different stages of development—Tonegawa and Hozumi used various restriction endonucleases to generate DNA fragments. The fragments were then separated by size and analyzed for their ability to hybridize with a radiolabeled mRNA probe. Two separate restriction fragments from the embryonic DNA hybridized with the mRNA, whereas only a single restriction fragment of the adult myeloma DNA hybridized with the same probe. Tonegawa and Hozumi suggested that, during differentiation of lymphocytes from the embryonic state to the fully differentiated plasma-cell stage (represented in their system by the

myeloma cells), the V and C genes undergo rearrangement. In the embryo, the V and C genes are separated by a large DNA segment that contains a restriction-endonuclease site; during differentiation, the V and C genes are brought closer together and the intervening DNA sequence is eliminated.

The pioneering experiments of Tonegawa and Hozumi employed a tedious and time-consuming procedure that has since been replaced by the much more powerful approach of Southern-blot analysis. This method, now universally used to investigate the rearrangement of immunoglobulin genes, eliminates the need to elute the separated DNA restriction fragments from gel slices prior to analysis by hybridization with an immunoglobulin gene segment probe. Figure 5-2 shows the detection of rearrangement at the κ light-chain locus by comparing the fragments produced by digestion of DNA from a clone of B-lineage cells with the pattern obtained by digestion of non-B cells (e.g., sperm or liver cells). The rearrangement of a V gene deletes an extensive section of germ-line DNA, thereby creating differences between rearranged and unrearranged Ig loci in the distribution and number of restriction sites. This results in the generation of

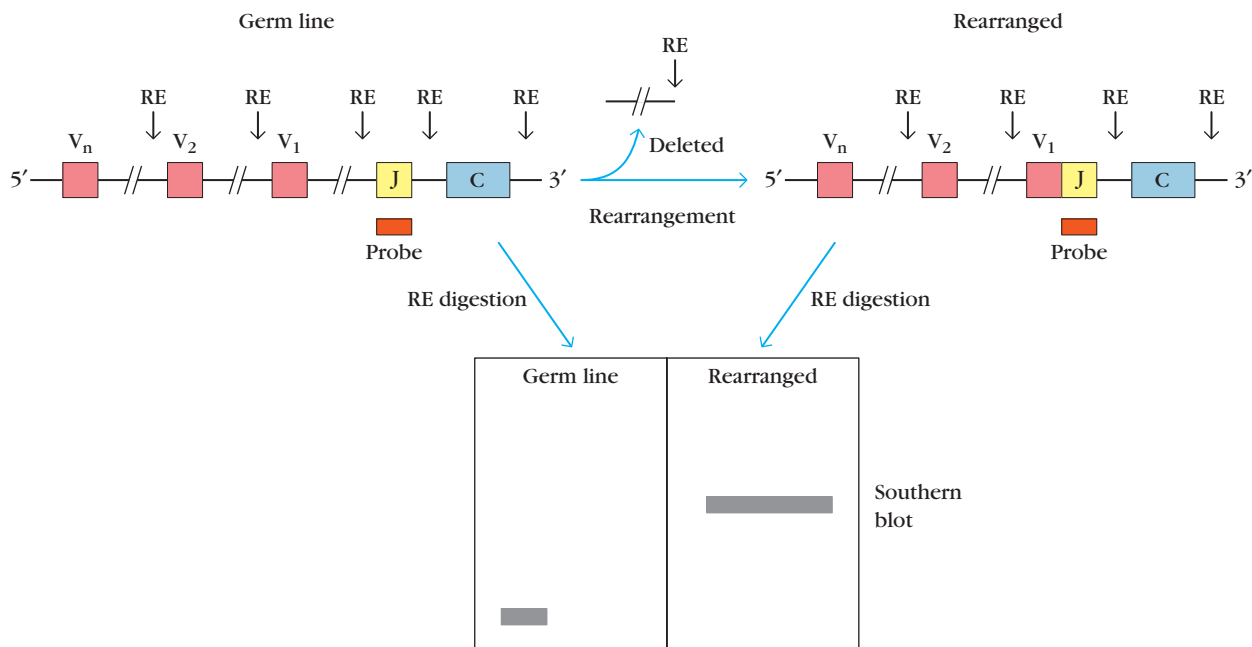


FIGURE 5-2 Experimental basis for diagnosis of rearrangement at an immunoglobulin locus. The number and size of restriction fragments generated by the treatment of DNA with a restriction enzyme is determined by the sequence of the DNA. The digestion of rearranged DNA with a restriction enzyme (RE) yields a pattern of restriction fragments that differ from those obtained by digestion of an unrearranged locus with the same RE. Typically, the fragments are analyzed by the technique of Southern blotting. In this example, a probe that includes a J gene segment is used to identify RE digestion fragments that include all or portions of this segment. As shown, rearrangement results in the deletion of a segment of germ-line DNA and the loss of the restriction sites that it includes. It also results in the joining of gene segments, in this case a V and a J segment, that

are separated in the germ line. Consequently, fragments dependent on the presence of this segment for their generation are absent from the restriction-enzyme digest of DNA from the rearranged locus. Furthermore, rearranged DNA gives rise to novel fragments that are absent from digests of DNA in the germ-line configuration. This can be useful because both B cells and non-B cells have two immunoglobulin loci. One of these is rearranged and the other is not. Consequently, unless a genetic accident has resulted in the loss of the germ-line locus, digestion of DNA from a myeloma or normal B-cell clone will produce a pattern of restriction that includes all of those in a germ-line digest plus any novel fragments that are generated from the change in DNA sequence that accompanies rearrangement. Note that only one of the several J gene segments present is shown.

different restriction patterns by rearranged and unrearranged loci. Extensive application of this approach has demonstrated that the Dreyer and Bennett two-gene model—one gene encoding the variable region and another encoding the constant region—applied to both heavy and light-chain genes.

Multigene Organization of Ig Genes

As cloning and sequencing of the light- and heavy-chain DNA was accomplished, even greater complexity was revealed than had been predicted by Dreyer and Bennett. The κ and λ light chains and the heavy chains are encoded by separate multigene families situated on different chromosomes (Table 5-1). In germ-line DNA, each of these multigene families contains several coding sequences, called **gene segments**, separated by noncoding regions. During B-cell maturation, these gene segments are rearranged and brought together to form functional immunoglobulin genes.

Each Multigene Family Has Distinct Features

The κ and λ light-chain families contain **V, J, and C gene segments**; the rearranged VJ segments encode the variable region of the light chains. The heavy-chain family contains **V, D, J, and C gene segments**; the rearranged VDJ gene segments encode the variable region of the heavy chain. In each gene family, C gene segments encode the constant regions. Each V gene segment is preceded at its 5' end by a small exon that encodes a short **signal** or **leader (L) peptide** that guides the heavy or light chain through the endoplasmic reticulum. The signal peptide is cleaved from the nascent light and heavy chains before assembly of the finished immunoglobulin molecule. Thus, amino acids encoded by this leader sequence do not appear in the immunoglobulin molecule.

λ -CHAIN MULTIGENE FAMILY

The first evidence that the light-chain variable region was actually encoded by two gene segments appeared when Tonegawa cloned the germ-line DNA that encodes the variable region of mouse λ light chain and determined its complete

nucleotide sequence. When the nucleotide sequence was compared with the known amino acid sequence of the λ -chain variable region, an unusual discrepancy was observed. Although the first 97 amino acids of the λ -chain variable region corresponded to the nucleotide codon sequence, the remaining 13 carboxyl-terminal amino acids of the protein's variable region did not. It turned out that many base pairs away a separate, 39-bp gene segment, called J for *joining*, encoded the remaining 13 amino acids of the λ -chain variable region. Thus, a functional λ variable-region gene contains two coding segments—a 5' V segment and a 3' J segment—which are separated by a noncoding DNA sequence in unrearranged germ-line DNA.

The λ multigene family in the mouse germ line contains three V_λ gene segments, four J_λ gene segments, and four C_λ gene segments (Figure 5-3a). The $J_{\lambda 4}$ is a **pseudogene**, a defective gene that is incapable of encoding protein; such genes are indicated with the psi symbol (ψ). Interestingly, $J_{\lambda 4}$'s constant region partner, $C_{\lambda 4}$, is a perfectly functional gene. The V_λ and the three functional J_λ gene segments encode the variable region of the light chain, and each of the three functional C_λ gene segments encodes the constant region of one of the three λ -chain subtypes ($\lambda 1$, $\lambda 2$, and $\lambda 3$). In humans, the lambda locus is more complex. There are 31 functional V_λ gene segments, 4 J_λ segments, and 7 C_λ segments. In addition to the functional gene segments, the human lambda complex contains many V_λ , J_λ , and C_λ pseudogenes.

κ -CHAIN MULTIGENE FAMILY

The κ -chain multigene family in the mouse contains approximately 85 V_κ gene segments, each with an adjacent leader sequence a short distance upstream (i.e., on the 5' side). There are five J_κ gene segments (one of which is a nonfunctional pseudogene) and a single C_κ gene segment (Figure 5-3b). As in the λ multigene family, the V_κ and J_κ gene segments encode the variable region of the κ light chain, and the C_κ gene segment encodes the constant region. Since there is only one C_κ gene segment, there are no subtypes of κ light chains. Comparison of parts *a* and *b* of Figure 5-3 shows that the arrangement of the gene segments is quite different in the κ and λ gene families. The κ -chain multigene family in humans, which has an organization similar to that of the mouse, contains approximately 40 V_κ gene segments, 5 J_κ segments, and a single C_κ segment.

HEAVY-CHAIN MULTIGENE FAMILY

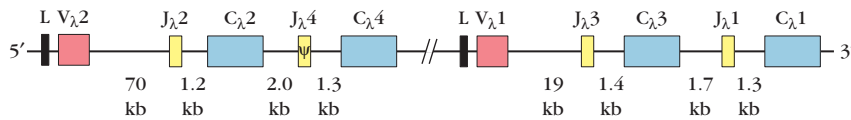
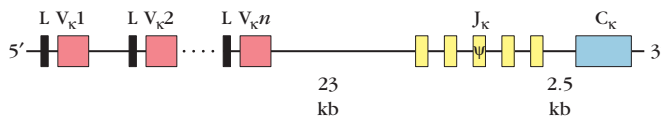
The organization of the immunoglobulin heavy-chain genes is similar to, but more complex than, that of the κ and λ light-chain genes (Figure 5-3c). An additional gene segment encodes part of the heavy-chain variable region. The existence of this gene segment was first proposed by Leroy Hood and his colleagues, who compared the heavy-chain variable-region amino acid sequence with the V_H and J_H nucleotide sequences. The V_H gene segment was found to encode amino acids 1 to 94 and the J_H gene segment

TABLE 5-1 Chromosomal locations of immunoglobulin genes in human and mouse

Gene	CHROMOSOME	
	Human	Mouse
λ Light chain	22	16
κ Light chain	2	6
Heavy chain	14	12



VISUALIZING CONCEPTS

(a) λ -chain DNA(b) κ -chain DNA $n \sim 85$ 

(c) Heavy-chain DNA

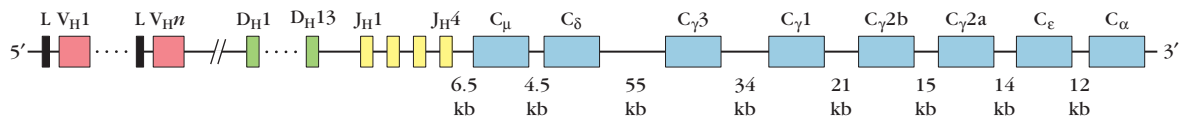
 $n \sim 134$ 

FIGURE 5-3 Organization of immunoglobulin germ-line gene segments in the mouse: (a) λ light chain, (b) κ light chain, and (c) heavy chain. The λ and κ light chains are encoded by V, J, and C gene segments. The heavy chain is encoded by V, D, J, and C gene

segments. The distances in kilobases (kb) separating the various gene segments in mouse germ-line DNA are shown below each chain diagram.

was found to encode amino acids 98 to 113; however, neither of these gene segments carried the information to encode amino acids 95 to 97. When the nucleotide sequence was determined for a rearranged myeloma DNA and compared with the germ-line DNA sequence, an additional nucleotide sequence was observed between the V_H and J_H gene segments. This nucleotide sequence corresponded to amino acids 95 to 97 of the heavy chain.

From these results, Hood and his colleagues proposed that a third germ-line gene segment must join with the V_H and J_H gene segments to encode the entire variable region of the heavy chain. This gene segment, which encoded amino acids within the third complementarity-determining region (CDR3), was designated D for *diversity*, because of its contribution to the generation of antibody diversity. Tonegawa and his colleagues located the D gene segments within mouse germ-line DNA with a cDNA probe complementary to the D region, which hybridized with a stretch of DNA lying between the V_H and J_H gene segments.

The heavy-chain multigene family on human chromosome 14 has been shown by direct sequencing of DNA to contain 51 V_H gene segments located upstream from a cluster of 27 functional D_H gene segments. As with the light-chain genes, each V_H gene segment is preceded by a leader

sequence a short distance upstream. Downstream from the D_H gene segments are six functional J_H gene segments, followed by a series of C_H gene segments. Each C_H gene segment encodes the constant region of an immunoglobulin heavy-chain isotype. The C_H gene segments consist of coding exons and noncoding introns. Each exon encodes a separate domain of the heavy-chain constant region. A similar heavy-chain gene organization is found in the mouse.

The conservation of important biological effector functions of the antibody molecule is maintained by the limited number of heavy-chain constant-region genes. In humans and mice, the C_H gene segments are arranged sequentially in the order C_{μ} , C_{δ} , C_{γ} , C_{ϵ} , C_{α} (see Figure 5-3c). This sequential arrangement is no accident; it is generally related to the sequential expression of the immunoglobulin classes in the course of B-cell development and the initial IgM response of a B cell to its first encounter with an antigen.

Variable-Region Gene Rearrangements

The preceding sections have shown that functional genes that encode immunoglobulin light and heavy chains are

assembled by recombinational events at the DNA level. These events and the parallel events involving T-receptor genes are the only known site-specific DNA rearrangements in vertebrates. Variable-region gene rearrangements occur in an ordered sequence during B-cell maturation in the bone marrow. The heavy-chain variable-region genes rearrange first, then the light-chain variable-region genes. At the end of this process, each B cell contains a single functional variable-region DNA sequence for its heavy chain and another for its light chain.

The process of variable-region gene rearrangement produces mature, immunocompetent B cells; each such cell is committed to produce antibody with a binding site encoded by the particular sequence of its rearranged V genes. As described later in this chapter, rearrangements of the heavy-chain constant-region genes will generate further changes in the immunoglobulin class (isotype) expressed by a B cell, but those changes will not affect the cell's antigenic specificity.

The steps in variable-region gene rearrangement occur in an ordered sequence, but they are random events that result in the random determination of B-cell specificity. The order, mechanism, and consequences of these rearrangements are described in this section.

Light-Chain DNA Undergoes V-J Rearrangements

Expression of both κ and λ light chains requires rearrangement of the variable-region V and J gene segments. In humans, any of the functional V_λ genes can combine with any of the four functional J_λ - C_λ combinations. In the mouse, things are slightly more complicated. DNA rearrangement can join the $V_\lambda 1$ gene segment with either the $J_\lambda 1$ or the $J_\lambda 3$ gene segment, or the $V_\lambda 2$ gene segment can be joined with the $J_\lambda 2$ gene segment. In human or mouse κ light-chain DNA, any one of the V_κ gene segments can be joined with any one of the functional J_κ gene segments.

Rearranged κ and λ genes contain the following regions in order from the 5' to 3' end: a short leader (L) exon, a non-coding sequence (intron), a joined VJ gene segment, a second intron, and the constant region. Upstream from each leader gene segment is a promoter sequence. The rearranged light-chain sequence is transcribed by RNA polymerase from the L exon through the C segment to the stop signal, generating a light-chain primary RNA transcript (Figure 5-4). The introns in the primary transcript are removed by RNA-processing enzymes, and the resulting light-chain messenger

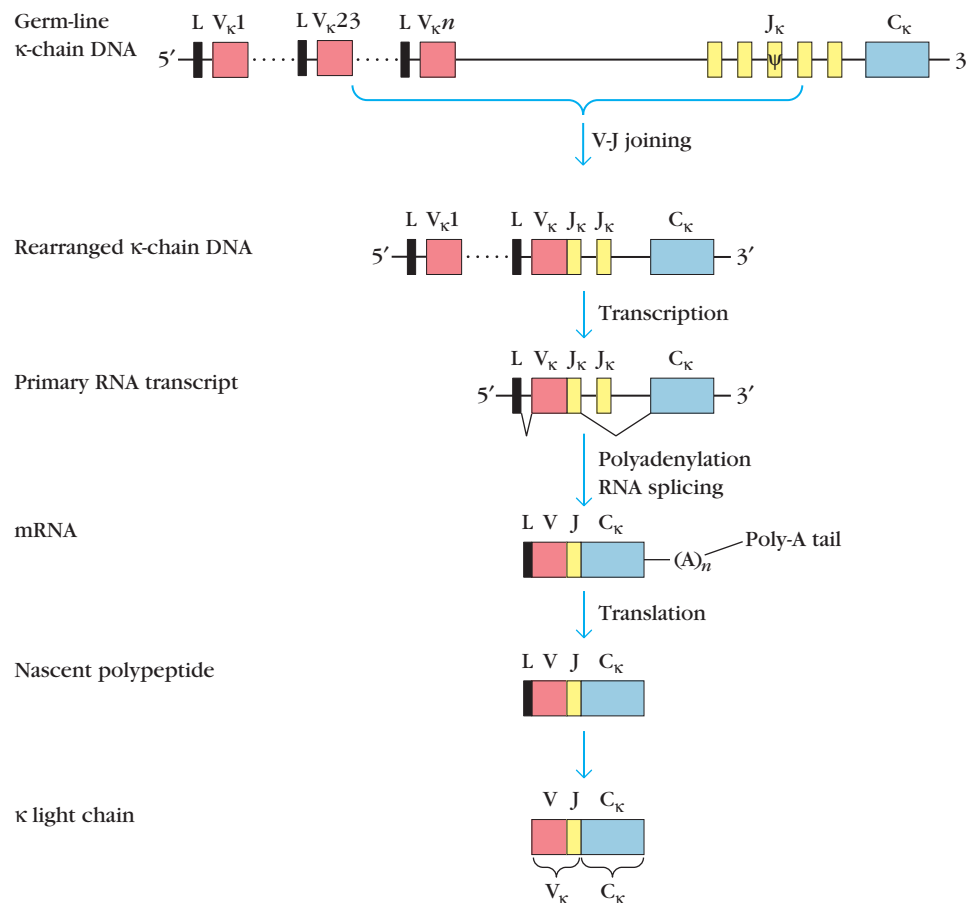


FIGURE 5-4 Kappa light-chain gene rearrangement and RNA processing events required to generate a κ light-chain protein. In this example, rearrangement joins $V_\kappa 23$ and $J_\kappa 4$.

RNA then exits from the nucleus. The light-chain mRNA binds to ribosomes and is translated into the light-chain protein. The leader sequence at the amino terminus pulls the growing polypeptide chain into the lumen of the rough endoplasmic reticulum and is then cleaved, so it is not present in the finished light-chain protein product.

Heavy-Chain DNA Undergoes V-D-J Rearrangements

Generation of a functional immunoglobulin heavy-chain gene requires two separate rearrangement events within the variable region. As illustrated in Figure 5-5, a D_H gene segment first joins to a J_H segment; the resulting D_HJ_H segment then moves next to and joins a V_H segment to generate a $V_HD_HJ_H$ unit that encodes the entire variable region. In heavy-chain DNA, variable-region rearrangement produces a rearranged gene consisting of the following sequences,

starting from the 5' end: a short L exon, an intron, a joined VDJ segment, another intron, and a series of C gene segments. As with the light-chain genes, a promoter sequence is located a short distance upstream from each heavy-chain leader sequence.

Once heavy-chain gene rearrangement is accomplished, RNA polymerase can bind to the promoter sequence and transcribe the entire heavy-chain gene, including the introns. Initially, both C_μ and C_δ gene segments are transcribed. Differential polyadenylation and RNA splicing remove the introns and process the primary transcript to generate mRNA including either the C_μ or the C_δ transcript. These two mRNAs are then translated, and the leader peptide of the resulting nascent polypeptide is cleaved, generating finished μ and δ chains. The production of two different heavy-chain mRNAs allows a mature, immunocompetent B cell to express both IgM and IgD with identical antigenic specificity on its surface.

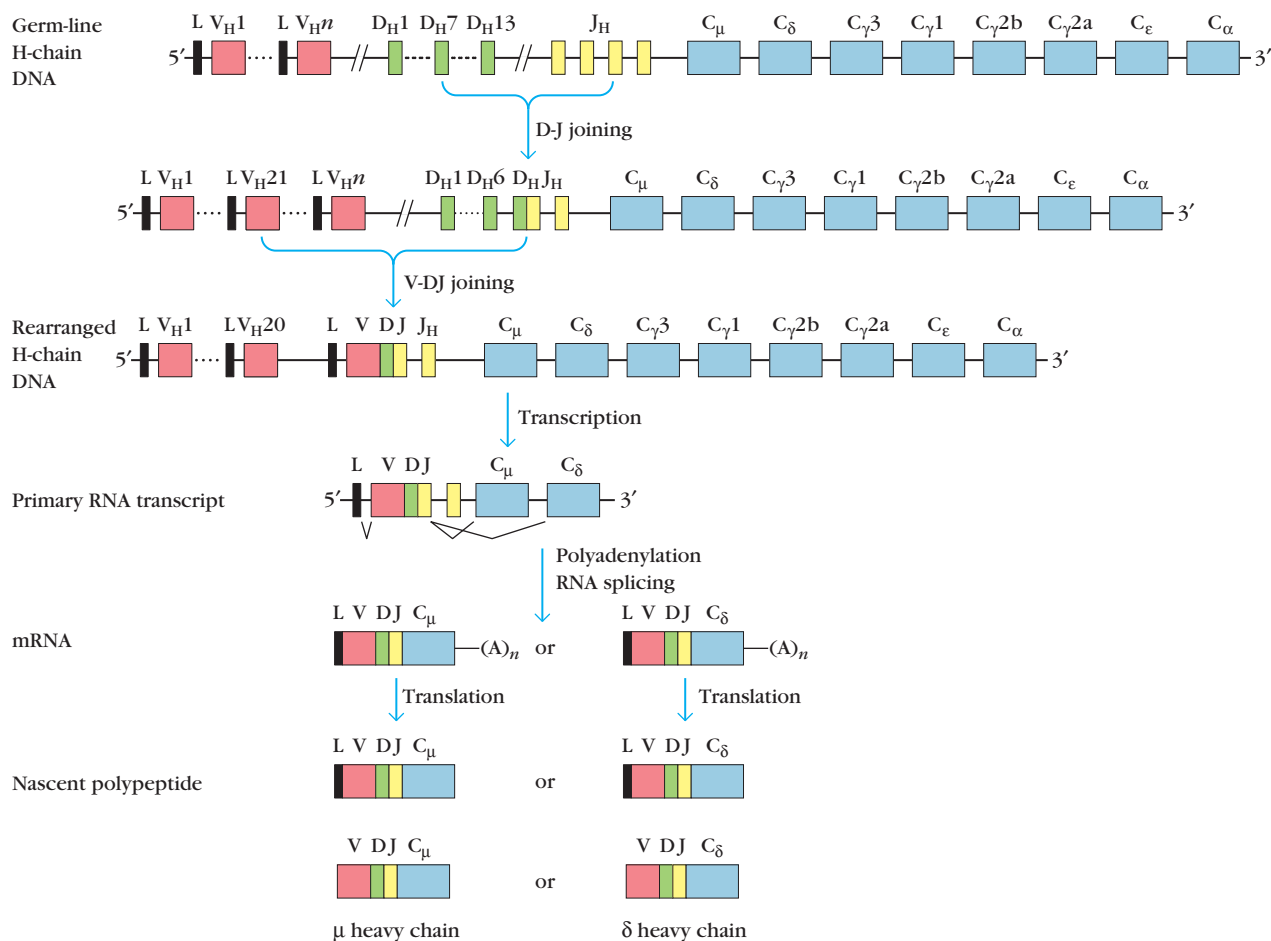


FIGURE 5-5 Heavy-chain gene rearrangement and RNA processing events required to generate finished μ or δ heavy-chain protein. Two DNA joinings are necessary to generate a functional heavy-chain gene: a D_H to J_H joining and a V_H to D_HJ_H joining. In this example, V_H21 , D_H7 , and J_H3 are joined. Expression of functional heavy-chain

genes, although generally similar to expression of light-chain genes, involves differential RNA processing, which generates several different products, including μ or δ heavy chains. Each C gene is drawn as a single coding sequence; in reality, each is organized as a series of exons and introns.

Mechanism of Variable-Region DNA Rearrangements

Now that we've seen the results of variable-region gene rearrangements, let's examine in detail how this process occurs during maturation of B cells.

Recombination Signal Sequences Direct Recombination

The discovery of two closely related conserved sequences in variable-region germ-line DNA paved the way to fuller understanding of the mechanism of gene rearrangements. DNA sequencing studies revealed the presence of unique **recombination signal sequences (RSSs)** flanking each germ-line V, D, and J gene segment. One RSS is located 3' to each V gene segment, 5' to each J gene segment, and on both sides of each D gene segment. These sequences function as signals for the recombination process that rearranges the genes. Each RSS contains a conserved palindromic heptamer and a conserved AT-rich nonamer sequence separated by an intervening sequence of 12 or 23 base pairs (Figure 5-6a). The intervening 12- and 23-bp sequences correspond, respectively, to one and two turns of the DNA helix; for this reason the sequences are called **one-turn recombination signal sequences** and **two-turn signal sequences**.

The V_{κ} signal sequence has a one-turn spacer, and the J_{κ} signal sequence has a two-turn spacer. In λ light-chain DNA, this order is reversed; that is, the V_{λ} signal sequence has a two-turn spacer, and the J_{λ} signal sequence has a one-turn

spacer. In heavy-chain DNA, the signal sequences of the V_H and J_H gene segments have two-turn spacers, the signals on either side of the D_H gene segment have one-turn spacers (Figure 5-6b). Signal sequences having a one-turn spacer can join only with sequences having a two-turn spacer (the so-called one-turn/two-turn joining rule). This joining rule ensures, for example, that a V_L segment joins only to a J_L segment and not to another V_L segment; the rule likewise ensures that V_H , D_H , and J_H segments join in proper order and that segments of the same type do not join each other.

Gene Segments Are Joined by Recombinases

V-(D)-J recombination, which takes place at the junctions between RSSs and coding sequences, is catalyzed by enzymes collectively called **V(D)J recombinase**.

Identification of the enzymes that catalyze recombination of V, D, and J gene segments began in the late 1980s and is still ongoing. In 1990 David Schatz, Marjorie Oettinger, and David Baltimore first reported the identification of two **recombination-activating genes**, designated **RAG-1** and **RAG-2**, whose encoded proteins act synergistically and are required to mediate V-(D)-J joining. The RAG-1 and RAG-2 proteins and the enzyme **terminal deoxynucleotidyl transferase (TdT)** are the only lymphoid-specific gene products that have been shown to be involved in V-(D)-J rearrangement.

The recombination of variable-region gene segments consists of the following steps, catalyzed by a system of recombinase enzymes (Figure 5-7):

- Recognition of recombination signal sequences (RSSs) by recombinase enzymes, followed by synapsis in which

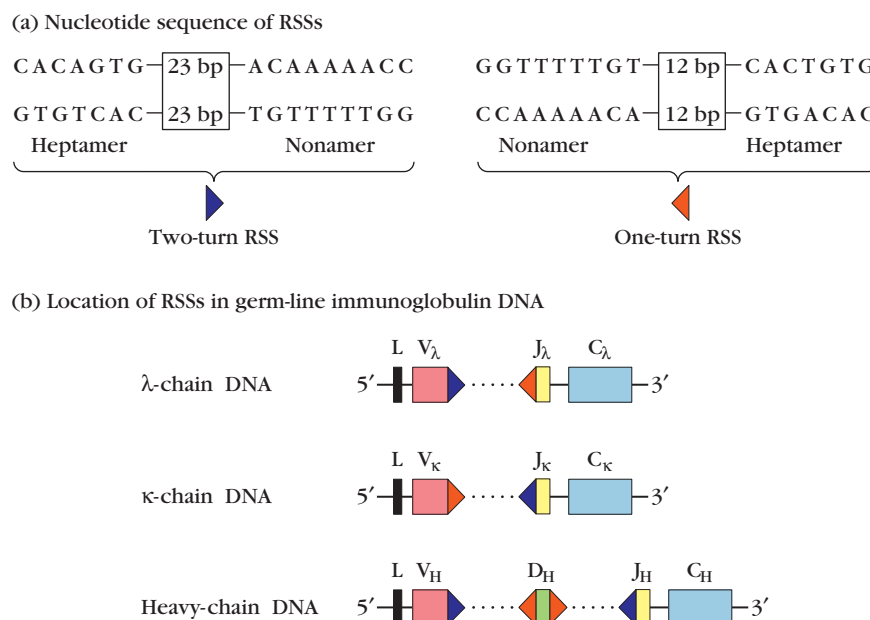


FIGURE 5-6 Two conserved sequences in light-chain and heavy-chain DNA function as recombination signal sequences (RSSs). (a) Both signal sequences consist of a conserved palindromic heptamer and conserved AT-rich nonamer; these are separated by nonconserved spacers of 12 or 23 base pairs. (b) The two types of

RSS—designated one-turn RSS and two-turn RSS—have characteristic locations within λ -chain, κ -chain, and heavy-chain germ-line DNA. During DNA rearrangement, gene segments adjacent to the one-turn RSS can join only with segments adjacent to the two-turn RSS.

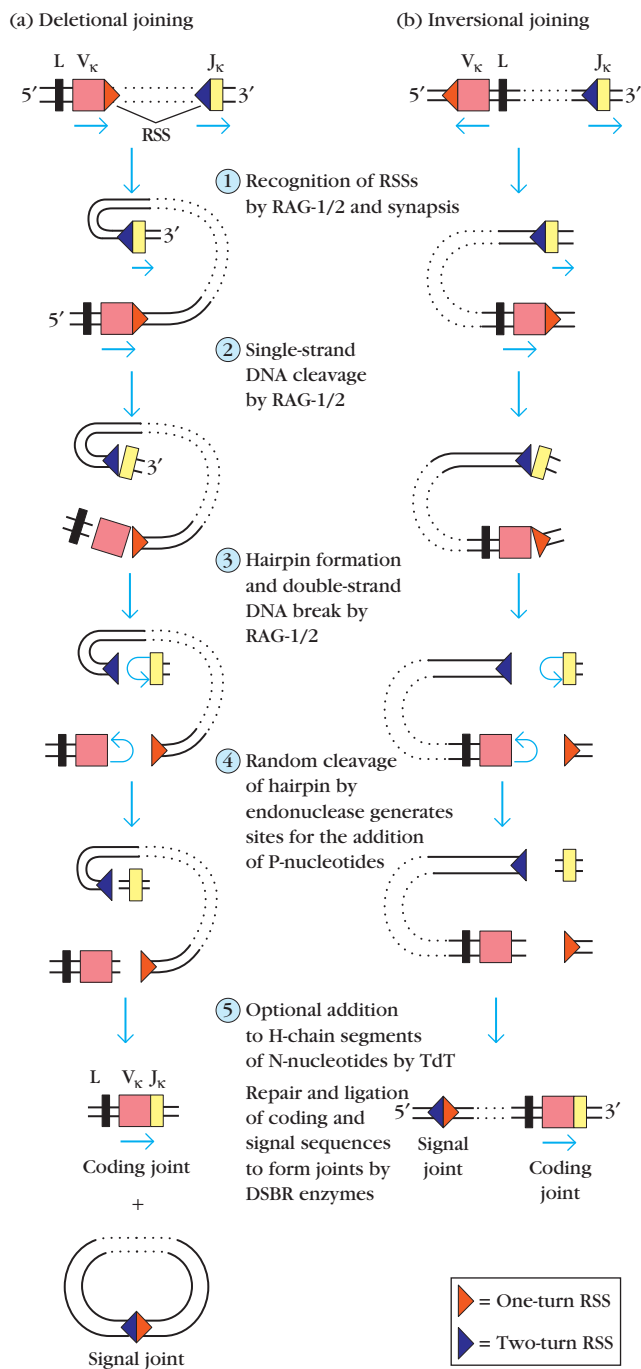


FIGURE 5-7 Model depicting the general process of recombination of immunoglobulin gene segments is illustrated with V_K and J_K. (a) Deletional joining occurs when the gene segments to be joined have the same transcriptional orientation (indicated by horizontal blue arrows). This process yields two products: a rearranged VJ unit that includes the coding joint, and a circular excision product consisting of the recombination signal sequences (RSSs), signal joint, and intervening DNA. (b) Inversional joining occurs when the gene segments have opposite transcriptional orientations. In this case, the RSSs, signal joint, and intervening DNA are retained, and the orientation of one of the joined segments is inverted. In both types of recombination, a few nucleotides may be deleted from or added to the cut ends of the coding sequences before they are rejoined.

two signal sequences and the adjacent coding sequences (gene segments) are brought into proximity

- Cleavage of one strand of DNA by RAG-1 and RAG-2 at the junctures of the signal sequences and coding sequences
- A reaction catalyzed by RAG-1 and RAG-2 in which the free 3'-OH group on the cut DNA strand attacks the phosphodiester bond linking the opposite strand to the signal sequence, simultaneously producing a hairpin structure at the cut end of the coding sequence and a flush, 5'-phosphorylated, double-strand break at the signal sequence
- Cutting of the hairpin to generate sites for the addition of **P-region nucleotides**, followed by the trimming of a few nucleotides from the coding sequence by a single-strand endonuclease
- Addition of up to 15 nucleotides, called **N-region nucleotides**, at the cut ends of the V, D, and J coding sequences of the heavy chain by the enzyme terminal deoxynucleotidyl transferase
- Repair and ligation to join the coding sequences and to join the signal sequences, catalyzed by normal double-strand break repair (DSBR) enzymes

Recombination results in the formation of a **coding joint**, falling between the coding sequences, and a **signal joint**, between the RSSs. The transcriptional orientation of the gene segments to be joined determines the fate of the signal joint and intervening DNA. When the two gene segments are in the same transcriptional orientation, joining results in deletion of the signal joint and intervening DNA as a circular excision product (Figure 5-8). Less frequently, the two gene segments have opposite orientations. In this case joining occurs by inversion of the DNA, resulting in the retention of

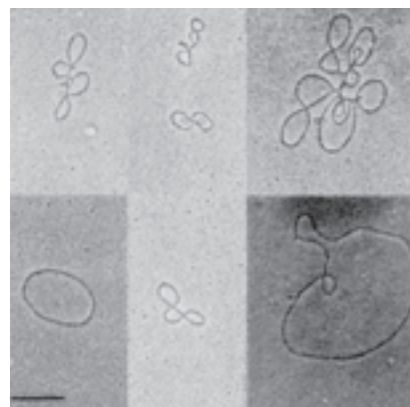


FIGURE 5-8 Circular DNA isolated from thymocytes in which the DNA encoding the chains of the T-cell receptor (TCR) undergoes rearrangement in a process like that involving the immunoglobulin genes. Isolation of this circular excision product is direct evidence for the mechanism of deletional joining shown in Figure 5-7. [From K. Okazaki et al., 1987, *Cell* 49:477.]

both the coding joint and the signal joint (and intervening DNA) on the chromosome. In the human κ locus, about half of the V_κ gene segments are inverted with respect to J_κ and their joining is thus by inversion.

Ig-Gene Rearrangements May Be Productive or Nonproductive

One of the striking features of gene-segment recombination is the diversity of the coding joints that are formed between any two gene segments. Although the double-strand DNA breaks that initiate V-(D)-J rearrangements are introduced precisely at the junctions of signal sequences and coding sequences, the subsequent joining of the coding sequences is imprecise. Junctional diversity at the V-J and V-D-J coding joints is generated by a number of mechanisms: variation in cutting of the hairpin to generate P-nucleotides, variation in trimming of the coding sequences, variation in N-nucleotide addition, and flexibility in joining the coding sequences. The introduction of randomness in the joining process helps generate antibody diversity by contributing to the hypervariability of the antigen-binding site. (This phenomenon is covered in more detail below in the section on generation of antibody diversity.)

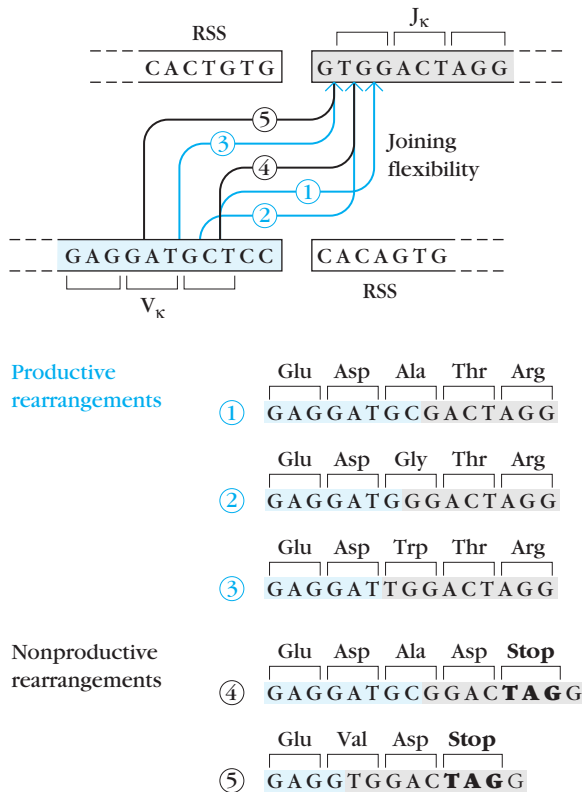


FIGURE 5-9 Junctional flexibility in the joining of immunoglobulin gene segments is illustrated with V_κ and J_κ . In-phase joining (arrows 1, 2, and 3) generates a productive rearrangement, which can be translated into protein. Out-of-phase joining (arrows 4 and 5) leads to a nonproductive rearrangement that contains stop codons and is not translated into protein.

Another consequence of imprecise joining is that gene segments may be joined out of phase, so that the triplet reading frame for translation is not preserved. In such a **nonproductive rearrangement**, the resulting VJ or VDJ unit is likely to contain numerous stop codons, which interrupt translation (Figure 5-9). When gene segments are joined in phase, the reading frame is maintained. In such a **productive rearrangement**, the resulting VJ or VDJ unit can be translated in its entirety, yielding a complete antibody.

If one allele rearranges nonproductively, a B cell may still be able to rearrange the other allele productively. If an in-phase rearranged heavy-chain and light-chain gene are not produced, the B cell dies by apoptosis. It is estimated that only one in three attempts at V_L - J_L joining, and one in three subsequent attempts at V_H - D_H - J_H joining, are productive. As a result, less than 1/9 (11%) of the early-stage pre-B cells in the bone marrow progress to maturity and leave the bone marrow as mature immunocompetent B cells.

Allelic Exclusion Ensures a Single Antigenic Specificity

B cells, like all somatic cells, are diploid and contain both maternal and paternal chromosomes. Even though a B cell is

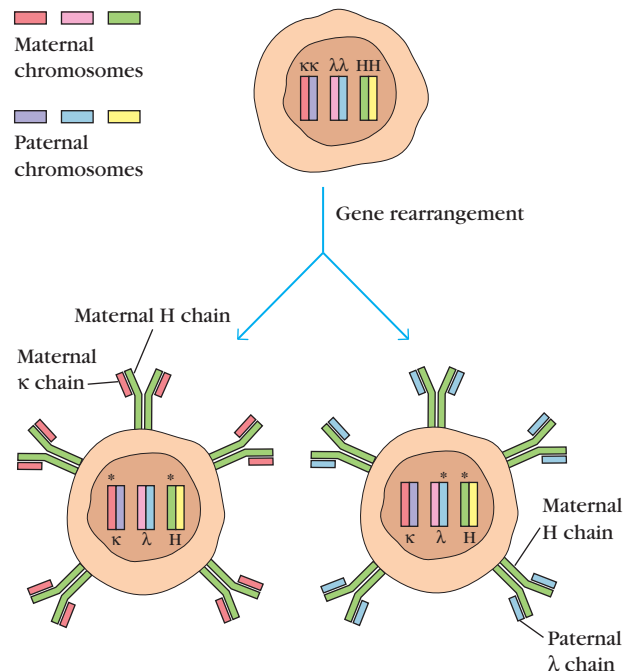


FIGURE 5-10 Because of allelic exclusion, the immunoglobulin heavy- and light-chain genes of only one parental chromosome are expressed per cell. This process ensures that B cells possess a single antigenic specificity. The allele selected for rearrangement is chosen randomly. Thus the expressed immunoglobulin may contain one maternal and one paternal chain or both chains may derive from only one parent. Only B cells and T cells exhibit allelic exclusion. Asterisks (*) indicate the expressed alleles.

diploid, it expresses the rearranged heavy-chain genes from only one chromosome and the rearranged light-chain genes from only one chromosome. The process by which this is accomplished, called **allelic exclusion**, ensures that functional B cells never contain more than one $V_H D_H J_H$ and one $V_L J_L$ unit (Figure 5-10). This is, of course, essential for the antigenic specificity of the B cell, because the expression of both alleles would render the B cell multispecific. The phenomenon of allelic exclusion suggests that once a productive $V_H D_H J_H$ rearrangement and a productive $V_L J_L$ rearrangement have occurred, the recombination machinery is turned off, so that the heavy- and light-chain genes on the homologous chromosomes are not expressed.

G. D. Yancopoulos and F. W. Alt have proposed a model to account for allelic exclusion (Figure 5-11). They suggest that once a productive rearrangement is attained, its encoded protein is expressed and the presence of this protein acts as a signal to prevent further gene rearrangement. According to their model, the presence of μ heavy chains signals the

maturing B cell to turn off rearrangement of the other heavy-chain allele and to turn on rearrangement of the κ light-chain genes. If a productive κ rearrangement occurs, κ light chains are produced and then pair with μ heavy chains to form a complete antibody molecule. The presence of this antibody then turns off further light-chain rearrangement. If κ rearrangement is nonproductive for both κ alleles, rearrangement of the λ -chain genes begins. If neither λ allele rearranges productively, the B cell presumably ceases to mature and soon dies by apoptosis.

Two studies with transgenic mice have supported the hypothesis that the protein products encoded by rearranged heavy- and light-chain genes regulate rearrangement of the remaining alleles. In one study, transgenic mice carrying a rearranged μ heavy-chain transgene were prepared. The μ transgene product was expressed by a large percentage of the B cells, and rearrangement of the endogenous immunoglobulin heavy-chain genes was blocked. Similarly, cells from a transgenic mouse carrying a κ light-chain transgene did not

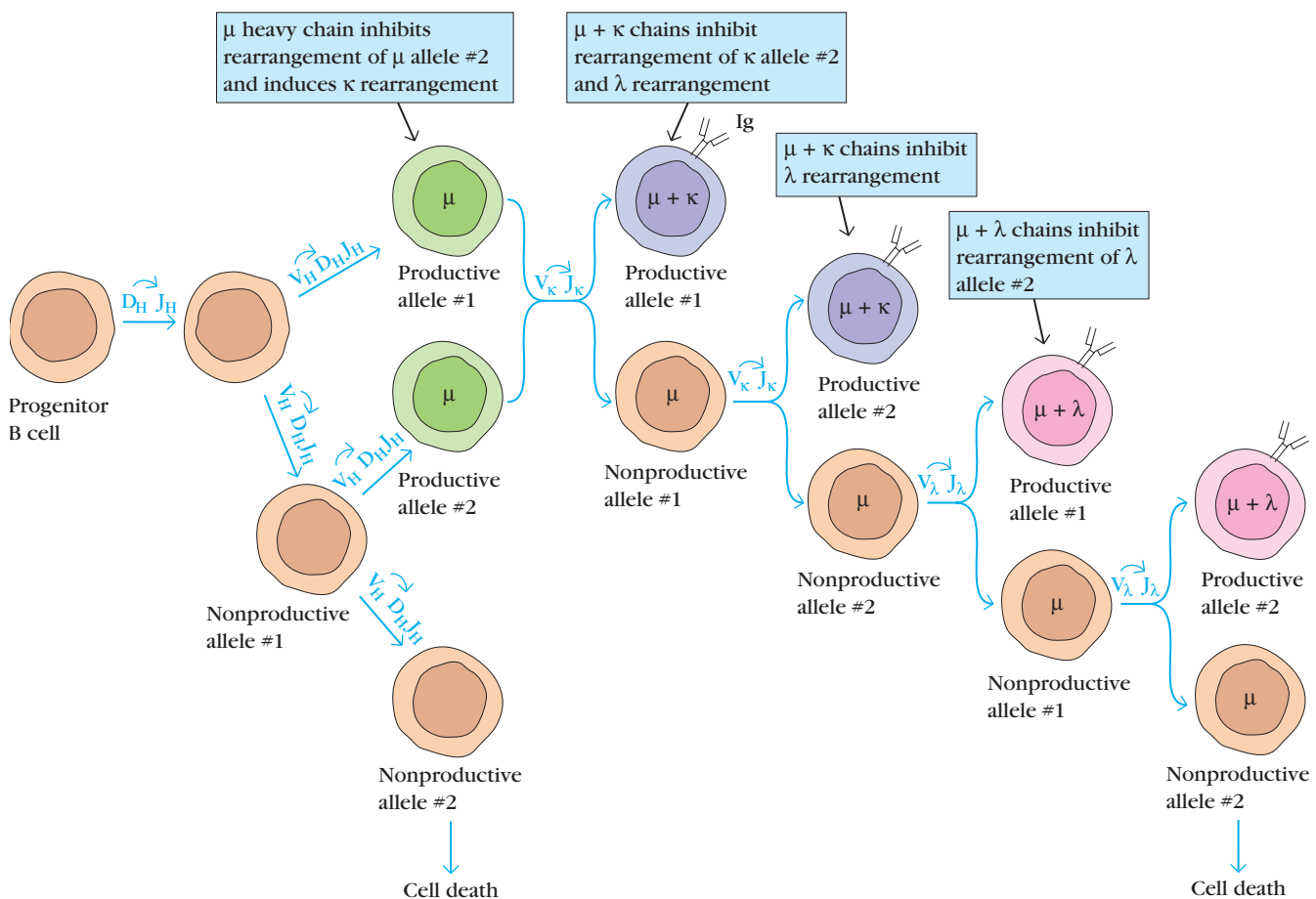


FIGURE 5-11 Model to account for allelic exclusion. Heavy-chain genes rearrange first, and once a productive heavy-chain gene rearrangement occurs, the μ protein product prevents rearrangement of the other heavy-chain allele and initiates light-chain gene rearrangement. In the mouse, rearrangement of κ light-chain genes precedes rearrangement of the λ genes, as shown here. In humans,

either κ or λ rearrangement can proceed once a productive heavy-chain rearrangement has occurred. Formation of a complete immunoglobulin inhibits further light-chain gene rearrangement. If a nonproductive rearrangement occurs for one allele, then the cell attempts rearrangement of the other allele. [Adapted from G. D. Yancopoulos and F. W. Alt, 1986, *Annu. Rev. Immunol.* **4**:339.]

rearrange the endogenous κ -chain genes when the κ trans-gene was expressed and was associated with a heavy chain to form complete immunoglobulin. These studies suggest that expression of the heavy- and light-chain proteins may indeed prevent gene rearrangement of the remaining alleles and thus account for allelic exclusion.

Generation of Antibody Diversity

As the organization of the immunoglobulin genes was deciphered, the sources of the vast diversity in the variable region began to emerge. The germ-line theory, mentioned earlier, argued that the entire variable-region repertoire is encoded in the germ line of the organism and is transmitted from parent to offspring through the germ cells (egg and sperm). The somatic-variation theory held that the germ line contains a limited number of variable genes, which are diversified in the somatic cells by mutational or recombinational events during development of the immune system. With the cloning and sequencing of the immunoglobulin genes, both models were partly vindicated.

To date, seven means of antibody diversification have been identified in mice and humans:

- Multiple germ-line gene segments
- Combinatorial V-(D)-J joining
- Junctional flexibility
- P-region nucleotide addition (P-addition)
- N-region nucleotide addition (N-addition)
- Somatic hypermutation
- Combinatorial association of light and heavy chains

Although the exact contribution of each of these avenues of diversification to total antibody diversity is not known, they each contribute significantly to the immense number of distinct antibodies that the mammalian immune system is capable of generating.

There Are Numerous Germ-Line V, D, and J Gene Segments

An inventory of functional V, D, and J gene segments in the germ-line DNA of one human reveals 51 V_H , 25 D, 6 J_H , 40 V_κ , 5 J_κ , 31 V_λ , and 4 J_λ gene segments. In addition to these functional segments, there are many pseudogenes. It should be borne in mind that these numbers were largely derived from a landmark study that sequenced the DNA of the immunoglobulin loci of a single individual. The immunoglobulin loci of other individuals might contain slightly different numbers of particular types of gene segments.

In the mouse, although the numbers are known with less precision than in the human, there appear to be about 85 V_κ gene segments and 134 V_H gene segments, 4 functional J_H , 4

functional J_κ , 3 functional J_λ , and an estimated 13 D_H gene segments, but only three V_λ gene segments. Although the number of germ-line genes found in either humans or mice is far fewer than predicted by early proponents of the germ-line model, multiple germ-line V, D, and J gene segments clearly do contribute to the diversity of the antigen-binding sites in antibodies.

Combinatorial V-J and V-D-J Joining Generates Diversity

The contribution of multiple germ-line gene segments to antibody diversity is magnified by the random rearrangement of these segments in somatic cells. It is possible to calculate how much diversity can be achieved by gene rearrangements (Table 5-2). In humans, the ability of any of the 51 V_H gene segments to combine with any of the 27 D_H segments and any of the 6 J_H segments allows a considerable amount of heavy-chain gene diversity to be generated ($51 \times 27 \times 6 = 8262$ possible combinations). Similarly, 40 V_κ gene segments randomly combining with 5 J_κ segments has the potential of generating 200 possible combinations at the κ locus, while 30 V_λ and 4 J_λ gene segments allow up to 120 possible combinations at the human λ locus. It is important to realize that these are minimal calculations of potential diversity. Junctional flexibility and P- and N-nucleotide addition, as mentioned above, and, especially, somatic hypermutation, which will be described shortly, together make an enormous contribution to antibody diversity. Although it is not possible to make an exact calculation of their contribution, most workers in this field agree that they raise the potential for antibody combining-site diversity in humans to well over 10^{10} . This does not mean that, at any given time, a single individual has a repertoire of 10^{10} different antibody combining sites. These very large numbers describe the set of possible variations, of which any individual carries a subset that is smaller by several orders of magnitude.

Junctional Flexibility Adds Diversity

The enormous diversity generated by means of V, D, and J combinations is further augmented by a phenomenon called **junctional flexibility**. As described above, recombination involves both the joining of recombination signal sequences to form a signal joint and the joining of coding sequences to form a coding joint (see Figure 5-7). Although the signal sequences are always joined precisely, joining of the coding sequences is often imprecise. In one study, for example, joining of the $V_\kappa 21$ and $J_\kappa 1$ coding sequences was analyzed in several pre-B cell lines. Sequence analysis of the signal and coding joints revealed the contrast in junctional precision (Figure 5-12).

As illustrated previously, junctional flexibility leads to many nonproductive rearrangements, but it also generates productive combinations that encode alternative amino acids at each coding joint (see Figure 5-9), thereby increasing antibody diversity. The amino acid sequence variation gener-

TABLE 5-2 Combinatorial antibody diversity in humans and mice

Multiple germ-line segments	Heavy chain	LIGHT CHAINS	
		κ	λ
ESTIMATED NUMBER OF SEGMENTS IN HUMANS*			
V	51	40	30
D	27	0	0
J	6	5	4
Combinatorial V-D-J and V-J joining (possible number of combinations)	$51 \times 27 \times 6 = 8262$	$40 \times 5 = 200$	$30 \times 4 = 120$
Possible combinatorial associations of heavy and light chains [†]	$8262 \times (200 \times 120) = 2.64 \times 10^6$		
ESTIMATED NUMBER OF SEGMENTS IN MICE*			
V	134	85	2
D	13	0	0
J	4	4	3
Combinatorial V-D-J and V-J joining (possible number of combinations)	$134 \times 13 \times 4 = 6968$	$85 \times 4 = 340$	$2 \times 3 = 6$
Possible combinatorial associations of heavy and light chains [†]	$6968 \times (340 + 6) = 2.41 \times 10^6$		

*These numbers have been determined from studies of single subjects; slight differences may be seen among different individuals. Also, in the human case, only the functional gene segments have been listed. The genome contains additional segments that are incapable of rearrangement or contain stop codons or both. In the mouse case, the figures contained in the table are only best estimates, because the locus has not been completely sequenced.

†Because of the diversity contributed by junctional flexibility, P-region nucleotide addition, N-region nucleotide addition, and somatic mutation, the actual potential exceeds these estimates by several orders of magnitude.

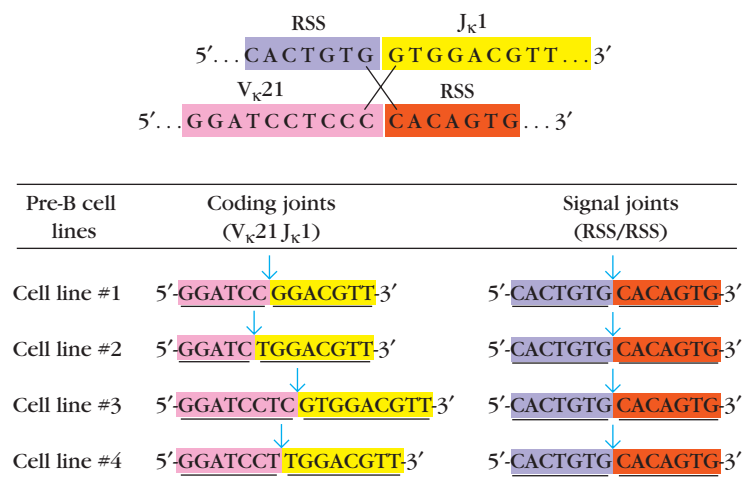


FIGURE 5-12 Experimental evidence for junctional flexibility in immunoglobulin-gene rearrangement. The nucleotide sequences flanking the coding joints between V_κ21 and J_κ1 and the corresponding signal joint sequences were determined in four pre-B cell lines. The

sequence constancy in the signal joints contrasts with the sequence variability in the coding joints. Pink and yellow shading indicate nucleotides derived from V_κ21 and J_κ1, respectively, and purple and orange shading indicate nucleotides from the two RSSs.

ated by junctional flexibility in the coding joints has been shown to fall within the third hypervariable region (CDR3) in immunoglobulin heavy-chain and light-chain DNA (Table 5-3). Since CDR3 often makes a major contribution to antigen binding by the antibody molecule, amino acid changes generated by junctional flexibility are important in the generation of antibody diversity.

P-Addition Adds Diversity at Palindromic Sequences

As described earlier, after the initial single-strand DNA cleavage at the junction of a variable-region gene segment and attached signal sequence, the nucleotides at the end of the coding sequence turn back to form a hairpin structure (see Figure 5-7). This hairpin is later cleaved by an endonuclease. This second cleavage sometimes occurs at a position that leaves a short single strand at the end of the coding sequence. The subsequent addition of complementary nucleotides to this strand (**P-addition**) by repair enzymes generates a palindromic sequence in the coding joint, and so these nucleotides are called **P-nucleotides** (Figure 5-13a). Variation in the position at which the hairpin is cut thus leads to variation in the sequence of the coding joint.

N-Addition Adds Considerable Diversity by Addition of Nucleotides

Variable-region coding joints in rearranged heavy-chain genes have been shown to contain short amino acid sequences that are not encoded by the germ-line V, D, or J gene segments. These amino acids are encoded by nucleotides added during the D-J and V to D-J joining process by a terminal deoxynucleotidyl transferase (TdT) catalyzed reaction

(Figure 5-13b). Evidence that TdT is responsible for the addition of these **N-nucleotides** has come from transfection studies in fibroblasts. When fibroblasts were transfected with the *RAG-1* and *RAG-2* genes, V-D-J rearrangement occurred but no N-nucleotides were present in the coding joints. However, when the fibroblasts were also transfected with the gene encoding TdT, then V-D-J rearrangement was accompanied by addition of N-nucleotides at the coding joints.

Up to 15 N-nucleotides can be added to both the D_H - J_H and V_H - D_H - J_H joints. Thus, a complete heavy-chain variable region is encoded by a V_H - N - D_H - N - J_H unit. The additional heavy-chain diversity generated by N-region nucleotide addition is quite large because N regions appear to consist of wholly random sequences. Since this diversity occurs at V-D-J coding joints, it is localized in CDR3 of the heavy-chain genes.

Somatic Hypermutation Adds Diversity in Already-Rearranged Gene Segments

All the antibody diversity described so far stems from mechanisms that operate during formation of specific variable regions by gene rearrangement. Additional antibody diversity is generated in rearranged variable-region gene units by a process called **somatic hypermutation**. As a result of somatic hypermutation, individual nucleotides in VJ or VDJ units are replaced with alternatives, thus potentially altering the specificity of the encoded immunoglobulins.

Normally, somatic hypermutation occurs only within germinal centers (see Chapter 11), structures that form in secondary lymphoid organs within a week or so of immunization with an antigen that activates a T-cell-dependent B-cell response. Somatic hypermutation is targeted to rearranged V-regions located within a DNA sequence containing about 1500 nucleotides, which includes the whole of the VJ or VDJ segment. Somatic hypermutation occurs at a frequency approaching 10^{-3} per base pair per generation. This rate is at least a hundred thousand-fold higher (hence the name *hypermutation*) than the spontaneous mutation rate, about 10^{-8} /bp/generation, in other genes. Since the combined length of the H-chain and L-chain variable-region genes is about 600 bp, one expects that somatic hypermutation will introduce at least one mutation per every two cell divisions in the pair of V_H and V_L genes that encode an antibody.

The mechanism of somatic hypermutation has not yet been determined. Most of the mutations are nucleotide substitutions rather than deletions or insertions. Somatic hypermutation introduces these substitutions in a largely, but not completely, random fashion. Recent evidence suggests that certain nucleotide motifs and palindromic sequences within V_H and V_L may be especially susceptible to somatic hypermutation.

Somatic hypermutations occur throughout the VJ or VDJ segment, but in mature B cells they are clustered within the CDRs of the V_H and V_L sequences, where they are most likely to influence the overall affinity for antigen. Following exposure to antigen, those B cells with higher-affinity receptors will be preferentially selected for survival. This result of this

TABLE 5-3 Sources of sequence variation in complementarity-determining regions of immunoglobulin heavy- and light-chain genes

Source of variation	CDR1	CDR2	CDR3
Sequence encoded by:	V segment	V segment	V_L - J_L junction; V_H - D_H - J_H junctions
Junctional flexibility	—	—	+
P-nucleotide addition	—	—	+
N-nucleotide addition*	—	—	+
Somatic hypermutation	+	+	+

*N-nucleotide addition occurs only in heavy-chain DNA.

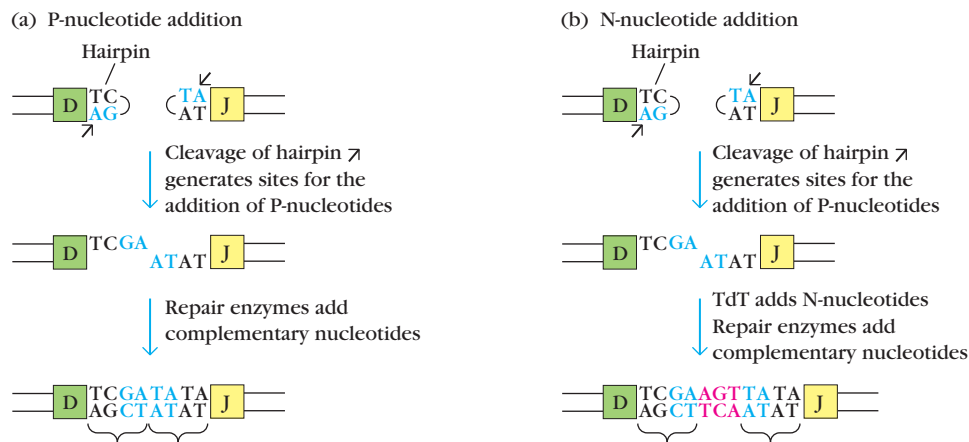


FIGURE 5-13 P-nucleotide and N-nucleotide addition during joining. (a) If cleavage of the hairpin intermediate yields a double-stranded end on the coding sequence, then P-nucleotide addition does not occur. In many cases, however, cleavage yields a single-stranded end. During subsequent repair, complementary nucleotides are added, called P-nucleotides, to produce palin-

dromic sequences (indicated by brackets). In this example, four extra base pairs (blue) are present in the coding joint as the result of P-nucleotide addition. (b) Besides P-nucleotide addition, addition of random N-nucleotides (light red) by a terminal deoxynucleotidyl transferase (TdT) can occur during joining of heavy-chain coding sequences.

differential selection is an increase in the antigen affinity of a population of B cells. The overall process, called **affinity maturation**, takes place within germinal centers, and is described more fully in Chapter 11.

Claudia Berek and Cesar Milstein obtained experimental evidence demonstrating somatic hypermutation during the course of an immune response to a hapten-carrier conjugate. These researchers were able to sequence mRNA that encoded antibodies raised against a hapten in response to primary, secondary, or tertiary immunization (first, second, or third exposure) with a hapten-carrier conjugate. The hapten they chose was 2-phenyl-5-oxazolone (phOx), coupled to a protein carrier. They chose this hapten because it had previously been shown that the majority of antibodies it induced were encoded by a single germ-line V_H and V_K gene segment. Berek and Milstein immunized mice with the phOx-carrier conjugate and then used the mouse spleen cells to prepare hybridomas secreting monoclonal antibodies specific for the phOx hapten. The mRNA sequence for the H chain and κ light chain of each hybridoma was then determined to identify deviations from the germ-line sequences.

The results of this experiment are depicted in Figure 5-14. Of the 12 hybridomas obtained from mice seven days after a primary immunization, all used a particular V_H , the V_H Ox-1 gene segment, and all but one used the same V_L gene segment, V_K Ox-1. Moreover, only a few mutations from the germ-line sequence were present in these hybridomas. By day 14 after primary immunization, analysis of eight hybridomas revealed that six continued to use the germ-line V_H Ox-1 gene segment and all continued to use the V_K Ox-1 gene segment. Now, however, all of these hybridomas

included one or more mutations from the germ-line sequence. Hybridomas analyzed from the secondary and tertiary responses showed a larger percentage utilizing germ-line V_H gene segments other than the V_H Ox-1 gene. In those hybridoma clones that utilized the V_H Ox-1 and V_K Ox-1 gene segments, most of the mutations were clustered in the CDR1 and CDR2 hypervariable regions. The number of mutations in the anti-phOx hybridomas progressively increased following primary, secondary, and tertiary immunizations, as did the overall affinity of the antibodies for phOx (see Figure 5-14).

A Final Source of Diversity Is Combinatorial Association of Heavy and Light Chains

In humans, there is the potential to generate 8262 heavy-chain genes and 320 light-chain genes as a result of variable-region gene rearrangements. Assuming that any one of the possible heavy-chain and light-chain genes can occur randomly in the same cell, the potential number of heavy- and light-chain combinations is 2,644,240. This number is probably higher than the amount of combinatorial diversity actually generated in an individual, because it is not likely that all V_H and V_L will pair with each other. Furthermore, the recombination process is not completely random; not all V_H , D, or V_L gene segments are used at the same frequency. Some are used often, others only occasionally, and still others almost never.

Although the number of different antibody combining sites the immune system can generate is difficult to calculate with precision, we know that it is quite high. Because the very large number of new sequences created by junctional

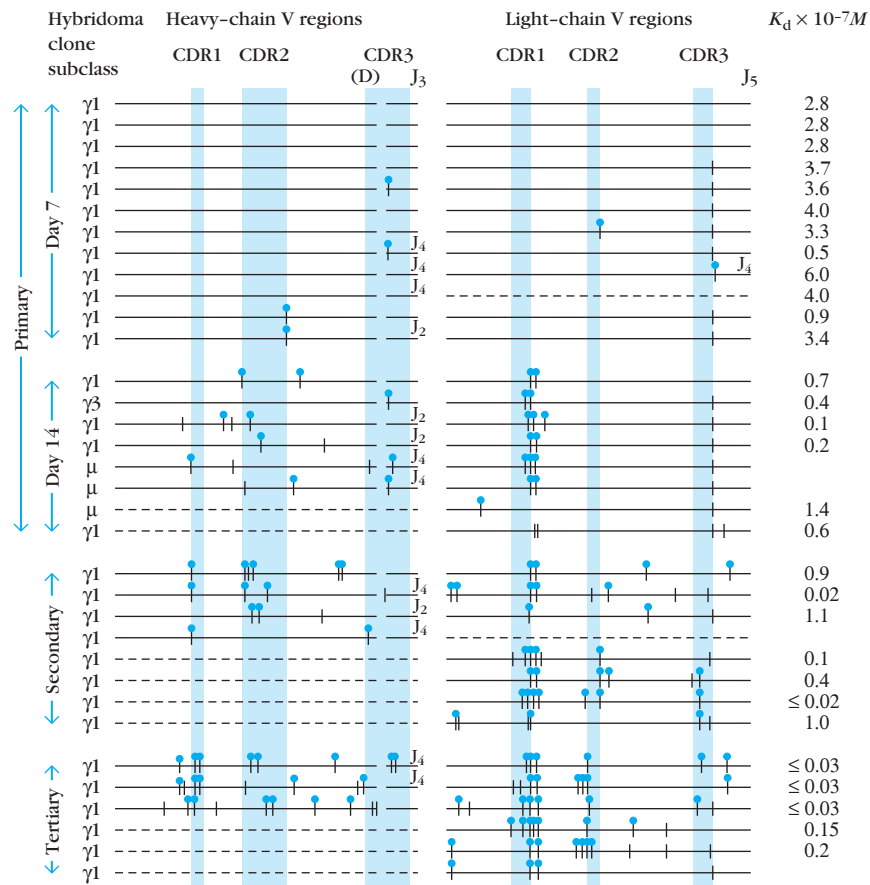


FIGURE 5-14 Experimental evidence for somatic mutation in variable regions of immunoglobulin genes. The diagram compares the mRNA sequences of heavy chains and of light chains from hybridomas specific for the phOx hapten. The horizontal solid lines represent the germ-line V_H and V_K Ox-1 sequences; dashed lines represent sequences derived from other germ-line genes. Blue shading shows the areas where mutations clustered; the blue circles with vertical lines indicate locations of mutations that encode a different amino acid than the germ-line sequence. These data show that the fre-

quency of mutation (1) increases in the course of the primary response (day 7 vs. day 14) and (2) is higher after secondary and tertiary immunizations than after primary immunization. Moreover, the dissociation constant (K_d) of the anti-phOx antibodies decreases during the transition from the primary to tertiary response, indicating an increase in the overall affinity of the antibody. Note also that most of the mutations are clustered within CDR1 and CDR2 of both the heavy and the light chains. [Adapted from C. Berek and C. Milstein, 1987, *Immunol. Rev.* **96**:23.]

flexibility, P-nucleotide addition, and N-nucleotide addition are within the third CDR, they are positioned to influence the structure of the antibody binding site. In addition to these sources of antibody diversity, the phenomenon of somatic hypermutation contributes enormously to the repertoire after antigen stimulation.

Class Switching among Constant-Region Genes

After antigenic stimulation of a B cell, the heavy-chain DNA can undergo a further rearrangement in which the $V_H D_H J_H$

unit can combine with any C_H gene segment. The exact mechanism of this process, called **class switching** or **isotype switching**, is unclear, but it involves DNA flanking sequences (called **switch regions**) located 2–3 kb upstream from each C_H segment (except C_{δ}). These switch regions, though rather large (2 to 10 kb), are composed of multiple copies of short repeats (GAGCT and TGGGG). One hypothesis is that a protein or system of proteins that constitute the switch recombinase recognize these repeats and upon binding carry out the DNA recombination that results in class switching. Intercellular regulatory proteins known as cytokines act as “switch factors” and play major roles in determining the particular immunoglobulin class that is expressed as a consequence of switching. Interleukin 4 (IL-4),

segments are organized as a series of coding exons and noncoding introns. Each exon of a C_H gene segment corresponds to a constant-region domain or a hinge region of the heavy-chain polypeptide. The primary transcript must be processed to remove the intervening DNA sequences, and the remaining exons must be connected by a process called RNA splicing. Short, moderately conserved splice sequences, or splice sites, which are located at the intron-exon boundaries within a primary transcript, signal the positions at which splicing occurs. Processing of the primary transcript in the nucleus removes each of these intervening sequences to yield the final mRNA product. The mRNA is then exported from the nucleus to be translated by ribosomes into complete H or L chains.

Heavy-Chain Primary Transcripts Undergo Differential RNA Processing

Processing of an immunoglobulin heavy-chain primary transcript can yield several different mRNAs, which explains how a single B cell can produce secreted or membrane-bound forms of a particular immunoglobulin and simultaneously express IgM and IgD.

EXPRESSION OF MEMBRANE OR SECRETED IMMUNOGLOBULIN

As explained in Chapter 4, a particular immunoglobulin can exist in either membrane-bound or secreted form. The two forms differ in the amino acid sequence of the heavy-chain carboxyl-terminal domains (C_{H3}/C_{H3} in IgA, IgD, and IgG and C_{H4}/C_{H4} in IgE and IgM). The secreted form has a hydrophilic sequence of about 20 amino acids in the carboxyl-terminal domain; this is replaced in the membrane-bound form with a sequence of about 40 amino acids containing a hydrophilic segment that extends outside the cell, a hydrophobic transmembrane segment, and a short hydrophilic segment at the carboxyl terminus that extends into the cytoplasm (Figure 5-16a). For some time, the existence of these two forms seemed inconsistent with the structure of germ-line heavy-chain DNA, which had been shown to contain a single C_H gene segment corresponding to each class and subclass.

The resolution of this puzzle came from DNA sequencing of the C_μ gene segment, which consists of four exons ($C_{\mu 1}$, $C_{\mu 2}$, $C_{\mu 3}$, and $C_{\mu 4}$) corresponding to the four domains of the IgM molecule. The $C_{\mu 4}$ exon contains a nucleotide sequence (called S) at its 3' end that encodes the hydrophilic sequence in the C_{H4} domain of secreted IgM. Two additional exons called M1 and M2 are located just 1.8 kb downstream from the 3' end of the $C_{\mu 4}$ exon. The M1 exon encodes the transmembrane segment, and M2 encodes the cytoplasmic segment of the C_{H4} domain in membrane-bound IgM. Later DNA sequencing revealed

that all the C_H gene segments have two additional downstream M1 and M2 exons that encode the transmembrane and cytoplasmic segments.

The primary transcript produced by transcription of a rearranged μ heavy-chain gene contains two polyadenylation signal sequences, or poly-A sites, in the C_μ segment. Site 1 is located at the 3' end of the $C_{\mu 4}$ exon, and site 2 is at the 3' end of the M2 exon (Figure 5-16b). If cleavage of the primary transcript and addition of the poly-A tail occurs at site 1, the M1 and M2 exons are lost. Excision of the introns and splicing of the remaining exons then produces mRNA encoding the secreted form of the heavy chain. If cleavage and polyadenylation of the primary transcript occurs instead at site 2, then a different pattern of splicing results. In this case, splicing removes the S sequence at the 3' end of the $C_{\mu 4}$ exon, which encodes the hydrophilic carboxyl-terminal end of the secreted form, and joins the remainder of the $C_{\mu 4}$ exon with the M1 and M2 exons, producing mRNA for the membrane form of the heavy chain.

Thus, differential processing of a common primary transcript determines whether the secreted or membrane form of an immunoglobulin will be produced. As noted previously, mature naive B cells produce only membrane-bound antibody, whereas differentiated plasma cells produce secreted antibodies. It remains to be determined precisely how naive B cells and plasma cells direct RNA processing preferentially toward the production of mRNA encoding one form or the other.

SIMULTANEOUS EXPRESSION OF IgM AND IgD

Differential RNA processing also underlies the simultaneous expression of membrane-bound IgM and IgD by mature B cells. As mentioned already, transcription of rearranged heavy-chain genes in mature B cells produces primary transcripts containing both the C_μ and C_δ gene segments. The C_μ and C_δ gene segments are close together in the rearranged gene (only about 5 kb apart), and the lack of a switch site between them permits the entire $VDJ C_\mu C_\delta$ region to be transcribed into a single primary RNA transcript about 15 kb long, which contains four poly-A sites (Figure 5-17a). Sites 1 and 2 are associated with C_μ , as described in the previous section; sites 3 and 4 are located at similar places in the C_δ gene segment. If the heavy-chain transcript is cleaved and polyadenylated at site 2 after the C_μ exons, then the mRNA will encode the membrane form of the μ heavy chain (Figure 5-17b); if polyadenylation is instead further downstream at site 4, after the C_δ exons, then RNA splicing will remove the intervening C_μ exons and produce mRNA encoding the membrane form of the δ heavy chain (Figure 5-17c).

Since the mature B cell expresses both IgM and IgD on its membrane, both processing pathways must occur simultaneously. Likewise, cleavage and polyadenylation of the primary heavy-chain transcript at poly-A site 1 or 3 in

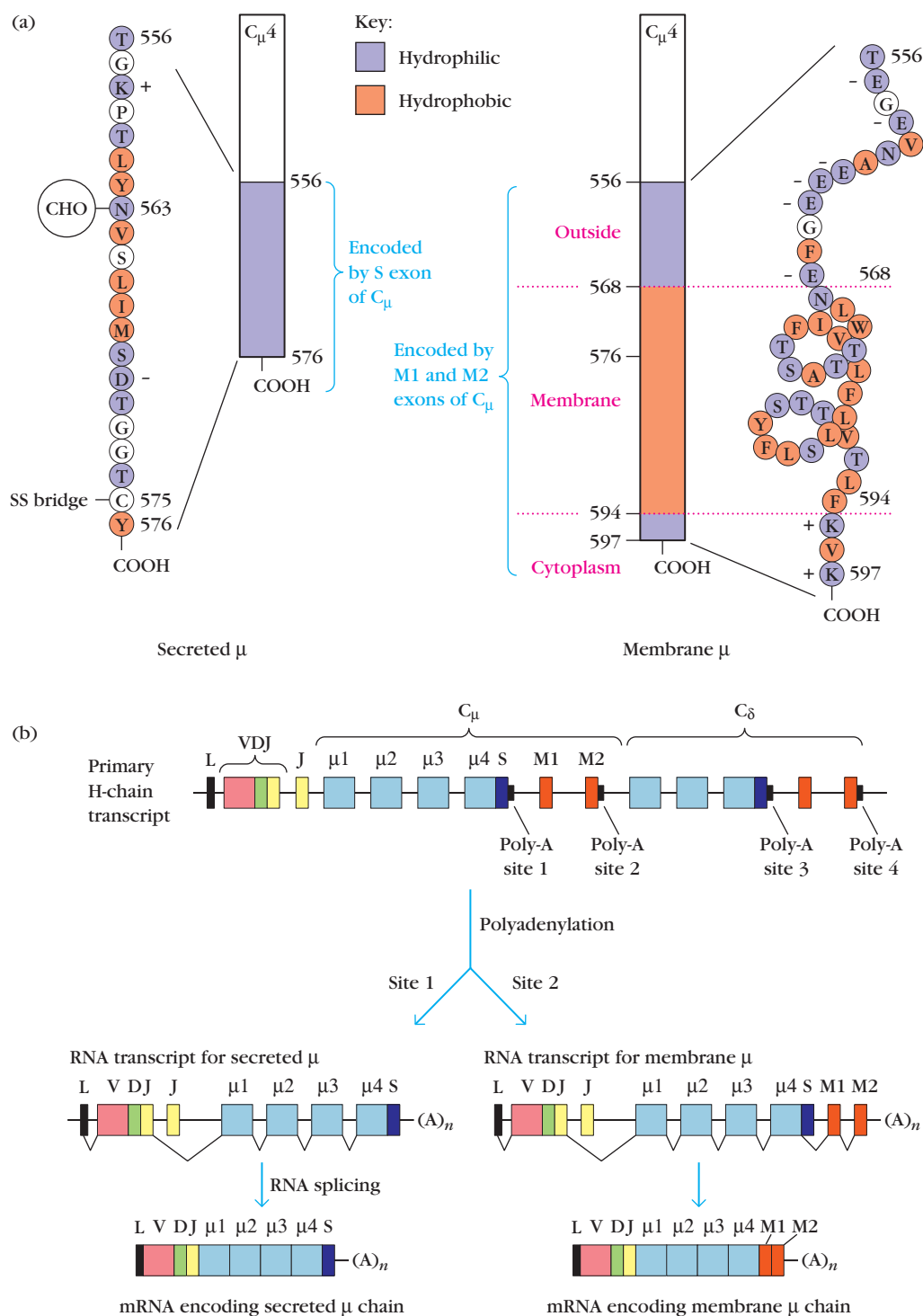


FIGURE 5-16 Expression of secreted and membrane forms of the heavy chain by alternative RNA processing. (a) Amino acid sequence of the carboxyl-terminal end of secreted and membrane μ heavy chains. Residues are indicated by the single-letter amino acid code. Hydrophilic and hydrophobic residues and regions are indicated by purple and orange, respectively, and charged amino acids are indicated with a + or -. The white regions of the

sequences are identical in both forms. (b) Structure of the primary transcript of a rearranged heavy-chain gene showing the C_μ exons and poly-A sites. Polyadenylation of the primary transcript at either site 1 or site 2 and subsequent splicing (indicated by V-shaped lines) generates mRNAs encoding either secreted or membrane μ chains.

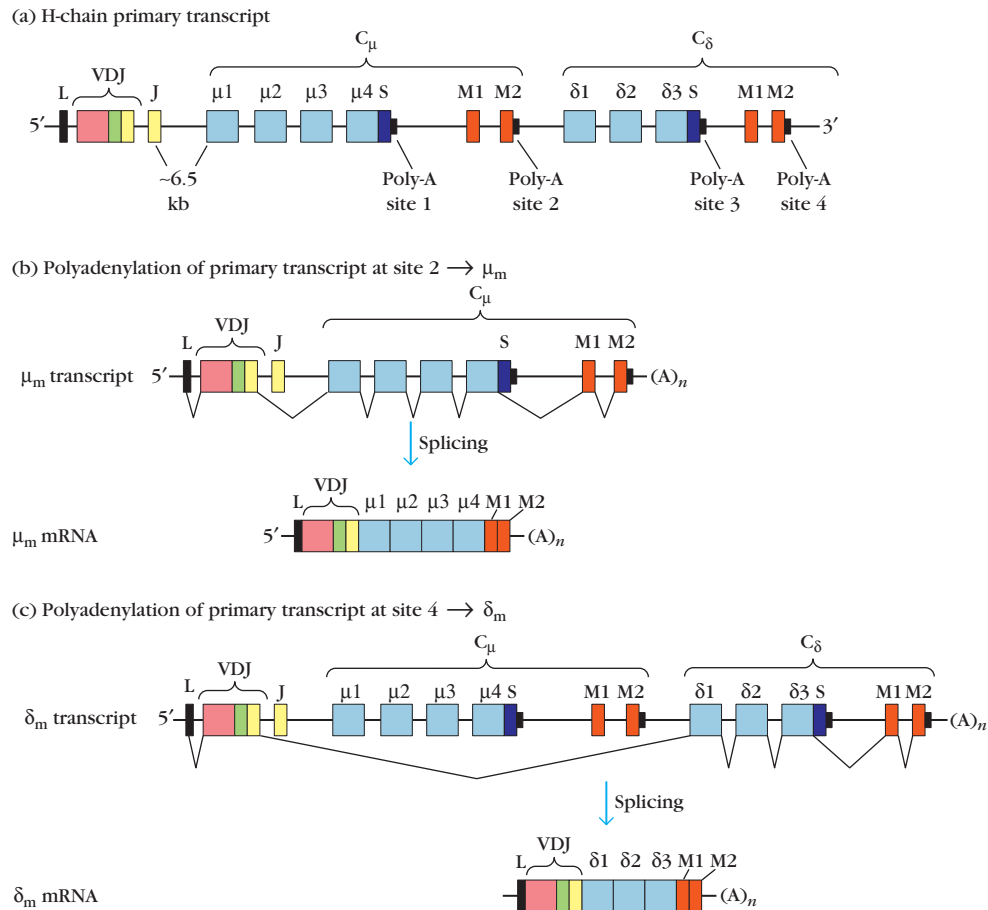


FIGURE 5-17 Expression of membrane forms of μ and δ heavy chains by alternative RNA processing. (a) Structure of rearranged heavy-chain gene showing C_μ and C_δ exons and poly-A sites. (b) Structure of μ_m transcript and μ_m mRNA resulting from poly-

adenylation at site 2 and splicing. (c) Structure of δ_m transcript and δ_m mRNA resulting from polyadenylation at site 4 and splicing. Both processing pathways can proceed in any given B cell.

plasma cells and subsequent splicing will yield the secreted form of the μ or δ heavy chains, respectively (see Figure 5-16b).

Synthesis, Assembly, and Secretion of Immunoglobulins

Immunoglobulin heavy- and light-chain mRNAs are translated on separate polyribosomes of the rough endoplasmic reticulum (RER). Newly synthesized chains contain an amino-terminal leader sequence, which serves to guide the chains into the lumen of the RER, where the signal sequence is then cleaved. The assembly of light (L) and heavy (H) chains into the disulfide-linked and glycosylated immunoglobulin molecule occurs as the chains pass through the cisternae of the RER. The complete molecules are transported to the Golgi apparatus and then into

secretory vesicles, which fuse with the plasma membrane (Figure 5-18).

The order of chain assembly varies among the immunoglobulin classes. In the case of IgM, the H and L chains assemble within the RER to form half-molecules, and then two half-molecules assemble to form the complete molecule. In the case of IgG, two H chains assemble, then an H₂L intermediate is assembled, and finally the complete H₂L₂ molecule is formed. Interchain disulfide bonds are formed, and the polypeptides are glycosylated as they move through the Golgi apparatus.

If the molecule contains the transmembrane sequence of the membrane form, it becomes anchored in the membrane of a secretory vesicle and is inserted into the plasma membrane as the vesicle fuses with the plasma membrane (see Figure 5-18, insert). If the molecule contains the hydrophilic sequence of secreted immunoglobulins, it is transported as a free molecule in a secretory vesicle and is released from the cell when the vesicle fuses with the plasma membrane.

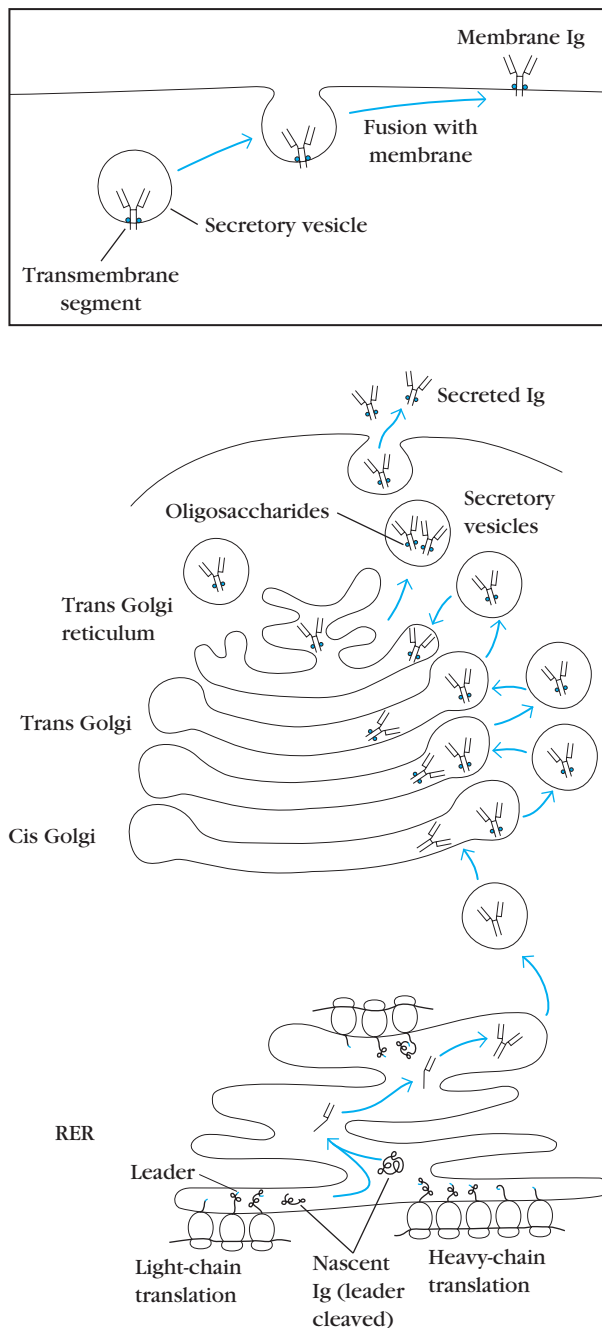


FIGURE 5-18 Synthesis, assembly, and secretion of the immunoglobulin molecule. The heavy and light chains are synthesized on separate polyribosomes (polysomes). The assembly of the chains to form the disulfide-linked immunoglobulin molecule occurs as the chains pass through the cisternae of the rough endoplasmic reticulum (RER) into the Golgi apparatus and then into secretory vesicles. The main figure depicts assembly of a secreted antibody. The inset depicts a membrane-bound antibody, which contains the carboxyl-terminal transmembrane segment. This form becomes anchored in the membrane of secretory vesicles and then is inserted into the cell membrane when the vesicles fuse with the membrane.

Regulation of Ig-Gene Transcription

The immunoglobulin genes are expressed only in B-lineage cells, and even within this lineage, the genes are expressed at different rates during different developmental stages. As with other eukaryotic genes, three major classes of cis regulatory sequences in DNA regulate transcription of immunoglobulin genes:

- **Promoters:** relatively short nucleotide sequences, extending about 200 bp upstream from the transcription initiation site, that promote initiation of RNA transcription in a specific direction
- **Enhancers:** nucleotide sequences situated some distance upstream or downstream from a gene that activate transcription from the promoter sequence in an orientation-independent manner
- **Silencers:** nucleotide sequences that down-regulate transcription, operating in both directions over a distance.

The locations of the three types of regulatory elements in germ-line immunoglobulin DNA are shown in Figure 5-19. All of these regulatory elements have clusters of sequence motifs that can bind specifically to one or more nuclear proteins.

Each V_H and V_L gene segment has a promoter located just upstream from the leader sequence. In addition, the J_K cluster and each of the D_H genes of the heavy-chain locus are preceded by promoters. Like other promoters, the immunoglobulin promoters contain a highly conserved AT-rich sequence called the TATA box, which serves as a site for the binding of a number of proteins that are necessary for the initiation of RNA transcription. The actual process of transcription is performed by RNA polymerase II, which starts transcribing DNA from the initiation site, located about 25 bp downstream of the TATA box. Ig promoters also contain an essential and conserved octamer that confers B-cell specificity on the promoter. The octamer binds two transcription factors, oct-1, found in many cell types, and oct-2, found only in B cells.

While much remains to be learned about the function of enhancers, they have binding sites for a number of proteins, many of which are transcription factors. A particularly important role is played by two proteins encoded by the *E2A* gene which can undergo alternate splicing to generate two collaborating proteins, both of which bind to the μ and κ intronic enhancers. These proteins are essential for B-cell development and *E2A* knockout mice make normal numbers of T cells but show a total absence of B cells. Interestingly, transfection of these enhancer-binding proteins into a T cell line resulted in a dramatic increase in the transcription of μ chain mRNA and even induced the T cell to undergo $D_H + J_H \rightarrow D_HJ_H$ rearrangement. Silencers may inhibit the activity of Ig

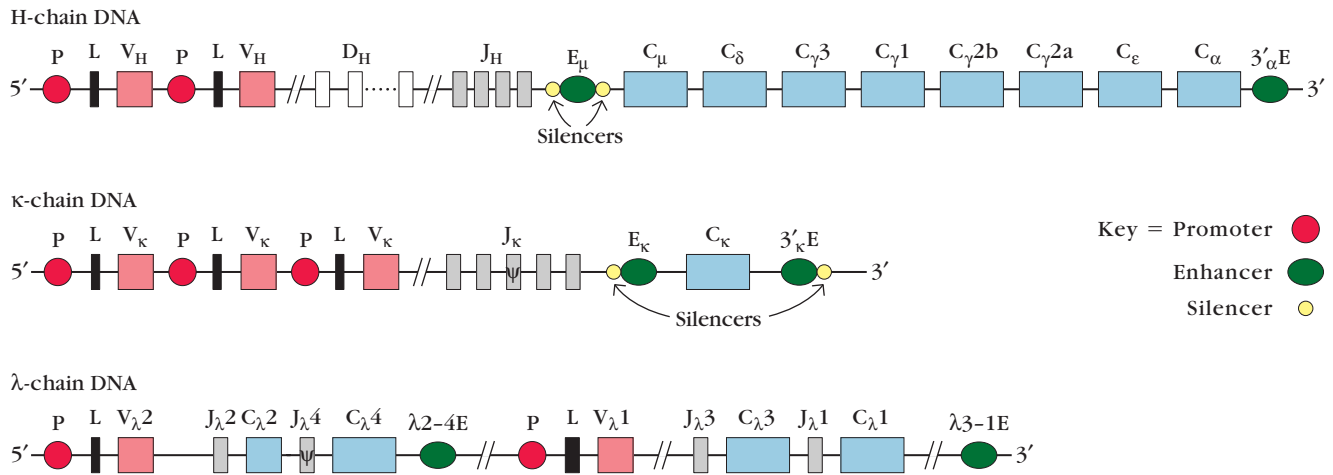


FIGURE 5-19 Location of promoters (dark red), enhancers (green), and silencers (yellow) in mouse heavy-chain, κ light-chain, and λ light-chain germ-line DNA. Variable-region DNA rearrangement moves an enhancer close enough to a promoter that the en-

hancer can activate transcription from the promoter. The promoters that precede the DH cluster, a number of the C genes and the J_λ cluster are omitted from this diagram for the sake of clarity.

enhancers in non-B cells. If so, they could be important contributors to the high levels of Ig gene transcription that are characteristic of B cells but absent in other cell types.

One heavy-chain enhancer is located within the intron between the last (3') J gene segment and the first (5') C gene segment (C_μ), which encodes the μ heavy chain. Because this heavy-chain enhancer (E_μ) is located 5' of the S_μ switch site near C_μ, it can continue to function after class switching has occurred. Another heavy-chain enhancer (3'_αE) has been detected 3' of the C_α gene segment. One κ light-chain enhancer (E_κ) is located between the J_κ segment and the C_κ segment, and another enhancer (3'_κE) is located 3' of the C_κ segment. The λ light-chain enhancers are located 3' of C_{λ4} and 3' of C_{λ1}. Silencers have been identified in heavy-chain and κ-chain DNA, adjacent to enhancers, but not in λ-chain DNA.

DNA Rearrangement Greatly Accelerates Transcription

The promoters associated with the immunoglobulin V gene segments bind RNA polymerase II very weakly, and the variable-region enhancers in germ-line DNA are quite distant from the promoters (about 250–300 kb), too remote to significantly influence transcription. For this reason, the rate of transcription of V_H and V_L coding regions is negligible in unrearranged germ-line DNA. Variable-region gene rearrangement brings a promoter and enhancer within 2 kb of each other, close enough for the enhancer to influence transcription from the nearby promoter. As a result, the rate of transcription of a rearranged V_L J_L or V_H D_H J_H unit is as much as 10⁴ times the rate of transcription of unrearranged V_L or V_H segments. This effect was demonstrated directly in a study in which B

cells transfected with rearranged heavy-chain genes from which the enhancer had been deleted did not transcribe the genes, whereas B cells transfected with similar genes that contained the enhancer transcribed the transfected genes at a high rate. These findings highlight the importance of enhancers in the normal transcription of immunoglobulin genes.

Genes that regulate cellular proliferation or prohibit cell death sometimes translocate to the immunoglobulin heavy- or light-chain loci. Here, under the influence of an immunoglobulin enhancer, the expression of these genes is significantly elevated, resulting in high levels of growth promoting or cell death inhibiting proteins. Translocations of the *c-myc* and *bcl-2* oncogenes have each been associated with malignant B-cell lymphomas. The translocation of *c-myc* leads to constitutive expression of c-Myc and an aggressive, highly proliferative B-cell lymphoma called Burkitt's lymphoma. The translocation of *bcl-2* leads to suspension of programmed cell death in B cells, resulting in follicular B-cell lymphoma. These cancer-promoting translocations are covered in greater detail in Chapter 22.

Ig-Gene Expression Is Inhibited in T Cells

As noted earlier, germ-line DNA encoding the T-cell receptor (TCR) undergoes V-(D)-J rearrangement to generate functional TCR genes. Rearrangement of both immunoglobulin and TCR germ-line DNA occurs by similar recombination processes mediated by RAG-1 and RAG-2 and involving recombination signal sequences with one-turn or two-turn spacers (see Figure 5-7). Despite the similarity of the processes, complete Ig-gene rearrangement of H and L chains occurs only in B cells and complete TCR-gene rearrangement is limited to T cells.

Hitoshi Sakano and coworkers have obtained results suggesting that a sequence within the κ -chain 3' enhancer (3' κ E) serves to regulate the joining of V_κ to J_κ in B and T cells. When a sequence known as the PU.1 binding site within the 3' κ -chain enhancer was mutated, these researchers found that V_κ - J_κ joining occurred in T cells as well as B cells. They propose that binding of a protein expressed by T cells, but not B cells, to the unmutated κ -chain enhancer normally prevents V_κ - J_κ joining in T cells. The identity of this DNA-binding protein in T cells remains to be determined. Similar processes may prevent rearrangement of heavy-chain and λ -chain DNA in T cells.

Antibody Genes and Antibody Engineering

There are many clinical applications in which the exquisite specificity of a mouse monoclonal antibody would be useful. However, when mouse monoclonal antibodies are introduced into humans they are recognized as foreign and evoke an antibody response that quickly clears the mouse monoclonal antibody from the bloodstream. In addition, circulating complexes of mouse and human antibodies can cause allergic reactions. In some cases, the buildup of these complexes in organs such as the kidney can cause serious and even life-threatening reactions. Clearly, one way to avoid these undesirable reactions is to use human monoclonal antibodies for clinical applications. However, the preparation of human monoclonal antibodies has been hampered by numerous technical problems. In response to the difficulty of producing human monoclonal antibodies and the complications resulting from the use of mouse monoclonal antibodies in humans, there is now a major effort to engineer monoclonal antibodies and antibody binding sites with recombinant DNA technology.

The growing knowledge of antibody gene structure and regulation has made possible what Cesar Milstein, one of the inventors of monoclonal antibody technology, has called "man-made antibodies." It is now possible to design and construct genes that encode immunoglobulin molecules in which the variable regions come from one species and the constant regions come from another. New genes have been created that link nucleotide sequences coding nonantibody proteins with sequences that encode antibody variable regions specific for particular antigens. These molecular hybrids or **chimeras** may be able to deliver powerful toxins to particular antigenic targets, such as tumor cells. Finally, by replacement of the immunoglobulin loci of one species with that of another, animals of one species have been endowed with the capacity to respond to immunization by producing antibodies encoded by the donor's genetically transplanted Ig genes. By capturing a significant sample of all of the immunoglobulin heavy- and light-chain variable-region genes via incorporation into libraries of bacteriophage, it has been

possible to achieve significant and useful reconstructions of the entire antibody repertoires of individuals. The next few sections describe each of these types of antibody genetic engineering.

Chimeric and Hybrid Monoclonal Antibodies Have Potent Clinical Potential

One approach to engineering an antibody is to clone recombinant DNA containing the promoter, leader, and variable-region sequences from a mouse antibody gene and the constant-region exons from a human antibody gene (Figure 5-20). The antibody encoded by such a recombinant gene is a mouse-human chimera, commonly known as a **humanized antibody**. Its antigenic specificity, which is determined by the variable region, is derived from the mouse DNA; its isotype, which is determined by the constant region, is derived from the human DNA. Because the constant regions of these chimeric antibodies are encoded by human genes, the anti-

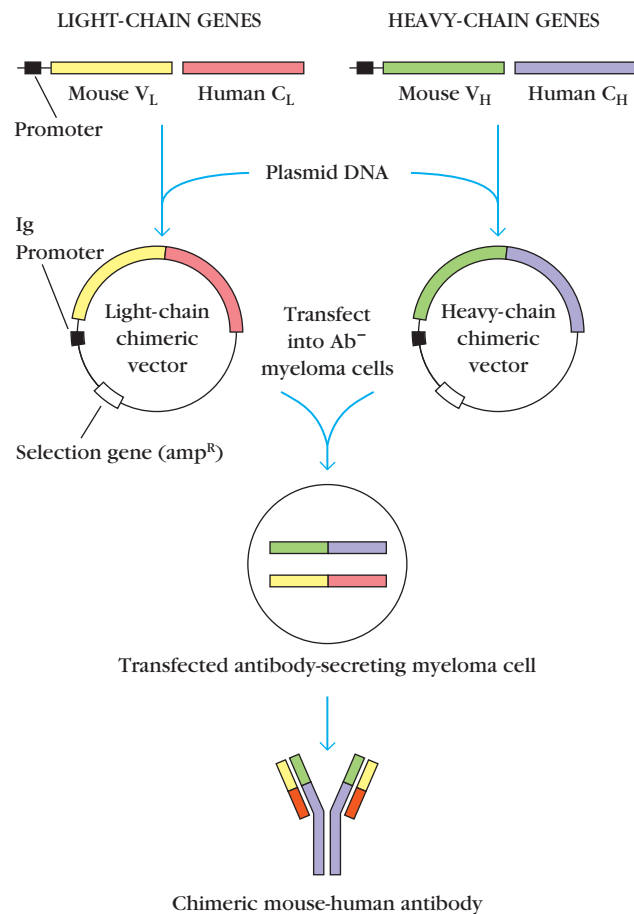


FIGURE 5-20 Production of chimeric mouse-human monoclonal antibodies. Chimeric mouse-human heavy- and light-chain expression vectors are produced. These vectors are transfected into Ab⁻ myeloma cells. Culture in ampicillin medium selects for transfected myeloma cells that secrete the chimeric antibody. [Adapted from M. Verhoeven and L. Reichmann, 1988, *BioEssays* 8:74.]

bodies have fewer mouse antigenic determinants and are far less immunogenic when administered to humans than mouse monoclonal antibodies (Figure 5-21a). The ability of the mouse variable regions remaining in these humanized antibodies to provide the appropriate binding site to allow specific recognition of the target antigen has encouraged further exploration of this approach. It is possible to produce chimeric human-mouse antibodies in which only the sequences of the CDRs are of mouse origin (Figure 5-21b). Another advantage of humanized chimeric antibodies is that they retain the biological effector functions of human antibody and are more likely to trigger human complement activation or Fc receptor binding. One such chimeric human-mouse antibody has been used to treat patients with B-cell varieties of non-Hodgkin's lymphoma (see Clinical Focus).

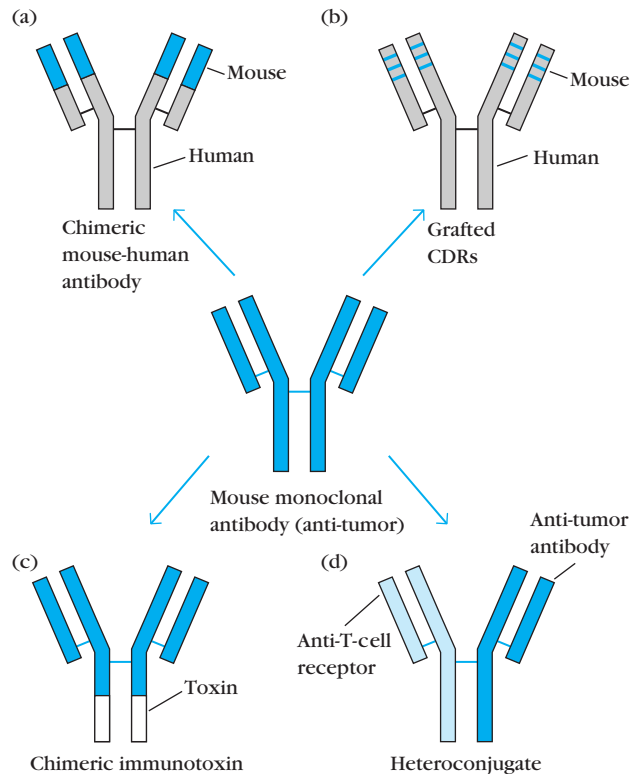


FIGURE 5-21 Chimeric and hybrid monoclonal antibodies engineered by recombinant DNA technology. (a) Chimeric mouse-human monoclonal antibody containing the V_H and V_L domains of a mouse monoclonal antibody (blue) and C_L and C_H domains of a human monoclonal antibody (gray). (b) A chimeric monoclonal antibody containing only the CDRs of a mouse monoclonal antibody (blue bands) grafted within the framework regions of a human monoclonal antibody is called a “humanized” monoclonal antibody. (c) A chimeric monoclonal antibody in which the terminal Fc domain is replaced by toxin chains (white). (d) A heteroconjugate in which one-half of the mouse antibody molecule is specific for a tumor antigen and the other half is specific for the CD3/T-cell receptor complex.

Chimeric monoclonal antibodies that function as immunotoxins (see Figure 4-23) can also be prepared. In this case, the terminal constant-region domain in a tumor-specific monoclonal antibody is replaced with toxin chains (Figure 5-21c). Because these immunotoxins lack the terminal Fc domain, they are not able to bind to cells bearing Fc receptors. These immunotoxins can bind only to tumor cells, making them highly specific as therapeutic reagents.

Heteroconjugates, or bispecific antibodies, are hybrids of two different antibody molecules (Figure 5-21d). They can be constructed by chemically crosslinking two different antibodies or by synthesizing them in hybridomas consisting of two different monoclonal-antibody-producing cell lines that have been fused. Both of these methods generate mixtures of monospecific and bispecific antibodies from which the desired bispecific molecule must be purified. Using genetic engineering to construct genes that will encode molecules only with the two desired specificities is a much simpler and more elegant approach. Several bispecific molecules have been designed in which one half of the antibody has specificity for a tumor and the other half has specificity for a surface molecule on an immune effector cell, such as an NK cell, an activated macrophage, or a cytotoxic T lymphocyte (CTL). Such heteroconjugates have been designed to activate the immune effector cell when it is crosslinked to the tumor cell so that it begins to mediate destruction of the tumor cell.

Monoclonal Antibodies Can Be Constructed from Ig-Gene Libraries

A quite different approach for generating monoclonal antibodies employs the polymerase chain reaction (PCR) to amplify the DNA that encodes antibody heavy-chain and light-chain Fab fragments from hybridoma cells or plasma cells. A promoter region and *EcoRI* restriction site (see Chapter 23) are added to the amplified sequences, and the resulting constructs are inserted into bacteriophage λ , yielding separate heavy- and light-chain libraries. Cleavage with *EcoRI* and random joining of the heavy- and light-chain genes yield numerous novel heavy-light constructs (Figure 5-22).

This procedure generates an enormous diversity of antibody specificities—libraries with $>10^{10}$ unique members have been obtained—and clones containing these random combinations of H + L chains can be rapidly screened for those secreting antibody to a particular antigen. The level of diversity is comparable to the human *in vivo* repertoire, and it is possible to demonstrate that specificities against a wide variety of antigens can be obtained from these libraries. Such a combinatorial library approach opens the possibility of obtaining specific antibodies without any need whatsoever for immunization.

However, the real challenge to bypassing *in vivo* immunization in the derivation of useful antibodies of high affinity lies in finding ways to mimic the biology of the humoral



CLINICAL FOCUS

Therapy for Non-Hodgkin's Lymphoma and Other Diseases by Genetically Engineered Antibodies

Lymphomas are cancers of lymphatic tissue in which the tumor cells are of lymphocytic origin. There are two major forms of lymphoma: Hodgkin's lymphoma and non-Hodgkin's lymphoma. The less common form is Hodgkin's lymphoma, named for its discoverer, Thomas Hodgkin, an English physician. This unusually gifted early pathologist, who worked without the benefit of a microscope, recognized this condition in several patients and first described the anatomical features of the disease in 1832. Because many tissue specimens taken from patients Hodgkin suspected of harboring the disease were saved in the Gordon Museum of Guy's Hospital in London, it has been possible for later generations to judge the accuracy of his diagnoses. Hodgkin has fared well. Studies of these preserved tissues confirm that he was right in about 60% of the cases, a surprising achievement, considering the technology of the time. Actually, most lymphoma is non-Hodgkin's type and includes about 10 different types of disease. B-cell lymphomas are an important fraction of these.

For some years now, the major therapies directed against lymphomas have been radiation, chemotherapy, or a combination of both. While these therapies benefit large numbers of patients by increasing survival, relapses after treatment are common, and many treated patients experience debilitating side effects. The side effects are an expected consequence of these therapies, because the agents used kill or severely damage a broad spectrum of normal cells as well as tumor cells. One of the holy grails of cancer treatment is the discovery of therapies

that will affect only the tumor cells and completely spare normal cells. If particular types of cancer cells had antigens that were tumor specific, these antigens would be ideal targets for immune attack. Unfortunately, there are few such molecules known. However, a number of antigens are known that are restricted to the cell lineage in which the tumor originated and are expressed on the tumor cells.

Many cell-lineage-specific antigens have been identified for B lymphocytes and B lymphomas, including immunoglobulin, the hallmark of the B cell, and CD20, a membrane-bound phosphoprotein. CD20 has emerged as an attractive candidate for antibody-mediated immunotherapy because it is present on B lymphomas, and antibody-mediated crosslinking does not cause it to down-regulate or internalize. Indeed, some years ago, mouse monoclonal antibodies were raised against CD20, and one of these has formed the basis for an anti-B-cell lymphoma immunotherapy. This approach appears ready to take its place as an adjunct or alternative to radiation and chemotherapy. The development of this anti-tumor antibody is an excellent case study of the combined application of immunological insights and molecular biology to engineer a novel therapeutic agent.

The original anti-CD20 antibody was a mouse monoclonal antibody with murine γ heavy chains and κ light chains. The DNA sequences of the light- and heavy-chain variable regions of this antibody were amplified by PCR. Then a chimeric gene was created by replacing the CDR gene sequences of a human $\gamma 1$ heavy chain with those from the murine heavy chain. In a similar maneuver, CDRs from the mouse κ were ligated into a human κ

gene. The chimeric genes thus created were incorporated into vectors that permitted high levels of expression in mammalian cells. When an appropriate cell line was co-transfected with both of these constructs, it produced chimeric antibodies containing CDRs of mouse origin together with human variable-region frameworks and constant regions. After purification, the biological activity of the antibody was evaluated, first in vitro and then in a primate animal model.

The initial results were quite promising. The grafted human constant region supported effector functions such as the complement-mediated lysis or antibody-dependent cell-mediated cytotoxicity (ADCC) of human B lymphoid cells. Furthermore, weekly injections of the antibody into monkeys resulted in the rapid and sustained depletion of B cells from peripheral blood, lymph nodes, and even bone marrow. When the anti-CD20 antibody infusions were stopped, the differentiation of new B cells from progenitor populations allowed B-cell populations eventually to recover and approach normal levels. From these results, the hope grew that this immunologically active chimeric antibody could be used to clear entire B cell populations, including B lymphoma cells, from the body in a way that spared other cell populations. This led to the trial of the antibody in human patients.

The human trials enrolled patients with B-cell lymphoma who had a relapse after chemotherapy or radiation treatment. These trials addressed three important issues: efficacy, safety, and immunogenicity. While not all patients responded to treatment with anti-CD20, close to 50% exhibited full or partial remission. Thus, efficacy was demonstrated, because this level of response is comparable to the success rate with traditional approaches that employ highly cytotoxic drugs or radiation—it offers a truly alternative therapy. Side effects such as nausea, low blood pressure, and shortness of breath were seen in some patients (usually during or shortly after the initiation of therapy); these were, for the most part, not serious or life-threatening. Consequently, treatment with the

chimeric anti-CD20 appears safe. Patients who received the antibody have been observed closely for the appearance of human anti-mouse-Ig antibodies (HAMA) and for human anti-chimeric antibody (HACA) responses. Such responses were not observed. Therefore, the antibody was not immunogenic. The absence of such responses demonstrate that antibodies can be genetically engineered to minimize, or even avoid, untoward immune reactions. Another reason for humanizing mouse antibodies arises from the very short half life (a few hours) of mouse IgG antibodies in humans compared with the three-week half lives of their human or humanized counterparts.

Antibody engineering has also contributed to the therapy of other malignancies such as breast cancer, which is diagnosed in more than 180,000 American women each year. A little more than a quarter of all breast cancer patients have

cancers that over-express a growth factor receptor called HER2 (human epidermal growth factor receptor 2). Many tumors that over-express HER2 grow faster and pose a more serious threat than those with normal levels of this protein on their surface. A chimeric anti-HER2 monoclonal antibody in which all of the protein except the CDRs are of human origin was created by genetic engineering. Specifically, the DNA sequences for the heavy-chain and light-chain CDRs were taken from cloned mouse genes encoding an anti-HER2 monoclonal antibody. As in the anti-CD20 strategy described above, each of the mouse CDR gene segments were used to replace the corresponding human CDR gene segments in human genes encoding the human IgG₁ heavy chain and the human κ light chain. When this engineered antibody is used in combination with a chemotherapeutic drug, it is highly effective against metastatic breast cancer. The

effects on patients who were given only a chemotherapeutic drug were compared with those for patients receiving both the chemotherapeutic drug and the engineered anti-HER2 antibody. The combination anti-HER2/chemotherapy treatment showed significantly reduced rates of tumor progression, a higher percentage of responding patients, and a higher one-year survival rate. Treatment with Herceptin, as this engineered monoclonal antibody is called, has become part of the standard repertoire of breast cancer therapies.

The development of engineered and conventional monoclonal antibodies is one of the most active areas in the pharmaceutical industry. The table provides a partial compilation of monoclonal antibodies that have received approval from the Food and Drug Administration (FDA) for use in the treatment of human disease. Many more are in various stages of development and testing.

Some monoclonal antibodies in clinical use

Monoclonal antibody [mAB] (Product Name)	Nature of antibody	Target (antibody specificity)	Treatment for
Muromonab-CD3 (Orthoclone OKT3)	Mouse mAB	T cells (CD3, a T cell antigen)	Acute rejection of liver, heart and kidney transplants
Abciximab (ReoPro)	Human-mouse chimeric	Clotting receptor of platelets (GP IIb/IIIa)	Blood clotting during angioplasty and other cardiac procedures
Daclizumab (Zenapax)	Humanized mAB	Activated T cells (IL-2 receptor alpha subunit)	Acute rejection of kidney transplants
Infliximab (Remicade)	Human-mouse chimeric	Tumor necrosis factor, (TNF) a mediator of inflammation. (TNF)	Rheumatoid arthritis and Crohn's disease
Palivizumab (Synagis)	Humanized mAB	Respiratory Syncytial Virus (RSV) (F protein, a component of RSV)	RSV infection in children, particularly infants
Gemtuzumab (Mylotarg)	Humanized mAB	Many cells of the myeloid lineage (CD33, an adhesion molecule)	Acute myeloid leukemia (AML)
Alemtuzumab (Campath)	Humanized mAB	Many types of leukocytes (CD52 a cell surface antigen)	B cell chronic lymphocytic leukemia
Trastuzumab (Herceptin)	Humanized mAB	An epidermal growth factor receptor (HER2 receptor)	HER2 receptor-positive advanced breast cancers
Rituximab (Rituxan)	Humanized mAB	B cells (CD20 a B cell surface antigen)	Relapsed or refractory non-Hodgkins lymphoma
Ibritumomab (Zevalin)	Mouse mAB	B cells (CD20, a B cell surface antigen)	Relapsed or refractory non-Hodgkins lymphoma

SOURCE: Adapted from P. Carter. 2001. Improving the efficacy of antibody-based cancer therapies. *Nature Reviews/Cancer* 1:118.

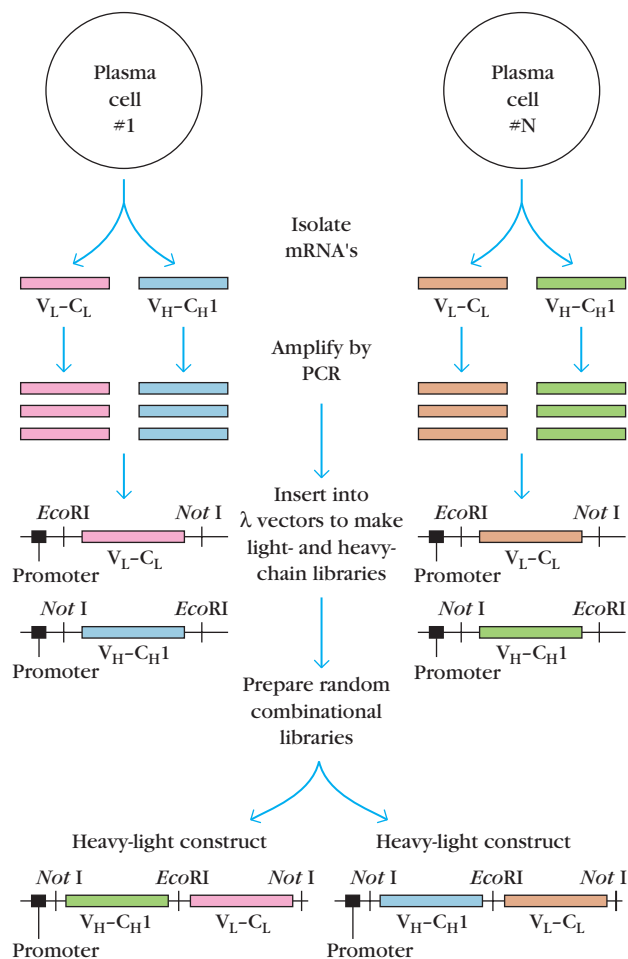


FIGURE 5-22 General procedure for producing gene libraries encoding Fab fragments. In this procedure, isolated mRNA that encodes heavy and light chains is amplified by the polymerase chain reaction (PCR) and cloned in λ vectors. Random combinations of heavy- and light-chain genes generate an enormous number of heavy-light constructs encoding Fab fragments with different antigenic specificity. [Adapted from W. D. Huse et al., 1989, *Science* **246**:1275.]

immune response. As we shall see in Chapter 11, the *in vivo* evolution of most humoral immune responses produces two desirable outcomes. One is class switching, in which a variety of antibody classes of the same specificity are produced. This is an important consideration because the class switching that occurs during an immune response produces antibodies that have the same specificity but different effector functions and hence, greater biological versatility. The other is the generation of antibodies of higher and higher affinity as the response progresses. A central goal of Ig-gene library approaches is the development of strategies to produce antibodies of appropriate affinity *in vitro* as readily as they are generated by an *in vivo* immune response. When the formidable technical obstacles to the achievement of these goals are overcome, combinatorial approaches based on phage

libraries will allow the routine and widespread production of useful antibodies from any desired species without the limitations of immunization and hybridoma technology that currently complicate the production of monoclonal antibodies.

Mice Have Been Engineered with Human Immunoglobulin Loci

It is possible to functionally knock out, or disable, the heavy- and light-chain immunoglobulin loci in mouse embryonic stem (ES) cells. N. Lonberg and his colleagues followed this procedure and then introduced large DNA sequences (as much as 80 kb) containing human heavy- and light-chain gene segments. The DNA sequences contained constant-region gene segments, J segments, many V-region segments, and, in the case of the heavy chain, D_H segments. The ES cells containing these miniature human Ig gene loci (miniloci) are used to derive lines of transgenic mice that respond to antigenic challenge by producing antigen-specific *human* antibodies (Figure 5-23). Because the human heavy- and light-chain miniloci undergo rearrangement and all the other diversity-generating processes, such as N-addition, P-addition, and even somatic hypermutation after antigenic challenge, there is an opportunity for the generation of a great deal of diversity in these mice. The presence of human heavy-chain minilocus genes for more than one isotype and their accompanying switch sites allows class switching as well. A strength of this method is that these completely human antibodies are made in cells of the mouse B-cell lineage, from which antibody-secreting hybridomas are readily derived by cell fusion. This approach thus offers a solution to the problem of producing human monoclonal antibodies of any specificity desired.

SUMMARY

- Immunoglobulin κ and λ light chains and heavy chains are encoded by three separate multigene families, each containing numerous gene segments and located on different chromosomes.
- Functional light-chain and heavy-chain genes are generated by random rearrangement of the variable-region gene segments in germ-line DNA.
- V(D)J joining is catalyzed by the recombinase activating genes, *RAG-1* and *RAG-2*, and the participation of other enzymes and proteins. The joining of segments is directed by recombination signal sequences (RSS), conserved DNA sequences that flank each V, D, and J gene segment.
- Each recombination signal sequence contains a conserved heptamer sequence, a conserved nonamer sequence, and either a 12-bp (one-turn) or 23-bp (two-turn) spacer. During rearrangement, gene segments flanked by a one-turn spacer join only to segments flanked by a two-turn spacer, assuring proper V_L-J_L and V_H-D_H-J_H joining.

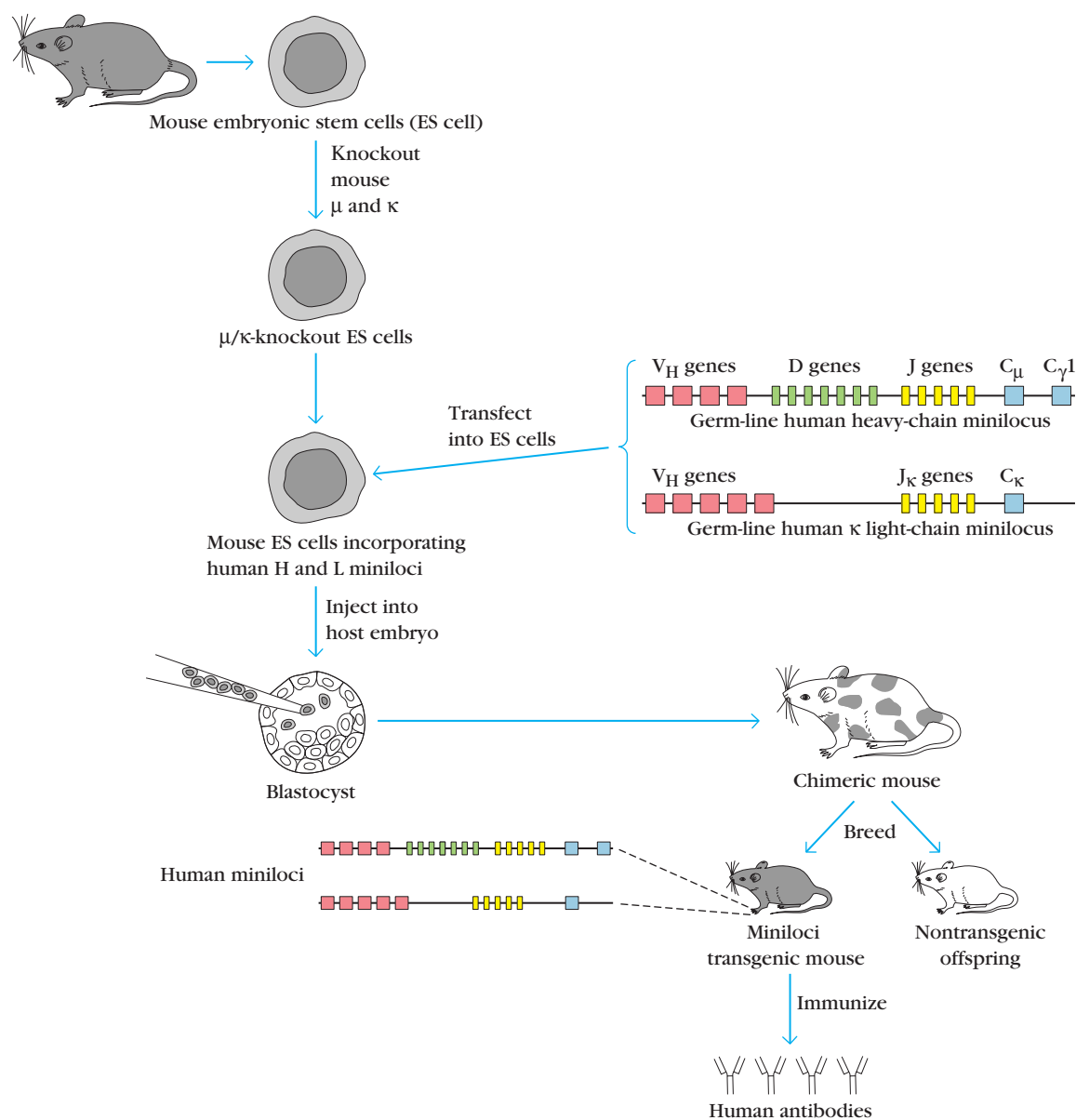


FIGURE 5-23 Grafting human heavy- and light-chain miniloci into mice. The capacity of mice to rearrange Ig heavy- and light-chain gene segments was disabled by knocking out the C_μ and C_κ loci. The antibody-producing capacity of these mice was reconstituted by introducing long stretches of DNA incorporating a large part of the human germ-line κ and heavy-chain loci (miniloci).

Chimeric mice were then bred to establish a line of transgenic mice bearing both heavy- and light-chain human miniloci. Immunization of these mice results in the production of human antibody specific for the target antigen. [N. Lonberg *et al.*, 1994, *Nature* **368**:856.]

- Immunoglobulin gene rearrangements occur in sequential order, heavy-chain rearrangements first, followed by light-chain rearrangements. Allelic exclusion is a consequence of the functional rearrangement of the immunoglobulin DNA of only one parental chromosome and is necessary to assure that a mature B cell expresses immunoglobulin with a single antigenic specificity.
- The major sources of antibody diversity, which can generate $>10^{10}$ possible antibody combining sites, are: random

joining of multiple V, J, and D germ-line gene segments; random association of heavy and light chains; junctional flexibility; P-addition; N-addition; and somatic mutation.

- After antigenic stimulation of mature B cells, class switching results in expression of different classes of antibody (IgG, IgA, and IgE) with the same antigenic specificity.
- Differential RNA processing of the immunoglobulin heavy-chain primary transcript generates membrane-bound antibody in mature B cells, secreted antibody in



plasma cells, and the simultaneous expression of IgM and IgD by mature B cells.

- Transcription of immunoglobulin genes is regulated by three types of DNA regulatory sequences: promoters, enhancers, and silencers.
- Growing knowledge of the molecular biology of immunoglobulin genes has made it possible to engineer antibodies for research and therapy. The approaches include chimeric antibodies, bacteriophage-based combinatorial libraries of Ig-genes, and the transplantation of extensive segments of human Ig loci into mice.

References

- Chen, J., Y. Shinkai, F. Young, and F. W. Alt. 1994. Probing immune functions in RAG-deficient mice. *Curr. Opin. Immunol.* **6**:313.
- Cook, G. P., and I. M. Tomlinson. 1995. The human immunoglobulin V_H repertoire. *Immunol. Today* **16**:237.
- Dreyer, W. J., and J. C. Bennett. 1965. The molecular basis of antibody formation. *Proc. Natl. Acad. Sci. U.S.A.* **54**:864.
- Fugmann, S. D., I. L. Lee, P. E. Shockett, I. J. Villey, and D. G. Schatz. 2000. The RAG proteins and V(D)J recombination: Complexes, ends and transposition. *Annu. Rev. Immunol.* **18**:495.
- Gavilondo, J. V., and J. W. Larrick. 2000. Antibody engineering at the millennium. *Biotechniques* **29**:128.
- Hayden, M. S., L. K. Gilliland, and J. A. Ledbetter. 1997. Antibody engineering. *Curr. Opin. Immunol.* **9**:201.
- Hesslein, D. G., and D. G. Schatz. 2001. Factors and forces controlling V(D)J recombination. *Adv. Immunol.* **78**:169.
- Hozumi, N., and S. Tonegawa. 1976. Evidence for somatic rearrangement of immunoglobulin genes coding for variable and constant regions. *Proc. Natl. Acad. Sci. U.S.A.* **73**:3628.
- Maloney, D. G., et al. 1997. IDEC-C2B8 (Rituximab) anti-CD20 monoclonal antibody therapy in patients with relapsed low-grade non-Hodgkin's lymphoma. *Blood* **90**:2188.
- Manis, J. P., M. Tian, and F. W. Alt. 2002. Mechanism and control of class-switch recombination. *Trends Immunol.* **23**:31.
- Matsuda, F., K. Ishii, P. Bourvagnet, Ki Kuma, H. Hayashida, T. Miyata, and T. Honjo. 1998. The complete nucleotide sequence of the human immunoglobulin heavy chain variable region locus. *J. Exp. Med.* **188**:2151.
- Max, E. E. 1998. Immunoglobulins: molecular genetics. In *Fundamental Immunology*, 4th ed., W. E. Paul, ed. Lippincott-Raven, Philadelphia.
- Mills, F. C., N. Harindranath, M. Mitchell, and E. E. Max. 1997. Enhancer complexes located downstream of both human immunoglobulin C alpha genes. *J. Exp. Med.* **186**:845.
- Oettinger, M. A., et al. 1990. RAG-1 and RAG-2, adjacent genes that synergistically activate V(D)J recombination. *Science* **248**:1517.
- Tonegawa, S. 1983. Somatic generation of antibody diversity. *Nature* **302**:575.
- Van Gent, D. C., et al. 1995. Initiation of V(D)J recombination in a cell-free system. *Cell* **81**:925.
- Winter, G., and C. Milstein. 1990. Man-made antibodies. *Nature* **349**:293.



USEFUL WEB SITES

<http://www.mrc-cpe.cam.ac.uk/imt-doc/public/INTRO.html#maps>

V BASE: This database and informational site is maintained at the MRC Centre for Protein Engineering in England. It is an excellent and comprehensive directory of information on the human germ-line variable region.

<http://www.mgen.uni-heidelberg.de/SD/SDscFvSite.html>

The Recombinant Antibody Page: This site has a number of links that provide interesting opportunities to explore the potential of genetic engineering of antibodies.

<http://www.ebi.ac.uk/imgt/hla/intro.html>

The IMGT site contains a collection of databases of genes relevant to the immune system. The IMGT/LIGM database houses sequences belonging to the immunoglobulin superfamily and of T cell antigen receptor sequences.

Study Questions

CLINICAL FOCUS QUESTION The Clinical Focus section includes a table of monoclonal antibodies approved for clinical use. Two, Rituxan and Zevalin, are used for the treatment of non-Hodgkins lymphoma. Both target CD20, a B-cell surface antigen. Zevalin is chemically modified by attachment of radioactive isotopes (yttrium-90, a β emitter or indium-111, a high energy γ emitter) that lethally irradiate cells to which the monoclonal antibody binds. Early experiments found that Zevalin without a radioactive isotope attached was an ineffective therapeutic agent, whereas unlabeled Rituxan, a humanized mAb, was effective. Furthermore, Rituxan with a radioactive isotope attached was too toxic; Zevalin bearing the same isotope in equivalent amounts was far less toxic. Explain these results. (Hint: The longer a radioactive isotope stays in the body, the greater the dose of radiation absorbed by the body.)

1. Indicate whether each of the following statements is true or false. If you think a statement is false, explain why.
 - a. V _{κ} gene segments sometimes join to C _{λ} gene segments.
 - b. With the exception of a switch to IgD, immunoglobulin class switching is mediated by DNA rearrangements.
 - c. Separate exons encode the transmembrane portion of each membrane immunoglobulin.
 - d. Although each B cell carries two alleles encoding the immunoglobulin heavy and light chains, only one allele is expressed.

- e. Primary transcripts are processed into functional mRNA by removal of introns, capping, and addition of a poly-A tail.
 - f. The primary transcript is an RNA complement of the coding strand of the DNA and includes both introns and exons.
2. Explain why a V_H segment cannot join directly with a J_H segment in heavy-chain gene rearrangement.
 3. Considering only combinatorial joining of gene segments and association of light and heavy chains, how many different antibody molecules potentially could be generated from germ-line DNA containing 500 V_L and 4 J_L gene segments and 300 V_H , 15 D_H , and 4 J_H gene segments?
 4. For each incomplete statement below (a–g), select the phrase(s) that correctly completes the statement. More than one choice may be correct.
 - a. Recombination of immunoglobulin gene segments serves to
 - (1) promote Ig diversification
 - (2) assemble a complete Ig coding sequence
 - (3) allow changes in coding information during B-cell maturation
 - (4) increase the affinity of immunoglobulin for antibody
 - (5) all of the above
 - b. Somatic mutation of immunoglobulin genes accounts for
 - (1) allelic exclusion
 - (2) class switching from IgM to IgG
 - (3) affinity maturation
 - (4) all of the above
 - (5) none of the above
 - c. The frequency of somatic mutation in Ig genes is greatest during
 - (1) differentiation of pre-B cells into mature B cells
 - (2) differentiation of pre-T cells into mature T cells
 - (3) generation of memory B cells
 - (4) antibody secretion by plasma cells
 - (5) none of the above
 - d. Kappa and lambda light-chain genes
 - (1) are located on the same chromosome
 - (2) associate with only one type of heavy chain
 - (3) can be expressed by the same B cell
 - (4) all of the above
 - (5) none of the above
 - e. Generation of combinatorial diversity among immunoglobulins involves
 - (1) mRNA splicing
 - (2) DNA rearrangement
 - (3) recombination signal sequences
 - (4) one-turn/two-turn joining rule
 - (5) switch sites
 - f. A B cell becomes immunocompetent
 - (1) following productive rearrangement of variable-region heavy-chain gene segments in germ-line DNA
 - (2) following productive rearrangement of variable-region heavy-chain and light-chain gene segments in germ-line DNA
 - (3) following class switching
 - (4) during affinity maturation
 - (5) following binding of T_H cytokines to their receptors on the B cell
 - g. The mechanism that permits immunoglobulins to be synthesized in either a membrane-bound or secreted form is
 - (1) allelic exclusion
 - (2) codominant expression
 - (3) class switching
 - (4) the one-turn/two-turn joining rule
 - (5) differential RNA processing
 5. What mechanisms generate the three hypervariable regions (complementarity-determining regions) of immunoglobulin heavy and light chains? Why is the third hypervariable region (CDR3) more variable than the other two (CDR1 and CDR2)?
 6. You have been given a cloned myeloma cell line that secretes IgG with the molecular formula $\gamma_2\lambda_2$. Both the heavy and light chains in this cell line are encoded by genes derived from allele 1. Indicate the form(s) in which each of the genes listed below would occur in this cell line using the following symbols: G = germ line form; R = productively rearranged form; NP = nonproductively rearranged form. State the reason for your choice in each case.
 - a. Heavy-chain allele 1 d. κ -chain allele 2
 - b. Heavy-chain allele 2 e. λ -chain allele 1
 - c. κ -chain allele 1 f. λ -chain allele 2
 7. You have a B-cell lymphoma that has made nonproductive rearrangements for both heavy-chain alleles. What is the arrangement of its light-chain DNA? Why?
 8. Indicate whether each of the class switches indicated below can occur (Yes) or cannot occur (No).
 - a. IgM to IgD d. IgA to IgG
 - b. IgM to IgA e. IgM to IgG
 - c. IgE to IgG
 9. Describe one advantage and one disadvantage of N-nucleotide addition during the rearrangement of immunoglobulin heavy-chain gene segments.
 10. X-ray crystallographic analyses of many antibody molecules bound to their respective antigens have revealed that the CDR3 of both the heavy and light chains make contact with the epitope. Moreover, sequence analyses reveal that the variability of CDR3 is greater than that of either CDR1 or CDR2. What mechanisms account for the greater diversity in CDR3?
 11. How many chances does a developing B cell have to generate a functional immunoglobulin light-chain gene?
 12. Match the terms below (a–h) to the description(s) that follow (1–11). Each description may be used once, more than once, or not at all; more than one description may apply to some terms.

Terms

- | | |
|--|------------------------|
| a. _____ RAG-1 and RAG-2 | e. _____ P-nucleotides |
| b. _____ Double-strand break repair (DSBR) enzymes | f. _____ N-nucleotides |
| c. _____ Coding joints | g. _____ Promoters |
| d. _____ RSSs | h. _____ Enhancers |

Descriptions

- (1) Junctions between immunoglobulin gene segments formed during rearrangement
- (2) Source of diversity in antibody heavy chains
- (3) DNA regulatory sequences
- (4) Conserved DNA sequences, located adjacent to V, D, and J segments, that help direct gene rearrangement
- (5) Enzymes expressed in developing B cells
- (6) Enzymes expressed in mature B cells
- (7) Nucleotide sequences located close to each leader segment in immunoglobulin genes to which RNA polymerase binds
- (8) Product of endonuclease cleavage of hairpin intermediates in Ig-gene rearrangement
- (9) Enzymes that are defective in SCID mice
- (10) Nucleotide sequences that greatly increase the rate of transcription of rearranged immunoglobulin genes compared with germ-line DNA
- (11) Nucleotides added by TdT enzyme

- 13.** Many B-cell lymphomas express surface immunoglobulin on their plasma membranes. It is possible to isolate this lymphoma antibody and make a high affinity, highly specific mouse monoclonal anti-idiotypic antibody against it. What steps should be taken to make this mouse monoclonal antibody most suitable for use in the patient. Is it highly likely that, once made, such an engineered antibody will be generally useful for lymphoma patients?

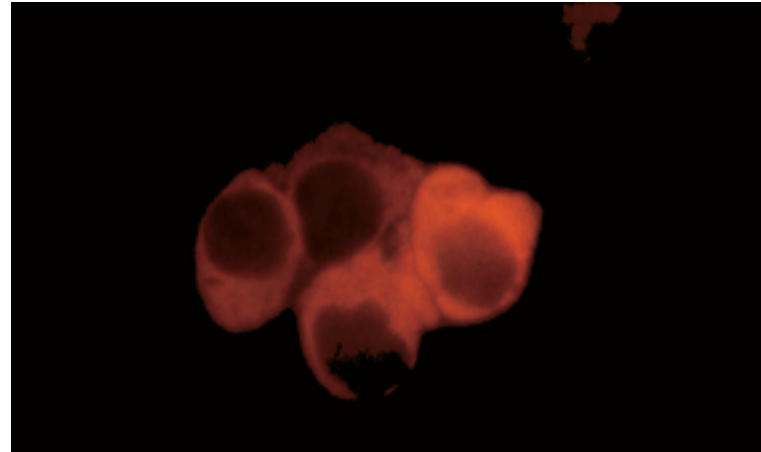
chapter 6

Antigen-Antibody Interactions: Principles and Applications

THE ANTIGEN-ANTIBODY INTERACTION IS A BIMOLECULAR association similar to an enzyme-substrate interaction, with an important distinction: it does not lead to an irreversible chemical alteration in either the antibody or the antigen. The association between an antibody and an antigen involves various noncovalent interactions between the antigenic determinant, or epitope, of the antigen and the variable-region (V_H/V_L) domain of the antibody molecule, particularly the hypervariable regions, or complementarity-determining regions (CDRs). The exquisite specificity of antigen-antibody interactions has led to the development of a variety of immunologic assays, which can be used to detect the presence of either antibody or antigen. Immunoassays have played vital roles in diagnosing diseases, monitoring the level of the humoral immune response, and identifying molecules of biological or medical interest. These assays differ in their speed and sensitivity; some are strictly qualitative, others are quantitative. This chapter examines the nature of the antigen-antibody interaction, and it describes various immunologic assays that measure or exploit this interaction.

Strength of Antigen-Antibody Interactions

The noncovalent interactions that form the basis of antigen-antibody (Ag-Ab) binding include hydrogen bonds, ionic bonds, hydrophobic interactions, and van der Waals interactions (Figure 6-1). Because these interactions are individually weak (compared with a covalent bond), a large number of such interactions are required to form a strong Ag-Ab interaction. Furthermore, each of these noncovalent interactions operates over a very short distance, generally about 1×10^{-7} mm (1 angstrom, Å); consequently, a strong Ag-Ab interaction depends on a very close fit between the antigen and antibody. Such fits require a high degree of complementarity between antigen and antibody, a requirement that underlies the exquisite specificity that characterizes antigen-antibody interactions.

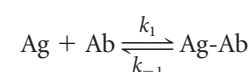


Fluorescent Antibody Staining Reveals Intracellular Immunoglobulin

- Strength of Antigen-Antibody Interactions
- Cross-Reactivity
- Precipitation Reactions
- Agglutination Reactions
- Radioimmunoassay
- Enzyme-Linked Immunosorbent Assay
- Western Blotting
- Immunoprecipitation
- Immunofluorescence
- Flow Cytometry and Fluorescence
- Alternatives to Antigen-Antibody Reactions
- Immunoelectron Microscopy

Antibody Affinity Is a Quantitative Measure of Binding Strength

The combined strength of the noncovalent interactions between a *single* antigen-binding site on an antibody and a *single* epitope is the **affinity** of the antibody for that epitope. Low-affinity antibodies bind antigen weakly and tend to dissociate readily, whereas high-affinity antibodies bind antigen more tightly and remain bound longer. The association between a binding site on an antibody (Ab) with a monovalent antigen (Ag) can be described by the equation





VISUALIZING CONCEPTS

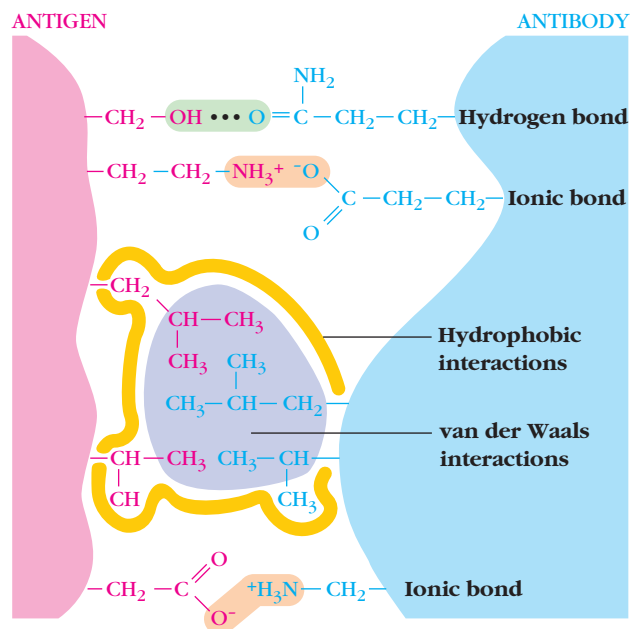


FIGURE 6-1 The interaction between an antibody and an antigen depends on four types of noncovalent forces: (1) hydrogen bonds, in which a hydrogen atom is shared between two electronegative atoms; (2) ionic bonds between oppositely charged residues; (3) hydrophobic interactions, in which water forces hy-

drophobic groups together; and (4) van der Waals interactions between the outer electron clouds of two or more atoms. In an aqueous environment, noncovalent interactions are extremely weak and depend upon close complementarity of the shapes of antibody and antigen.

where k_1 is the forward (association) rate constant and k_{-1} is the reverse (dissociation) rate constant. The ratio k_1/k_{-1} is the association constant K_a (i.e., $k_1/k_{-1} = K_a$), a measure of affinity. Because K_a is the equilibrium constant for the above reaction, it can be calculated from the ratio of the molar concentration of bound Ag-Ab complex to the molar concentrations of unbound antigen and antibody at equilibrium as follows:

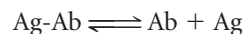
$$K_a = \frac{[\text{Ag-Ab}]}{[\text{Ab}][\text{Ag}]}$$

The value of K_a varies for different Ag-Ab complexes and depends upon both k_1 , which is expressed in units of liters/mole/second (L/mol/s), and k_{-1} , which is expressed in units of 1/second. For small haptens, the forward rate constant can be extremely high; in some cases, k_1 can be as high as 4×10^8 L/mol/s, approaching the theoretical upper limit of diffusion-limited reactions (10^9 L/mol/s). For larger protein antigens, however, k_1 is smaller, with values in the range of 10^5 L/mol/s.

The rate at which bound antigen leaves an antibody's binding site (i.e., the dissociation rate constant, k_{-1}) plays a major role in determining the antibody's affinity for an antigen. Table 6-1 illustrates the role of k_{-1} in determining

the association constant K_a for several Ag-Ab interactions. For example, the k_1 for the DNP-L-lysine system is about one fifth that for the fluorescein system, but its k_{-1} is 200 times greater; consequently, the affinity of the anti-fluorescein antibody K_a for the fluorescein system is about 1000-fold higher than that of anti-DNP antibody. Low-affinity Ag-Ab complexes have K_a values between 10^4 and 10^5 L/mol; high-affinity complexes can have K_a values as high as 10^{11} L/mol.

For some purposes, the dissociation of the antigen-antibody complex is of interest:



The equilibrium constant for that reaction is K_d , the reciprocal of K_a

$$K_d = [\text{Ab}][\text{Ag}]/[\text{Ab-Ag}] = 1/K_a$$

and is a quantitative indicator of the stability of an Ag-Ab complex; very stable complexes have very low values of K_d , and less stable ones have higher values.

The affinity constant, K_a , can be determined by **equilibrium dialysis** or by various newer methods. Because equilibrium dialysis remains for many the standard against which